

A Conditional Sixth-Order Lie-Group FullVA Integrator for Lower-Pair Mechanisms

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Abstract

Lower-pair mechanisms require Lie-group integrators that synchronize position, velocity, and acceleration constraints. This paper proposes Gauss6/FullVA, a three-stage Gauss collocation Lie-group integrator. The contribution is a monolithic FullVA stage object: stage kinematics, Newton–Euler balance, and lower-pair position-, velocity-, and acceleration-level constraints are coupled in a square nonlinear system with Lie-group endpoint reconstruction. The main theorem proves a conditional sixth-order same-initial-state reported-grid estimate for the selected smooth branch by transferring the classical three-stage Gauss local defect through the implemented FullVA stage root, endpoint-closure perturbation, and inexact-Newton perturbation. The retained interfaces are compact-tube regularity, endpoint/stability, implementation binding, and branch-selected solver scale $\eta_h^{\text{tube}} \leq c_\eta h^7$. The theorem’s residual-value input is discharged by a direct $96 + 36$ bridge on the lifted Gauss stage: the 96 non-dynamic FullVA rows have an $O(h^7)$ row certificate and the 36 Newton–Euler rows vanish by direct substitution. This is neither an empirical residual fit nor primitive/Taylor closure; the latter remains non-active, with five open lift/reduction obligations and no accepted 162-subterm bound. On a smooth cylindrical-chain benchmark, the implementation records position/velocity orders 7.161/7.066 as branch-consistency diagnostics. Four ASME-style mechanisms exercise the same residual vocabulary: single- and double-pendulum rows provide dynamic-order evidence, while four-link and slider-crank rows provide closed-loop constraint/reaction consistency and coarse-window mechanism diagnostics. Fixed production

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tolerances remain finite-run diagnostics; residual-to-error promotion, full external same-test reproduction, and complete source-paper TFE residual replacement are not claimed. The comparison to TFE is kept at the formal-order and diagnostic levels.

Keywords: Lie group integrator, index-3 DAE, multibody dynamics, lower-pair mechanisms, Gauss collocation, temporal finite elements

1. Introduction

Implicit integration of constrained multibody systems must keep position, velocity, and acceleration constraints synchronized while respecting the geometry of finite rotations. This is especially important for absolute- or lower-pair formulations, where a step update that looks accurate in position can still carry endpoint velocity drift or inconsistent reaction dynamics. Recent time finite element methods on Lie groups have shown that high-order integration of frictional index-3 differential-algebraic equations is feasible and practically valuable [1]. The present work asks a more specific numerical-methods question: can a Gauss-collocation Lie-group FullVA residual be organized so that lower-pair position, velocity, and acceleration constraints are enforced monolithically while retaining a theorem-level conditional sixth-order claim on the retained smooth branch?

The resulting method is Gauss6/FullVA, a Gauss collocation FullVA integrator whose sixth-order claim is tied to the branch smoothness, residual-defect, endpoint/stability, implementation-binding, and nonlinear-solver contracts stated below. Its novelty is not the existence of Gauss collocation itself; it is the monolithic Lie-group FullVA residual organization for lower-pair mechanisms, where Gauss stage kinematics, $\text{SO}(3)$ retraction variables, Newton–Euler dynamics, and position/velocity/acceleration lower-pair rows are solved in one square nonlinear system. Temporal finite elements enter this paper as a formal-order comparator only after the method and theorem have been fixed. Their role is to mark the local $m = 3$ order-five formula target and the diagnostic boundary for separate same-test reproduction, not to supply an implementation superiority claim, a full-TFE replacement, or a theorem premise. Accordingly, the central mathematical object is the branch-selected Gauss6/FullVA residual together with its retained-interface order theorem. The comparison and reproducibility material states the formal-order, external same-test, and source-package scope without modifying the theorem.

The proof organization follows the Lie-group constrained-BDF literature only as an obligation ordering: fix the discrete object, insert the smooth branch, prove the local defect, and then propagate it globally; all constants and residual inputs used below are supplied by the FullVA lemmas and certificates in this manuscript. In particular, the Wieloch–Arnold BLieDF proof is invoked as a proof-order template, not as an estimate source: its BDF order theorem, hidden-constraint and multiplier estimates, and Lie-algebra recurrence constants are not imported into the present Gauss/FullVA proof.

The contributions are as follows.

1. a monolithic square 132-row Lie-group FullVA stage residual for lower-pair mechanisms;
2. a complete one-step algorithm and conditional sixth-order same-initial-state reported-grid theorem obtained by local-defect-to-global-error transfer on the compact-tube Gauss-predictor branch under retained endpoint/stability, implementation-binding, and solver-scale interfaces, with finite-window smooth slopes 7.161/7.066 reported only as branch-consistency diagnostics rather than solver-scale proof discharge;
3. method-side dynamic-order evidence for the single- and double-pendulum examples, and mechanism-coverage evidence for the four-link and slider-crank examples under the same FullVA vocabulary;
4. a scope boundary that limits TFE material to the local order-five $m = 3$ formula target and keeps implemented source-paper comparisons diagnostic until the same-test promotion criteria are met.

A separate reproducibility appendix records external reproduction, full-TFE, primitive-route, and source-package boundaries. It is included to delimit non-claims and reproduce the accepted rows, not as an additional method contribution or theorem input.

Table 1 fixes the hierarchy used throughout the paper. The proof contribution is the analytic residual-to-local-defect transfer for the implemented Gauss6/FullVA map. Numerical rows are downstream evidence for the accepted path. External same-test reproduction, source-package, and full-TFE replacement records are boundary evidence; they keep stronger external claims out of the manuscript and are not premises of the theorem. The load-bearing proof is the direct local-defect-to-global-error argument for that same-initial-state reported-grid theorem.

Table 1: Claim hierarchy used in the manuscript.

Evidence layer	Role in the paper	Claim role
Gauss6/FullVA residual and theorem	Defines the 132-row Lie-group FullVA stage solve and the conditional sixth-order same-initial-state reported-grid proof obtained from compact-branch local-defect-to-global-error transfer.	Main contribution.
Smooth cylindrical-chain and pendulum rows	Check the accepted smooth path and provide method-side dynamic-order evidence for the examples whose trajectory-order policy is accepted.	Primary numerical evidence.
Four-link and slider-crank rows	Exercise closed-loop lower-pair constraints, reaction consistency, and local coarse dynamics under the same residual vocabulary.	Mechanism coverage, not standalone asymptotic order proof.
TFE and common-reference comparisons	Place the method against the local $m = 3$ order-five formula target and record bounded diagnostic comparisons.	Formal-order and diagnostic comparison only.
External reproduction, source-package, and auxiliary primitive-route records	Record why external same-test superiority, full-TFE replacement, primitive-route closure, and residual-to-error promotion are outside the accepted theorem and numerical claim.	Claim-scope and reproducibility boundary, not a method contribution or theorem input.

Throughout the paper, “conditional order” has this narrow meaning: a direct 132-row residual-defect and local-to-global consistency transfer under the stated compact-tube, endpoint-closure, and branch-solver assumptions. It is an analytic perturbation statement, not an empirical order fit: the smooth trajectory slopes are reported only after the theorem-level estimates have been stated. It is not a residual-to-error transfer theorem for mechanism-coverage rows, not a proof that fixed production tolerances are asymptotically sufficient, and not a full source-paper same-test TFE repro-

duction. The P7 residual-to-error boundary is therefore a downstream output nonclaim, not one of the retained branch-domain hypotheses used to prove the smooth-branch theorem. The one-step map is defined only for transitions that remain on the selected Gauss-predictor branch and satisfy the compact-tube, endpoint-closure, and scaled Newton residual conditions in the theorem. Transitions outside that domain are not assigned a theorem-level order by residual tables, fixed tolerances, or external comparison rows. The scaled Newton condition is a proof-norm residual condition imposed before endpoint or trajectory errors are evaluated; it is not an assumed local-defect or grid-error conclusion.

Proof structure. The proof starts from the classical Gauss defect, passes through the implemented 132-row residual bridge with the 96-row $O(h^7)$ non-dynamic certificate plus the direct Newton–Euler substitution certificate, under which the 36 Newton–Euler rows vanish, adds the endpoint-closure and inexact-Newton perturbations, and then applies local-to-global transfer and the reporting map.

Section 7 formalizes this handoff as a four-term local-defect estimate on the selected branch: Gauss defect, residual-bridge perturbation, endpoint-closure perturbation, and inexact-Newton perturbation. The reported order statement then uses a fifth step—the first-exit/discrete-Gronwall transfer followed by the reporting map—rather than introducing another local-defect source. The Newton–Euler certificate discharges the 36 dynamic rows, while the 96-row non-dynamic certificate supplies the row-local residual input under the fixed branch, norm, and implementation binding. Residual-only mechanism rows, external comparison records, and primitive/Taylor fragments are recorded separately because they are not terms in this estimate. The theorem-facing tables below are reading maps for this lemma chain: they identify hypotheses, row-local certificates, and forbidden promotions, but the estimates themselves are supplied only by the stated lemmas and equations. In particular, residual tables, solver logs, external reproduction rows, and auxiliary primitive-route fragments are not theorem inputs unless the corresponding theorem interface is explicitly closed. The proof below separates domain hypotheses from proved algebraic residual inputs. The domain side fixes the compact branch, local inverse, endpoint map, stability scale, implementation binding, and solver-scale envelope with h -independent constants. The residual side identifies the exact implemented 132-row residual used by Newton, evaluates the lifted Gauss stages in that residual, combines the 96-row non-dynamic certificate with the 36-row direct Newton–Euler sub-

stitution, and then passes through the Kantorovich perturbation step. Once those objects are fixed, the proof derives the h^7 local defect and the h^6 same-initial-state reported-grid estimate forward from the Gauss defect and the implemented residual bridge. Numerical evidence and appendix records do not substitute for this local-defect argument.

Reference proof usage. The non-TFE proof-style reference is Wieloch–Arnold [8], not the TFE comparison paper. Its role is limited to proof ordering for Lie-group constrained DAEs: define the discrete Lie-group object, insert the smooth exact branch into that object, prove the local truncation/defect estimate, and only then write the global error recursion and reported-output consequence. The present proof instantiates that order with a different object, namely the row-scaled 132-row Gauss/FullVA residual $F_{A,h}$, the accepted Gauss-predictor branch, the endpoint-output map, and the compact-tube solver envelope. Thus the active Taylor object is the full scaled residual map used by Lemma 39, not the separate 162 primitive Newton–Euler subterm inventory. No BLieDF constant, BDF $p = k$ estimate, hidden-constraint/multiplier estimate, or primitive FullVA subterm bound is transferred into the theorem. The reference paper’s technical neighborhood and start-value assumptions are not suppressed in this translation: their role is played here by the retained compact-tube/branch domain, endpoint and solver-scale interfaces, and the same-reduced-initial-state clause used in the local-to-global transfer. These domain clauses are fixed before diagnostics are read; finite residual tables, smooth h -sweeps, or source-policy rows do not verify them retroactively.

Table 2 is the introductory reader-facing map only. The proof-level correspondence is Table 11, where the same ordering discipline is tied to the FullVA lemmas and the explicit non-import rules are restated at theorem-interface level.

How to read the proof layers. The proof has one theorem route. The direct Gauss/FullVA residual bridge assembles the 96 non-dynamic rows and the 36 direct Newton–Euler rows into the 132-row stage-residual input used by Theorem 1. The primitive/Taylor material asks a stronger term-by-term question for the same Newton–Euler rows using primitive state, acceleration, and multiplier lifts; it is an alternative sufficient route, not a premise of the theorem. Residual and external-comparison rows are treated separately because they would require their own trajectory-error or source-policy promotion arguments. Thus the proof strength rests on the direct theorem route, while the auxiliary material delimits what is deliberately not being

Table 2: How the non-TFE reference proof is invoked.

Reference-proof step	FullVA instantiation	Explicit non-transfer
Fix the discrete Lie-group object before any estimate.	Fix the row-scaled 132-row residual $F_{A,h}$, the accepted Gauss-predictor branch, row order, proof norm, and endpoint-output map.	No BLieDF equation or source-paper weak row is imported.
Insert the smooth exact branch into the fixed object.	Insert the smooth Gauss lift Z_G into the implemented FullVA residual and evaluate the certified $96 + 36$ row split on that same branch.	No hidden-constraint, multiplier-recursion, or primitive lift estimate is borrowed.
Prove the local defect before global propagation.	Combine the Gauss defect, the direct 132-row residual bridge, endpoint closure, and inexact-Newton perturbation into the four-term local-defect bound.	No finite h -sweep, residual table, or separate primitive/Taylor route closes a separate theorem interface.
Only then derive the global reported error.	Apply the same-branch local-to-global transfer and reporting-map estimate to the accepted Gauss6/FullVA map.	No residual-to-error, external-superiority, source-policy, or full-TFE claim follows without its separate theorem or source-policy criterion.

claimed.

The route is a one-way conditional implication, not a backfitted interpretation of numerical evidence. The compact-tube hypotheses choose the branch and constants before any reported run is inspected. On that fixed object, the stage-row defect on Z_G supplies the residual input, endpoint closure and inexact Newton add controlled perturbations, and the local-to-global transfer then gives the same-initial-state reported-grid estimate. Finite step sweeps may identify the reported branch, but they do not discharge the theorem-level solver hypothesis and do not promote residual-only mechanism or external comparison rows. The retained interfaces delimit where the theorem may be invoked.

The remainder of the paper follows the evidence chain in that order. Section 5 defines the accepted one-step map and the complete execution algorithm. Section 6 fixes the validation policy, including the rule that external comparisons remain diagnostic unless the same-test promotion criteria are satisfied. Section Appendix A records the detailed same-test promotion boundary. Section 7 gives the conditional sixth-order theorem and separates stage-residual inputs from theorem-domain interfaces. Sections 8 and 9 report the numerical evidence. The final diagnostic and reproducibility sections document what has been checked without turning those checks into broader claims.

2. Related work

The method sits at the intersection of geometric integration, constrained multibody dynamics, collocation methods for differential-algebraic equations (DAEs), variational integration, nonsmooth contact dynamics, and time finite elements. The relevant literature separates into six clusters, each of which leaves a different gap.

First, Lie-group integrators preserve the rotation manifold by updating in a local tangent chart rather than advancing redundant orientation coordinates and then repairing them. Runge–Kutta–Munthe-Kaas methods provide a general high-order construction on manifolds [5, 6]. Multibody-specific work has adapted this geometric viewpoint to constrained mechanics through Lie-group generalized-alpha schemes [7], BDF-type schemes on Lie groups [8], and quaternion-based singularity-free rotational integration [9, 10]. These methods establish how to evolve rotations geometrically, but they do not by themselves prescribe a stage residual that simultaneously enforces lower-pair position, velocity, and acceleration consistency in one Newton solve.

Second, constrained multibody systems in absolute or redundant coordinates are index-3 DAEs. Their numerical order is affected not only by the nominal order of the time integrator, but also by how the algebraic constraints and their differentiated forms are enforced [3]. The absolute-coordinate Lie-group formulations of Kissel, Taves, Fang, Zhang, and Negrut clarify the connection between small body-frame rotation increments and the first-order sensitivities needed in Newton iterations [11, 12, 13, 14, 15]. That line of work is important for constructing matrix rows for constraints and reaction forces, and it also supplies a public-code benchmark suite covering the single pendulum, double pendulum, slider-crank, four-link, and scaling tests. The present paper uses the same body-frame perturbation viewpoint to build a stage system in which the constraint row, velocity row, and acceleration row are all part of the accepted residual; it cannot be read as a fair external comparison until those published benchmark definitions are matched exactly.

Third, high-order collocation and index-reduction methods are a standard route to high-order DAE integration when the constrained flow is regular and the algebraic variables are solved consistently [4, 3]. Symplectic partitioned Runge–Kutta methods also show how holonomic constraints can be preserved within a structure-preserving Runge–Kutta framework [16]. This places the present construction near constrained collocation, but with a different emphasis on the lower-pair residual rows that are exposed to Newton’s method.

The three-stage Gauss path therefore provides a natural sixth-order reference construction on the smooth reduced problem. What is less standard in a lower-pair multibody setting is the concrete residual organization: the accepted path here couples Gauss stage kinematics, Newton–Euler balance, Lie retraction variables, and FullVA lower-pair rows into a square nonlinear system. This is the level at which the manuscript’s contribution is intended to sit; it is not a claim that Gauss collocation itself is new.

Fourth, variational and symplectic integrators derive the discrete equations from a discrete action principle and are valued for their momentum, symplectic, and long-time energy behavior [17, 18, 19]. Constrained variational integrators are therefore a close conceptual neighbor: they treat holonomic constraints at the discrete level and can reduce the constrained solve through null-space coordinates. Their objective, however, is not the same as the FullVA residual used here. The present method does not construct a discrete variational principle; it builds an absolute-coordinate stage residual that enforces lower-pair position, velocity, and acceleration consistency simultaneously and then evaluates the resulting one-step map under the validation policy below.

Fifth, nonsmooth contact and friction solvers form a separate branch of multibody dynamics. Time-stepping and complementarity formulations handle unilateral contact, impact, and Coulomb friction without resolving every event explicitly [20, 21, 22]. Those methods are essential for large contact-rich systems, but their central problem is a contact complementarity or cone-complementarity solve. The validation problems studied here are bilateral lower-pair mechanisms with smooth driving and reaction reconstruction; the paper therefore does not claim a new nonsmooth contact integrator or a replacement for complementarity-based frictional dynamics.

Sixth, time finite elements provide a different route to high-order direct integration. Early space-time and time finite-element methods were developed for structural dynamics and constrained rigid-body dynamics [23, 24, 25]. Later direct integration families based on time finite elements and weighted residuals showed high-order behavior and controllable numerical damping [2, 26, 27]. The frictional Lie-group time finite-element work of Chaturvedi, Sandu, and Sandu extends this direction to index-3 multibody DAEs with nonlinear friction and provides the local paper-style temporal finite-element comparator used here [1]. Its Gauss–Lobatto family uses the order convention $2m - 1$, so its $m = 3$ member is an order-five formula target, although DAE order reduction can still appear in the reported pendulum experiments.

The present manuscript is intentionally different from a full source-paper reimplementation. It reports method-side evidence for the accepted Gauss collocation FullVA residual on lower-pair mechanisms and compares its conditional sixth-order path against the local order-five temporal finite-element formula target. Complete substitution of all source-paper temporal finite-element weak rows remains outside the accepted claim. The gap addressed here is narrower but concrete: an implemented Lie-group lower-pair stage residual that carries position, velocity, and acceleration consistency through the same nonlinear solve and is checked on the cylindrical-chain benchmark plus four ASME-style mechanisms, with dynamic-order, mechanism-coverage, and diagnostic evidence kept separate.

3. Mathematical setting and definitions

Consider a constrained multibody system with generalized position q , velocity v , acceleration a , and Lagrange multipliers λ . The absolute-coordinate index-3 form may be written abstractly as

$$M(q)\dot{v} + G(q, v) + \Phi_q(q)^T \lambda = F(q, v, t), \quad (1)$$

$$\dot{q} = \mathcal{K}(q)v, \quad (2)$$

$$\Phi(q) = 0. \quad (3)$$

Equations (1), (2), and (3) fix the abstract constrained dynamics, kinematics, and holonomic constraint notation used throughout the paper. Here $\mathcal{K}$ includes the local Lie-group kinematic map for finite rotations. For orientations, the step update is represented by a retraction

$$Q_{n+1} = Q_n \text{Exp}(\eta_{n+1}), \quad (4)$$

with η_{n+1} in the local rotation algebra. The implementation uses this local representation to keep finite rotations on the group while solving a Newton system in vector coordinates.

Definition 1 (FullVA lower-pair residual). *In this manuscript, FullVA denotes full position-, velocity-, and acceleration-level lower-pair enforcement. The FullVA residual is the square nonlinear stage system that enforces lower-pair position constraints, their velocity consistency equations, and their acceleration consistency equations in the same Newton solve, together with the collocation dynamics and the Lie-group update equations.*

Definition 2 (Accepted method path). *The accepted method path is the Gauss6/FullVA one-step map generated by the 132-row FullVA residual in the implemented residual path. This is the only method path used for the accepted dynamic-order rows and four-example mechanism-coverage validation claims in this manuscript.*

Definition 3 (Full-TFE stage replacement). *In this study, full-TFE stage replacement means replacing all 132 Gauss/FullVA stage rows inside Newton by paper-derived temporal finite-element weak rows, without endpoint source data, terminal-row replacement, output projection, or target-direction oracle information. This is a stronger source-paper reproduction criterion and is not accepted by the evidence reported here.*

4. Nomenclature

5. Accepted one-step integrator

Let $x_n = (q_n, v_n, a_n)$ denote the endpoint position, velocity, and acceleration variables. The accepted method defines a one-step map

$$x_{n+1} = \Psi_h^{\text{G6FVA}}(x_n), \quad (5)$$

Equation (5) is the method-level map induced by one accepted FullVA stage solve, where the stage variables Z solve the nonlinear residual

$$R_{\text{G6FVA}}(q_n, v_n, a_n, Z; h) = 0. \quad (6)$$

For the cylindrical-chain benchmark used for the main convergence evidence, the residual has two bodies, two lower-pair joints, and three Gauss stages. Each body stage stores translational position, local rotation increment, translational velocity, angular velocity, translational acceleration, and angular acceleration. This gives 18 body unknowns per body per stage. Each lower-pair joint contributes four multiplier unknowns per stage. Thus the Newton stage vector has

$$3\{2(18) + 2(4)\} = 132 \quad (7)$$

scalar unknowns by Eq. (7), and the accepted residual is square. The closed-loop four-link and slider-crank examples use their own fully constrained kinematic FullVA systems, but they preserve the same position, velocity, and acceleration-level lower-pair residual vocabulary.

Table 3: Main symbols used in the manuscript.

Symbol	Meaning
q, v, a	Generalized position, velocity, and acceleration variables.
Q	Rotation variable on the Lie group.
η	Local tangent-space rotation increment used in the retraction.
λ	Lagrange multipliers for the lower-pair constraints.
$M(q)$	Configuration-dependent mass and inertia operator.
$F(q, v, t)$	Applied and generalized force vector.
$\Phi(q), \Phi_q(q)$	Constraint function and constraint Jacobian.
$\mathcal{K}(q)$	Kinematic map from generalized velocity to configuration rate.
h	Time-step size.
Z	Stacked stage unknown vector in the nonlinear one-step solve.
R_{G6FVA}	Accepted Gauss6/FullVA residual solved inside Newton.
m	Polynomial index for the paper-style Gauss–Lobatto TFE comparator.
p_{TFE}	Expected order of the local paper-style temporal finite-element target.

The residual is assembled from six row families. For stage $i = 1, 2, 3$ with Gauss abscissa c_i , weights b_i , and Runge–Kutta matrix A_{ij} , the stage variables are denoted in Eq. (8):

$$Z_i = (r_i, \eta_i, v_i, \omega_i, a_i, \alpha_i, \lambda_i), \quad (8)$$

where r_i is the translational position, η_i is the local Lie-algebra rotation increment, v_i, ω_i are translational and angular velocities, a_i, α_i are translational and angular accelerations, and λ_i collects lower-pair multipliers. The implementation evaluates rotations by $Q_i = Q_n \text{Exp}(\eta_i)$ and uses the right-trivialized differential of the exponential map in the rotational kinematic rows.

The complete one-step equation can be read as the following square stage system plus endpoint reconstruction. With $t_i = t_n + c_i h$, define $F_{b,i}^r = f_{b,i} + \Phi_{r,b}^T \lambda_i$, $T_{b,i}^\theta = \tau_{b,i} + \Phi_{\theta,b}^T \lambda_i$, and $A_{\Phi,i} = \Phi_q(q_i, t_i) a_i + \dot{\Phi}_q(q_i, v_i, t_i) v_i$. Each stage solves

$$\begin{aligned} 0 &= r_i - r_n - h \sum_{j=1}^3 A_{ij} v_j, & 0 &= \eta_i - h \sum_{j=1}^3 A_{ij} J_r^{-1}(\eta_j) \omega_j, \\ 0 &= v_i - v_n - h \sum_{j=1}^3 A_{ij} a_j, & 0 &= \omega_i - \omega_n - h \sum_{j=1}^3 A_{ij} \alpha_j, \\ 0 &= m_b a_{b,i} - F_{b,i}^r, \\ 0 &= J_b \alpha_{b,i} + \omega_{b,i} \times J_b \omega_{b,i} - T_{b,i}^\theta, \\ 0 &= \Phi(q_i, t_i) - g_i^{(0)}, & 0 &= \Phi_q(q_i, t_i) v_i - g_i^{(1)}, \\ 0 &= A_{\Phi,i} - g_i^{(2)}. \end{aligned} \quad (9)$$

Here $g_i^{(0)}, g_i^{(1)}, g_i^{(2)}$ are the position-, velocity-, and acceleration-level driven right-hand sides; they are zero for passive lower-pair constraints. Equation (9) is the method equation solved by Newton in Algorithm 1. The next paragraphs unpack the same equation by row family and define the endpoint map.

The first two row families are the collocation kinematic defects. In com-

pact notation,

$$r_i - r_n - h \sum_{j=1}^3 A_{ij} v_j = 0, \quad (10)$$

$$\eta_i - h \sum_{j=1}^3 A_{ij} J_r^{-1}(\eta_j) \omega_j = 0, \quad (11)$$

where J_r^{-1} is the inverse right Jacobian of $\text{SO}(3)$. The next two families enforce velocity collocation,

$$v_i - v_n - h \sum_{j=1}^3 A_{ij} a_j = 0, \quad (12)$$

$$\omega_i - \omega_n - h \sum_{j=1}^3 A_{ij} \alpha_j = 0. \quad (13)$$

The fifth family is the Newton–Euler balance row,

$$m_b a_{b,i} - f_{b,i} - \Phi_{r,b}(q_i)^T \lambda_i = 0, \quad (14)$$

$$J_b \alpha_{b,i} + \omega_{b,i} \times J_b \omega_{b,i} - \tau_{b,i} - \Phi_{\theta,b}(q_i)^T \lambda_i = 0, \quad (15)$$

for each body b . The sixth family is the FullVA lower-pair block. It contains position, velocity, and acceleration-level constraint rows,

$$\Phi(q) = 0, \quad \Phi_q(q)v = 0, \quad \Phi_q(q)a + \dot{\Phi}_q(q,v)v = 0, \quad (16)$$

with nonzero right-hand sides included for driven mechanisms. The lower-pair library covers coordinate-difference (CD), dot-product-one (DP1), dot-product-two (DP2), and distance (D) constraints. The method is therefore a FullVA constrained stage solve, not a post-step position-only projection.

After Newton convergence, the endpoint is reconstructed by the Gauss weights,

$$r_{n+1} = r_n + h \sum_{i=1}^3 b_i v_i, \quad v_{n+1} = v_n + h \sum_{i=1}^3 b_i a_i, \quad (17)$$

$$Q_{n+1} = Q_n \text{Exp} \left(h \sum_{i=1}^3 b_i J_r^{-1}(\eta_i) \omega_i \right), \quad \omega_{n+1} = \omega_n + h \sum_{i=1}^3 b_i \alpha_i. \quad (18)$$

The Newton linearization solves

$$\frac{\partial R_{\text{G6FVA}}}{\partial Z} \Delta Z = -R_{\text{G6FVA}}, \quad (19)$$

with automatic differentiation used to assemble the dense reference Jacobian. The sparse backend uses the same residual and Jacobian entries; it is tracked as an implementation caveat rather than as a primary accuracy claim.

5.1. Stage Jacobian structure

This subsection records the block structure behind Eq. (19). It is not a separate hand-coded Jacobian; the implementation differentiates the same residual by automatic differentiation. The formulas below specify the row-to-unknown dependencies used to verify the AD Jacobian and to define the square Newton system.

For stage i , write the differential of the stage pose as Eq. (20):

$$\delta q_i = (\delta r_i, \delta \theta_i), \quad \delta \theta_i = J_r(\eta_i) \delta \eta_i, \quad (20)$$

so that $Q_i = Q_n \text{Exp}(\eta_i)$ is perturbed by the corresponding right-trivialized rotation vector. Let $G_i = J_r^{-1}(\eta_i)$ and let $D_\eta(G_i \omega_i)[\delta \eta_i]$ denote the directional derivative of $J_r^{-1}(\eta_i) \omega_i$. The four collocation blocks have the exact stage-coupling derivatives in Eq. (24):

$$\delta R_i^r = \delta r_i - h \sum_{j=1}^3 A_{ij} \delta v_j, \quad (21)$$

$$\delta R_i^\eta = \delta \eta_i - h \sum_{j=1}^3 A_{ij} (D_\eta(G_j \omega_j)[\delta \eta_j] + G_j \delta \omega_j), \quad (22)$$

$$\delta R_i^v = \delta v_i - h \sum_{j=1}^3 A_{ij} \delta a_j, \quad (23)$$

$$\delta R_i^\omega = \delta \omega_i - h \sum_{j=1}^3 A_{ij} \delta \alpha_j. \quad (24)$$

Thus the only off-diagonal stage couplings in the kinematic rows are the Gauss matrix entries A_{ij} ; all other nonlinear stage coupling enters through dynamics and constraints evaluated at each stage.

For body b at stage i , the translational and rotational balance linearizations can be written

$$\delta R_{b,i}^T = M_b \delta a_{b,i} - F_{q,b,i} \delta q_i - F_{\dot{q},b,i} \delta \dot{q}_i - \delta(\Phi_{r,b,i}^T) \lambda_i - \Phi_{r,b,i}^T \delta \lambda_i, \quad (25)$$

$$\delta R_{b,i}^R = J_b \delta \alpha_{b,i} + \mathcal{G}_{b,i} \delta \omega_{b,i} - T_{q,b,i} \delta q_i - T_{\dot{q},b,i} \delta \dot{q}_i - \delta(\Phi_{\theta,b,i}^T) \lambda_i - \Phi_{\theta,b,i}^T \delta \lambda_i. \quad (26)$$

Here Eq. (27) defines the gyroscopic linearization operator:

$$\mathcal{G}_{b,i} \delta \omega = [\delta \omega]_{\times} J_b \omega_{b,i} + [\omega_{b,i}]_{\times} J_b \delta \omega, \quad (27)$$

and $F_q, F_{\dot{q}}, T_q, T_{\dot{q}}$ collect load derivatives, including the velocity-dependent Brown–McPhee friction terms in the cylindrical-chain benchmark [28]. The terms $\delta(\Phi_{r,b,i}^T) \lambda_i$ and $\delta(\Phi_{\theta,b,i}^T) \lambda_i$ are the geometric stiffness blocks from the lower-pair Jacobian.

For a scalar lower-pair row $\Phi(q, t) = 0$, the FullVA row linearization at a stage is

$$\delta R_i^{\Phi} = \Phi_q \delta q_i, \quad (28)$$

$$\delta R_i^{\dot{\Phi}} = D_q(\Phi_q \dot{q} - \nu)[\delta q_i] + \Phi_q \delta \dot{q}_i, \quad (29)$$

$$\delta R_i^{\ddot{\Phi}} = D_q(\Phi_q \ddot{q} + \dot{\Phi}_q \dot{q} - \gamma)[\delta q_i] + D_{\dot{q}}(\Phi_q \ddot{q} + \dot{\Phi}_q \dot{q} - \gamma)[\delta \dot{q}_i] + \Phi_q \delta \ddot{q}_i. \quad (30)$$

The CD/DP1/DP2/D formulas in Section 10 supply the explicit Φ , $\dot{\Phi}$, and $\ddot{\Phi}$ expressions used in these directional derivatives.

With row groups ordered as Eq. (31)

$$(R^r, R^\eta, R^v, R^\omega, R^T, R^R, R^\Phi, R^{\dot{\Phi}}, R^{\ddot{\Phi}}) \quad (31)$$

and unknown groups ordered as Eq. (32)

$$(\delta r, \delta \eta, \delta v, \delta \omega, \delta a, \delta \alpha, \delta \lambda), \quad (32)$$

define $K_{ik}^{p,x} = \partial R_i^p / \partial \chi_k$. The nonzero block pattern is summarized in Table 4. The symbols M and J denote block-diagonal mass and inertia matrices, and the $\Phi_\eta, \Phi_\omega, \Phi_\alpha$ columns are obtained from Eq. (20). This is the paper-level block pattern corresponding to the 132-row dense AD Jacobian used by the accepted cylindrical-chain solve.

The endpoint closure certificate is part of the accepted Gauss6/FullVA evidence path. It does not convert the method into a source-paper TFE residual reproduction. A full-TFE replacement would instead replace all stage rows in Eq. (6) by paper-derived weak rows and would pass the same rank, residual, endpoint, and convergence checks.

Table 4: Nonzero block pattern of the accepted stage Newton matrix.

Row group	Direct stage blocks	Cross-stage or nonlinear blocks
R^r	$K_{ii}^{r,r} = I$	$K_{ij}^{r,v} = -hA_{ij}I.$
R^η	$K_{ii}^{\eta,\eta} = I - hA_{ii}D_\eta(G_i\omega_i)$	$K_{ij}^{\eta,\eta} = -hA_{ij}D_\eta(G_j\omega_j)$ and $K_{ij}^{\eta,\omega} = -hA_{ij}G_j.$
R^v	$K_{ii}^{v,v} = I$	$K_{ij}^{v,a} = -hA_{ij}I.$
R^ω	$K_{ii}^{\omega,\omega} = I$	$K_{ij}^{\omega,\alpha} = -hA_{ij}I.$
R^T	$K_{ii}^{T,a} = M, K_{ii}^{T,\lambda} = -\Phi_r^T$	Load derivatives and geometric-stiffness blocks in Eq. (26).
R^R	$K_{ii}^{R,\alpha} = J, K_{ii}^{R,\lambda} = -\Phi_\theta^T$	Gyroscopic, load-derivative, and geometric-stiffness blocks in Eq. (26).
R^Φ	$K_{ij}^{\Phi,q} = \Phi_q$	Rotation columns mapped by Eq. (20).
$R^{\dot{\Phi}}$	$K_{ij}^{\Phi,\dot{q}} = \Phi_q$	$D_q(\Phi_q\dot{q} - \nu)$ blocks.
$R^{\ddot{\Phi}}$	$K_{ii}^{\Phi,\ddot{q}} = \Phi_q$	D_q and $D_{\dot{q}}$ blocks from Eq. (30).

Table 5: Operational definition of the accepted method path.

Item	Accepted definition
Production method	Gauss6/FullVA one-step map.
Nominal order	Conditional sixth-order Gauss collocation path under the retained interfaces, including solver-scale control, of Theorem 1.
Stage residual	Square 132-row FullVA lower-pair nonlinear system.
Endpoint closure	Controlled by P2 local right-inverse and raw-defect hypotheses; KKT/projection diagnostics are recorded separately.
Full-TFE replacement	Not accepted; this stronger source-paper residual replacement is outside the present claim.

Table 6: Algorithmic form of one Gauss6/FullVA step.

Step	Operation
1	Build the lower-pair constraint graph, mass/inertia blocks, loads, and FullVA stage unknown vector for (q_n, v_n, a_n) and step size h .
2	Assemble the 132-row residual $R_{\text{G6FVA}}(q_n, v_n, a_n, Z; h)$, including dynamics, Lie-group kinematics, and position/velocity/acceleration constraint rows.
3	Solve the nonlinear stage system by Newton iteration with automatic differentiation for the residual Jacobian.
4	Recover the endpoint by Gauss quadrature and the Lie-group retraction in Eq. (4).
5	Check endpoint position, velocity, residual, rank, order, and four-example quantities using the reproducibility checks summarized in the reproducibility appendix.

5.2. Complete execution algorithm

The implemented Gauss6/FullVA path used for the cylindrical-chain and four-example validation rows is the trajectory algorithm below. The listing is deliberately written as an execution protocol: it starts from mechanism data and ends with the error/order records used in the reproduced results.

Operationally, for an example e and a step size h , the accepted run is the composition in Eq. (33):

$$\begin{aligned} Z_{n,h,e} &= \text{NewtonSolve}_{\tau_N, k_{\max}} \left(R_{\text{G6FVA}}(x_{n,h,e}, Z; h) = 0; Z_{n,h,e}^{(0)} \right), \\ \tilde{x}_{n+1,h,e} &= \mathcal{E}_h(x_{n,h,e}, Z_{n,h,e}), \\ x_{n+1,h,e} &= \mathcal{C}_h(\tilde{x}_{n+1,h,e}), \\ \mathcal{R}_e(h) &= \mathcal{P}_e \left(\{x_{n,h,e}\}_{n=0}^{N_h}, \{D_{n,h,e}\}_{n=0}^{N_h-1}, x_e^{\text{ref}} \right). \end{aligned} \quad (33)$$

Here $\mathcal{E}_h$ is the endpoint reconstruction in Eqs. (17)–(18), $\mathcal{C}_h$ is the accepted endpoint closure map, $D_{n,h,e}$ stores Newton and constraint diagnostics, x_e^{ref} is the analytic or nested local reference specified by the validation policy, and $\mathcal{P}_e$ is the postprocessing map that writes the reproduced error, order, diagnostic, and comparison-scope records.

Algorithm 1 (complete accepted Gauss6/FullVA run).

1. Choose the validation example e , its mechanism data, its step-size set H_e , final time T_e , Newton tolerance τ_N , Newton cap $k_{\max}$, Jacobian backend, and reference policy.
2. Build the body mass and inertia blocks, lower-pair graph, joint frames, axes, CD/DP1/DP2/D constraint bases, driven right-hand sides, loads, and friction data.
3. Construct the initial state $x_{0,h,e}$ on $\mathbb{R}^3 \times \text{SO}(3)$ for every body and project the initial velocity onto the velocity-level lower-pair constraints.
4. For each $h \in H_e$, first check the accepted-grid condition $|T_e/h - \text{round}(T_e/h)| \leq \varepsilon_{\text{grid}}$. Accepted order rows use only grids satisfying this condition; non-integer final-time grids are recorded as external-comparison diagnostics rather than promoted to reported order rows. This is a validation/reporting policy for the finite order rows, not an additional theorem condition; the proof theorem below states its grid-point estimate on the reported grid and does not require exact final-time divisibility. Set $N_h = \text{round}(T_e/h)$, precompute the three-stage Gauss coefficients and the fixed FullVA stage-vector layout, and initialize empty trajectory and diagnostic records.

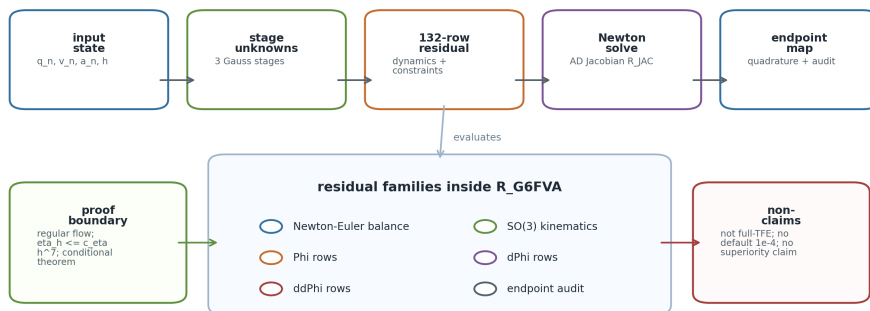
5. For $n = 0, \dots, N_h - 1$, form the predictor $Z_{n,h,e}^{(0)}$ from the endpoint state, constant-acceleration motion, zero initial Lie increments, and force-balance multiplier estimates.
6. Assemble $R_{\text{G6FVA}}(x_{n,h,e}, Z; h)$ by evaluating stage poses, right-Jacobian terms, lower-pair positions, velocities, accelerations, Newton–Euler balance rows, and FullVA position-, velocity-, and acceleration-level constraint rows.
7. Iterate Newton: evaluate the residual, form or apply $\partial R_{\text{G6FVA}}/\partial Z$, solve Eq. (19), update Z , and stop only after the residual or update norm satisfies τ_N ; reject the step if k_{max} is reached.
8. Reconstruct $\tilde{x}_{n+1,h,e}$ with the Gauss endpoint formulas and the Lie-group retraction, then evaluate raw endpoint position and velocity constraint residuals.
9. Apply the accepted endpoint closure map $\mathcal{C}_h$ when the velocity-level closure threshold is exceeded, and record raw residuals, corrected residuals, projection rank, condition number, and correction norm.
10. Store $x_{n+1,h,e}$ and the diagnostics $D_{n,h,e}$, including Newton iterations, residual norms, linear-solve residuals, Jacobian or colored-derivative timings, endpoint constraint norms, stage acceleration constraint norms, and quaternion unit error.
11. After all steps and all $h \in H_e$ are complete, compare each trajectory with x_e^{ref} , compute the paper’s error norms, error ratios, observed orders, mechanism-coverage rows, and reaction residual summaries.
12. Write the CSV, JSON, plot, metadata, and comparison-scope records used in the manuscript. These postprocessing checks reproduce the evidence boundary; they do not alter the accepted numerical trajectory.

All rows in Algorithm 1 are the accepted Gauss/FullVA rows in Eq. (6). A full source-paper TFE replacement would replace the stage residual assembled in the residual-assembly step by paper-derived weak rows; that stronger algorithm is explicitly outside the accepted claim in this manuscript. Algorithm–theorem tolerance boundary: Algorithm 1 records the executable finite-run protocol, including the reported Newton tolerance τ_N . Theorem 1 uses the same implemented residual map only after the accepted branch also satisfies the scaled proof policy $\eta_h^{\text{tube}} \leq c_\eta h^7$ in the fixed proof norm. A fixed τ_N run is therefore theorem-admissible only for those steps on which its accepted branch-selected residuals meet that h^7 bound; otherwise it is finite-run diagnostic evidence and does not discharge P6. The correspondence between

Algorithm 1 and the supplied source package is summarized in Appendix Table B.62; that table is a reproducibility map, not an additional numerical claim.

Accepted Gauss6/FullVA one-step architecture

The figure separates the mathematical one-step map from open source-paper TFE replacement and external-superiority claims.



Accepted claim: a conditional sixth-order Gauss6/FullVA path for lower-pair mechanisms. Open gates remain explicit rather than hidden in the method diagram.

Figure 1: Accepted Gauss6/FullVA one-step architecture. The schematic separates the 132-row FullVA residual and conditional order boundary from non-claims: it is not a full-TFE replacement, does not require a default 10^{-4} step-size sweep, and does not assert external superiority without same-test evidence.

The reproducibility record is described in the reproducibility appendix. The main text uses that record only to keep the claim boundary precise; the scientific argument rests on the residual definition, the conditional sixth-order theorem, and the numerical evidence below.

Formal-order comparator guardrail.

This short guardrail fixes only the scalar formal-order comparator used later in the paper. The comparison to Lie-group TFE methods is formal-order-level unless same-test implementation evidence is provided. The source-paper comparison used in this manuscript is therefore local and explicit. It is placed before validation only to define the comparator convention; it is not benchmark evidence, not a theorem input, and not a claim that the TFE implementation in the source paper has been reproduced. The main evidence chain remains method definition, conditional FullVA proof, accepted dynamic-order rows, and mechanism-coverage rows. The

paper-style temporal finite-element family indexed by polynomial degree m uses the expected order convention

$$p_{\text{TFE}}(m) = 2m - 1. \quad (34)$$

The $m = 3$ Gauss–Lobatto temporal finite-element target therefore has expected order five. The source paper should not be read as reporting uniform fifth-order DAE convergence in every experiment: in its pendulum study it observes that the $m = 3$ TFE implementation can lose order on the DAE problem, even though the underlying Kim–Reddy TFE construction has the expected $2m - 1$ formula order [2, 1]. This manuscript therefore uses the paper-style formula target $p_{\text{TFE}}(3) = 5$ as the comparator boundary, not a weakened observed-order target. The accepted Gauss6/FullVA path has a conditional method-order claim of six. The manuscript-level order comparison is the formal-order comparison of that conditional Gauss6/FullVA path against the local order-five target. It is not a claim that the present implementation is faster or more accurate than the source-paper TFE implementation on the original source-paper tests.

External-comparison guardrail.

The comparison policy used in this paper is deliberately conservative. A statement about implemented accuracy or cost against a source-paper method is accepted only after a same-test row satisfies all entries in Table 8. Until then, the TFE comparison remains the formal-order comparison in Eq. (34), and work/precision plots are candidate or common-reference diagnostics.

Consequently the manuscript does not say that Gauss6/FullVA is faster or more accurate than an implemented source-paper TFE method. The stronger sentence would require the Table 8 promotion criterion to close on the same benchmark. The accepted claim is narrower: Gauss6/FullVA is a conditional sixth-order Lie-group Gauss-collocation FullVA path under the retained branch-smoothness, residual-defect, endpoint/stability, implementation-binding, and solver-scale interfaces later stated in Theorem 1.

6. Validation protocol

The validation protocol organizes the numerical evidence into three layers: accepted dynamic-order evidence for the Gauss6/FullVA branch,

Table 7: Comparison boundary used in the manuscript.

Question	Boundary
What is claimed?	Under the stated compact-branch, endpoint/stability, implementation-binding, residual-defect, and solver-scale contracts, Gauss6/FullVA has a conditional sixth-order Gauss-collocation FullVA path. Its formal-order contrast is order six versus the local order-five $m = 3$ Gauss-Lobatto TFE formula target; no implementation superiority follows from this contrast.
What is the comparator?	A local paper-style temporal finite-element formula target with expected order $2m - 1 = 5$; source-paper implementation superiority requires same-test evidence.
What is not claimed?	Complete source-paper residual reproduction and accepted independent full-TFE stage replacement; faster or more accurate implemented source-paper TFE performance on the original tests is also not claimed.
Where is the reproducibility record?	The reproducibility appendix lists the source package, comparison-scope files, and read-only checks used to reproduce the reported evidence.

Table 8: Guardrail for promoting a baseline row to an implemented same-test comparison.

Promotion item	Required evidence	Evidence boundary
Mechanism and parameters	The source-paper frictionless pendulum or revolute-friction benchmark is run with the same masses, geometry, loads, friction law, final time, and output variables.	Open for external same-test promotion.
Methods	The same script or verified harness reports Newmark- β , trapezoidal, TFE $m = 1, 2, 3$, and Gauss6/FullVA rows.	Candidate diagnostics only.
Grid and solver policy	The rows use the same step sizes, same reference solution, same Newton tolerance policy, and same accepted-grid handling.	Not fully same-test closed.
Error norms	Position, velocity, orientation, angular-velocity, constraint, and reaction or multiplier errors are evaluated with the same norm definitions.	Partly available in common-reference rows.
Work metric	Wall time, Newton iterations, Jacobian assembly time, and linear-solve time are reported so that an accuracy claim is not separated from cost.	Diagnostic compendium only.

mechanism-coverage evidence for lower-pair examples, and diagnostic comparison/reproducibility boundaries. It also fixes the error norms and reference policies used in the paper, so that the numerical tables are not merely unsupported assertions. For a trajectory sampled at accepted output times t_k , translational errors use the maximum-in-time Euclidean or infinity norms recorded in the CSV files. Orientation errors use the Lie-group distance

$$d_R(Q_h, Q_{\text{ref}}) = \|\log(Q_{\text{ref}}^T Q_h)\|_2. \quad (35)$$

Equation (35) defines the rotation component of the reported trajectory error. Angular-velocity errors use the corresponding vector norm. The observed order reported in the tables is the least-squares slope of $\log e(h)$ against $\log h$ over the stated step sizes. The fit protocol is fixed before the reported row is inspected: step window, reference policy, reported components, error norm, and inclusion or exclusion of floor-limited rows are the policy stated for that evidence row. Recomputing a slope after dropping a coarse or fine row, swapping position, velocity, or orientation components, changing a norm, or moving a row between dynamic-order, mechanism-coverage, and diagnostic status defines a different evidence row. Such a post-hoc row may be recorded as a new diagnostic, but it cannot upgrade the theorem claim or replace the predeclared row in the acceptance policy. A result can support the main manuscript claim only if it satisfies the following acceptance criteria.

1. The method identifier is Gauss6/FullVA, and the order claim is the conditional sixth-order path of Theorem 1, not an unconditional property of fixed-tolerance or source-policy rows.
2. The smooth cylindrical-chain h-sweep uses at least three step sizes and reports position and velocity orders above the local order-five comparator.
3. The four ASME-style examples are accepted under the same method-side policy, but their roles differ: only the single and double pendula are used as accepted dynamic-order examples, while the closed-loop examples provide mechanism coverage, constraint closure, reaction residuals, and local coarse dynamic evidence.
4. The source-paper comparator remains a local $m = 3$ Gauss-Lobatto temporal finite-element target with expected order five.
5. Any claim against a source-paper TFE implementation requires same-test evidence with matched parameters, step sizes, tolerances, reference

solution, error norm, and work metric; this manuscript does not make that source-paper superiority claim.

6. Complete source-paper residual reproduction and independent full-TFE stage replacement remain outside the accepted method claim.
7. Routine reproducibility checks are documented separately in the reproducibility appendix.

Table 9: Reproducibility policy for the numerical evidence.

Evidence block	Step sizes	Reference	Primary reported quantities
Smooth cylindrical chain	0.04, 0.02, 0.01	$h = 0.005$ trajectory	Pos./vel. errors; constraints; Newton iterations.
Sharp-friction cylindrical chain	0.04, 0.02, 0.01 and refined diagnostics	$h = 0.005$ or finer reference by diagnostic check	Order reduction, endpoint constraints, refinement cost.
Driven single pendulum	0.2, 0.1, 0.05	Analytic $\theta(t)$ trajectory	Position, velocity, orientation, and residual errors.
Double pendulum	0.04, 0.02, 0.01	Nested local FullVA $h = 0.005$ reference	Orientation and angular-velocity errors plus endpoint constraints.
Four-link and slider-crank	0.02, 0.01, 0.005	Local kinematic FullVA $h = 0.001$ reference	Constraints; rank; reaction residuals.

The paper uses three evidence types throughout the results. A dynamic-order row is used to support an order statement for the accepted one-step map. A mechanism-coverage row checks that the same FullVA position, velocity, and acceleration vocabulary closes a lower-pair mechanism and produces consistent reaction dynamics, but it is not by itself an asymptotic

order row. A diagnostic row clarifies a boundary condition, failure mode, or external-comparison gap and is never promoted to a theorem or superiority claim. This separation is important for the four-link and slider-crank examples, where the closed-loop constraint/reaction checks and coarse-window trajectory traces are reported only as coverage/consistency diagnostics; they do not become accepted asymptotic order proof, residual-to-error evidence, or source-policy comparison closure.

The accepted cylindrical-chain h-sweep is reported with endpoint velocity closure, and the closure itself is checked explicitly rather than hidden inside the error table. Two endpoint records are retained. The first is the actual projected endpoint path used for the main finite-window order row. The second is a square endpoint KKT check in which endpoint velocity variables, least-squares stationarity, and endpoint velocity constraints are solved as an augmented closure residual. Table 10 reports the smooth branch scaling. The endpoint functional E used in Lemma 6 is the velocity-level/KKT closure subsystem actually corrected by $\mathcal{C}_h$, not every raw endpoint diagnostic in the table. The raw endpoint position defect is retained as a separate constraint monitor: it remains small and approximately sixth order over the reported window, but it is not the $\|E(z_u)\| \leq C_{E,\text{raw}}h^7$ hypothesis. The raw endpoint velocity defect and the KKT correction norm both have finite-window fitted slopes near seven over the reported window, and the corrected endpoint velocity residual is at roundoff scale. This evidence is a finite-run consistency check for the endpoint-closure perturbation scale in Lemma 6; it is not a proof of that lemma’s retained P2 endpoint-ball, raw-defect, or uniform right-inverse hypotheses, and it neither closes the full-TFE replacement nor the source-policy boundaries. The sharp-branch coarse h-sweep remains a limitation, not theorem evidence.

Table 10: Smooth cylindrical-chain endpoint closure scaling diagnostic.

h	Raw pos. defect	Raw vel. defect	KKT correction	Closed vel. defect	Newton iters.
0.04	5.883×10^{-12}	9.517×10^{-11}	7.926×10^{-11}	2.431×10^{-16}	12
0.02	9.196×10^{-14}	7.427×10^{-13}	6.395×10^{-13}	1.573×10^{-16}	24
0.01	1.635×10^{-15}	6.505×10^{-15}	5.484×10^{-15}	3.634×10^{-16}	48
Obs. order	5.907	6.918	6.909	roundoff limited	—

7. Conditional sixth-order theorem

Under the retained compact-branch, endpoint, stability, implementation-binding, and solver-scale interfaces stated below, the theorem proves a sixth-order same-initial-state reported-grid error bound by a conditional local-defect-to-global-error argument on the accepted compact-tube Gauss-predictor branch. It starts from the standard three-stage Gauss order on a smooth reduced constrained flow and then accounts for the perturbations introduced by the implemented FullVA residual, endpoint closure, and Newton tolerance. It is not a derivation of the complete source-paper temporal finite-element weak form. The branch-selected map on the accepted Gauss-predictor domain is the only theorem object in this section. The theorem below is a conditional numerical-analysis theorem: under the fixed compact-branch, endpoint, implementation, stability, and solver-scale hypotheses on that branch, the h^6 same-initial-state reported-grid bound follows from a proved h^7 local defect, not from fitted slopes or downstream diagnostic records.

Proof spine. The load-bearing proof route is a direct residual-to-local-defect argument. After the smooth reduced Gauss stage Z_G is lifted into the implemented stage variables, the proof evaluates the fixed row-scaled 132-row residual at that same Z_G . The 96 position-, velocity-, acceleration-, endpoint-, and lower-pair rows are supplied by the non-dynamic FullVA certificate, and the 36 Newton–Euler rows are supplied only as a same-branch base-point zero block by the D5 direct-substitution route: D1/D2 balance identities, the D3 virtual-work multiplier-wrench identity, D4 smooth-force lift, and D6 row-ordering/scaling/AD binding together give zero dynamic residual on Z_G . Together these two row families, under the retained branch, norm, and implementation-binding tuple, give the $O(h^7)$ implemented residual input consumed by Lemma 39: the single T2 input is $\|F_{A,h}(Z_G; y)\| \leq C_R h^7$ before Taylor expansion in $\Delta = Z - Z_G$. The D5 identity supplies only the base-point dynamic block of that same residual; it does not supply derivative bounds, primitive subterm bounds, or an additional local-defect term. This is the active route used by the theorem. The primitive/Taylor decomposition of the dynamic rows is organized as a separate candidate sufficient route for verifying the same dynamic block; until its primitive subterm bounds are proved on the same compact branch, it stays outside the theorem and contributes no additional summand to the local-defect estimate below.

Proof contract. The theorem below should be read as the branch-local implication in Eq. (36), with all constants fixed before numerical evidence is used:

$$\begin{aligned} \mathcal{I}_h(K) \quad \wedge \quad \mathbf{R}_{96}(Z_G) = O(h^7) \quad \wedge \quad \mathbf{D}_{36}(Z_G) = 0 \quad \wedge \quad \eta_h^{\text{tube}} \leq c_\eta h^7 \\ \implies \max_{0 \leq n \leq N_h} \|\mathcal{R}(y_n) - \mathcal{R}(y(t_n))\| \leq C_{qv} h^6. \end{aligned} \quad (36)$$

The residual part of Eq. (36) is the same direct bridge later used in the theorem’s residual-certificate slot:

$$\begin{aligned} \mathbf{B}_h^{96+36}(Z_G; y) &:= \{\mathbf{R}_{96}(Z_G) = O(h^7), \mathbf{D}_{36}(Z_G) = 0\}, \\ \mathbf{B}_h^{96+36}(Z_G; y) &\implies \|F_{A,h}(Z_G; y)\| \leq C_R h^7. \end{aligned} \quad (37)$$

Equation (37) is the early notation for the $\mathbf{R}_{\text{dir}}$ certificate fixed in Eq. (253); it is not a second residual route and it is not combined with the primitive/Taylor replacement template. Here $\mathcal{I}_h(K)$ denotes the retained compact-tube, inverse, endpoint, stability, implementation-binding, and proof-norm interfaces on the accepted Gauss-predictor branch. The predicate $\mathbf{R}_{96}(Z_G) = O(h^7)$ is the proved non-dynamic FullVA row certificate, and $\mathbf{D}_{36}(Z_G) = 0$ is the proved direct Newton–Euler row identity on the lifted Gauss stage. Both are formula-level statements on the same branch-selected, row-scaled residual object, not finite-run logged residuals, and they are fixed before the stage-root perturbation is invoked. Thus the theorem does not assume the implemented residual defect as a black box: the 96 + 36 row bridge supplies it before the Taylor/Kantorovich step. At the same time, $\mathcal{I}_h(K)$ and the solver-scale predicate remain domain and branch hypotheses; they are not inferred from observed slopes, solver logs, external-comparison diagnostics, or the separate primitive/Taylor route. The theorem closure below refers only to the direct residual-bridge/Kantorovich argument under the stated branch, endpoint, implementation, and solver-scale hypotheses; it does not prove a separate theorem for fixed production tolerances, a theorem converting mechanism residual histories into trajectory errors, or the separate primitive/Taylor route. The Taylor step used in this implication is the full-map Taylor/Kantorovich step for the fixed 132-row residual $F_{A,h}$, not a term-by-term expansion of the separate primitive Newton–Euler inventory. The direct D5 identity supplies the dynamic block of $F_{A,h}(Z_G; y)$ at the base point before that full-map Taylor step is applied. A separate primitive 162-term proof, if supplied on the same branch and proof norm, would therefore be an alternative sufficient certificate for that dynamic-block residual value only; it would

not replace the full-map inverse, endpoint closure, solver-scale condition, or local-to-global slots of the theorem.

Residual-bridge handoff. The theorem uses the direct dynamic-row identity, together with the already certified 96 non-dynamic rows, to form the 132-row residual bridge at Z_G . The compact smooth branch, local inverse, endpoint ball/right inverse, implementation binding, and branch-solver scale are theorem-domain hypotheses rather than empirical closures; they are fixed before the local defect is transferred. The P7 residual-to-error route is deliberately not used: residual-only mechanism rows, reaction rows, common-reference rows, and external-comparison rows cannot enter the sixth-order theorem unless a separate transfer theorem is proved. This is the sense in which the theorem is strong but conditional: the local residual-defect input is proved for the implemented FullVA residual, while the domain and solver-scale hypotheses are stated instead of being hidden in numerical evidence.

Fixed theorem object. Before any numerical row is read, fix the compact proof tube K , the reduced chart y , the proof residual norm, the row ordering and scaling of the implemented 132-row residual, and the Gauss-predictor branch. For $y \in K$, let $\varphi_h(y)$ be the exact reduced constrained flow, $Z_G(y, h)$ the isolated three-stage Gauss stage in the reduced chart, $F_{A,h}(Z; y)$ the implemented Gauss6/FullVA residual written in the same unknowns and row scaling, $Z_A(y, h)$ the local zero selected from the Gauss-predictor branch, $\tilde{Z}_A(y, h)$ the inexact Newton iterate, $\mathcal{E}_h$ the Lie endpoint reconstruction, $\mathcal{C}_h$ the accepted endpoint-closure map, and $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$. The one-step map whose order is proved is Eq. (38):

$$\Psi_h^{\text{G6FVA}}(y) = \mathcal{P}_h(\tilde{Z}_A(y, h)). \quad (38)$$

The proof evaluates the residual bridge at the lifted Gauss stage Z_G , uses the local branch inverse to obtain Z_A , then applies endpoint closure and the inexact-Newton perturbation. Thus the proof object is fixed analytically before h -sweeps, Newton logs, residual tables, mechanism coverage rows, external comparison rows, or source-package records are interpreted. Those records can identify or corroborate an accepted finite branch, but they cannot define the map, choose constants, or supply a separate theorem input. This fixes the theorem object before numerical evidence is introduced, including the retained P2 interface: the retained domain hypotheses and discharged residual inputs are stated explicitly before the lemmas are used.

Data for theorem invocation. The theorem is invoked only after the following four inputs are fixed on the same accepted branch. First, the compact tube, reduced chart, endpoint map, row ordering, row scaling, proof norm, and Gauss-predictor branch define the one-step object and its local inverse constants. Second, the direct residual bridge supplies the stage-residual value at the lifted Gauss stage: the 96 non-dynamic rows contribute $O(h^7)$, and the 36 Newton–Euler rows vanish by direct substitution. Third, the endpoint-closure and stability interfaces provide the right-inverse and local-to-global constants used after the stage perturbation. Fourth, the P6 solver-scale envelope $\eta_h^{\text{tube}} \leq c_\eta h^7$ is retained for the selected Newton branch. These inputs are not interchangeable. The theorem input is not a separate primitive/Taylor subterm closure, a residual-to-error promotion, a fixed-tolerance log, or a source-policy comparison row; those records are outside this invocation unless their own theorem interfaces are separately proved.

Positive and retained theorem inputs. The invocation contains two logically different kinds of theorem input. The positive residual input is the proved $96 + 36$ row bridge at Z_G : the P4 row-local estimate and the P5 direct dynamic-row identity together provide the residual value consumed by Lemma 39. The retained interfaces are the compact chart/tube, local inverse, endpoint, stability, implementation-binding, and P6 solver-envelope hypotheses; they define the admissible branch and constants, but they are not outputs of P5 or of numerical evidence. The proof uses the positive residual input only after the retained interfaces have fixed the branch and norm, and it uses the retained interfaces only as hypotheses. No retained interface is promoted to a proved residual row, and no proved residual row is reused as a proof of an interface.

Load-bearing proof skeleton. The theorem has five analytic moves, all on the same accepted branch: (i) the classical three-stage Gauss defect gives the reduced $C_G h^7$ local truncation term; (ii) the $96 + 36$ direct residual bridge plus the Kantorovich inverse gives the accepted stage-root perturbation $C_A h^7$; (iii) the endpoint-closure right-inverse gives $C_E h^7$; (iv) the retained P6 branch-solver scale gives the inexact-Newton term $C_N c_\eta h^7$; and (v) the first-exit/discrete-Gronwall transfer plus the reporting map converts the resulting one-step $C_{\text{loc}} h^7$ bound into the h^6 same-initial-state reported-grid estimate. The surrounding tables are only signposts for this sequence: they identify the common theorem object, constant dependencies, and excluded claims, but they are not additional local-defect estimates and not substitutes for any of the five analytic moves.

One-step perturbation decomposition. For a fixed retained-tube state y and step h , the proof uses the four maps in Eq. (39) on the same branch:

$$\begin{aligned}\Psi_h^G(y) &= \mathcal{E}_h(Z_G(y, h)), \\ \Psi_h^A(y) &= \mathcal{E}_h(Z_A(y, h)), \\ \Psi_h^E(y) &= \mathcal{P}_h(Z_A(y, h)), \\ \Psi_h^{\text{G6FVA}}(y) &= \mathcal{P}_h(\tilde{Z}_A(y, h)).\end{aligned}\tag{39}$$

After the smooth FullVA lift identifies the reduced Gauss endpoint with $\mathcal{E}_h(Z_G)$, the four comparisons in the theorem are, in order: Gauss truncation $\Psi_h^G - \varphi_h$, accepted stage-root perturbation $\Psi_h^A - \Psi_h^G$, endpoint closure $\Psi_h^E - \Psi_h^A$, and inexact Newton perturbation $\Psi_h^{\text{G6FVA}} - \Psi_h^E$. Each term is evaluated for the same y , same h , same compact tube, same row-scaled residual, and same endpoint-output map. Thus the local $O(h^7)$ defect is one triangle inequality on a single branch-selected one-step object, not a patchwork of Gauss, endpoint, Newton, or diagnostic quantities from different runs.

Reference-paper proof-method alignment. The proof organization follows the Lie-group constrained-BDF convergence analysis of Wieloch and Arnold [8], not the temporal finite-element comparator used later for formal-order context. The constrained-BDF proof paper and the comparator paper therefore have disjoint roles: the former organizes the proof-obligation order, whereas the latter supplies only the order-five formula target and external comparison boundary used outside the theorem. The constrained-BDF reference first fixes the Lie-group update and increment variables, then proves a Taylor-expansion local truncation estimate, defines global configuration error in the Lie algebra, uses Baker–Campbell–Hausdorff perturbation estimates for the exponential composition, and finally closes the constrained problem through holonomic-constraint differences, a multiplier estimate, and a coupled error recursion. We invoke the reference as a proof-obligation ordering template, not as a literal proof transfer: the BLieDF hidden-constraint multiplier estimate and coupled multistep recursion are not repeated below. More specifically, the reference is invoked only for the Section 6.2/Theorems 2–4 style of proof sequencing: Theorem 2 local truncation estimates, Theorem 3 configuration and velocity error recursion, and Theorem 4 hidden-constraint difference approximation, multiplier-error estimate, and coupled-recursion closure. It supplies no FullVA residual Taylor constant, no D5 primitive lift bound, and no one-step Gauss theorem for the present residual. Adapted

to one-step Gauss collocation, the stage vector Z , FullVA derivative rows, lower-pair constraints, Newton–Euler rows, row scaling, proof norm, and endpoint reconstruction are fixed first as one algebraic object. The formal Gauss order is used only for the reduced smooth Gauss branch that supplies $\Psi_h^G - \varphi_h$. FullVA enters the next proof slot through the fixed 132-row residual bridge evaluated at Z_G : the direct-route $O(h^7)$ stage-residual input is discharged as the residual value of that bridge, while the compact-tube inverse, branch, endpoint, and solver-scale interfaces remain theorem hypotheses before any h -sweep, residual table, work/precision row, or source-policy row is interpreted. This is the manuscript’s reference-style Taylor slot T2: fix the discrete object, insert the smooth branch into the fixed $F_{A,h}$, then use the full-map Taylor/Kantorovich perturbation. The T3 primitive Taylor inventory is a separate alternative certificate specification for the dynamic-block residual value; because its primitive lift hypotheses are not theorem inputs here, it is not the Taylor layer required by the accepted theorem route. If a future primitive/Taylor proof is completed, its mathematical role is only to replace the dynamic-row residual-value certificate in the same 132-row residual bridge before the T2 full-map Taylor/Kantorovich step; it would not create a second Taylor layer, add a second residual summand, or strengthen the accepted theorem by combination with the direct certificate. Thus the reference paper informs the order of proof obligations—define the Lie-group reconstruction, prove the local defect, transfer the defect through the constrained algebraic solve, and only then report numerical diagnostics after the theorem boundary is fixed—while the load-bearing sixth-order estimate here is supplied by the direct residual-bridge and perturbation lemmas below, not by importing the BDF $p = k$ theorem. The reference-paper correspondence is therefore structural and stepwise, not an estimate transfer. The reference local-truncation step corresponds only to the proof slot occupied below by the four-term local defect sum; its hidden-constraint and multiplier steps correspond only to the proof slots occupied here by the retained compact FullVA chart, endpoint, lower-pair right-inverse, and branch-stability interfaces; the Newton–Euler row identities are separate FullVA residual-bridge inputs, not imported constraint/multiplier estimates. Its final coupled error recursion corresponds only to the accepted-branch stability recurrence used after Lemma 44. The estimates themselves are not reused: every constant in Theorem 1 is fixed in Assumption 1 or proved by the FullVA lemmas below. The present proof uses the same discipline—insert the smooth exact branch into the fixed discrete equations before proving global error—but

replaces the multistep BDF recursion by a one-step Gauss-collocation residual bridge on the 132-row FullVA system. In particular, the reference Appendix’s coupled recursion for stacked history variables such as E_y, E_z , and its zero-stability norms for the BDF companion matrices T_y, T_z , are not theorem inputs here. They mark the proof slot occupied in this manuscript by the retained one-step stability hypothesis for Ψ_h^{G6FVA} and the scalar/vector Gronwall recurrence for the reduced-chart error e_n . This replacement is legitimate only because the FullVA stage multipliers, lower-pair constraints, endpoint map, and Newton–Euler rows have already been consumed inside the same 132-row stage-root object before the recurrence is written; it does not prove multiplier-history, reaction-output, or residual-output order. The reference theorem also carries multistep start-value assumptions and a separate multiplier-error recursion. Those are not silently omitted here: there is no $2k - 1$ start-up layer for the one-step accepted map, and the lower-pair multipliers are stage unknowns inside $F_{A,h}$, consumed through the same residual bridge and endpoint perturbation rather than reported as a separate global-error component. A separate trajectory-order theorem for multiplier histories or residual-only mechanism rows would require additional outputs and assumptions; it is not part of Theorem 1. This is an output-scope choice backed by the retained theorem interfaces, not a deletion of the constrained part of the proof. The recurrence below is written for the reduced-chart state y after the lower-pair algebraic variables have been absorbed into the accepted stage-root map on the compact branch. That absorption is admissible only under the same compact inverse, endpoint right-inverse, branch-ball, and stability hypotheses that appear in Assumption 1. It therefore supports the stated position–velocity grid estimate, but it does not by itself give a multiplier history, reconstructed reaction-force, or residual-output order estimate. To claim those outputs one would have to augment the reporting map and prove a separate multiplier/reaction stability or residual-to-error theorem on the same branch.

Reference proof-order correspondence. Table 11 records how the Lie-group constrained-BDF proof architecture of Wieloch and Arnold [8] is invoked in this manuscript: as an ordering discipline, not as a theorem transfer. This table is the canonical reader-facing map. The following no-estimate-transfer and translation-map blocks are guardrails on that same map, not additional proof routes or theorem inputs.

Reference no-estimate-transfer table. The correspondence above is intentionally one-way: it fixes the order in which proof obligations are dis-

Table 11: Reference proof-order correspondence. The reference proof is used only to organize the FullVA proof obligations; none of its BLieDF estimates, BDF order theorem, multiplier-history estimates, or source-paper comparison claims are imported.

Reference proof step	Present FullVA analogue	What is not imported
Fix Lie-group increments and reconstruction before the error proof	Fix the stage vector Z , $\text{SO}(3)$ retractions, FullVA rows, row scaling, proof norm, and endpoint reconstruction before any estimate.	No BLieDF multistep update, no hidden-coordinate convention, and no post-hoc residual-row selection.
Taylor local truncation estimate for the fixed discrete equations	Insert the smooth lifted Gauss stage Z_G into the implemented 132-row residual, use Lemma 40 to obtain the $O(h^7)$ residual defect from the 96-row non-dynamic certificate and the D5 direct-substitution certificate, and then use Lemma 39 to transfer that residual defect to the accepted local stage-root and endpoint perturbation.	No BLieDF Taylor algebra, hidden-constraint or multiplier estimate, BDF $p = k$ theorem, source-paper/TFE estimate, FullVA residual constant, or primitive D5 Taylor subterm is imported. No primitive D5 Taylor subterm is certified by the reference correspondence. The primitive/Taylor route is a separate sufficient specification for verifying the dynamic-block residual value; no primitive D5 subterm bound is consumed by the accepted theorem unless that separate route is independently proved on the same branch and reassembled with the same 96-row non-dynamic block into a same-tuple 132-row residual certificate.
Constraint and multiplier estimates before global propagation	Retain the compact FullVA chart, lower-pair right inverse, endpoint closure, and branch stability as theorem-domain interfaces; the Newton–Euler row identities enter separately as FullVA-specific residual-bridge inputs.	No source-policy row, residual table, or fixed-tolerance run supplies these interfaces.
Multistep start values and multiplier-error recursion	For the one-step accepted map, use the single reduced initial state and absorb stage multipliers into the 132-row stage solve and endpoint perturbation chain.	No BLieDF start-up theorem, multiplier-history order theorem, or residual-row trajectory-error theorem is imported.
Coupled error recursion and Lie-algebra error transfer	Use the accepted-branch stability recurrence and the reporting-map perturbation after the local defect has been proved for the same branch.	No BDF $p = k$ theorem, no source-paper superiority claim, and no residual-to-error transfer theorem for mechanism-coverage rows.

charged, but it supplies no numerical, asymptotic, or hidden-constraint estimate for the present method. The following table states which reference estimates are not imported and gives the replacement argument used in this proof for each corresponding role.

Table 12: No-estimate-transfer table for the Wieloch–Arnold constrained-BDF proof.

Reference ingredient	Forbidden estimate transfer	Replacement in this proof
BLieDF local-truncation algebra	No BLieDF Taylor constant, BDF $p = k$ conclusion, or multistep defect bound is a premise for Theorem 1.	The local h^7 input is the three-stage Gauss defect plus the implemented 132-row residual bridge evaluated at the lifted Gauss stage.
BCH perturbation and Lie-algebra recursion	The BCH constants and global Lie-algebra recursion constants from the reference paper are not imported.	SO(3) chart and reporting constants are retained compact-tube constants in Assumption 1 and the reporting-map lemmas.
Hidden-constraint and multiplier estimates	The reference multiplier-history estimate and hidden-constraint estimate are not used to control lower-pair multipliers or endpoint closure here.	Lower-pair and endpoint effects enter through the FullVA residual rows, endpoint/KKT closure lemma, and retained right-inverse interfaces.
Reference correspondence table	The table is not a theorem input, not a certificate for primitive Taylor subterms, and not a source-policy or full-TFE replacement argument.	It is only an ordering rule enforcing the proof order: fixed object, smooth-branch insertion, local defect, and local-to-global propagation.

Reference-to-FullVA translation map. The reference proof is used only as a partial order on obligations, and each translated obligation below is discharged or retained by a present manuscript object. R1 fixes the discrete Lie-group object before expansion; here this is the accepted Gauss-predictor branch, $F_{A,h}$, Z_G , $\mathcal{E}_h$, C_h , the row scaling, and the fixed proof norm. R2 inserts the smooth exact branch into that same object; here this produces the Gauss defect $C_G h^7$ and the full residual value $\|F_{A,h}(Z_G; y)\| \leq C_R h^7$ from the direct $96 + 36$ bridge. R3 converts the fixed-object defect to a nearby accepted algebraic solution; here the conversion is Lemma 39 together with the endpoint-closure and inexact-Newton perturbation lemmas, using constants C_A , C_E , and C_N fixed on the compact tube. R4 applies a stability recursion only after the local defect is fixed; here this is the first-exit/local-to-global transfer and the reporting-map consequence. This translation is one-way: every right-hand object is proved or retained independently in the present FullVA proof. If any right-hand lemma or retained interface failed, the constrained-BDF reference theorem would not repair it. The reference therefore supplies a proof-order dependency graph, not a theorem premise,

source-policy substitute, primitive Taylor bound, solver-policy discharge, or residual-to-error promotion.

Reference constrained-DAE slot map. The constrained part of the reference proof has two distinct DAE slots: the local truncation estimate for the fixed Lie-group equations and the later hidden-constraint/multiplier estimate obtained from constraint differences before the coupled recursion is closed. In the present proof these slots are not merged. The local-defect slot is occupied by the Gauss defect plus the full 132-row residual bridge and Lemma 39. The constrained-DAE slot is occupied by the retained compact FullVA chart, lower-pair row certificate, endpoint KKT/right-inverse hypotheses, and branch-stability interfaces, together with the fact that multipliers are stage unknowns in $F_{A,h}$, not separate reported histories. Consequently the reference multiplier argument is not imported as a multiplier-order theorem, and the FullVA proof does not bypass the constraint layer: it records exactly where those DAE hypotheses are consumed before Lemma 44 and the reporting-map estimate are applied. The slot analogy is not symmetric: the BDF hidden-constraint estimate cannot replace the FullVA lower-pair position/velocity/acceleration row certificate or endpoint KKT interface, and the FullVA direct residual bridge cannot be read as proving the reference paper’s multiplier-history theorem.

Reference-style constrained-slot contract. At the theorem-interface level the reference analogy is the following four-slot contract, with every slot discharged or retained by a present FullVA object. C1, the local-error slot, is filled by the smooth Gauss defect, the implemented 132-row residual bridge, and Lemma 39. C2, the constrained algebraic slot, is filled by the lower-pair position-, velocity-, and acceleration-row certificate, the retained compact chart/right-inverse interfaces, the endpoint KKT closure, and the fact that multipliers are solved as stage unknowns inside $F_{A,h}$. C3, the one-step perturbation slot, is filled by the Kantorovich stage-root, endpoint, and inexact-Newton perturbation lemmas on the same compact branch and proof norm. C4, the global-recursion slot, is filled only after C1–C3 by the accepted-branch stability recurrence and reporting-map estimate. No slot imports a BLieDF constant, multiplier-history estimate, BDF zero-stability theorem, finite residual log, or source-policy row. Conversely, if any of C1–C4 failed for the present FullVA object, the Wieloch–Arnold theorem would not close that slot for us; the contract is a proof-order guard, not an extra theorem premise or an estimate-transfer device.

Typed-slot causality rule. The C1–C4 labels above are not merely a

narrative outline. They fix the causal order in which estimates may be consumed. A later slot may use the output of an earlier slot only after the earlier slot has been proved or retained on the same FullVA object; it may not be read backward to prove that earlier slot. Thus the global recursion in C4 cannot certify the local residual bridge in C1, a finite reported order row cannot certify the P6 solver-scale envelope in C3, and a mechanism residual or reaction table cannot certify the P7 residual-to-error theorem outside the stated output scope. Likewise, the separate primitive 162-term Taylor route cannot borrow the T2 full-map Kantorovich conclusion to prove its own primitive lift hypotheses. This slot-local causality is the present proof's operational use of the constrained-BDF reference discipline: it preserves a forward chain from fixed discrete object, to smooth-branch insertion, to local defect, to global/reporting consequence, without importing any estimate from the reference theorem or from downstream diagnostics.

Lemma 1 (Reference-order contract and no-estimate transfer). *The use of the Wieloch–Arnold constrained-BDF proof in this section is a one-way proof-order contract. It allows the following typed ordering rule and no other theorem input: first fix the discrete Lie-group object, then insert the smooth branch into that fixed object, then prove the local defect for that object, then propagate the accepted-branch error. This is an expository ordering rule, not a mathematical premise of Theorem 1; deleting the reference comparison would leave the theorem dependent only on the FullVA assumptions and lemmas stated here. The reference theorem's technical-neighborhood and start-value clauses are not imported either; in the present FullVA theorem they are replaced by the retained compact-tube/branch domain, endpoint/right-inverse, solver-scale, and same-reduced-initial-state clauses, all fixed before any diagnostic table is read. In the present FullVA theorem these four slots are instantiated respectively by the fixed row-scaled residual $F_{A,h}$ and endpoint map, the lifted Gauss stage Z_G , Lemmas 40 and 39 together with the endpoint and inexact-Newton perturbation lemmas, and finally Lemma 44 plus the reporting-map consequence. No BLieDF local-truncation constant, BDF $p = k$ theorem, BCH recurrence constant, hidden-constraint estimate, multiplier-history estimate, start-value theorem, primitive Newton–Euler Taylor sub-term bound, source-policy row, solver log, or residual-to-error transfer theorem is imported from the reference paper.*

Proof. Each estimate used in Theorem 1 is typed by the object on which it is proved. The constrained-BDF reference estimates are estimates for a

BLieDF multistep update, its Lie-algebra increments, its hidden constraints, and its multiplier recursion. The estimates needed here are estimates for the one-step Gauss/FullVA stage vector Z , the row-scaled 132-row map $F_{A,h}$, the endpoint-output map $\mathcal{P}_h$, and the accepted compact-tube branch. These objects have different unknowns, residual rows, endpoint maps, start-up structure, and global recursions, so a numerical constant or asymptotic conclusion attached to the BLieDF object cannot be substituted for a FullVA theorem interface.

The only invariant common to the two proofs is the order in which typed objects must be fixed and used. The present proof fixes $F_{A,h}$, its row ordering, row scaling, proof norm, endpoint map, and branch before any Taylor or stability estimate is invoked. It then inserts Z_G into that same object and obtains the local residual defect from the $96 + 36$ direct bridge. The residual defect is converted to a local stage-root and endpoint defect by the compact-tube Taylor/Kantorovich and endpoint lemmas, and only after those local estimates are available is the stability recursion applied. Thus the reference proof is used only to order obligations; the actual assumptions, constants, and conclusions are exactly those stated or proved for the FullVA object in this manuscript.

Consequently, if any FullVA slot above failed, the Wieloch–Arnold theorem would not close it: the BDF proof is not a replacement for the P1/P2 compact branch interfaces, the P6 solver-scale envelope, the P7 residual-to-error transfer theorem, or the separate primitive/Taylor lift obligations. This proves both directions needed for the contract: the reference-order analogy prevents estimates from being used out of order, and the no-estimate-transfer rule prevents reference-paper constants or conclusions from being used as missing FullVA hypotheses. $\square$

Reference-order Taylor discipline. The Taylor-related lesson from the Wieloch–Arnold proof is structural ordering discipline only; no Taylor estimate, primitive lift bound, or multiplier estimate is imported. Their Theorem 2 first inserts the analytical solution into fixed BLieDF equations and proves the local defect estimates; their Theorem 4 then uses constraint differences and a multiplier estimate before closing the coupled recursion. In the present one-step FullVA proof, the analogue of the first slot is fully occupied by T1 and T2 below: the smooth Gauss defect and the Taylor/Kantorovich expansion of the fixed 132-row map $F_{A,h}$. The reference proof separates this local-defect slot from the constrained multiplier/coupled-recursion slot. In

this manuscript, T1 and T2 fill only the local-defect slot for the fixed FullVA object; the later Gronwall step is a reduced-chart propagation step and imports no BLieDF multiplier, hidden-constraint, or coupled-recursion estimate. A literal primitive Newton–Euler Taylor proof would still have to prove, inside the present stage variables and row scaling, the non-circular root/direct primitive obligations P_{state} , P_{acc} , and P_{λ} . The actual P_{λ} rate is not borrowed from the reference paper: in the primitive dependency graph its PL4 propagation is instantiated only after independent P_{state} and P_{acc} lift inputs are available, and only then may the downstream P_{geom} and P_{gyro} reductions be used. Thus the active T2 full-map Taylor/Kantorovich estimate is established only for the fixed 132-row residual map $F_{A,h}$, where “established” means the direct T2 residual-defect input is discharged for that theorem object only, while the separate primitive subterm Taylor proof remains conditional at the analogous present FullVA lift/multiplier interfaces; the reference paper only motivates the ordering discipline, not those bounds.

Taylor-layer separation. There are three distinct Taylor-layer slots in this manuscript. T1 is the standard smooth Gauss collocation local-truncation layer: in the reduced chart the exact smooth branch gives the $C_G h^7$ defect for $\Psi_h^G - \varphi_h$. T2 is the theorem-level FullVA residual-map layer: after the implemented residual, row ordering, row scaling, proof norm, and branch have been fixed, Lemma 39 expands the single 132-row map $F_{A,h}$ in the full stage increment $\Delta = Z - Z_G$, uses the bridged value $\|F_{A,h}(Z_G; y)\| \leq C_R h^7$, and applies the compact-tube Kantorovich argument to transfer that formula defect to the accepted local stage root. T3 is the separate primitive Newton–Euler Taylor route: it decomposes the same 36 dynamic rows into state, acceleration, multiplier, geometry, and gyroscopic lift subterms. Only T1 and T2 are load-bearing in Theorem 1. T3 is a separate primitive residual-certificate specification for the same 36 dynamic rows: it is a stricter alternative factorization of the dynamic-row residual value and is useful as a finer verification route, but it is not the local truncation proof used by the theorem and it is not imported into the active residual-bridge estimate. The primitive subterm-bound inventory is therefore a conditional specification for that separate sufficient route, not evidence against the direct theorem route and not an additional summand in C_{loc} . Equivalently, the conditional Taylor templates and the actual primitive-route Taylor bounds are two separate mathematical statuses: the first gives a row-by-row reduction plan under named lift hypotheses, while the current absence of certified actual primitive-

route bounds prevents those conditional templates from being spliced into the direct residual-bridge proof. There is also no partial promotion by selecting apparently benign subterms: the theorem consumes one norm bound for the fixed dynamic residual block, and the primitive route contributes no termwise dynamic-block bound until the common lift and multiplier inputs are proved on the same compact tube. The active theorem already consumes the direct dynamic-row identity through the full 132-row residual bridge. Thus the Taylor analysis is not being omitted or replaced by a numerical record. The active Taylor argument is Lemma 39: Taylor’s formula with integral remainder expands the implemented 132-row residual about Z_G , the $O(h^7)$ residual value is supplied by the direct bridge, and the compact-tube inverse absorbs the quadratic remainder. Concretely, the Taylor variable is the full stage increment $\Delta = Z - Z_G$ of the scaled residual map $F_{A,h}$; the input is the single norm bound $\|F_{A,h}(Z_G; y)\| \leq C_R h^7$ after the row-block bridge has already supplied the 96-row non-dynamic $O(h^7)$ block and the 36 Newton–Euler zero rows. It is not a term-by-term Taylor expansion of the 162 primitive Newton–Euler subterms. The primitive Newton–Euler route is a stricter factorization of one block of that residual value, not a prerequisite for the theorem-level Kantorovich stage-root perturbation. This matches the reference-paper cadence at the level needed here: insert the smooth branch into the fixed discrete equations, prove the local defect in those equations, and only then propagate the accepted-branch error. The typed Taylor objects are therefore Eq. (40):

$$\begin{aligned}
\mathsf{T}_1 : \quad & \|\Psi_h^G(y) - \varphi_h(y)\| \leq C_G h^7, \\
\mathsf{T}_2 : \quad & \mathcal{I}_{\text{inv}}^{\mathsf{T}2}(K) \wedge \|F_{A,h}(Z_G; y)\| \leq C_R h^7 \implies \|\Psi_h^A(y) - \Psi_h^G(y)\| \leq C_A h^7, \\
\mathsf{T}_3 : \quad & (P_{\text{state}}, P_{\text{acc}}, P_\lambda, P_{\text{geom}}, P_{\text{gyro}}) \\
& \implies \mathcal{T}_{\text{D5}}^{162} = O(h^7) \text{ for all primitive D5 subterms.}
\end{aligned} \tag{40}$$

In this T_3 shorthand, P_{tube} is the already closed compact-tube constant slot; the five displayed antecedents are the remaining open primitive lift/reduction inputs. Here $\mathcal{I}_{\text{inv}}^{\mathsf{T}2}(K)$ denotes the retained full-map inverse, branch ball, and second-derivative bounds used by Lemma 39; the $\mathsf{T}2$ implication in Eq. (40) is not a residual-only implication without those compact-tube hypotheses. Theorem 1 consumes T_1 and T_2 . A completed T_3 would be a separate sufficient route only to the dynamic-row part of the same $C_R h^7$ bridge. It would not replace the full-map inverse, endpoint, inexact-Newton, or local-

to-global steps, and failure to complete T_3 is not the negation of the direct bridge. It only rules out treating primitive subterm bounds whose hypotheses have not been established as theorem inputs. Conversely, a separate completed T_3 certificate would not be an appendix shortcut that retroactively changes the present proof: it would be a replacement residual certificate for the dynamic block, after which the same full-map inverse, endpoint, inexact-Newton, and local-to-global steps would still have to be invoked on the same branch. Equivalently, a separate T_3 proof could only change the provenance of the 36-row dynamic-block contribution to the same $C_R h^7$ residual input. It would replace the direct P_5 zero-row certificate for that block, not add a second local-defect term, not bypass the T_2 full-map inverse/Kantorovich step, and not discharge the endpoint, P_6 solver-scale, P_7 residual-to-error, or global-transfer interfaces.

Table 13: Taylor-layer dependency map for the accepted proof route.

Slot	Fixed object and estimate	Theorem use	Explicit non-use
T1	Reduced smooth Gauss collocation defect in the retained chart, $\ \Psi_h^G(y) - \varphi_h(y)\ \leq C_G h^7$.	Positive local-defect summand before endpoint and solver perturbations.	Does not choose the FullVA branch, residual row scaling, endpoint map, or solver residual policy.
T2	Full implemented 132-row residual map $F_{A,h}$ in the stage increment $Z - Z_G$, with $\ F_{A,h}(Z_G; y)\ \leq C_R h^7$.	The load-bearing Taylor/Kantorovich step converting the bridged residual value to the accepted stage-root perturbation.	Does not decompose the dynamic block into primitive subterms and does not infer any primitive lift estimate.
T3	Separate 162-subterm primitive Newton–Euler Taylor inventory under $P_{\text{state}}, P_{\text{acc}}, P_\lambda, P_{\text{geom}}, P_{\text{gyro}}$.	No theorem use in the accepted route; it is only a separate alternative dynamic-block residual-certificate route for $C_R h^7$. It can replace the direct dynamic-block certificate only after the five antecedent primitives are proved on the same branch and the resulting dynamic block is assembled with the same 96-row non-dynamic certificate into the same 132-row residual bridge.	Cannot be added to T_2 , used to lower constants, remove the solver-scale condition, prove a residual-to-error theorem for reported mechanism rows, or claim primitive-route closure before those antecedents are proved.

Taylor-layer consumption rule: the reference-style ordering is fixed object first, smooth-branch insertion second, local defect third, and propagation last. In this manuscript that means T_2 consumes exactly one same-branch

dynamic-block residual certificate before the full-map Taylor/Kantorovich step is invoked. In the accepted theorem instance that certificate is the direct P5 substitution certificate together with the 96-row non-dynamic certificate, giving $\|F_{A,h}(Z_G; y)\| \leq C_R h^7$.

The D5 primitive route remains a replacement-certificate route, not a second Taylor layer. For the results claimed here it is only a template-level alternative: it supplies conditional templates for 162/162 subterms but no certified primitive Taylor bounds and five open lift/bilinear primitives. A future completed T3 proof on the same branch and proof norm could replace the dynamic-block certificate only inside the same 132-row residual bridge that feeds the T2 chain; it would not be added to T2, lower constants, discharge P6/P7, or shorten the endpoint and global-transfer steps. Definition 4 and Lemma 2 spell out this replacement certificate as a same-object closure target and use rule. The canonical status card for this separate route is Eq. (41):

$$S_{T3} = \left(\begin{array}{l} \text{conditional primitive templates 162/162,} \\ \text{certified primitive Taylor bounds 0/162,} \\ \text{open same-object primitive inputs 5,} \\ \text{theorem use: replacement only, not additive} \end{array} \right). \quad (41)$$

Here S_{T3} records the current status of the separate T3 route, not a new theorem assumption. Every later reference to primitive/Taylor nonclosure, replacement-only use, or non-additivity is tied to Eq. (41) unless a future manuscript version replaces it by a discharged same-object T3 certificate.

Definition 4 (T3 primitive-route certificate template). *For a fixed theorem instance, a completed T3 primitive-route certificate would be the same-object tuple in Eq. (42),*

$$\mathcal{T}_{T3} = (K, \chi, Z_G, F_{A,h}^{\text{dyn}}, \Pi_{D5}, \tau_h, \|\cdot\|_{\text{proof}}, C_{\text{state}}, C_{\text{acc}}, C_{\lambda}, C_{\text{geom}}, C_{\text{gyro}}, \Gamma_{D5}, h_0), \quad (42)$$

chosen before any finite residual log, observed slope, direct residual-bridge output, or source-policy row is read. Here K and χ are the same compact proof tube and chart used by Theorem 1, Z_G is the lifted smooth Gauss stage, $F_{A,h}^{\text{dyn}}$ is the 36-row Newton–Euler block of the implemented residual map, Π_{D5} is the fixed primitive D5 inventory map, and $\tau_h = (\tau_{1,h}, \dots, \tau_{162,h})$ is the ordered primitive Taylor-subterm vector. The tuple must certify, uniformly for $0 < h \leq h_0$ on the same branch, the two root lifts in Eq. (43),

$$P_{\text{state}} : \|\delta S\| \leq C_{\text{state}} h^7, \quad P_{\text{acc}} : \|\delta A\| \leq C_{\text{acc}} h^7, \quad (43)$$

with P_{acc} in the unweighted acceleration norm used by the D5 Newton–Euler rows. It must then propagate P_λ , P_{geom} , and P_{gyro} only forward from those root lifts, produce all 162 bounds in Eq. (44),

$$\|\tau_{j,h}(Z_G; y)\| \leq C_j h^7, \quad 1 \leq j \leq 162, \quad (44)$$

and use the fixed finite assembly bound in Eq. (45),

$$\|F_{A,h}^{\text{dyn}}(Z_G; y)\|_{\text{proof}} \leq \Gamma_{\text{D5}} \sum_{j=1}^{162} \|\tau_{j,h}(Z_G; y)\| \quad (45)$$

to obtain a dynamic-block residual certificate. Only after this dynamic-block certificate is assembled with the same 96-row non-dynamic certificate, row scaling, proof norm, endpoint map, and branch rule could it replace the direct P5 dynamic-block certificate in the T2 full-map Taylor/Kantorovich proof. The finite D5 inventory is the D5 term-budget record for this replacement certificate: it records 162 Taylor subterms with zero term-level bounds currently certified on the separate primitive/Taylor route, and the present theorem therefore asserts no term-level $O(h^7)$ bounds for those subterms. The tuple is therefore ordered like the Wieloch–Arnold constrained-BDF proof: fix the discrete object, insert the smooth branch, prove local primitive defects in that object, assemble them, and only then enter the residual-to-root and global-error steps. It is not allowed to use Lemma 39, the direct D5 zero-row certificate, the already assembled 132-row residual estimate, a reported grid error, a finite Newton log, or a source-policy row as an input to the primitive lifts. This certificate template is not discharged here: the available status is 162/162 conditional inventory coverage and 0/162 actual primitive Taylor bounds for the separate T3 route.

Lemma 2 (T3 replacement-certificate use rule). *Fix one theorem instance and the same compact branch, proof norm, row scaling, endpoint map, lifted Gauss stage Z_G , and implemented residual $F_{A,h}$ as in Theorem 1. If a completed T3 primitive-route certificate satisfying Definition 4 were available, its only admissible theorem-level use would be to replace the direct P5 dynamic-block certificate before the full 132-row residual bridge is formed. It could then be combined with the same 96-row non-dynamic certificate, branch rule, row scaling, and proof norm to produce a new same-tuple residual-value bound for $F_{A,h}(Z_G; y)$. It could not be appended to the direct certificate, averaged with it, used to lower constants, remove the P6 solver-scale envelope, prove*

a P7 residual-to-error boundary, or bypass the endpoint, inexact-Newton, or local-to-global steps. Until such a T3 tuple is proved, the primitive inventory supplies no theorem input. Equivalently, the active residual bridge and any future T3 certificate have a one-slot replacement type: they may produce the same full-residual input, but neither route is an inverse for the other route’s internal hypotheses.

Proof. The definition of a T3 certificate has the same fixed object as the accepted FullVA theorem instance and its output is only a dynamic-block residual certificate for the Newton–Euler rows evaluated at Z_G . The theorem-level residual bridge has a different type: it requires one dynamic-block certificate and the existing 96-row non-dynamic certificate to obtain a single full residual-value estimate for the row-scaled map $F_{A,h}$. Thus a completed T3 route would have the same input type as the direct P5 dynamic certificate and could occupy the same slot, but it would not create a second local-defect summand.

After that slot is filled, Lemma 39 still requires the full-map inverse and Taylor/Kantorovich hypotheses on the same branch, and the proof still requires the endpoint-closure, inexact-Newton, and local-to-global transfer lemmas. None of those hypotheses is a dynamic primitive subterm bound. The single residual-certificate slot in Eq. (253) therefore makes the two routes alternative provenance for the same full residual input, not cumulative evidence. Since the present manuscript records no completed T3 tuple and no actual primitive Taylor bounds, the primitive inventory remains a conditional replacement route only. The non-inference used here can be written as

$$\begin{aligned} R_{\text{dir}}(Z_G; y) &:= (\|F_{A,h}(Z_G; y)\| \leq C_R h^7) \text{ or } (D_{36}(Z_G) = 0), \\ \mathcal{P}_{\text{T3}} &:= (P_{\text{state}}, P_{\text{acc}}, P_{\lambda}, P_{\text{geom}}, P_{\text{gyro}}) \wedge \{\|\tau_{j,h}(Z_G; y)\| \leq C_j h^7\}_{j=1}^{162}, \\ R_{\text{dir}}(Z_G; y) &\not\Rightarrow \mathcal{P}_{\text{T3}}. \end{aligned} \tag{46}$$

Equation (46) is not an additional assumption; it is the type separation just proved. A full residual-value estimate is the input required by the T2 full-map perturbation lemma, whereas the displayed primitive lift and subterm bounds are sufficient antecedents for one possible future T3 proof of the dynamic-block part of that input. $\square$

Primitive-route obligation ladder. If the separate T3 route is pursued, its proof obligations have the same forward-only ordering discipline as the

Wieloch–Arnold constrained-BDF proof: define the fixed discrete object, insert the smooth branch, prove local defects in that object, and only then propagate them through the error recursion. For the present FullVA dynamic block the corresponding implication chain is Eq. (47):

$$\begin{aligned}
(P_{\text{state}}, P_{\text{acc}}) &\implies P_{\lambda} \implies (P_{\text{geom}}, P_{\text{gyro}}) \\
&\implies \{\tau_{j,h}(Z_G; y) = O(h^7)\}_{j=1}^{162} \\
&\implies \|F_{A,h}^{\text{dyn}}(Z_G; y)\| = O(h^7).
\end{aligned} \tag{47}$$

All arrows in this ladder are one-way sufficient implications on the same compact branch, with the same stage variables, row scaling, and proof norm as the theorem-level residual map. The last arrow is the finite-sum implication proved below; the earlier arrows are exactly the primitive lift and reduction obligations listed as open in the T3 status map. Therefore a primitive-route proof for that route cannot start from the assembled 132-row residual, from the direct D5 zero-row certificate, from a finite grid order, or from the Wieloch–Arnold theorem and then infer the primitive lifts backward. Conversely, once all arrows in this ladder were proved, the output would still be only a sufficient dynamic-block residual certificate to be inserted into the existing T2 full-map perturbation chain.

Taylor accounting rule. The useful methodological constraint inherited from the reference proof is not merely that a Taylor expansion appears, but that the scheme, variables, comparison solution, and constants are fixed before the expansion is used. In the present proof, after K , the reduced chart, the row-scaled residual $F_{A,h}$, the proof norm, the Gauss-predictor branch, and the endpoint map are fixed, all Taylor constants in the load-bearing lemmas are compact-tube constants of that fixed object. They are not calibrated from h -sweeps, Newton logs, source-policy rows, or mechanism residual diagnostics. The Taylor variable for T2 is $\Delta = Z - Z_G$, and the D5 direct certificate is a base-point residual identity at Z_G : the zero Newton–Euler rows enter before the full-map Taylor expansion, not as consequences of the stage-root perturbation. By contrast, a primitive D5 Taylor proof would have to prove the P_{state} , P_{acc} , and P_{λ} lift bounds and their downstream geometry/gyroscopic reductions independently before invoking the theorem-level local-defect result; it cannot use the T2 conclusion to prove those primitives. This is the anti-circularity rule that keeps the reference-style Taylor discipline strong while keeping the separate primitive route separate from the accepted direct route.

Full-map Taylor sufficiency. The load-bearing Taylor estimate is a statement about the differentiable family $F_{A,h} : Z \mapsto \mathbb{R}^{132}$ after row scaling and proof-norm selection, not about a coordinate projection onto primitive symbolic subterms. The finite-dimensional residual inverse in Lemma 39 maps the full bridged residual value $R_h = F_{A,h}(Z_G; y)$ to the stage increment on the same branch. Therefore the proof needs a uniform full-map inverse, a uniform full-map second derivative bound, and the bridged value $\|R_h\| \leq C_R h^7$; it does not need a right inverse from the 132 implemented rows to the 162 primitive Newton–Euler subterms. Such a primitive projection would be a strictly stronger verification route. It is deliberately not part of the theorem instance, and no theorem conclusion is used to infer it. The direct D5 zero-row identity is consumed only in the base-point value R_h : it supplies the dynamic block of $F_{A,h}(Z_G; y)$ before the Taylor expansion is taken. It is not used as a derivative estimate for $D_Z F_{A,h}$, not used as a bound on $D_Z^2 F_{A,h}$, and not used to choose the Kantorovich ball. Those quantities remain full-map compact-tube inputs of Lemma 39, fixed before the direct residual bridge is inserted.

This gives the active Taylor sufficiency statement used by the theorem: once the fixed scaled map $F_{A,h}$ has uniform J_h^{-1} and $D_Z^2 F_{A,h}$ bounds on the compact branch, and once the direct bridge gives $\|F_{A,h}(Z_G; y)\| \leq C_R h^7$, Taylor’s formula for the full map is enough to produce $\|Z_A - Z_G\| = O(h^7)$. A term-by-term primitive proof of the 162 Newton–Euler subterms would be a different way to certify part of the same base-point residual value; it is not an additional hypothesis of this full-map Taylor argument.

Lemma 3 (Conditional finite-sum Taylor implication for the D5 route). *Let $\tau_{j,h}(Z_G; y)$, $1 \leq j \leq 162$, denote the primitive Newton–Euler Taylor subterms recorded in the D5 primitive-route specification, grouped by the fixed row ordering into the 36 dynamic residual rows. Suppose that, on the same compact branch, with the same row scaling and proof norm used for $F_{A,h}$, the five primitive lift obligations in Eq. (48)*

$$P_{\text{state}}, \quad P_{\text{acc}}, \quad P_{\lambda}, \quad P_{\text{geom}}, \quad P_{\text{gyro}} \quad (48)$$

are proved with constants independent of h , and that each recorded primitive subterm satisfies Eq. (49),

$$\|\tau_{j,h}(Z_G; y)\| \leq C_j h^7, \quad 1 \leq j \leq 162. \quad (49)$$

Then the D5 dynamic block assembled from those primitive subterms satisfies Eq. (50),

$$\|F_{A,h}^{\text{dyn}}(Z_G; y)\| \leq C_{\text{D5,prim}} h^7 \quad (50)$$

with $C_{\text{D5,prim}}$ depending only on the finite row assembly operators, the fixed norm-equivalence constants, and $\sum_{j=1}^{162} C_j$. This is a sufficient primitive-route implication for the dynamic-row part of the theorem's stage-residual input. It is not an inverse projection from the already assembled 132-row residual estimate back to primitive subterms, and it is not additive proof evidence once the direct D5 zero-row certificate has already supplied the same dynamic block.

Proof. Each $\tau_{j,h}$ is already an expanded primitive Taylor summand: product and bilinear Taylor reductions, multiplier-geometry reductions, and gyroscopic mean-value terms have been assigned to the finite D5 inventory before this lemma is invoked. After the branch, row ordering, row scaling, and proof norm have been fixed, assembling those already-expanded summands into the 36 dynamic rows is only a fixed finite row aggregation. On a fixed finite-dimensional row space, all block norms are equivalent; hence there is a constant Γ_{D5} , independent of h and of the accepted step index, such that the dynamic block assembled from the primitive vector $\tau_h = (\tau_{1,h}, \dots, \tau_{162,h})$ obeys Eq. (51),

$$\|F_{A,h}^{\text{dyn}}(Z_G; y)\| \leq \Gamma_{\text{D5}} \sum_{j=1}^{162} \|\tau_{j,h}(Z_G; y)\|. \quad (51)$$

The assumed primitive bounds give Eq. (52),

$$\|F_{A,h}^{\text{dyn}}(Z_G; y)\| \leq \Gamma_{\text{D5}} \left(\sum_{j=1}^{162} C_j \right) h^7, \quad (52)$$

so $C_{\text{D5,prim}} = \Gamma_{\text{D5}} \sum_{j=1}^{162} C_j$ proves the displayed dynamic-block bound. The argument is one-way: a small assembled residual does not identify or bound each primitive subterm, because the finite assembly may contain cancellations or many-to-one projections. Therefore this lemma records the conditional primitive-route implication only. In the accepted theorem the dynamic block is already supplied by the direct D5 substitution certificate, so this conditional primitive implication is not combined with that direct certificate to strengthen C_R or to remove any retained interface. $\square$

Lemma 4 (Taylor-route separation). *For Theorem 1, the local Taylor object is the implemented scaled residual map $F_{A,h}$ on the full stage variable Z , expanded about the lifted smooth Gauss stage Z_G . In the theorem invocation this Taylor object is used only after Lemma 40 supplies the full-residual estimate in Eq. (53),*

$$\|F_{A,h}(Z_G; y)\| \leq C_R h^7 \quad (53)$$

on the accepted compact tube. The separate primitive Newton–Euler Taylor route is a different proof factorization: it would first prove uniform $O(h^7)$ lift bounds for P_{state} , P_{acc} , and P_λ , then use those bounds to close P_{geom} and P_{gyro} , and only after that certify the 162 primitive dynamic subterms. Consequently, once Lemma 40 independently supplies the full residual estimate, an open primitive lift obligation cannot invalidate the theorem-level Taylor/Kantorovich route; conversely the theorem-level route does not certify any primitive subterm bound. The only projection used by the theorem is the fixed finite-dimensional block projection separating the 96 non-dynamic rows from the 36 Newton–Euler rows before the full residual norm is formed; it is not a projection from the full residual estimate back onto the 162 primitive symbolic terms. Any primitive-route proof for that route would therefore be a replacement proof for the dynamic-block portion of the same full-residual hypothesis, not an additional theorem layer or a shortcut around the full-map perturbation lemma.

Proof. The reference BLieDF proof first fixes the reconstruction and discrete equations, inserts the smooth solution into that fixed scheme, estimates the local truncation defect, and only then derives the Lie-algebra/coupled error recursion. The analogue used here fixes the $\text{SO}(3)$ retractions, row scaling, FullVA stage variables, endpoint reconstruction, and accepted branch before any Taylor estimate is invoked. In the theorem invocation, after Lemma 40 has supplied the $C_R h^7$ residual input from the 96 non-dynamic rows and the 36 direct Newton–Euler rows on the same branch, Lemma 39 applies Taylor’s formula with integral remainder to the single smooth map $F_{A,h}$ in the full increment $\Delta = Z - Z_G$. The inverse and quadratic-remainder constants are compact-tube constants of this full map, so the stage-root perturbation uses only that full-residual defect and the retained branch/tube hypotheses. This inverse use is local-stage, not residual-to-error: it maps the fixed 132-row stage residual perturbation near Z_G to a nearby stage root on the accepted Newton branch before endpoint reporting. It does not map reported mechanism residual histories, reaction residuals, or constraint-monitor rows to tra-

jectory errors. Such an output-level transfer would require the separate P7 fixed-operator theorem and an h -independent constant C_{P7} for the reported residual/error objects. Therefore closing the T2 full-map Taylor/Kantorovich step supplies only the local $C_A h^7$ stage-root term in the four-term defect sum; it does not prove or replace the P7 boundary theorem. The contraction proof is therefore a full-map Taylor argument: it expands $F_{A,h}(Z_G + \Delta; y)$, inverts $D_Z F_{A,h}(Z_G; y)$ on the accepted branch, and controls the integral quadratic remainder in the same 132-row proof norm. It never projects the resulting stage increment back into primitive subterm errors, and it never treats aggregate cancellation in the 132-row residual as evidence that a primitive term-by-term Taylor proof is closed.

The primitive route rearranges only the proof of the 36 dynamic-row residual value. Its dependency graph starts from the stage state, acceleration, and multiplier lifts; the geometry and gyroscopic reductions are downstream of those lifts. The primitive-route reduction records rules for the 162 subterms but not the required uniform lift bounds, so no primitive subterm is used as a theorem input. This is a route-separation statement rather than a gap in the active theorem: the theorem consumes the full-residual bridge, while the primitive route remains a separate finer implication record for one input to that bridge. $\square$

Assumption 1 (Accepted-path regularity and perturbation scale). *On the smooth cylindrical-chain benchmark over a fixed interval $0 \leq t \leq T$, the active lower-pair constraint Jacobian has constant full row rank, the constrained mass matrix restricted to the tangent space is nonsingular, and the index-3 FullVA equations admit a smooth reduced Lie-group chart y with a C^7 vector field $\dot{y} = f(y, t)$. Write K for the retained compact reduced-state proof tube on this branch; the stage neighborhoods used below are contained in the corresponding compact stage tube over K . This is notation for the retained theorem domain, not a posteriori selection from the reported grids. The implemented three-stage accepted residual is considered on the Gauss-predictor branch neighborhood in the accepted compact proof tube, with the stage Jacobian at the lifted Gauss predictor uniformly invertible and with the local derivative bounds required by Lemma 39. The existence of the local accepted root Z_A is then the conclusion of that Kantorovich perturbation lemma, not an additional assumption here. This is a compact-tube inverse condition on the Gauss-predictor branch; finite rank probes or pointwise solved-stage Jacobian checks are not used as a compact-tube inverse proof. Compact-inverse*

quantifier discipline: the compact tube, Gauss-predictor branch, proof residual norm, and inverse constants are fixed before any h -sweep, rank probe, solver log, or observed convergence row is inspected. Shrinking h_0 after these objects are fixed is allowed; enlarging the inverse constant or changing the tube after diagnostics changes the retained P2 interface rather than proving it. The inexact-Newton residual-to-stage-error estimate is also part of this local branch contract: it uses the strong local residual inverse stated in Lemma 41, equivalently uniform invertibility of the averaged Jacobian along the segment from the accepted root to the computed iterate under the small-neighborhood perturbation bound. This condition is not inferred from pointwise Jacobian invertibility alone. The final accepted one-step maps on this branch also satisfy the stability scale in Eq. (54):

$$\|\Psi_h^{\text{G6FVA}}(y) - \Psi_h^{\text{G6FVA}}(\bar{y})\| \leq (1 + C_s h) \|y - \bar{y}\| \quad (54)$$

for points in the compact proof tube and sufficiently small h , with a nonnegative stability constant $C_s \geq 0$ after enlarging the bound if needed. This is the local-to-global stability input used below; ordinary compact-tube Lipschitz continuity alone is not enough for the stated fixed-time global order transfer. The exact accepted trajectory remains a positive distance from the boundary of the same compact tube, and the local-to-global transfer uses the tube-retention bootstrap in Lemma 44. This is a conditional compact-tube hypothesis, not an invariant-region proof for remote nonlinear roots or residual-only mechanism rows. The endpoint-closure map has a same-branch closed endpoint-anchor family on the accepted branch, written locally as $z_ = z_*(y, h)$ with $E(z_*) = 0$, and satisfies the endpoint-closure local right inverse condition, together with the endpoint-ball margin, uniformly in (y, h) as in Lemma 6; this is a compact-tube splitting assumption and is not inferred from a single endpoint Jacobian right inverse without the small-neighborhood perturbation bound. The smooth FullVA lift used in Lemma 7 exists on the same tube. The endpoint-closure defect satisfies the hypothesis of Lemma 6. Inexact Newton solves are treated by Lemma 41: for the asymptotic order statement the accepted residual tolerance must scale no larger than $O(h^7)$. The fixed tolerances used in the numerical section are reported separately as finite-run solver data, not as an asymptotic solver-error proof. More precisely, the theorem uses a per-step branch-selected residual bound $\eta_h^{\text{tube}} \leq c_\eta h^7$ for every branch-selected solve to which the compact-tube one-step map and stability estimates are applied. On a reported trajectory this policy is recorded as the maximum $\eta_h^{\text{reported}} = \max_{0 \leq n < N_h} \eta_{h,n} \leq c_\eta h^7$ over the reported branch-selected*

Newton steps $0 \leq n < N_h$, where the corresponding output grid satisfies $t_n = nh$, $0 \leq n \leq N_h$, and $N_h h \leq T$. It does not assert that arbitrary Newton initializations, fixed tolerances, or remote nonlinear roots select that branch; the theorem branch is initialized by the Gauss predictor on the same accepted branch. Dense and sparse backends evaluate the same residual and Jacobian entries. The proof row-scaling map and residual norm are fixed before h , N_h , backend choice, and diagnostic logs; they introduce no hidden h -dependent row weights beyond the explicit h -factors written in the residual formulas. **Backend-norm compatibility rule.** A backend termination record is theorem-grade P6 evidence only after a predeclared h -uniform conversion to the proof residual norm of the same row-scaled map $F_{A,h}$ has been fixed on the accepted branch. In particular, if a backend reports a norm or proxy $\eta_{h,n}^{\text{be}}$, the proof may consume it only through an h -independent constant C_{be} satisfying Eq. (55):

$$\|F_{A,h}(\tilde{Z}_{A,n}; y_n)\|_{\text{proof}} \leq C_{\text{be}} \eta_{h,n}^{\text{be}} \quad (55)$$

for the same branch-selected iterate, residual rows, and row scaling used by Lemma 41. Without this pre-run compatibility certificate, backend norms, relative residuals, correction norms, and mixed scaled/unscaled residual summaries remain diagnostics; they cannot supply η_h^{tube} , choose c_η , or absorb a solver term into C_{loc} . The reported maximum over accepted steps is theorem-grade only after this backend-to-proof-norm conversion and the compact-branch solver envelope have both been fixed before the run; taking a maximum over finite solver logs afterward records a consistency diagnostic, but it does not create the uniform P6 envelope required by Lemma 41. The theorem retains this as a uniform solver-scale hypothesis. Fixed-tolerance runs and backend stopping records provide finite diagnostics only unless they are prebound to the proof norm and the h^7 envelope on the same compact branch.

Discharged residual input, not an assumption. The implementation-specific residual step used by the theorem is supplied after Assumption 1 has fixed the branch, tube, chart, proof norm, row ordering, and row scaling. Lemma 40 then gives the discharged pre-perturbation residual-defect input $\|F_{A,h}(Z_G; y)\| \leq C_R h^7$ for the implemented accepted residual. The implementation part of the theorem is not a compact-tube regularity assumption; it is the 96-row non-dynamic certificate together with the D5 direct-substitution certificate for the 36 Newton–Euler rows. After

the mathematical row identities are expanded into the implemented row ordering, the lifted Gauss stage has the defect bound in Eq. (56):

$$\begin{aligned} \|F_{A,h}(Z_G; y)\| &\leq C_{\text{blk}}\|F_{\text{nd},h}(Z_G; y)\| + C_{\text{blk}}\|F_{\text{dyn},h}(Z_G; y)\| \\ &\leq C_R h^7, \quad F_{\text{dyn},h}(Z_G; y) = 0. \end{aligned} \tag{56}$$

This displayed handoff says that the lifted Gauss stage has a defect bounded by $C_R h^7$; it is the point where the proved 96-row non-dynamic certificate and the P5 direct Newton–Euler certificate enter the proof. It is a one-step accepted residual defect statement only: a consumed row-local certificate on Z_G , not a compact-tube regularity hypothesis, not an empirical residual fit, and not a promotion of mechanism residual-to-error rows, source-policy rows, or fixed-tolerance production residuals into theorem interfaces.

Proof skeleton. The theorem is proved by a perturbation chain. The lemmas and final proof carry the estimates; the tables below only make the interfaces and nonuse rules explicit. First, the classical three-stage Gauss construction on the smooth reduced chart gives a local reduced-flow truncation term $C_G h^7$. Second, Lemma 7 lifts the reduced Gauss stage to the FullVA variables, and Lemma 39 consumes the proved 132-row residual-defect input: the 96 non-dynamic rows and the 36 Newton–Euler rows give $\|F_{A,h}(Z_G; y)\| \leq C_R h^7$, hence the accepted local stage root changes the endpoint by at most $C_A h^7$. Third, Lemma 6 bounds the endpoint closure correction by $C_E h^7$, with the closed endpoint-anchor family, endpoint-ball constants, and local right inverse kept explicit. Fourth, Lemma 41 converts the branch-selected solver residual into a perturbation of the full endpoint-output map $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$, bounded by $C_N \eta_h^{\text{tube}}$, and the theorem assumes $\eta_h^{\text{tube}} \leq c_\eta h^7$, so this term is $C_N c_\eta h^7$. Finally, Lemma 44 applies the $1 + C_s h$ stability scale and tube-retention bootstrap, giving the fixed-time reduced-chart bound with the constant in Eq. (57):

$$C_{\text{red}} = (C_G + C_A + C_E + C_N c_\eta) \Gamma_s(T). \tag{57}$$

Lemma 45 then converts this reduced-chart estimate to the position–velocity quantities reported in the numerical section by a compact-tube chart-to-output derivative bound. The proof-interface tables that follow do not add another proof source. They only locate the assumptions, the consumed row-local certificates, and the diagnostic rows that are deliberately excluded from

the theorem input set. The load-bearing residual-defect step is therefore upstream of all empirical trajectory evidence: the implemented 132-row residual is evaluated at the lifted exact Gauss stage Z_G , before the accepted stage root, endpoint closure, inexact Newton iterate, local-to-global transfer, and reporting-map conclusion are invoked. Observed h -sweeps, residual/reaction tables, and source-policy rows cannot be used to justify this stage-local certificate.

Constant hierarchy for the local-defect chain. The constants used in Theorem 1 are fixed in the same order as the proof objects. The row-local non-dynamic certificate gives C_{nd} . The fixed finite-dimensional block norm and row scaling give C_{blk} , hence $C_R = C_{\text{blk}}C_{\text{nd}}$ after the D5 direct Newton–Euler rows give $F_{\text{dyn,h}}(Z_G; y) = 0$. The compact-tube inverse constants in Lemma 39 then give $C_Z = 2MC_R$ and $C_A = M_EC_Z$. Lemma 6 supplies the endpoint-closure constant C_E , and Lemma 41 supplies $C_N = 2M_AM_N$. Therefore the one-step constant consumed by the global theorem is Eq. (58):

$$C_{\text{loc}} = C_G + C_A + C_E + C_Nc_\eta, \quad (58)$$

where C_G is the classical three-stage Gauss local-defect constant on the same smooth reduced flow. Finally Eq. (59) records the constants after the global stability and reporting-map steps:

$$C_{\text{red}} = C_{\text{loc}}\Gamma_s(T), \quad C_{qv} = C_{\mathcal{R}}C_{\text{red}}, \quad (59)$$

where $C_{\mathcal{R}}$ is the compact-tube Lipschitz constant of the reported (q, v) map. This hierarchy is not an additional assumption: it records the proof’s quantifier order. Each constant may depend on the fixed compact tube, chart, row scaling, inverse margins, endpoint right-inverse, stability factor, and c_η , but none is selected from fitted slopes, finite residual logs, separate primitive-route status, source-policy rows, backend timing, or reproducibility package state. In particular, C_R is fixed before the inexact-Newton term is introduced: P6 enters only through the separate C_Nc_η contribution to C_{loc} , and P7 residual-only mechanism diagnostics do not enter either constant.

Same-object proof invariant: every estimate in the local-defect chain is taken for the same transition index, same accepted compact-tube branch, same reduced chart, same implemented 132-row residual, and same fixed proof norm. The Gauss truncation term, residual-to-stage perturbation, endpoint closure term, and inexact-Newton term are

not assembled from different runs, different row scalings, or different Newton roots. This invariant is what permits the four $O(h^7)$ terms to be added in one triangle inequality before the local-to-global estimate is invoked.

Order-accounting checkpoint. The theorem uses the standard one-step/global order accounting: the accepted branch-selected map has a local defect bounded by $C_{\text{loc}}h^7$, and the same-initial-state reduced-chart reported-grid estimate is $C_{\text{red}}h^6$ because $N_h = O(h^{-1})$ transitions are accumulated through the stability recurrence. The reporting-map estimate for (q, v) has the same reported-grid h^6 order. Therefore the word sixth-order in the theorem refers to the same-initial-state reported-grid h^6 bound, while the h^7 quantities are local-defect and residual-envelope inputs. The observed smooth-window slopes 7.161/7.066 are downstream consistency evidence for the selected branch; they are not a seventh-order theorem, not a global $O(h^7)$ error bound, and not an additional solver-policy discharge. A stronger global-order statement would require a new theorem with its own local defect, stability, solver, and reporting-map estimates, not just larger finite-window fitted slopes.

Analytic inputs versus numerical evidence. The theorem is therefore not an empirical order fit. Before any numerical h -sweep is cited, each load-bearing estimate in the proof is quantified uniformly on the accepted compact proof tube: the Gauss local defect, the 132-row FullVA residual defect on Z_G together with the local stage-root inverse, the endpoint-closure perturbation, the inexact Newton tolerance scale, and the $1 + C_s h$ stability/tube-retention transfer. Numerical dynamic-order rows are used only after this analytic chain, as consistency evidence for the smooth accepted path. Mechanism-coverage rows verify that the same FullVA lower-pair vocabulary, reaction reconstruction, and closure checks are exercised on closed-loop examples, but they do not replace a residual-to-error transfer theorem. Diagnostic rows, source-policy rows, finite solver probes, and the local TFE comparator likewise cannot be inserted as analytic inputs. If one analytic input is not discharged, the corresponding statement remains a retained condition; it is not promoted by a small residual table or by an observed order fit.

Theorem status. The word conditional is used in the standard numerical-analysis sense: the result is a full local-defect and fixed-time global-error theorem once the listed compact-branch, implementation-identity, and solver-

scale hypotheses are in force. It is not a weak empirical claim. The proof contribution in the present manuscript is to make the 132-row Gauss6/FullVA residual enter that theorem through a square FullVA stage-root perturbation, including the Newton–Euler rows by the P5 direct route, and then to carry endpoint closure and inexact Newton as explicit $O(h^7)$ perturbations before the local-to-global transfer. What remains retained is the automatic satisfaction of the compact-tube and branch-solver hypotheses for every possible initialization or fixed-tolerance run; these are domain-of-validity conditions, not substitutes for the theorem’s error estimate. Thus the conditional qualifier is attached to the domain on which the branch-selected map is defined and stable. It is not attached to the implemented 132-row residual-defect input itself: that input is supplied by the certified non-dynamic rows and the D5 direct Newton–Euler route before the endpoint, Newton, and global-transfer estimates are applied. The resulting theorem is therefore conditional but not evidentiary: retained interfaces are quantified domain and solver-scale hypotheses with fixed constants, while discharged interfaces are algebraic inputs proved in the manuscript. The conclusion follows for every step size in that fixed domain; it is not a finite-sweep acceptance rule and it is not weakened by keeping residual-to-error and source-policy promotion outside the theorem.

Proof-scope invariant. Throughout the proof section, “proved” refers only to a row-local or algebraic certificate after the branch, norm, row order, and implemented residual map have already been fixed. “Retained” refers to a quantified theorem interface that is assumed on that same compact branch; it is not empirical evidence and it is not treated as discharged. Thus the P4 binding interface is retained, while P4 also supplies a proved 96-row non-dynamic certificate. The theorem can consume that certificate only after P1–P3 have fixed the branch, norm, row ordering, and implementation binding, so the P4 theorem interface is not a free-standing compact-tube discharge. P5 is the only direct-route dynamic-row interface discharged at the theorem-input level. The P7 residual-to-error boundary, residual-only mechanism rows, and source-policy rows are neither theorem inputs nor theorem conclusions here; they are explicit scope boundaries. Consequently an open primitive/Taylor or P7 entry cannot weaken the direct residual-bridge estimate, and a closed row-local certificate cannot be reused to prove P6, prove P7, establish source-policy readiness, or establish a full-TFE replacement unless a separate theorem proves that implication.

To make the proof interfaces explicit, Assumption 1 is used through separate interfaces rather than as a single opaque hypothesis. The theorem-interface list is P1–P6. The final entry is displayed separately as the P7 residual-to-error boundary because it is an output nonclaim boundary, not a theorem premise; the P7 shorthand is used only so the proof tables, residual tables, and reproducibility records point to the same transfer-scope rule. For reader traceability, P7 output nonclaim/residual-to-error boundary denotes this displayed output boundary and not an additional theorem premise.

- P1. smooth reduced-chart and FullVA-lift regularity on the accepted trajectory tube;
- P2. uniform local invertibility of the accepted stage residual Jacobian at the lifted Gauss predictor on the compact-tube branch, together with the local derivative bounds used to obtain the accepted root, the endpoint-closure local right inverse with a closed anchor and endpoint-ball margin, the strong local residual inverse used by Lemma 41, and the one-step stability scale $1 + C_s h$ with tube-retention bootstrap;
- P3. row identity between the mathematical 132-row residual and the residual differentiated and solved by the implemented residual path;
- P4. two distinct subslots: the branch/norm/implementation-binding side is retained as a theorem interface, while the 96-row non-dynamic implementation-defect certificate on the lifted Gauss branch is proved and consumed only through the P1–P3 branch, norm, and implementation binding fixed for the theorem;
- P5. the discharged direct Newton–Euler implementation-defect certificate that supplies the 36 dynamic rows on the lifted Gauss branch;
- P6. the theorem-level nonlinear-solver scale $\eta_h^{\text{tube}} \leq c_\eta h^7$ uniformly on the compact-tube branch-selected solver hypothesis, with the reported-grid maximum $\eta_h^{\text{reported}} = \max_{0 \leq n < N_h} \eta_{h,n}$ as its trajectory specialization, distinct from fixed-tolerance production runs. The theorem retains this as a uniform solver-scale hypothesis; fixed-tolerance runs provide finite diagnostics only. Changing the solver scale changes the theorem hypothesis and the local-defect constant, so this finite diagnostic status does not close P6;

boundary (P7 boundary). a separate residual-to-error promotion condition for residual-only mechanism rows. P7 is not used to prove the accepted smooth-branch theorem below; it is kept as an explicit non-claim boundary for four-link, slider-crank, and source-policy rows. No residual-to-error transfer the-

orem is assumed here; such a theorem would require a fixed reported residual/error operator on the same reported grid and an h -independent transfer constant, so the seven transfer obligations remain outside this theorem.

The notation is a compact proof-domain partition, not a list of empirical acceptance tests. Interpreted as a theorem instance, P1/P2/P3, the P4 binding side, and P6 are the retained admissibility interfaces: they fix the compact branch, proof norm, local inverse, endpoint closure, stability scale, implementation path, and solver scale on which the argument is allowed to run. P5 is different: it is the proved stage-local Newton–Euler identity on Z_G , supplying the 36 dynamic rows by direct substitution. The P4 notation has two theorem roles, distinct from the four-layer theorem-output reading guide below: the branch/norm/implementation-binding side is part of the retained admissibility interface, while the 96-row $O(h^7)$ non-dynamic residual estimate itself is proved and discharged. P4 supplies the proved 96-row non-dynamic part, but only after the same branch, norm, and implemented residual map have been fixed; it is not a standalone 132-row residual proof without the P5 dynamic handoff. The PC1–PC2 local residual route therefore refers to this particular direct residual-defect route under the retained P6 solver-scale condition, while PC4 records the residual-to-error exclusion. The output boundary (P7 boundary) is a declared transfer-scope clause: residual-only mechanism rows require a separate residual-to-error theorem before they can become trajectory-order evidence. The primitive dynamic symbolic oracle, the primitive/Taylor symbolic-defect route, external same-test claims, and reproducibility package claims likewise remain outside the theorem-level PC2 route until their own implications are proved.

With that admissibility map fixed, the proof is a one-way conditional implication rather than a backfilled interpretation of numerical evidence. The compact-tube hypotheses choose the branch and constants before h ; the direct residual bridge supplies the stage-row defect on Z_G ; endpoint closure and inexact Newton add controlled $O(h^7)$ perturbations; and the local-to-global transfer then gives the $O(h^6)$ position–velocity grid estimate. Finite step sweeps may identify the reported branch and check consistency, but they do not prove automatic branch selection for arbitrary initializations, do not discharge the theorem-level solver hypothesis from fixed production tolerances, and do not promote residual-only mechanism or external comparison rows through a residual-to-error theorem. Those retained interfaces delimit where

the theorem may be invoked; they are not hidden residual terms and they are not inferred from the theorem conclusion. Using the theorem-interface labels below: P1/P2/P3, the retained P4 binding side, and P6 describe the branch, implementation, and solver-scale interfaces under which the theorem is invoked; P4 also contains the proved 96-row non-dynamic certificate; P5 is the proved stage-local Newton–Euler identity; the direct residual bridge is used under the retained solver-scale condition, and the residual-to-error clause records only a transfer-scope exclusion; and the primitive dynamic symbolic oracle and primitive/Taylor symbolic-defect route remain outside the active route. The implication direction is fixed throughout, so no stage-row, solver-scale, or residual-to-error premise is inferred backward from observed error.

Theorem invocation criterion. Theorem 1 may be invoked only after retained P1/P2/P3 interfaces, the P4 binding interface, and the P6 solver-scale interface are fixed and the discharged P5 direct-route certificate is bound on the same accepted compact branch. P1–P2 supply the smooth FullVA lift, compact-tube inverse and derivative bounds, endpoint-closure, and stability/tube-retention hypotheses; P3 binds the implemented residual path, row ordering, row scaling, and proof norm; P4 contributes the proved 96-row non-dynamic certificate only under that binding; P5 contributes the 36 Newton–Euler rows through the direct substitution route; and P6 supplies the branch-selected solver scale $\eta_h^{\text{tube}} \leq c_\eta h^7$. The theorem output is only the reported-grid order-six position–velocity error on that same branch. It does not prove a solver-policy theorem, a P7 residual-to-error transfer theorem, source-policy reproduction, closed-loop residual promotion, or a full-TFE stage replacement. Therefore numerical support rows may corroborate the accepted branch after the theorem has been invoked, but they cannot backfill a separate invocation condition.

Retained/discharged input rule. The retained domain hypotheses are the compact-branch regularity, endpoint, stability, inverse, implementation-binding, and P6 solver-scale interfaces; the certified stage-residual input is assembled only after those retained interfaces are fixed, from the 96-row P4 non-dynamic certificate and the discharged P5 direct-substitution certificate for the 36 Newton–Euler rows. Thus the theorem does not merely assume the accepted implementation residual defect as an unproved premise; it consumes the implemented direct-route certificate for that input while retaining the compact-tube and solver-scale interfaces. Thus the direct PC2 residual-defect input is discharged under the retained P1/P2/P3 interfaces, the P4 binding interface, and the P6 solver-scale interface: the exact Gauss defect,

the 96-row non-dynamic lift, the 36-row Newton–Euler direct substitution, endpoint closure, inexact Newton perturbation, and local-to-global transfer are chained in this order. The theorem remains conditional on those domain and solver interfaces; it does not complete the primitive/Taylor route, prove a P7 residual-to-error boundary, or establish source-policy or full-TFE readiness.

Assumption admissibility note. Assumption 1 does not assume the theorem conclusion. In particular, P2 supplies a compact-tube stability scale and tube-retention bootstrap for the branch-selected map; it is independent of, and logically prior to, the discrete Gronwall step. It is not an assumed local defect estimate, not an assumption that Ψ_h^{G6FVA} is already $O(h^7)$ -close to the exact flow, and not inferred from observed sixth-order slopes, residual/reaction tables, or successful Newton logs. The local $O(h^7)$ defect is proved later from Gauss truncation, the certified 132-row stage-root perturbation, endpoint closure, and the retained P6 inexact-Newton scale. The stability/tube-retention input only permits that one-step defect to be iterated on the same compact tube; it does not choose C_{loc} , does not prove a theorem-level solver-policy result, and does not prove a P7 residual-to-error boundary.

Non-vacuity/admissibility note. Assumption admissibility note: Assumption 1 and the retained solver-scale hypotheses define the theorem domain rather than a shortcut to the theorem conclusion. The retained P1–P3 conditions, P4 binding condition, and P6 solver-scale condition define a local same-branch domain of applicability, not the error estimate itself. They require a smooth compact tube, Gauss-predictor branch selection, local inverse/endpoint/stability structure, implementation-path binding, and a branch-solver residual scale fixed before the asymptotic limit. They do not assume the $O(h^7)$ local defect, the $O(h^6)$ grid estimate, or the observed position–velocity orders. Once that domain is fixed, the proof still derives the local h^7 defect from Gauss truncation, the proved 132-row direct residual bridge, endpoint closure, and the retained P6 solver-scale term, and then derives the global grid bound by the stability/tube-retention transfer. Finite numerical rows may identify the reported branch and check consistency, but they are not part of this admissibility claim and do not turn the conditional theorem into an unconditional source-policy result.

Proof-domain partition. The Proof-domain partition is therefore explicit: P5 is the only stage-local dynamic identity proved by direct substitution. P1/P2/P3, the retained P4 binding side, and P6 describe the branch, imple-

Table 14: Assumptions and theorem scope.

Interface	Theorem role	Meaning	Explicit exclusion
P1–P2	Domain interface	Smooth FullVA lift, compact-tube inverse, endpoint-ball/right-inverse, and tube-retention stability remain theorem interfaces.	Finite sweeps, rank probes, and endpoint checks diagnose the selected branch only.
P3–P4	Implementation interface; P4 row certificate proved	Runtime formula/AD binding identifies the implemented direct residual path, and P4’s row-local certificate supplies the proved 96-row non-dynamic part of PC2.	Primitive symbolic oracle completion remains a separate verification route, not an input to the active direct-route 132-row bridge; P3/P4 also require the separate P5 dynamic-row handoff.
P5	Stage-local identity proved	The D5 direct-substitution certificate proves the 36 Newton–Euler residual rows vanish on Z_G .	This is stage-local and does not promote finite trajectories, source-policy rows, or residual-to-error claims.
P6	Solver-scale condition	The theorem uses the compact-tube branch-solver hypothesis $\eta_h^{\text{tube}} \leq c_\eta h^7$.	Finite solver probes do not prove a theorem-level solver-policy theorem or turn fixed tolerances into an asymptotic proof.
P7 boundary	Residual-output nonclaim	Residual-to-error promotion for mechanism-coverage and source-policy rows requires a separate theorem.	Residual tables remain diagnostics until all seven residual-to-error obligations close.
Stage-residual bridge / solver-scale hypothesis / residual exclusion	The direct bridge supplies the local residual route under the retained solver-scale hypothesis; residual-to-error promotion is not used.	This separation does not promote theorem-domain interfaces into proved theorem inputs.	

mentation, and solver-scale interfaces on which the theorem is invoked, with P4’s row-local certificate proving the 96 non-dynamic rows used in the PC2 stage-residual condition. P7 remains a separate residual-to-error boundary rather than a local-defect theorem premise. PC1–PC2 supply the local residual route under the retained P6 solver-scale hypothesis, while PC4 records the residual-to-error exclusion. These interfaces do not turn domain conditions or the P7 boundary into a proved solver-policy theorem, residual-to-error promotion, source-policy rows, or reproducibility package readiness claims. Thus PC2 is discharged only as a route-level stage-residual statement: the accepted stage-residual input is the 96-row non-dynamic certificate together with the 36-row P5 direct Newton–Euler certificate. It is not a statement that P4 by itself has become a full 132-row theorem input, and it is not a primitive symbolic-oracle or source-paper residual-replacement closure. No residual-to-error theorem is assumed in this route. The seven residual-to-error obligations remain outside the local-defect proof, so small mechanism residuals and reaction rows cannot enter PC2 or P6 as theorem inputs.

Table 15: Route and interface separation table for Theorem 1.

Route requirement	Satisfaction mode	What it supplies	What remains outside it
PC1 balance closure	D1–D2 weak-balance identities and row binding.	The Newton–Euler balance identities used by the direct residual route.	It does not prove compact-tube regularity, source-policy rows, or package readiness.
PC2 stage-residual condition	96-row $O(h^7)$ non-dynamic certificate plus the 36-row P5 direct-substitution zero-residual certificate.	The accepted 132-row stage-residual defect on the smooth Gauss lift.	It does not complete the primitive symbolic route or prove residual-to-error transfer.
PC3 solver-scale retention	P6 is retained as the theorem condition $\eta_h^{\text{tube}} \leq c_\eta h^7$.	The inexact-Newton perturbation is absorbed into the local h^7 defect.	It does not prove a solver-policy theorem or convert finite tolerances into an asymptotic proof.
PC4 residual demotion	Residual-to-error promotion is not used for the accepted theorem.	Residual-only four-link, slider-crank, and source-policy rows stay diagnostic.	It does not prove a separate residual-to-error transfer theorem or promote residual rows to trajectory-error evidence.

Route and interface separation. The separation table has a narrow mathematical role: it separates the PC1–PC2 local residual route together with the retained P6 solver-scale condition from the PC4 residual-to-error exclusion. PC3 is therefore a retained-condition reference, not an empirical proof and not an extra certificate. PC4 is recorded only as a retained

transfer-scope exclusion: residual-to-error promotion is not made. The term PC4 residual-to-error exclusion denotes precisely this non-use of residual-to-error promotion. These modes do not prove P1/P2/P3 theorem interfaces, the retained P4 binding interface, or the P6 solver-scale interface, do not prove a solver-policy theorem, do not prove a separate residual-to-error transfer theorem, and do not supply source-policy reproduction evidence or reproducibility package readiness.

Lemma 5 (Route-to-local-defect sufficiency on the same accepted branch). *Suppose P1/P2 fix the compact reduced-chart tube, P3/P4 fix the implemented residual path and the 96-row non-dynamic certificate, P5 supplies the 36-row Newton–Euler direct substitution, and P6 supplies the retained branch-solver scale on the same accepted branch. Then the PC1–PC2 local residual route, together with the retained P6 solver-scale hypothesis, supplies exactly the local-defect input used by Theorem 1: there is a compact-branch constant $C_{\text{loc}} = C_G + C_A + C_E + C_{Nc_\eta}$, independent of h , the reported grid length, backend, finite solver logs, and source-policy rows, such that Eq. (60) holds:*

$$\|\Psi_h^{\text{G6FVA}}(y) - \varphi_h(y)\| \leq C_{\text{loc}} h^7 \quad (60)$$

for every accepted branch-selected transition in the retained proof tube. PC4 is not a local-defect ingredient; it is the retained transfer-scope exclusion for residual-only mechanism and source-policy rows.

Proof. The proof uses the same proof-obligation ordering discipline as the reference convergence analysis, not an estimate transfer: first insert the smooth branch into the already fixed discrete equations, then estimate the one-step local defect, then move that defect through the constrained algebraic solve and endpoint map, and only afterwards apply the stability recursion. The variables and estimates differ from the BLieDF proof—this is a one-step Gauss FullVA residual rather than a multistep hidden-constraint recursion—so only the causal order of the proof is shared. Under P1/P2, the smooth reduced constrained flow and the fixed compact tube give the classical three-stage Gauss local defect in Eq. (61):

$$\|\Psi_h^G(y) - \varphi_h(y)\| \leq C_G h^7 \quad (61)$$

with C_G chosen from reduced-flow regularity before any implementation diagnostic is read. On the same transition define $\Psi_h^G = \mathcal{E}_h(Z_G)$, $\Psi_h^A = \mathcal{E}_h(Z_A)$,

$\Psi_h^E = \mathcal{C}_h(\Psi_h^A)$, and $\Psi_h^{\text{G6FVA}} = \mathcal{P}_h(\tilde{Z}_A)$, with $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$. These maps use the same state y , branch, chart, endpoint-output map, and proof norm; no table row or computed trajectory is used to identify them. Endpoint-map identity rule: the same local closure map $\mathcal{C}_h$, endpoint norm, endpoint-anchor family, and endpoint ball are used in $\Psi_h^E = \mathcal{C}_h(\mathcal{E}_h(Z_A))$ and $\Psi_h^{\text{G6FVA}} = \mathcal{C}_h(\mathcal{E}_h(\tilde{Z}_A))$. The inexact-Newton term therefore compares two outputs of one fixed map $\mathcal{P}_h$, not an unclosed exact endpoint against a closed computed endpoint and not two separately fitted projection operators. If the endpoint correction rule, endpoint norm, or endpoint anchor were changed between the exact-stage and inexact-stage terms, the four-term triangle would refer to different one-step maps and would require a separate equivalence lemma before the constants could be added. The PC2 condition supplies the implemented residual estimate in Eq. (62):

$$\|F_{A,h}(Z_G; y)\| \leq C_R h^7 \quad (62)$$

for the implemented 132-row residual on the lifted Gauss stage. The certificate combines the 96 non-dynamic rows identified by P3 and proved by P4 with the 36 Newton–Euler rows discharged by P5, all in the same chart, row order, row scaling, and proof norm fixed by P1–P3 plus the retained P4 binding side. Equivalently, after the fixed block norm and row-scaling map are chosen, write $F_{\text{nd}} = F_{\text{nd},h}(Z_G; y)$ and $F_{\text{dyn}} = F_{\text{dyn},h}(Z_G; y)$. Then Eq. (63) gives the fixed block-norm bridge:

$$\begin{aligned} \|F_{A,h}(Z_G; y)\| &\leq C_{\text{blk}}(\|F_{\text{nd}}\| + \|F_{\text{dyn}}\|) \\ &\leq C_{\text{blk}}(C_{\text{nd}} h^7 + 0) \leq C_R h^7. \end{aligned} \quad (63)$$

Here F_{nd} is the 96-row non-dynamic block and $F_{\text{dyn}} = 0$ is the 36-row Newton–Euler block supplied by the D5 direct-substitution certificate. This displayed bridge is stage-local at Z_G , before Newton termination, endpoint closure, and local-to-global transfer; it is not a finite-probe, primitive/Taylor, or residual-to-error promotion statement. Lemma 39 therefore gives the exact accepted-stage perturbation $C_A h^7$. Lemma 6 adds the endpoint contribution $C_E h^7$, and Lemma 41 adds the solver term in Eq. (64):

$$C_N \eta_h^{\text{tube}} \leq C_N c_\eta h^7 \quad (64)$$

under the retained P6 solver-scale condition. Thus the local sum consumed by the theorem is Eq. (65):

$$C_G h^7 + C_A h^7 + C_E h^7 + C_N c_\eta h^7, \quad (65)$$

with constants chosen on the compact branch before the h -limit. PC4 is a negative route rule: no residual-only mechanism, source-policy, or reaction row is inserted into that decomposition. Explicitly, the same-branch triangle inequality is

$$\begin{aligned} \|\Psi_h^{\text{G6FVA}}(y) - \varphi_h(y)\| &\leq \|\Psi_h^G(y) - \varphi_h(y)\| + \|\Psi_h^A(y) - \Psi_h^G(y)\| \\ &\quad + \|\Psi_h^E(y) - \Psi_h^A(y)\| + \|\Psi_h^{\text{G6FVA}}(y) - \Psi_h^E(y)\| \quad (66) \\ &\leq (C_G + C_A + C_E + C_N c_\eta) h^7. \end{aligned}$$

Equation (66) is a single same-branch triangle inequality; it does not splice quantities from different runs, endpoint policies, row scalings, or Newton roots. Read as a sufficiency implication, this statement is not an equivalence or a necessity theorem for all possible sixth-order routes. It does not choose constants from finite slopes, Newton logs, source-policy rows, or reproducibility package records. Hence every term used by the local-defect proof is either a theorem-domain interface or a proved row-local certificate on the same accepted branch, and the route is sufficient for the stated conditional theorem without promoting residual-only evidence. $\square$

Assumption-anchor map. The proof records where the compact-tube interfaces, the implemented residual path, the direct Newton–Euler identity, the solver-scale condition, and the residual-to-error boundary enter the manuscript before any dependency is consumed. Anchor presence is not a proof discharge: it does not prove compact-tube regularity, the solver hypothesis, primitive symbolic-oracle work, residual-to-error promotion, source-policy rows, full-TFE replacement, or reproducibility package completeness. Its purpose is to make the theorem domain explicit without replacing the local-defect proof, which still proceeds through the residual bridge, endpoint perturbation, inexact Newton perturbation, and fixed-time transfer.

Theorem input-output flow. The displayed inputs record the proof-input flow together with explicit exclusions. P1/P2 are consumed by the compact-tube and local-perturbation lemmas; P3/P4 identify the implemented residual path; P5 supplies the direct Newton–Euler rows; P6 enters only through the retained branch-solver scale; and P7 is recorded only in the exclusion column as a residual-to-error boundary, not as a theorem premise. The input-output flow does not prove a theorem-level solver-policy result, a residual-to-error transfer, source-policy readiness, full-TFE readiness, or external reproducibility package claims. In short, the input-output table lists theorem inputs and explicit exclusions only; it is not an additional proof argument.

Table 16: Theorem-interface and output-boundary anchors for Theorem 1.

Interface	Manuscript anchor	Theorem role	Consumption rule
P1	Assumption 1 and Table 19	Domain interface	Defines the smooth reduced-chart and FullVA-lift tube consumed by every local estimate. It is not discharged by finite trajectories or row checks.
P2	Assumption 1, Lemma 6, Lemma 41, Theorem 1, and Table 19	Domain interface	Provides compact-tube inverse, endpoint-ball/right-inverse, stability, and branch-solver hypotheses; rank probes and endpoint diagnostics do not close it.
P3	Tables 25 and 20	Implementation interface	Binds the implemented residual/Jacobian path; primitive symbolic oracle completion remains a separate non-active certificate track and is not used by PC2; this is also not source-paper residual replacement.
P4	Lemma 7 and Lemma 39, plus Table 20	Implementation interface; row-local certificate proved	Supplies the proved 96 non-dynamic rows only after the accepted branch, proof norm, and implementation binding have been fixed; the full 132-row defect also requires P5.
P5	Tables 23 and 24, plus Lemma 39 and Table 21	Stage-local identity proved	Supplies the 36 Newton–Euler rows through the D5 direct route on Z_G ; it does not promote P6, P7, or source-policy readiness.
P6	Lemma 41 and Theorem 1, plus Table 18	Solver-scale condition	Consumes only the compact-tube $\eta_h^{\text{tube}} \leq c_\eta h^7$ solver condition; finite solver probes are consistency diagnostics and do not instantiate theorem-level solver-policy proof.
P7	Table 41 and Table 30	Output nonclaim boundary	Records the residual-to-error boundary; residual, reaction, and common-reference/source-policy rows are scope records only. No residual-to-error transfer theorem is assumed here; seven transfer obligations remain outside the theorem.

Table 17: Theorem input-output flow for Theorem 1.

Dependency	First consumed by	Later theorem output	Explicit non-use
P1/P2 compact-tube data	Endpoint closure, local stage-root perturbation, inexact Newton, and local-to-global transfer lemmas.	Accepted-branch local defect, tube retention, and reduced-chart grid estimate.	Finite sweeps, endpoint checks, rank probes, or remote nonlinear roots.
P3/P4 implementation path	Formula-row/AD binding and the 96-row non-dynamic certificate before the stage-residual perturbation step.	Implemented direct residual path for the non-dynamic part of the 132-row defect.	Primitive symbolic oracle closure, source-paper residual replacement, or dynamic-row closure by P4 alone.
P5 direct dynamic rows	D5 direct-substitution identity evaluated on the lifted Gauss stage Z_G .	The 36 Newton–Euler rows needed for the direct stage-residual route.	Residual control at inexact Newton iterates, finite trajectories, or residual-to-error promotion.
P6 branch-solver scale	Inexact Newton endpoint contribution $C_{Nc_\eta}h^7$.	The accepted local defect remains $O(h^7)$ under the retained compact-tube solver hypothesis.	Fixed production tolerances or finite solver probes as theorem-level solver-policy proof.
P7 residual-to- error boundary	Only the theorem-use and exclusion tables, not the order proof.	Residual-only mechanism rows stay diagnostic until a separate transfer theorem closes.	Four-link, slider-crank, source-policy, or reaction residual tables as trajectory-error evidence.
PC1–PC2 route checks plus retained P6 and PC4 boundary	The conditional direct residual-defect route and residual-to-error exclusion.	PC1–PC2 close the local residual route under the stated theorem interfaces and retained P6 solver-scale hypothesis; PC4 records residual-to-error exclusion only.	Promotion of domain interfaces into proved theorem inputs.

Route separation. The PC1–PC2 labels are local route checks for the accepted residual-defect proof route under the retained P6 solver-scale hypothesis, while PC4 labels the residual-to-error exclusion; they are not replacements for retained P1/P2/P3 theorem interfaces, the retained P4 binding interface, the P6 solver-scale interface, the discharged P5 direct-route certificate, or the P7 output boundary and do not assert that every auxiliary proof route has closed. The active route uses the implemented residual/Jacobian path, the non-dynamic implementation certificate, and the D5 direct-substitution certificate at the lifted Gauss stage Z_G . The primitive symbolic/Taylor decomposition is retained only as a separate residual-certificate route. Likewise, finite AD-expansion probes and scaled-tolerance probes check the recorded implementation path and solver scale on finite windows, while Theorem 1 uses only the explicit branch-selected $\eta_h^{\text{tube}} \leq c_\eta h^7$ condition for its asymptotic claim. Thus PC3 is interpreted only through the retained P6 theorem condition: the finite scaled-tolerance probes are consistency checks, not a discharge of the theorem-level nonlinear-solver hypothesis. P6 is the retained theorem-level solver-policy condition. P6 solver boundary: finite one-step, short-trajectory, and h -scaled tolerance probes are finite diagnostics of branch-scale consistency only. P6 remains the explicit theorem condition $\eta_h^{\text{tube}} \leq c_\eta h^7$ on the compact-tube branch-selected solver hypothesis; these probes do not prove a theorem-level solver-policy theorem, do not turn fixed tolerances into an asymptotic proof, and do not assert global Newton convergence or remote-root selection.

Definition 5 (P6 solver-policy certificate template). *For a fixed theorem instance, a theorem-level P6 solver-policy certificate would be the same-branch tuple in Eq. (67),*

$$\mathcal{T}_{\text{P6}} = (\mathcal{N}_h, \mathcal{S}_h, \|\cdot\|_{\text{proof}}, c_\eta, h_0, r_N, M_N, \mathcal{B}_N). \quad (67)$$

and this tuple is fixed before any residual log or error table is inspected. Here $\mathcal{N}_h$ is the nonlinear-solver map initialized from the accepted Gauss predictor, $\mathcal{S}_h$ is the stopping rule, and $\|\cdot\|_{\text{proof}}$ is the same proof residual norm used in Lemma 41. The tuple must certify the following uniform facts on the same compact branch for every $0 < h \leq h_0$ and every accepted transition in the theorem domain:

1. *the Newton iterates selected by $\mathcal{N}_h$ remain in the branch ball $\mathcal{B}_N$ of radius r_N attached to the strong local residual inverse bound M_N ;*

Table 18: P6 solver-policy interface table for Theorem 1.

Retained object	Consumed by	Current diagnostic	Non-discharge
Branch residual scale $\eta_h^{\text{tube}} \leq c_\eta h^7$	Theorem 1 and Lemma 41.	Finite one-step and short-trajectory scaled-tolerance probes.	No finite window proves a theorem-level solver-policy theorem for every accepted step. Logged residuals do not create a second residual budget, change the proof norm, or prove P6 as a uniform solver theorem.
Proof residual norm and implementation binding	The residual norm used in Lemma 41 after the accepted chart, row ordering, and residual scaling are fixed.	Implementation residual logs bind to this norm only through the accepted implementation path and compact-tube norm equivalence.	No proxy stopping rule, machine-precision floor, or adaptive backend flag proves P6 by itself.
Stopping-proxy rejection	Only the fixed proof-norm inequality $\ F_{A,h}(\tilde{Z}_A; y)\ \leq c_\eta h^7$ is consumed.	Correction-norm, relative-residual, iteration-cap, or backend success flags are usable only after a predeclared conversion to that same absolute residual bound on the accepted branch.	AD Jacobian probes and branch-selected Newton diagnostics.
Strong local residual inverse	The inexact-Newton perturbation contribution to the local defect.	Recorded reduced-chart initialization and Gauss-predictor branch selection.	These diagnostics do not assert global Newton convergence or remote-root selection. Arbitrary Newton initializations and remote nonlinear roots remain outside the theorem.
Same accepted branch and initial state	The PC3 perturbation route and accepted reduced-chart branch.	Tolerance-regime sweeps and finite solver residual summaries.	Fixed tolerances are not promoted into an asymptotic proof.
Fixed production tolerances	Finite reported diagnostics and reproducibility checks only.	The finite one-step and short-trajectory probes are recorded diagnostics, not theorem evidence.	The uniform all-transition scaled trajectory solver theorem is not proved here and is not needed to state the conditional theorem. It would be needed only to replace P6 as a retained hypothesis by a proved solver theorem.
Uniform all-transition solver-policy theorem	Additional theorem evidence needed only to prove P6 as a separate uniform all-transition solver-policy result; the present theorem remains conditional on P6.		

2. the stopping rule $\mathcal{S}_h$ is stated in, or is converted in advance to, the same absolute proof residual norm;
3. the terminated iterate $\tilde{Z}_{A,n}$ satisfies Eq. (68),

$$\|F_{A,h}(\tilde{Z}_{A,n}; y_n)\|_{\text{proof}} \leq c_\eta h^7, \quad 0 \leq n < N_h; \quad (68)$$

4. the constants c_η, h_0, r_N, M_N are independent of h , of the reported grid length, and of finite residual/error tables.

If the implementation stops in a backend residual norm rather than directly in $\|\cdot\|_{\text{proof}}$, the conversion must be part of $\mathcal{T}_{P6}$ before the run. In particular, the certificate must include an h -independent constant κ_N on the same branch ball such that Eq. (69) holds,

$$\|F_{A,h}(Z; y)\|_{\text{proof}} \leq \kappa_N \|r_h^{\text{backend}}(Z; y)\|_{\text{backend}}, \quad Z \in \mathcal{B}_N. \quad (69)$$

Only after Eq. (69) is proved in advance can a backend stopping target be interpreted as the proof-norm bound in Eq. (68). Without this predeclared norm conversion, a logged backend residual remains a finite diagnostic and does not instantiate P6. Such a certificate would promote P6 from a retained branch-solver hypothesis to a discharged solver-policy theorem for that fixed instance. No such certificate is proved in this manuscript. The one-step scaled-tolerance probe, short-trajectory probe, tolerance-regime sweep, fixed production tolerances, and Newton logs are finite diagnostics only; they do not instantiate $\mathcal{T}_{P6}$ unless the branch, norm, stopping conversion, and constants above were fixed before the run.

P6 solver-scale interpretation. The displayed separation is not a solver-policy assumption discharge. It separates finite one-step probes, finite short-trajectory probes, tolerance-regime sweeps, AD Jacobian probes, and branch-selected Newton diagnostics from the retained theorem-level solver-policy condition. The proof residual norm is fixed before reading implementation logs; those logs enter only through the accepted implementation-path binding and an h -independent compact-tube norm equivalence. It does not prove P6, does not promote fixed tolerances into asymptotic proof, and does not assert global Newton convergence, remote-root selection, a P7 residual-to-error boundary, source-policy/full-TFE readiness, or reproducibility package readiness evidence.

P6 admissibility boundary. The retained $\eta_h^{\text{tube}} \leq c_\eta h^7$ condition is a residual-tolerance contract, not a restatement of the theorem conclusion.

It is stated in terms of the computable 132-row residual norm after chart, scaling, row-ordering, and branch conventions are fixed, so an h -dependent stopping rule can be applied before endpoint error or observed order slopes are measured. For a trajectory-level invocation, the same c_η bounds the maximum selected branch residual over all accepted transitions before the local-to-global bootstrap is applied; it is not a stepwise constant fitted after individual Newton logs are inspected. The condition is not inferred from observed slopes, fixed-tolerance logs, or post-hoc trajectory errors, and it does not assume access to the exact trajectory. What remains retained is the compact-tube branch-membership and strong-local-inverse hypothesis that turns this computable residual bound into the $C_N c_\eta h^7$ perturbation term in Lemma 41. The compact-tube branch-membership and strong-local-inverse hypothesis remains retained.

Solver-envelope quantifier discipline. The scalar solver-envelope constant, proof residual norm, row scaling, and branch rule are fixed before the step-size window and before any residual logs are read. Increasing that constant after inspecting logs changes the theorem hypothesis and the local-defect constant; it is not a proof of P6 and does not convert finite diagnostics into theorem-level solver-policy proof.

P6 evidence-grade rule. P6 has only one theorem-grade input: a predeclared same-branch residual envelope in the fixed proof norm, $\eta_h^{\text{tube}} \leq c_\eta h^7$, with the branch rule, row scaling, proof norm, and c_η fixed before the accepted transition is reported. Finite one-step probes, short-trajectory probes, tolerance-regime sweeps, fixed-tolerance production logs, and observed order slopes are diagnostic-grade evidence only. They may check whether a reported run is consistent with a predeclared P6 instance, but they cannot define c_η , change the proof norm or row scaling, certify branch-ball membership, or upgrade a fixed tolerance into theorem-level solver-policy proof. Thus a solver log can specialize a P6 hypothesis that has already been fixed; it cannot create that hypothesis after the fact.

P6 conditional-instantiation checkpoint. The retained P6 condition has an operational sufficient instantiation, but that instantiation is not the same as the current fixed-tolerance evidence. A trajectory instantiates P6 for Theorem 1 only after the proof norm, row scaling, branch rule, and c_η have been fixed before the run, consistently with Definition 5; every accepted transition certifies the same Gauss-predictor branch and the branch-ball/strong-local-inverse hypotheses of Lemma 41; and the nonlinear solve enforces the

proof-norm stopping target in Eq. (70):

$$\|F_{A,h}(\tilde{Z}_{A,n}; y_n)\| \leq c_\eta h^7, \quad 0 \leq n < N_h. \quad (70)$$

Under those already fixed choices, the reported-grid specialization $\eta_h^{\text{reported}} = \max_{0 \leq n < N_h} \eta_{h,n} \leq c_\eta h^7$ is a consequence of the solver hypothesis and can be consumed by Lemma 41. This is a sufficient instantiation rule, not a hidden empirical discharge: it does not prove a theorem-level solver-policy theorem for the current fixed-tolerance runs, does not choose c_η from residual ratios, does not prove global Newton convergence or remote-root exclusion, and does not prove P7, source-policy, full-TFE, or reproducibility package readiness. A transition obtained only after changing the proof norm, row scaling, branch rule, or c_η in response to the computed residual log is therefore not an instantiation of P6 for this theorem. Likewise, P1 remains a smooth reduced-chart and FullVA-lift existence condition on the compact proof tube: finite h -sweeps and local mechanism-coverage rows diagnose branch plausibility but do not prove or remove the theorem’s uniform smooth-lift hypothesis. Similarly, P2 remains a uniform compact-tube inverse, endpoint-ball, and tube-retention interface: finite rank probes, endpoint-Jacobian checks, and ordinary Lipschitz estimates may diagnose the branch but do not eliminate those hypotheses.

P1/P2 compact-tube boundary. P1 and P2 remain retained compact-tube theorem interfaces. Finite h -sweeps, local mechanism rows, finite rank probes, endpoint-Jacobian checks, and ordinary Lipschitz estimates diagnose the selected branch only; they do not prove or remove smooth-lift regularity, uniform compact-tube invertibility, endpoint-ball/right-inverse hypotheses, or tube-retention stability.

Compact-inverse quantifier discipline. The compact tube, Gauss-predictor branch, proof residual norm, and inverse constants must be selected before numerical diagnostics are read. Shrinking the admissible step-size window after those objects are fixed is part of the theorem hypothesis; refitting the tube or inverse constant from diagnostics is not proof closure. A rank probe, solver log, per-step projection log, and endpoint norm may diagnose the selected run, but none of them fixes the retained tube, inverse constant, or right-inverse margin after the asymptotic theorem instance has been chosen.

Definition 6 (P1/P2 compact-branch certificate template). *A theorem-level certificate discharging the retained P1/P2 compact-branch interfaces, rather*

Table 19: P1/P2 compact-tube interfaces for Theorem 1.

Retained object	Consumed by	Current diagnostic	Non-discharge
P1 smooth lift and chart tube	Gauss-stage lift, local defect, and reporting-map estimates.	Accepted smooth-path construction and finite branch diagnostics.	No finite trajectory row proves a uniform smooth FullVA lift on the whole compact proof tube.
P2 stage inverse on the Gauss-predictor branch	Stage-root perturbation and accepted-branch consistency.	Formula-row/AD binding and finite rank probes diagnose the selected branch.	Rank probes do not prove a uniform compact-tube inverse or remote-root exclusion.
P2 endpoint-ball and local right inverse	Lemma 6 and the endpoint contribution to the one-step defect.	Endpoint anchor and local correction diagnostics diagnose the branch.	Endpoint checks do not discharge the closed-anchor, ball-margin, or right-inverse hypotheses.
P2 stability and tube-retention bootstrap	Lemma 44.	Observed smooth-window order and bounded local rows diagnose plausibility.	Ordinary Lipschitz estimates and finite h -sweeps do not prove tube retention for every retained state.
P2 strong local residual inverse	Lemma 41.	AD Jacobian probes and branch-selected Newton diagnostics identify the implemented local path.	They do not replace the compact-tube residual inverse or the separate P6 $\eta_h^{\text{tube}} \leq c_\eta h^7$ policy.

than assuming them as in Assumption 1, would be a fixed same-branch tuple

$$\mathcal{T}_{P12} = (K, \chi, \mathcal{L}, \mathcal{G}_h, \mathcal{E}_h, \mathcal{C}_h, \mathcal{P}_h, M_Z, L_Z, r_Z, M_E, r_E, C_s, \Gamma_s, C_{\mathcal{R}}, h_0). \quad (71)$$

chosen before any finite trajectory, rank probe, residual log, or observed order is read. Equation (71) fixes the P1/P2 certificate object before any diagnostic is inspected. Here K is the compact reduced-chart tube, χ is the chart, $\mathcal{L}$ is the smooth FullVA lift, $\mathcal{G}_h$ is the Gauss-predictor branch, $\mathcal{E}_h$ is Lie endpoint reconstruction, $\mathcal{C}_h$ is the local endpoint-closure map, and $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$ is the reported endpoint map. The tuple would have to certify, uniformly for $0 < h \leq h_0$, all retained states in K , and all accepted branch-selected transitions, the following facts:

1. a C^7 reduced constrained vector field and C^7 FullVA lift on K , with the lifted Gauss stages lying in the chart domain;
2. a unique local stage root on the Gauss-predictor branch, a uniform inverse bound M_Z for $D_Z F_{A,h}$, a quadratic-remainder bound L_Z , and a branch radius r_Z ;

3. a closed endpoint-anchor family, endpoint-ball radius r_E , raw-defect bound, and local right inverse for endpoint closure with uniform bound M_E ;
4. a strong local residual inverse on the same branch for the inexact-Newton perturbation, compatible with the proof residual norm used in Lemma 41;
5. a one-step stability estimate with factor $1 + C_s h$, a tube-retention/first-exit bootstrap over $N_h h \leq T$, and the resulting fixed-time transfer constant Γ_s ;
6. a reporting-map norm-equivalence constant C_R for the displayed position-velocity grid norm.

Such a certificate would promote P1/P2 from retained compact-branch interfaces to a constructed compact-branch theorem for the fixed FullVA object. No such certificate is proved in this manuscript. Finite smooth branch runs, rank probes, endpoint checks, ordinary Lipschitz estimates, and mechanism-coverage rows may diagnose compatibility with $\mathcal{T}_{P12}$, but they do not instantiate it unless the tube, branch, chart, inverses, endpoint anchors, stability constants, and reporting norm were fixed before those diagnostics were read.

P1/P2 compact-tube interpretation. The displayed separation is not an assumption discharge. It separates smooth-lift regularity, compact-tube inverse, endpoint-ball/right-inverse, strong local residual inverse, and tube-retention stability from finite h -sweeps, rank probes, endpoint checks, local mechanism rows, ordinary Lipschitz estimates, residual/reaction tables, source-policy rows, and reproducibility package evidence. It does not prove P1/P2, a solver-policy theorem, or a P7 residual-to-error transfer, and it does not supply source-policy/full-TFE readiness or reproducibility package readiness evidence. The load-bearing P1/P2 data are not pointwise diagnostics. The theorem requires a single compact tube, an h -independent radius and constants, and a branch selection map fixed before the asymptotic limit: every accepted predictor, stage, endpoint-correction, and reported reduced-chart state must stay in that tube; the stage residual must be written in the same row/column chart; the stage Jacobian and endpoint-closure Jacobian must have uniform right inverses on the tube; and the one-step map must keep the accepted trajectory inside the tube under the stated bootstrap. Finite rank probes, endpoint checks, ordinary Lipschitz estimates, and observed convergence rows sample this structure on visited states only. They

do not establish tube retention, a uniform lower margin for all nearby accepted states, or branch-jump exclusion; the constants used in the endpoint, inexact-Newton, and local-to-global lemmas therefore cannot be fitted from observed convergence rows. The uniform right inverses on the tube are retained hypotheses, not finite-run conclusions. Equivalently, the manuscript retains P1/P2 as Assumption 1 interfaces rather than proving the certificate in Definition 6.

PC1/P3 are therefore read as implementation-path checks: the implemented formula-row checks and the AD-expanded implementation-path certificate identify the direct-route residual/Jacobian path. They do not close the primitive dynamic symbolic oracle, the primitive/Taylor symbolic-defect route, source-paper residual reproduction, or a discharge of the retained compact-tube hypotheses. Thus a still-open primitive dynamic symbolic oracle is not a missing theorem input for the accepted route; it is only the alternative primitive/Taylor lane that would have to be completed before replacing the direct residual certificate.

P3/P4 implementation-defect boundary. P3 and P4 remain retained theorem-interface inputs rather than standalone discharged interfaces. Formula-row checks and AD-expanded runtime binding identify the implemented direct-route residual/Jacobian path; the AD-expanded implementation-path certificate is closed for that active route and is consumed by the same-branch direct residual bridge. They do not prove the primitive dynamic symbolic oracle or the primitive/Taylor symbolic-defect route, and they are not source-paper residual reproduction. They also do not claim primitive/Taylor symbolic-defect route closure. The 96-row non-dynamic certificate covers only kinematic and lower-pair rows; it does not by itself prove the full 132-row stage residual, which uses the separate 36-row D5 direct-substitution certificate for Newton–Euler rows. The full 132-row stage residual required for the order argument is supplied only after this dynamic-row handoff is combined with the non-dynamic certificate through Lemma 40. The separate 36-row D5 direct-substitution phrase refers only to this dynamic-row handoff, not to P3/P4 closure. Within the theorem, P3 is used only as a fixed map-identity interface: before the local perturbation argument, the mathematical row formulas, row ordering, row scaling, and AD-differentiated implementation path are fixed as one residual map. The proof then evaluates this fixed map on the lifted Gauss stage. Finite runtime probes and AD checks identify that implementation path; they are not used backward to infer the h^7 defect from reported trajectories

or Newton residuals.

Table 20: P3/P4 implementation-defect interface table for Theorem 1.

Retained object	Consumed by	Current evidence	Non-discharge
P3 runtime formula/AD binding	The implemented direct-route residual/Jacobian path.	Formula-row checks and AD-expanded runtime binding.	This does not prove the primitive dynamic symbolic oracle or primitive/Taylor symbolic-defect route.
Excluded full-TFE source-paper residual identity	The stronger full-TFE replacement and source-policy route only.	The current manuscript records source-policy rows separately.	This is not source-paper residual reproduction or full-TFE readiness.
P4 96-row non-dynamic certificate	The non-dynamic part of the accepted local defect argument.	The kinematic and lower-pair rows are certified on the smooth lift.	This does not by itself prove the full 132-row stage residual.
P4 handoff to dynamic rows	The direct residual route for PC2 together with P5.	The separate 36-row D5 direct-substitution certificate supplies the Newton–Euler rows.	The handoff does not convert P3/P4 alone into a standalone 132-row direct-route certificate.
P3/P4 implementation-domain interface	The conditional order theorem’s implementation-defect premise.	Dependency tables map the implemented runtime path and non-dynamic rows.	It does not prove P3/P4 as standalone primitive/source-policy closures, a solver-policy theorem, a P7 residual-to-error boundary, source-paper comparison, full-TFE replacement, or external reproducibility package claims.

P3/P4 implementation-defect boundary. This is not a standalone implementation-defect proof. It separates formula-row checks, AD-expanded runtime binding, the 96-row non-dynamic certificate, and the separate P5 direct Newton–Euler certificate from primitive dynamic symbolic oracle closure, primitive/Taylor symbolic-defect route closure, source-paper residual reproduction, full 132-row stage residual promotion by P3/P4 alone, source-paper comparison, full-TFE replacement, and external reproducibility package evidence. It does not close P3/P4 as standalone primitive/source-policy closures or promote P3/P4 alone into an independent direct-route theorem input.

P3/P4 evidence convention. Their theorem use is through a specified same-branch interface. P3 is backed by the implemented formula-row and AD-expanded runtime binding, and P4 is backed by the 96-row non-dynamic certificate. Their retained status means that these checks are consumed

through the same-branch direct residual bridge and still require the separate P5 dynamic-row handoff; they are not, by themselves, a primitive dynamic symbolic-oracle closure, a source-paper residual replacement, or a full 132-row certificate independent of P5.

P5 direct-route boundary: P5. P5 is proved only as a stage-local Newton–Euler residual identity on the smooth Gauss lift. It supplies the 36 dynamic rows required by PC2 through the D5 direct-substitution certificate; it does not assert residual control at the inexact Newton iterate, finite-run trajectories, closed-loop mechanism residual tables, source-policy rows, or residual-to-error promotion.

P5 inverse-boundary. The direct Newton–Euler zero rows close the dynamic block of the residual vector on Z_G , but they do not provide a stage-Jacobian inverse, contraction radius, branch-selection theorem, endpoint right inverse, stability factor, or η_h solver hypothesis. Those inputs enter through P1/P2/P6 and Lemmas 39, 6, 41, and 44.

Table 21: P5 direct-route satisfaction for Theorem 1.

Satisfied object	Consumed by	Current mathematical input	Non-promotion rule
P5 stage-local dynamic rows	PC2 and Lemma 39.	The D5 direct-substitution certificate evaluates the 36 Newton–Euler rows on the smooth Gauss lift Z_G .	This is not residual control at the inexact Newton iterate.
P5 smooth-lift residual identity	The direct residual route only.	The row-target map and dynamic residual-identity table bind the accepted Newton–Euler rows.	It is not a finite-run trajectory statement or a closed-loop residual table.
P5 handoff from P3/P4	The accepted 132-row local-defect route.	The 96-row non-dynamic certificate plus the 36-row P5 certificate cover the implemented direct route.	This handoff does not close P3/P4 as primitive dynamic symbolic oracle evidence.
P5 scope/exclusion rule	The theorem-interface table.	P5 is the only stage-local dynamic identity proved by direct substitution.	It does not prove P1/P2/P3 theorem interfaces, the retained P4 binding interface, the P6 solver-scale interface, the P7 residual-to-error boundary interface, source-paper comparison, full-TFE replacement, or external reproducibility package claims.

P5 direct-route satisfaction rule: the displayed evidence is stage-local, not a global residual or trajectory certificate. It separates the D5 direct-substitution certificate on the smooth Gauss lift, the 36 Newton–Euler row identity, and the P3/P4 handoff from inexact-Newton residual control, finite-run trajectories, closed-loop mechanism residual tables, source-policy rows, residual-to-error promotion, primitive dynamic symbolic oracle closure, and external reproducibility package evidence. It does not promote P5 beyond the direct stage-local Newton–Euler route.

Table 22: Proof causality for Theorem 1.

Mathematical input	Consumed before	Allowed source	Explicit non-source
P1–P2 compact tube	The local stage-root, endpoint-closure, inexact Newton, and local-to-global estimates.	Smooth lift, compact-tube inverse, endpoint-ball/right-inverse, strong residual inverse, and tube-retention stability hypotheses.	Finite h -sweeps, rank probes, endpoint checks, or ordinary Lipschitz diagnostics.
Direct stage residual	The stage-root perturbation bound and the four-term local defect sum.	The 96-row non-dynamic certificate plus the 36-row D5 direct-substitution identity evaluated on Z_G .	Computed trajectory error, inexact Newton iterates, residual/reaction tables, or source-policy rows.
P6 solver scale	The $C_N c_\eta h^7$ inexact-Newton endpoint contribution.	The compact-tube branch-solver hypothesis $\eta_h^{\text{tube}} \leq c_\eta h^7$.	Fixed production tolerances or finite scaled-tolerance probes as theorem evidence.
P7 and boundary records	The theorem-use and reproducibility-boundary exclusions.	Residual-to-error obligations, source-policy records, and reproducibility package checks as downstream boundary records.	Inputs to PC2, a theorem-level solver-policy theorem, or source-policy/full-TFE readiness.

Proof-causality rule: each mathematical input is consumed before the theorem conclusion that depends on it. The D5 direct-substitution certificate is evaluated on the lifted Gauss stage Z_G before the local perturbation argument; the local defect bound is established before the discrete Gronwall transfer; and the position–velocity reporting map is used only after the reduced-chart grid estimate. Finite h -sweeps, inexact Newton iterates, residual/reaction tables, source-policy rows, and local reproducibility records are downstream diagnostics or reproducibility-boundary evidence. They do not feed PC2, P6, or P7, and they do not prove a theorem-level solver-policy the-

orem, a separate residual-to-error transfer theorem, source-policy readiness, or full-TFE readiness.

Lemma 6 (Endpoint-closure perturbation). *Let z_u be the same-branch unclosed endpoint to which the accepted endpoint-closure map is applied. In the theorem application $z_u = \Psi_h^A = \mathcal{E}_h(Z_A)$, the endpoint reconstructed from the exact accepted local stage root before endpoint closure; it is not the Gauss predictor endpoint Ψ_h^G . Let $E(z) = 0$ denote the square endpoint closure subsystem actually corrected by the accepted endpoint-closure map—the velocity-level/KKT closure equations in the reported implementation. This E is not the set of every raw endpoint diagnostic reported in Table 10; in particular, the raw position residual in that table is a separate monitor and not the local $E(z_u)$ input below. For each retained transition (y, h) , let $z_* = z_*(y, h)$ be the same-branch closed endpoint anchor with $E(z_*) = 0$. Assume that E is C^2 on a ball $B_{r_E}(z_*)$, that $z_u = z_u(y, h) \in B_{r_E/2}(z_*)$ for all sufficiently small h , and that $DE(z_*)$ has a right inverse $B_* = B_*(y, h)$ with $\|B_*\| \leq M_{E,ri}$. The constants r_E , $M_{E,ri}$, L_E , $C_{E,raw}$, and the second-derivative bound used below are uniform over retained $y \in K$ and the admissible small-step window. **Endpoint-time binding.** If the endpoint-closure rows are nonautonomous because prescribed constraints, loads, or drive functions enter the closure equations, then E , C_h , z_u , and z_* in this lemma are fixed-transition abbreviations for E_{h,t_n} , C_{h,t_n} , $z_u(y, t_n, h)$, and $z_*(y, t_n, h)$. All endpoint data are evaluated at the endpoint time $t_n + h$ and with the stage history of that same transition. The right-inverse and derivative constants are uniform over the retained time interval, but no row in E may be evaluated at a neighboring transition time, a resampled output time, or an unreported terminal T and then used in this endpoint perturbation bound. Assume also the derivative perturbation and smallness conditions in Eq. (72):*

$$\|DE(z) - DE(z_*)\| \leq L_E \|z - z_*\|, \quad M_{E,ri} L_E r_E \leq \frac{1}{2}, \quad (72)$$

and that the unprojected endpoint closure defect has the explicit local constant in Eq. (73):

$$\|E(z_u)\| \leq C_{E,raw} h^7. \quad (73)$$

This raw endpoint defect is a same-object local statement: z_u is the unclosed endpoint on the same accepted branch and is evaluated by the same endpoint functional E used by the accepted closure map. In the theorem’s four-map

chain this is precisely the $\Psi_h^E - \Psi_h^A$ comparison; it is a retained P2 endpoint-closure hypothesis for this lemma and Theorem 1, not a post-Newton projection log, not a terminal-time correction, and not a constraint/reaction residual table. Endpoint diagnostics may record this scale on reported runs, but they do not discharge the compact-tube endpoint raw-defect/right-inverse hypothesis. The endpoint-anchor family, uniform ball radius, right inverse, derivative constants, and endpoint norm are fixed on the compact proof tube before any reported endpoint residual, projection diagnostic, step-size sweep, or per-step projection log is inspected. **Endpoint diagnostic-scope checkpoint.** The order-fit row in Table 10 is not used to infer the hypothesis $\|E(z_u)\| \leq C_{E,\text{raw}} h^7$. For the theorem invocation, E , the endpoint norm, the endpoint-anchor family $z_*(y, h)$, r_E , $B_*(y, h)$, $M_{E,\text{ri}}$, and $C_{E,\text{raw}}$ are fixed uniform compact-tube data before that table is read. Thus the raw position monitor may have a different fitted slope without contradicting the lemma, because it is not the local closure functional E ; conversely, a small raw position monitor would not discharge the velocity-level/KKT defect bound, the endpoint ball, or the right-inverse condition. The table is a consistency diagnostic for the reported closure subsystem, not an empirical source of the endpoint perturbation hypothesis. If the accepted endpoint correction Δz_E is obtained by first solving the minimum-norm linearized closure equation in Eq. (74)

$$DE(z_u) \delta z_1 = -E(z_u), \quad (74)$$

and then, if needed, refined by the same local Newton/right-inverse closure solve in Eq. (75)

$$\begin{aligned} z_1 &= z_u + \delta z_1, & r_j &= E(z_j), \\ DE(z_j) \delta z_{j+1} &= -r_j, & z_{j+1} &= z_j + \delta z_{j+1}, \end{aligned} \quad (75)$$

with each later correction chosen no larger than a local right-inverse particular solution while the iterates remain in $B_{r_E}(z_*)$, then, after possibly shrinking the admissible step-size window, the correction obeys Eq. (76):

$$\|\Delta z_E\| \leq C_E h^7, \quad C_E = 4M_{E,\text{ri}} C_{E,\text{raw}}. \quad (76)$$

In the four-map local-defect chain this endpoint correction is the map difference in Eq. (77):

$$\begin{aligned} \Psi_h^A(y) &= z_u, \\ \Psi_h^E(y) &= \mathcal{C}_h(z_u) = z_u + \Delta z_E, \\ \|\Psi_h^E(y) - \Psi_h^A(y)\|_{\text{end}} &\leq C_E h^7. \end{aligned} \quad (77)$$

The endpoint norm $\|\cdot\|_{\text{end}}$ in Eq. (77) is the fixed endpoint-output component of the proof norm used in the local-defect sum; any finite-dimensional conversion between the closure-solve coordinates and the reduced-chart endpoint norm is fixed on the compact tube and absorbed into C_E before Theorem 1 chooses its constants. Thus the corrected endpoint $z_u + \Delta z_E$ differs from the same-branch unclosed endpoint by only this explicit local $O(h^7)$ perturbation.

Proof. Let $A_* = DE(z_*)$ and $A_u = DE(z_u)$. The assumed perturbation bound gives $\|(A_u - A_*)B_*\| \leq 1/2$. Hence $A_u B_* = I + (A_u - A_*)B_*$ is invertible, and the induced right inverse is Eq. (78):

$$B_u = B_*(A_u B_*)^{-1} \quad (78)$$

is a right inverse for A_u with $\|B_u\| \leq 2M_{E,\text{ri}}$. The minimum-norm solution of the linearized closure equation is no larger than the particular solution $-B_u E(z_u)$, so Eq. (79) gives

$$\|\delta z_1\| \leq 2M_{E,\text{ri}}\|E(z_u)\| \leq 2M_{E,\text{ri}}C_{E,\text{raw}}h^7. \quad (79)$$

Since $z_u \in B_{r_E/2}(z_*)$, choosing h small enough gives $2\|\delta z_1\| < r_E/2$, and hence $z_u + \delta z_1 \in B_{r_E}(z_*)$. This endpoint-ball margin keeps the linear correction and the following local Newton refinement inside the same ball on which the derivative and right-inverse perturbation bounds are valid. The same endpoint-ball constants and endpoint norm are used for every retained-tube state. The admissible h_0 may shrink after these constants are fixed, but the proof does not choose the endpoint-anchor family, ball radius, or right inverse from the computed endpoint correction or from a per-step projection log. Let $C_{E,2} = \sup_{B_{r_E}(z_*)} \|D^2 E\|$. Since $A_u \delta z_1 = -E(z_u)$, Taylor's formula gives the explicit post-linear closure residual in Eq. (80):

$$r_1 = E(z_u + \delta z_1), \quad \|r_1\| \leq \frac{1}{2}C_{E,2}\|\delta z_1\|^2 \leq 2C_{E,2}M_{E,\text{ri}}^2C_{E,\text{raw}}^2h^{14}. \quad (80)$$

For every later iterate that remains in the ball, the same Neumann-series right-inverse argument gives a local right inverse of $DE(z_j)$ with norm at most $\overline{M}_E = 2M_{E,\text{ri}}$. Hence the chosen correction satisfies Eq. (81):

$$\|\delta z_{j+1}\| \leq \overline{M}_E\|r_j\|, \quad \|r_{j+1}\| \leq \frac{1}{2}C_{E,2}\overline{M}_E^2\|r_j\|^2. \quad (81)$$

After shrinking the step-size window, r_1 is small enough that the residuals decrease at least geometrically, while $\overline{M}_E C_{E,2} \|\delta z_1\| \leq 1$. Consequently the total refinement tail is bounded by Eq. (82):

$$\sum_{j \geq 1} \|\delta z_{j+1}\| \leq 2\overline{M}_E \|r_1\| \leq \overline{M}_E C_{E,2} \|\delta z_1\|^2 \leq \|\delta z_1\|. \quad (82)$$

The same bound also applies to any finite accepted refinement prefix. Thus the cumulative endpoint displacement from z_u is at most $2\|\delta z_1\| < r_E/2$, so all iterates stay in $B_{r_E}(z_*)$ by the endpoint-ball margin, and Eq. (83) closes the endpoint perturbation estimate:

$$\|\Delta z_E\| \leq 2\|\delta z_1\| \leq 4M_{E,\text{ri}} C_{E,\text{raw}} h^7 = C_E h^7. \quad (83)$$

Thus endpoint closure contributes at most the explicit $C_E h^7$ local perturbation to the one-step map. $\square$

Lemma 6 does not by itself prove the full Assumption 1. It isolates the endpoint-closure part of the perturbation budget: once the raw endpoint closure defect has the same local scale as the Gauss truncation defect and the closure Jacobian remains right-invertible under the stated local splitting bound, the endpoint-closure correction cannot reduce the asymptotic order. Pointwise right-invertibility at one endpoint is not used as a shortcut; the closed endpoint-anchor family $E(z_*(y, h)) = 0$, the uniform endpoint-ball margin, and the compact-tube perturbation bound are all part of the condition. This endpoint-closure lemma by itself does not address the stage-row residual obligation. The companion direct-substitution certificate verifies the accepted stage residual on the lifted branch; the separate symbolic/primitive route and any residual-to-error transfer theorem remain separate non-active/open proof routes rather than endpoint-closure consequences or external proof-completion claims. Endpoint-refinement boundary: the refinement in Lemma 6 is a local closure solve inside the same endpoint ball. It is not an extra terminal solve, not a global projection proof, and not a fitted KKT correction norm; its only theorem-level use is the $O(h^7)$ endpoint-increment bound under the stated local right-inverse and endpoint-ball assumptions. This is a local right-inverse perturbation bound for endpoint closure only. The refinement is branch-local and endpoint-norm fixed; it cannot define a new terminal output time or calibrate an h -dependent projection constant from finite runs. Endpoint-output nonfeedback rule: the

accepted endpoint closure $\mathcal{C}_h$ is applied after the stage residual solve and Lie endpoint reconstruction. Its correction changes only the endpoint-output value $\mathcal{P}_h(Z) = \mathcal{C}_h(\mathcal{E}_h(Z))$ used in the one-step map; it does not redefine the stage vector Z , the lifted Gauss base point Z_G , the exact accepted stage root Z_A , the implemented residual $F_{A,h}$, or the D5 Newton–Euler row identity. Consequently endpoint closure cannot repair, re-center, or weaken the 132-row residual bridge: the bridge is evaluated before endpoint closure on the fixed stage object, and any method that fed the endpoint correction back into the stage equations would be a different residual map requiring a separate proof. Endpoint-time nonfeedback rule: for a nonautonomous transition the closure map in this paragraph is $\mathcal{C}_{h,t_n}$, evaluated at the endpoint time $t_n + h$ for the same transition that produced Z_A or $\tilde{Z}_A$. It is not a dense-output interpolant, not a terminal-time postprocessor for an unreported T , and not a way to change the time slice of the residual bridge or the subsequent Gronwall recurrence.

Lemma 7 (FullVA stage-row lift consistency). *Assume the smooth reduced chart in Assumption 1 admits a smooth FullVA lift, written in Eq. (84),*

$$\mathcal{L}(y, t) = (q, v, a, \lambda) \quad (84)$$

for which the kinematic identities, Newton–Euler balance equations, and the position-, velocity-, and acceleration-level lower-pair constraints in Eqs. (10)–(16) hold pointwise on the trajectory tube. Let Z_G be the lifted stage tuple obtained from the three-stage Gauss collocation solution of the reduced equation $\dot{y} = f(y, t)$, using the same Gauss abscissae, transition input, local Lie chart, and endpoint-output convention that will be used in the implemented residual bridge. Then the mathematical accepted stage residual R_{G6FVA} in Eq. (6), evaluated in exact arithmetic at Z_G , satisfies the ideal row identities exactly in all six row families. Equivalently, its mathematical residual defect is zero, and in particular it has the $O(h^7)$ scale required by Lemma 39 before implementation binding is considered. This statement concerns the ideal target-equation residual only; it is not a statement about the row-scaled implemented residual $F_{A,h}$. The implemented 132-row bridge consumed by the theorem is supplied later, after P3/P4/P5 bind row order, row scaling, the AD/formula path, and the direct Newton–Euler substitution. Thus the ideal zero statement is not used as a shortcut around the implemented PC2 discharge: P3 fixes the formula path and row scaling, P4 supplies the 96-row non-dynamic bound, and P5 supplies the 36 dynamic zero rows consumed

by Lemma 40. The object Z_G in this lemma is the same lifted Gauss object consumed by that bridge; it is not replaced by an accepted Newton root, an inexact Newton iterate, or an endpoint-closed postprocessed state.

Proof. The reduced Gauss stage solution satisfies the collocation equations for $\dot{y} = f(y, t)$. Applying the smooth lift $\mathcal{L}$ gives stage positions, orientations, velocities, accelerations, and multipliers on the same smooth constraint manifold. The translational and rotational position rows (10)–(11) are exactly the integral form of the lifted kinematic identities $\dot{r} = v$ and $\dot{Q} = Q\hat{\omega}$ written in the local Lie chart; the velocity rows (12)–(13) are the corresponding integral form of $\dot{v} = a$ and $\dot{\omega} = \alpha$. The Newton–Euler rows (14)–(15) vanish because the lift supplies the stage accelerations and multipliers satisfying force and moment balance. Finally, the FullVA lower-pair rows (16) vanish because the same lift lies on the position constraint manifold and its first two time derivatives. Hence each row family of R_{G6FVA} is the lifted reduced Gauss stage equation or a pointwise algebraic consequence of the lifted constrained dynamics. At this ideal mathematical level the residual defect is therefore zero, which is stronger than $O(h^7)$; the subsequent implemented bridge is still needed to bind this row identity to the residual path actually differentiated and solved. $\square$

Lemma 7 proves only the ideal mathematical lift identity for the stage-row equations. It does not itself discharge the implemented PC2 residual bridge; that discharge is the later 132-row bridge under P3/P4/P5. This zero statement is not the same object as the endpoint truncation estimate $\|\Psi_h^G - \varphi_h\| \leq C_G h^7$: the former says that the lifted reduced Gauss stage satisfies the fixed stage equations, while the latter compares the endpoint reconstructed from that Gauss stage with the exact reduced flow. The two estimates are therefore consumed in different terms of the four-map local defect decomposition. The remaining implementation question is row identity: the residual appearing in Eq. (6) must be the same residual differentiated and solved in the implemented residual path, with the same 132-row ordering, scaling, and chart conventions. The source-identity and row-layout records fix this correspondence. With that correspondence fixed, the proof certificates supply the accepted direct-route residual/Jacobian binding used by the bridge: the non-dynamic certificate covers the 96 collocation and lower-pair rows on the smooth FullVA lift, the D5 direct-substitution certificate covers the 36 Newton–Euler weak-balance rows on the lifted Gauss stage, and the AD-expanded implementation-path derivative-cell certificate

records the derivative identities used by that binding for those 36 dynamic rows. These certificates are used only to bind the implemented residual to the mathematical row identities in this proof; they do not convert the method into a source-paper temporal finite-element residual replacement. Mathematical-lift versus implemented-residual split: the zero-defect statement in Lemma 7 is an ideal row identity after the smooth FullVA lift has been defined. It is not, by itself, the theorem’s implemented residual-defect certificate. The theorem consumes the implemented residual only after P3 fixes the formula/AD path and row ordering, P4 supplies the 96-row non-dynamic bound, and P5 supplies the 36-row D5 direct-substitution identity. Thus the ideal lift does not bypass P3/P4/P5, does not choose the Newton branch, and does not discharge the compact-tube inverse or solver-scale interfaces. Ideal lift is not implementation discharge: the ideal zero statement is a target-equation identity for the smooth lift, while the implemented stage-residual input is the later P3/P4/P5 bridge on the code-facing 132-row residual path. There is no contradiction between the ideal zero-defect lift and the later $O(h^7)$ implemented bridge. The theorem deliberately consumes the code-facing implemented $O(h^7)$ bridge statement because the residual used by the code must first be tied to one formula path, row ordering, and row scaling. The zero mathematical lift identifies the target equations; the P3/P4/P5 bridge proves that the implemented residual actually used in Newton has the required residual value on Z_G . This bridge is also the only place where implementation-specific row binding enters the local-defect proof; it is upstream of the endpoint closure, inexact Newton, and global error recursion, and downstream of the reduced Gauss truncation theorem. Thus the D5 certificate is a stage-local identity on Z_G . It is not evidence about residuals evaluated at the inexact Newton iterate $\tilde{Z}_A$, finite-run diagnostic trajectories, or closed-loop mechanism residual tables; those objects enter only through Lemma 41, numerical evidence, or the still-open residual-to-error boundary, respectively. Equivalently, the static implementation certificate and source-identity record identify one residual/Jacobian implementation path, while the row-layout records fix the 132-row ordering used in the proof.

For the dynamic rows, the active proof route is the direct-substitution certificate, with five mathematical sub-obligations feeding the residual identity. The older symbolic-defect certificate remains a provenance record for the primitive/Taylor route. In the direct route, each of the 36 dynamic rows is expanded into its body-specific proximal/distal force and torque terms. The multiplier contribution is fixed by the D3 virtual-work identity, includ-

ing the six row-expanded lower-pair multiplier identities in the implemented site ordering. The force and torque terms are evaluated on the same smooth C^7 compact proof tube, the D1/D2 balance-identity certificate proves the translational and rotational Newton–Euler balance identities, and the D6 row-ordering/scaling/AD binding certificate fixes the implemented row ordering and scaling. Substituting the smooth Gauss lift Z_G into the accepted dynamic residual therefore leaves only identity terms and a lifted-stage remainder; the direct-substitution step reduces that remainder row-locally to zero. Hence the 36 dynamic rows satisfy the $O(h^7)$ stage-residual condition without invoking the primitive/Taylor lift route.

Table 24 is the compact form of the row-level target inventory used by the direct certificate. Target inventory complete; the active D5 direct-substitution certificate closes the dynamic defect on the smooth Gauss lift. For stage $s \in \{0, 1, 2\}$ and local dynamic-row offset $\rho \in \{0, \dots, 11\}$, the implemented global row index is Eq. (85):

$$i_{\text{row}}(s, \rho) = 44s + 24 + \rho. \quad (85)$$

Offsets 0–2 and 6–8 are the two-body translational balances; offsets 3–5 and 9–11 are the corresponding body-frame rotational balances. This target map indexes the direct D5 certificate and the closed D1/D2/D3/D4/D6 sub-obligations for the 36 implemented dynamic rows. The map alone is not the proof of the defect bound; the proof is the row-local substitution of Z_G into the accepted residual, recorded by the D5 direct-substitution certificate.

The row map separates dynamic-row correspondence into two mathematical checks. First, each target row is mapped to the corresponding translational or rotational template, so the force, moment, multiplier, and inertia terms occur in the intended balance equation. Second, after role-anchoring the proximal and distal joint forces, multiplier torques, inertia terms, and external loads, the scalar Newton–Euler templates and the body-specific symbolic expansions simplify to zero difference. In compressed notation the checked scalar templates are Eq. (86):

$$\begin{aligned} T_0^{\text{tr}} &= ma - mg - f_e - (F_0 - F_1), & T_1^{\text{tr}} &= ma - mg - f_e - F_1, \\ T_0^{\text{rot}} &= I - \tau_e - (P + D + A - B), & T_1^{\text{rot}} &= I - \tau_e - (P + A), \end{aligned} \quad (86)$$

where I abbreviates the body-frame inertial term, P, D are proximal and distal moment-arm torques, and A, B are the corresponding axis-torque contributions. The runtime scalar template and the body-specific symbolic template give identical expressions after this role anchoring for every stage, body,

Table 23: Direct-route Newton–Euler row obligations used by the P5 stage-residual bridge; the primitive symbolic-defect route is tracked separately.

ID	Row scope	Required mathematical statement	Proof state
D1	Translational balance rows	After row binding and role anchoring, substituting the lifted Gauss stage Z_G gives zero residual for the translational force-balance rows, hence the required $O(h^7)$ bound.	D1 balance identity closed; D5 direct residual certificate closed.
D2	Rotational balance rows	After row binding and role anchoring, substituting the lifted Gauss stage Z_G gives zero residual for the body-frame angular balance, Lie-retraction increment, and inertia rows, hence the required $O(h^7)$ bound.	D2 balance identity closed; D5 direct residual certificate closed.
D3	Multiplier wrench rows	The lower-pair multiplier reactions used in the position, velocity, and acceleration rows induce the same generalized wrench as the dynamic rows.	D3 row-expanded lower-pair multiplier identity closed; used in the D5 direct residual certificate.
D4	Force/friction smoothness	All applied force and friction laws used by the accepted smooth branch are C^7 on the trajectory tube and share the same FullVA lift.	Smooth C^7 force/friction lift closed on the accepted compact proof tube; sharp/small-stribeck branch remains diagnostic.
D5	Dynamic-row defect rate	Evaluating the assembled dynamic rows on the lifted Gauss stage gives a uniform $O(h^7)$ residual bound.	Closed by direct-substitution residual certificate.
D6	Implemented row ordering	The algebraic row order, scaling, and automatic-differentiation blocks in the implemented residual path are symbolically equivalent to D1–D5.	Row ordering, fixed residual row-scaling (the D6 unweighted scaling convention), and accepted AD binding closed; D5 direct residual certificate closed.

Table 24: Row-level target map for the Newton–Euler weak-balance rows.

Stage s	Global rows	Translational targets	Rotational targets	Residual role
0	24–35	Rows 24–26 and 30–32: bodies 0–1 and components x, y, z satisfy linear-momentum balance.	Rows 27–29 and 33–35: bodies 0–1 and components x, y, z satisfy angular-momentum balance.	D1/D2 balance identities give zero dynamic residual on Z_G .
1	68–79	Rows 68–70 and 74–76: bodies 0–1 and components x, y, z satisfy linear-momentum balance.	Rows 71–73 and 77–79: bodies 0–1 and components x, y, z satisfy angular-momentum balance.	D1/D2 balance identities give zero dynamic residual on Z_G .
2	112–123	Rows 112–114 and 118–120: bodies 0–1 and components x, y, z satisfy linear-momentum balance.	Rows 115–117 and 121–123: bodies 0–1 and components x, y, z satisfy angular-momentum balance.	D1/D2 balance identities give zero dynamic residual on Z_G .

and Cartesian component. This closes the template-level C2 subcheck and, together with the D6 row-ordering/scaling/AD certificate, closes the implemented row-order and unweighted residual-scaling equivalence for every dynamic row. The D1/D2 balance-identity certificate combines this template equivalence with the D3/D4/D6 closure evidence and closes the implemented Newton–Euler balance identities for all 36 rows. This supplies the E_{bal} term in the direct-substitution proof. The D5 direct-substitution certificate combines it with D3, D4, and D6 and certifies zero dynamic residual on Z_G for every dynamic row; the older symbolic defect certificate is retained only as row-level correspondence for the primitive/Taylor route. Representative row substitution: for any accepted stage s , body b , and Cartesian or body-frame component c , the direct D5 certificate reduces one translational row and one rotational row on the lifted Gauss stage to Eq. (87):

$$\begin{aligned}
 r_{sbc}^{\text{tr}}(Z_G) &= [m_b a_{s,b}^G - f_{s,b}^{\text{ext}}(Z_G) - G_{s,b}^T \lambda_s^G]_c = 0, \\
 r_{sbc}^{\text{rot}}(Z_G) &= [J_b \alpha_{s,b}^G + \omega_{s,b}^G \times J_b \omega_{s,b}^G - \tau_{s,b}^{\text{ext}}(Z_G) - H_{s,b}^T \lambda_s^G]_c = 0.
 \end{aligned} \tag{87}$$

The displayed zero is not an unexplained cancellation. The D1/D2 balance identities used in this substitution are the componentwise Newton–Euler

equations on the same lifted stage, shown in Eq. (88):

$$\begin{aligned} m_b a_{s,b}^G &= f_{s,b}^{\text{ext}}(Z_G) + G_{s,b}^T \lambda_s^G, \\ J_b \alpha_{s,b}^G + \omega_{s,b}^G \times J_b \omega_{s,b}^G &= \tau_{s,b}^{\text{ext}}(Z_G) + H_{s,b}^T \lambda_s^G. \end{aligned} \quad (88)$$

Substitution of these two identities into the two residual definitions above gives $r_{sbc}^{\text{tr}}(Z_G) = 0$ and $r_{sbc}^{\text{rot}}(Z_G) = 0$ for each component. D3 fixes the signs in $G_{s,b}^T \lambda_s^G$ and $H_{s,b}^T \lambda_s^G$ from the lower-pair virtual-work identities, while D6 states that the implemented unweighted scalar rows are exactly these component residuals after the accepted row ordering and scaling are applied. Here $G_{s,b}^T \lambda_s^G$ and $H_{s,b}^T \lambda_s^G$ denote the lower-pair multiplier force and torque components bound to the FullVA constraint rows by D3; D1/D2 provide the linear- and angular-momentum balance identities, D4 fixes the smooth force/torque evaluation on the same lift, and D6 binds the displayed component equations to the implemented row ordering and scaling. This display is the representative direct substitution used for all 36 mapped dynamic rows; it is not a primitive/Taylor subterm proof and does not promote the separate primitive route. The finite implementation check also links the 36 row slots to the full formula-row map: the independent formula vector covers all 132 residual rows, includes the `newton_euler_weak_balance` family, and its AD Jacobian matches the accepted residual-Jacobian path on three finite probes. This finite row-slice check is retained only as implementation consistency evidence; the proof of the row identity is the symbolic template reduction above and the direct substitution of Z_G . In the supporting source package, this is the row-template instantiation and template-level algebraic equivalence subcheck layer. It provides runtime row-slice evidence only, not a symbolic identity proof.

Table 25 records the inputs to the dynamic row proof. The PC2 stage-residual condition is discharged by the D5 direct-substitution certificate: for each Newton–Euler component row the accepted dynamic residual evaluated on the smooth Gauss lift Z_G reduces to Eq. (89),

$$R_{\text{dyn}}(Z_G) = E_{\text{bal}} + E_{\lambda} + E_{\text{smooth}} + E_{\text{bind}} + E_{\text{lift}}, \quad (89)$$

where $E_{\text{bal}} = 0$ is supplied by the closed D1/D2 balance-identity certificate, $E_{\lambda} = 0$ by the D3 virtual-work multiplier identity, $E_{\text{smooth}} = 0$ on the smooth lifted force/torque inputs by the D4 force-lift certificate, and $E_{\text{bind}} = 0$ by the D6 row-ordering, unweighted-scaling, and AD-binding certificate. The remaining $E_{\text{lift}} = 0$ clause is the P5 direct-substitution identity: after the

Table 25: Dynamic-row residual-identity table for the Newton–Euler rows.

Item	Verified input	Proof role and remaining non-use
D1–D2	The 36 dynamic rows are present in the row-family-major residual; the formula-row oracle matches the runtime residual and AD Jacobian on three finite probes; and the D1/D2 balance-identity certificate proves the implemented translational and rotational Newton–Euler identities.	No remaining D1–D2-specific identity test; use the closed balance identities inside the D5 lifted-stage $O(h^7)$ dynamic-defect proof.
D3	The lower-pair FullVA rows and the dynamic rows use the same multiplier variables in the accepted source path, and the package records a 36-row virtual-work sign-skeleton certificate covering the proximal/distal force and axis torque transfer sites plus two template triple-product identities for point and axis multiplier terms plus six row-expanded lower-pair multiplier identities in the implemented site ordering.	No remaining D3-specific closure test; the remaining dynamic proof work is the D5 direct-substitution residual certificate.
D4	The smooth benchmark branch records bounded force, torque, prescribed-motion, and friction inputs; the Brown–McPhee smooth-branch source structure, regularized normal-load map, stage-local force lift, and compact proof-tube C^7 derivative bound have been checked. The sharp branch is reported separately as a limitation.	No remaining D4-specific smoothness test; use the closed D4 certificate in the D5 direct-substitution residual certificate.
D5	The 96 non-dynamic rows have a proof-scope certificate, and the 36 dynamic Newton–Euler rows have a direct-substitution certificate on the smooth Gauss lift Z_G . After substituting Z_G , each dynamic residual reduces to the closed D1/D2 balance identities, the D3 multiplier-wrench identity, the D4 smooth force lift, and the D6 row-ordering/scaling/AD binding.	The direct stage-residual input is supplied by zero row-local dynamic residual on Z_G . This route does not use a P_{state} actual PS3 lift, P_{acc} , P_{λ} , finite probes, or residual-to-error promotion.
D6	The source-path certificate identifies the residual/Jacobian path and the 132-row layout; the D6 certificate independently proves the dynamic row ordering, unweighted residual scaling, and accepted AD binding for all 36 Newton–Euler rows.	No remaining D6-specific layout/scaling/AD test; use the closed D6 certificate in the D5 direct-substitution residual certificate.

smooth FullVA Gauss lift Z_G is inserted into the implemented dynamic rows, the lifted acceleration, multiplier-wrench, force, and gyroscopic terms match the continuous Newton–Euler balance row by row. Hence all 36 dynamic rows have zero residual on Z_G , which is stronger than the $O(h^7)$ residual required by Lemma 39. This is an identity in the 132-row implemented residual, not a cancellation argument over the separate 162 primitive Taylor subterms. The proof is non-circular: it does not use a P_{state} actual PS3 lift, P_{acc} , P_{λ} , finite numerical probes, or residual-to-error promotion.

Lemma 8 (Same-object instantiation for the direct D5 lift). *For each stage s , body b , component c , and translational or rotational row type, the E_{lift} term in Eq. (89) vanishes by same-object substitution on the accepted direct route. It is not an additional Taylor estimate, not a primitive-route assumption, and not inferred from finite-grid or residual-to-error evidence.*

Proof. Fix the accepted step data, the single lifted object

$$Z_G = (Q^G, V^G, A^G, \Lambda^G), \quad (90)$$

and a single implemented scalar row index in the 132-row residual. The row in the residual is then evaluated at the object in Eq. (90), with the same stage time, the same FullVA acceleration entry, the same lower-pair multiplier wrench, the same smooth force/torque lift, and the same unweighted row ordering and scaling used by the executable residual. D1–D2 give the Newton–Euler balance identity on this very lift; D3 identifies the lower-pair multiplier wrench in the same scalar row; D4 identifies the force and torque maps on the same smooth branch; and D6 identifies the row ordering, scaling, and AD-bound formula row with the implemented residual component. After those same-object identifications, the implemented row is literally the left-hand side of the continuous Newton–Euler balance displayed in Eq. (88), so the lift mismatch term vanishes. If any ingredient were taken from a different lift, row scaling, time slice, multiplier convention, or force evaluation, this same-object identity would fail and the argument would fall back to the separate primitive/Taylor route rather than to the accepted direct PC2 proof. $\square$

Table 26 is retained as a separate primitive-route specification for a more granular primitive/Taylor verification route. The theorem-use rule is deliberately one-way and is summarized as follows: the table records a conditional

D5 primitive/Taylor certificate schema, while the active direct residual bridge closes PC2 independently. Thus the primitive/Taylor route remains open and is non-active in the theorem: its unproved inputs define a separate dynamic-row replacement-certificate route, not an additional premise of the direct residual-bridge theorem. Equivalently, the primitive/Taylor route is non-active in the theorem and its obligations are primitive-route inputs only; this route does not discharge the PC2 stage-residual condition. Its conditional coverage is nevertheless more specific than an unstructured separate-route open gap: all 162/162 Newton–Euler primitive subterms have assigned Taylor or mean-value reduction templates and primitive dependencies. This is template coverage only; the certified count of actual primitive Taylor bounds is 0/162, and only the compact-tube constant primitive is discharged. The remaining non-active primitive-route inputs are the five open same-object lift/reduction estimates $P_{\text{state}}, P_{\text{acc}}, P_{\lambda}, P_{\text{geom}}, P_{\text{gyro}}$ needed to turn those templates into actual $O(h^7)$ subterm bounds; those bounds are not part of the accepted theorem route.

Reference Taylor boundary for D5. The Taylor local-error argument in the Lie-group BDF reference [8] serves here only as a proof-organization model: fix the Lie-group reconstruction, insert the smooth exact solution into the already defined discrete equations, estimate local defects, and only then propagate the constrained error recursion. It does not by itself supply any primitive FullVA Taylor bound for the present Gauss-stage residual. In particular, the reference-paper insertion step is typed by the BLieDF multistep equations themselves; its analogue here is not a free Taylor expansion of separate physical subterms, but the same-object evaluation $F_{A,h}(Z_G; y)$ after the FullVA stage vector, row ordering, scaling, and proof norm have been fixed. A primitive D5 subterm estimate can enter this analogue only after it has been transported to that same fixed residual map and after its root lift inputs are proved independently on the same compact tube. By contrast, the BLieDF acceleration and multiplier estimates are tied to a multistep increment representation, hidden-constraint differences, and a coupled BDF recursion, whereas the separate D5 primitive route here would need a non-circular unweighted bound on the acceleration components that enter the implemented Newton–Euler FullVA rows after the lower-pair constraints, Lie chart, row ordering, and residual scaling have already been fixed. Thus the reference proof architecture serves only to order the local obligations for $P_{\text{state}}, P_{\text{acc}}, P_{\lambda}, P_{\text{geom}},$ and P_{gyro} . It does not certify, import, or bound any of

Table 26: D5 primitive/Taylor route specification for the Newton–Euler rows; not invoked as a premise of Theorem 1.

Target	Primitive-route target statement	Proof state
Stage-lift substitution	Construct the smooth FullVA Gauss lift Z_G in the accepted Lie chart and substitute the lifted stage pose, velocity, acceleration, and multiplier variables into the implemented dynamic-row expressions.	Separate primitive-route target only; not invoked by the theorem. The direct residual bridge supplies the PC2 input.
Translational balance rows	For each body, stage, and Cartesian component, reduce the expanded row $m_b \hat{a}_{i,b} - f_b(\hat{q}_i, \hat{v}_i, t_i) - G_{i,b}^T \hat{\lambda}_i$ to the continuous Newton balance on the lift plus a Taylor remainder bounded by Ch^7 .	Separate primitive-route target only; not invoked by the theorem. The direct residual bridge supplies the PC2 input.
Rotational balance rows	For each body, stage, and body-frame component, reduce $J_b \hat{\alpha}_{i,b} + \hat{\omega}_{i,b} \times J_b \hat{\omega}_{i,b} - \tau_b(\hat{q}_i, \hat{v}_i, t_i) - H_{i,b}^T \hat{\lambda}_i$ to the lifted angular-momentum balance plus a Taylor remainder bounded by Ch^7 .	Separate primitive-route target only; not invoked by the theorem. The direct residual bridge supplies the PC2 input.
Closed support certificates	Use D3 multiplier-wrench consistency, D4 smooth force lift, and D6 row binding as zero or bounded inputs in the above reductions without changing the row scaling.	D3/D4/D6 support closed only; P_{state} , P_{acc} , P_{λ} , P_{geom} , and P_{gyro} remain open; no primitive-route Taylor bounds are certified.
Uniformity	Show the same constant family is bounded on the compact smooth proof tube, uniformly for all 36 dynamic component rows and all sufficiently small h .	Separate primitive-route target only; not invoked by the theorem. The direct residual bridge supplies the PC2 input.

the 162 D5 primitive Taylor subterms unless the required P_{state} and P_{acc} lift inputs are proved inside this stage-residual setting. No Taylor term is therefore certified as proved from the reference paper or from proof-architecture similarity.

The manuscript statement is deliberately narrower than the full primitive/Taylor route. The Newton–Euler symbolic-defect certificate is used here as a provenance record for the row-expansion and runtime-equivalence checks and as the 36-row inventory feeding Lemma 39. It is not the active dynamic-row closure route. For the direct residual-bridge route used by PC2, the dynamic-row residual input is supplied by the D5 direct-substitution certificate: every expanded Newton–Euler component row has zero residual after substituting the smooth Gauss lift Z_G . The symbolic-certificate C3 primitive/Taylor defect-bound lane remains open and is not the route consumed by the PC2 condition here. The reproducibility package includes both this direct certificate and the older row-local D5 primitive-route record. The latter lists the closed D1/D2/D3/D4/D6 inputs and the one open lifted-stage dynamic term for the alternative primitive/Taylor route; it is not the route that discharges the PC2 stage-residual condition. The D5 primitive inventory, recorded as a D5 term-budget record, further decomposes that open lifted-stage term into 162 Taylor subterms: four translational subterms per translational row and five rotational subterms per rotational row. The decomposition lists the available regularity and row-binding inputs, and it supplies conditional reduction rules for all 162/162 subterms. This is a conditional finite implication map rather than an active theorem premise: if the primitive lift inputs below are separately proved uniformly on the same tube, the table identifies which Taylor/mean-value estimate closes each subterm. At present, however, this is template coverage only: the certified count of actual primitive Taylor bounds is 0/162 on that route, and each subterm still needs an independent uniform $O(h^7)$ Taylor bound before the primitive/Taylor route can close. The obstruction is coupled, not cosmetic: all 162 conditional templates are anti-circular reduction templates, but none is a theorem input until its blocking primitive is proved; in particular the 36 linear acceleration subterms require unweighted acceleration $O(h^7)$, so the available h -weighted acceleration control cannot promote those terms. A companion primitive-bound reduction groups the 162 subterms into six primitive obligations: stage state, acceleration, multiplier, dynamic-row geometry, gyroscopic bilinear, and compact-tube uniformity bounds. This is an implication map, not a proof of the primitive bounds: all 162 subterms

have a recorded reduction rule, but it does not certify the lift-rate primitive bounds. The compact-tube constants primitive is discharged below, while no induced Taylor bounds are certified on the primitive/Taylor route. Only the compact-tube constant obligation P_{tube} is discharged below; the five lift and bilinear primitives remain open. The primitive-obligation closure plan is therefore explicit: prove P_{tube} , P_{state} , P_{acc} , P_{λ} , P_{geom} , and P_{gyro} uniformly on the same smooth proof tube. The first supplies compact-tube constants; the next three bound stage state, acceleration, and multiplier lifts; the last two bound dynamic-row geometry and the gyroscopic bilinear term. The compact-tube constant obligation P_{tube} is closed by Lemma 9; the five lift and bilinear obligations remain open. Primitive dependency graph for the separate Taylor route: P_{state} and P_{acc} are root lift obligations. The former still lacks the actual PS3 state-lift instantiation, and the latter has only the h -weighted acceleration control, not the unweighted $O(h^7)$ acceleration lift needed by the primitive route. The multiplier obligation P_{λ} has its rate-propagation mechanism available only as a conditional implication from those state and acceleration inputs. The geometry obligation P_{geom} is downstream of P_{state} and P_{λ} , and the gyroscopic obligation P_{gyro} is downstream of the angular-velocity component of P_{state} . Thus the two row-level reductions above are conditional sufficiency statements inside the Taylor route, and the admissible dependency order is Eq. (91),

$$\begin{aligned}
P_{\text{state}}, P_{\text{acc}} &\implies P_{\lambda}, \\
P_{\text{state}}, P_{\lambda} &\implies P_{\text{geom}}, \\
P_{\text{state}} &\implies P_{\text{gyro}}.
\end{aligned} \tag{91}$$

This dependency graph is used together with the already closed P_{tube} . The current primitive-route conditions are therefore: This table is a primitive-route obstruction map: no actual D5 primitive Taylor bound from this route is invoked by the theorem, and the separate primitive-route implication remains a separate sufficient route for the dynamic-row PC2 input. Here “conditional” means that the local reduction lemma is closed under the listed inputs, while the actual primitive $O(h^7)$ bound is not established on the accepted theorem branch. The two root targets are precise. To close P_{state} on this route one must prove a non-circular actual PS3 statement: using only the non-dynamic 96-row defect, the compact-tube PS2 inverse, and an acceleration input independent of the state perturbation being estimated, obtain $\|\delta S\| \leq C_S h^7$ uniformly on the tube. To close P_{acc} , one must prove the

Table 27: D5 primitive/Taylor retained-interface map; separate primitive-route specification, not a premise of Theorem 1.

Primitive	Primitive-route condition	Affected Taylor rows	Required primitive input	Scope limitation
P_{tube}	closed	162 template constants	Compact proof-tube constants only.	Does not by itself prove any primitive lift rate or term-level Taylor bound.
P_{state}	root not established	126	Non-circular actual PS3 state lift from the non-dynamic rows.	The direct full-residual route corollary does not close the primitive route.
P_{acc}	root not established	36	Unweighted $O(h^7)$ acceleration lift; current $h\ \delta A\ = O(h^7)$ gives only unweighted $O(h^6)$.	Cannot be read from dynamic balance or from the h -weighted control.
P_{λ}	conditional on roots	72	Actual P_{state} and P_{acc} inputs before PL4 propagates multiplier rates.	Closed PL4 propagation is conditional, not an actual multiplier lift.
P_{geom}	conditional downstream	36	Actual P_{state} and P_{λ} inputs.	Geometry reduction alone is not an actual term closure.
P_{gyro}	conditional downstream	18	Angular-rate part of the actual P_{state} lift.	Bilinear reduction alone is not an actual term closure.

unweighted estimate $\|\delta A\| \leq C_A h^7$ in the acceleration variables that enter the Newton–Euler rows. The currently recorded velocity-collocation algebra gives only $h\|\delta A\| = O(h^7)$, equivalently $\|\delta A\| = O(h^6)$, so a Taylor-route proof would need either $O(h^8)$ velocity/velocity-row inputs, an independent non-dynamic inverse controlling acceleration directly, or an explicit P_{acc} assumption. Once these two root targets are proved, PL4 supplies P_λ , and the row-level geometry and gyroscopic corollaries below supply P_{geom} and P_{gyro} . Without those inputs, the primitive/Taylor route remains a separate sufficient-route specification for the dynamic-block residual value, not the route that discharges PC2 in the theorem.

Lemma 9 (Compact proof-tube constants for D5). *Under Assumption 1, the accepted smooth branch has a compact proof tube K and a step threshold $h_0 > 0$ such that, for $0 < h < h_0$, the exact reduced trajectory, the lifted Gauss stages, and the local accepted Newton stage variables remain in K . On this tube all force, torque, friction, constraint-geometry, and gyroscopic maps that enter the 36 Newton–Euler dynamic rows have uniformly bounded derivatives through the order needed by the D5 Taylor reductions. Hence the primitive obligation P_{tube} holds.*

Proof. The image of the exact smooth reduced trajectory on the fixed interval is compact. Assumption 1 gives a smooth C^7 reduced vector field, a smooth FullVA lift on the same tube, and the Gauss-predictor inverse and derivative bounds consumed by Lemma 39. That lemma, not an isolated-root premise in this compact-tube argument, supplies the local accepted stage root $Z_A(h)$ for sufficiently small h on the selected branch. To make the tube selection explicit, take the compact image of the exact trajectory and its smooth Gauss-stage lift for $0 \leq c_i \leq 1$, choose a finite chart cover whose closures stay inside the regularity domain, and let $d_K > 0$ be the minimum distance from those closures to the chart, rank, and smooth-force boundaries. The Gauss lift is continuous in h , while the accepted-stage bound from Lemma 39 gives $\|Z_A - Z_G\| \leq C_Z h^7$; hence h_0 is chosen so that both the Gauss-stage displacement from the exact compact image and $C_Z h^7$ are less than $d_K/3$. The closed $d_K/3$ -neighborhood inside the finite cover is then one compact tube K containing the exact trajectory, the reduced Gauss stage reconstruction, and the lifted FullVA stage variables for every $0 < h < h_0$ on the selected branch.

The D4 smooth-branch certificate supplies C^7 force and friction maps on this fixed tube, and the constraint geometry, moment-arm maps, and the

bilinear gyroscopic map are smooth compositions of the same lifted pose and velocity variables. Since the list of row maps and D5 primitive subterms is finite, the maxima of the required derivative suprema over K are finite and row-independent. This proves only the uniform-constant primitive P_{tube} ; it does not prove the state, acceleration, multiplier, geometry, or gyroscopic lift rates. $\square$

Lemma 10 (Non-dynamic stage-map definition for P_{state}). *Let Eq. (92) collect the accepted stage pose and velocity block and the auxiliary acceleration block for all bodies in the accepted Lie chart:*

$$S = (r_i, \eta_i, v_i, \omega_i)_{i=1}^3, \quad A = (a_i, \alpha_i)_{i=1}^3 \quad (92)$$

Define $\mathcal{N}_h^{\text{nd}}(S, A)$ as the ordered vector of the five non-dynamic row families in Eq. (93),

$$\begin{aligned} r_i - r_n - h \sum_j A_{ij} v_j, & \quad \eta_i - h \sum_j A_{ij} J_r^{-1}(\eta_j) \omega_j, \\ v_i - v_n - h \sum_j A_{ij} a_j, & \quad \omega_i - \omega_n - h \sum_j A_{ij} \alpha_j, \\ \Phi(q_i, t_i) - g_i^{(0)}, & \quad \Phi_q(q_i, t_i) v_i - g_i^{(1)}, \quad \Phi_q(q_i, t_i) a_i + \dot{\Phi}_q(q_i, v_i, t_i) v_i - g_i^{(2)}. \end{aligned} \quad (93)$$

This map has 96 scalar rows in the cylindrical-chain stage layout: 18 translational position-collocation rows, 18 rotational Lie-collocation rows, 18 translational velocity-collocation rows, 18 angular velocity-collocation rows, and 24 lower-pair FullVA rows. It is exactly the non-dynamic map used by the 96-row certificate. This closes only the PS1 map-definition subproof for P_{state} . It does not assert a bounded inverse, an inf-sup estimate, an $O(h^7)$ state-lift rate, or PC2.

Proof. The row families listed above are the non-dynamic rows in Eqs. (10), (11), (12), (13), and (16). Their row counts are the same counts recorded in the kinematic row-defect certificate: four 18-row collocation families and one 24-row lower-pair FullVA family. The multiplier block is absent from $\mathcal{N}_h^{\text{nd}}$, and the Newton–Euler balance family is excluded. Therefore the map interface needed by PS1 is fixed in the accepted Lie chart and is bound to the same 96 certified rows. No estimate for (S, A) follows from this definition alone; a separate local inverse or inf-sup proof is still needed for PS2 before P_{state} can be closed. $\square$

The next primitive obligation, P_{state} , is deliberately kept open. The 96-row non-dynamic certificate proves that the kinematic and lower-pair residual rows vanish on the smooth FullVA lift, but a residual-row identity is not by itself a variable-lift estimate. The aggregate PS2 inverse needed by this route is proved below as a compact-tube estimate, but closing P_{state} still requires an actual PS3 instantiation: the non-dynamic residual defect and the h -weighted acceleration input must be combined to prove that the stage pose, translational velocity, and angular velocity variables differ from the smooth lifted Gauss variables by $O(h^7)$. This proof must not use Lemma 39, because that lemma already assumes the dynamic residual defect that P_{state} is meant to help prove.

Lemma 11 (Anti-circular route for P_{state}). *The separate PS2–PS3 proof route for P_{state} may use only the 96-row non-dynamic residual certificate, the stage-map definition $\mathcal{N}_h^{\text{nd}}$, and compact proof-tube smoothness constants. The D5 dynamic residual defect is disallowed as an input to P_{state} . Likewise Lemma 39 and residual identities alone are disallowed as inputs for converting row defects into state-variable lift rates. This closes the PS4 anti-circularity subproof, but it does not close PS2, PS3, P_{state} , any induced D5 Taylor bound, or PC2.*

Lemma 12 (Weighted PS2 target for P_{state}). *The PS2 inverse statement for P_{state} is a weighted PS2 target, not a literal inverse for all state and acceleration variables. In the cylindrical-chain chart, the state block $S = (r_i, \eta_i, v_i, \omega_i)_{i=1}^3$ has 72 scalar coordinates, the auxiliary acceleration block $A = (a_i, \alpha_i)_{i=1}^3$ has 36, and $\mathcal{N}_h^{\text{nd}}$ has 96 residual rows. Thus the correct finite-dimensional target is the state-block estimate in Eq. (94),*

$$\|\delta S\| \leq C_{\text{PS2}}(\|\delta \mathcal{N}_h^{\text{nd}}\| + h\|\delta A\|), \quad (94)$$

with acceleration treated as a parameter in the h -weighted norm $h\|\delta A\|$. This specification fixes the PS2 target, but it does not close PS2, does not close PS3, does not prove the uniform constant C_{PS2} , and does not close P_{state} or PC2.

Lemma 13 (Kinematic-block certificate for the weighted PS2 route). *For one scalar position–velocity channel, the three-stage Gauss kinematic subblock has the linearized form in Eq. (95),*

$$\delta R_q = \delta q - hA_G \delta v, \quad \delta R_v = \delta v - hA_G \delta a, \quad (95)$$

where A_G is the fixed three-stage Gauss matrix. Therefore Eq. (96) gives the corresponding block inverse:

$$\begin{bmatrix} \delta q \\ \delta v \end{bmatrix} = \begin{bmatrix} I & hA_G & hA_G^2 \\ 0 & I & A_G \end{bmatrix} \begin{bmatrix} \delta R_q \\ \delta R_v \\ h\delta a \end{bmatrix}. \quad (96)$$

For $0 < h \leq 0.04$ this matrix has a bounded Euclidean operator norm depending only on A_G and the step threshold. Kronecker lifting gives the same bound for all translational channels and for the angular-velocity collocation channels after the accepted chart norm equivalence is applied. Thus the 72 kinematic rows already provide a concrete analytic subblock for the weighted PS2 estimate; the 24 lower-pair rows are surplus for this subblock. This certificate records the algebraic constant needed by the PS2 route, but the full nonlinear PS2 proof still has to bind the implemented rotational nonlinear row and the 72-to-96 residual-row injection to this template. Consequently it does not by itself close PS2, PS3, P_{state} , any induced Taylor bound, or PC2.

Lemma 14 ($SO(3)$ chart binding for the weighted PS2 route). *On the compact proof tube, restrict the accepted rotational chart to $\|\eta\| \leq \rho$ with $\rho = 0.5 < \pi$. The $SO(3)$ right Jacobian and its inverse obey the uniform bounds in Eq. (97),*

$$\|J_r(\eta)\|_2 \leq 1, \quad \|J_r(\eta)^{-1}\|_2 \leq \frac{\rho}{2 \sin(\rho/2)} = 1.010493 \dots \quad (97)$$

Thus chart-coordinate perturbations and body-angular-velocity perturbations may be measured in equivalent Euclidean norms with an h -independent constant on the PS2 proof tube. This records the pure Lie-chart norm-equivalence input needed by the kinematic-block certificate. It does not supply the nonlinear mean-value estimate for $\eta_i - h \sum_j A_{ij} J_r^{-1}(\eta_j) \omega_j$, does not prove the 72-to-96 row injection/scaling estimate, and does not close PS2, PS3, P_{state} , any induced Taylor bound, or PC2.

Lemma 15 (72-to-96 row injection for the weighted PS2 route). *Let Π_{kin} select the four kinematic row families from the accepted 96-row non-dynamic residual $\mathcal{N}_h^{\text{nd}}$: translational position collocation, rotational Lie collocation, translational velocity collocation, and angular-velocity collocation. These four families have 72 scalar rows, and the remaining 24 scalar rows are the lower-pair FullVA rows. With the unweighted residual ordering used in the implemented residual and in the 96-row certificate, the injection bound is Eq. (98):*

$$\|\Pi_{\text{kin}}\|_2 = 1, \quad \|\Pi_{\text{kin}} \delta \mathcal{N}_h^{\text{nd}}\| \leq \|\delta \mathcal{N}_h^{\text{nd}}\|. \quad (98)$$

Consequently the unweighted non-dynamic residual norm dominates the kinematic residual norm used by the PS2 kinematic block. This closes only the 72-to-96 row injection/scaling subclaim. It does not supply the nonlinear mean-value estimate for the rotational row, does not close PS2, PS3, P_{state} , any induced Taylor bound, or PC2.

Lemma 16 (Nonlinear rotational-row binding for the weighted PS2 route). *Let the rotational chart velocity map be*

$$F(\eta, \omega) = J_r(\eta)^{-1}\omega. \quad (99)$$

On the accepted compact proof tube, the chart radius from Lemma 14 keeps J_r^{-1} smooth, and Lemma 9 supplies for the map in Eq. (99) a finite Lipschitz constant L_F for F . Hence, for two stage states in the tube,

$$\|\Delta F\| \leq L_F(\|\Delta\eta\| + \|\Delta\omega\|). \quad (100)$$

For the implemented rotational row

$$R_{\eta,i} = \eta_i - h \sum_j A_{ij} J_r(\eta_j)^{-1} \omega_j, \quad (101)$$

the nonlinear part therefore enters as an $h\|A_G\|_2 L_F$ perturbation of the triangular Gauss kinematic block by Eqs. (100) and (101). After shrinking the asymptotic step threshold, if necessary, so that $h\|A_G\|_2 L_F \leq 1/2$, this perturbation is absorbed on the left and gives the compact-tube mean-value estimate

$$\|\Delta\eta\| + \|\Delta\omega\| \leq C_{\text{rot}}(\|\Delta R_\eta\| + \|\Delta R_\omega\| + h\|\Delta\alpha\|). \quad (102)$$

Equation (102) is the small-step absorption that binds the implemented rotational Lie row to the weighted PS2 kinematic template without using the D5 dynamic residual theorem or Lemma 39. It closes the rotational nonlinear mean-value subclaim. It does not by itself promote the aggregate PS2 inverse, does not close PS3, P_{state} , any induced Taylor bound, or PC2.

Lemma 17 (Aggregate weighted PS2 promotion for P_{state}). *There are constants $h_0 > 0$ and $C_{\text{PS2}} > 0$, depending only on the compact proof tube and on the fixed Gauss matrix, such that for $0 < h < h_0$ every pair of accepted non-dynamic stage states in the tube satisfies the state-block estimate*

$$\|\delta S\| \leq C_{\text{PS2}}(\|\delta \mathcal{N}_h^{\text{nd}}\| + h\|\delta A\|). \quad (103)$$

Thus Eq. (103) promotes the component certificates to a uniform weighted PS2 inverse for the non-dynamic stage map. The finite linearization probe is diagnostic only and is not used in the theorem proof. This lemma does not instantiate PS3, does not prove an $O(h^7)$ state lift, does not certify any D5 Taylor term bound, and does not close P_{state} or PC2.

Proof. Write the selected kinematic residual as $\Pi_{\text{kin}}\mathcal{N}_h^{\text{nd}}$, where Π_{kin} selects the four 18-row kinematic families. The 72-to-96 selector has norm one by Lemma 15, hence

$$\|\Pi_{\text{kin}}\delta\mathcal{N}_h^{\text{nd}}\| \leq \|\delta\mathcal{N}_h^{\text{nd}}\|. \quad (104)$$

The lower-pair rows are therefore harmless surplus rows for the estimate in Eq. (104).

For the translational position-velocity channels, and for the angular-velocity collocation channels after the chart norm equivalence of Lemma 14, the linearized row map is the fixed three-stage triangular Gauss block of Lemma 13. Its inverse is explicit and has an h -independent bound for $0 < h \leq h_0$. Consequently those channels satisfy

$$\|\delta r\| + \|\delta v\| + \|\delta \omega\| \leq C_0(\|\delta R_r\| + \|\delta R_v\| + \|\delta R_\omega\| + h\|\delta a\| + h\|\delta \alpha\|) \quad (105)$$

with C_0 depending only on A_G , the chart-equivalence constant, and the chosen step threshold. Eq. (105) is the linear kinematic part that remains before the nonlinear rotational row is absorbed.

It remains to bind the nonlinear rotational Lie row. Let $F(\eta, \omega) = J_r(\eta)^{-1}\omega$. Since $\|\eta\| \leq \rho < \pi$ on the compact proof tube, F is smooth there. For two stage vectors in the tube, Taylor's formula with integral remainder, equivalently the finite-dimensional mean-value theorem, gives

$$F(\eta + \delta\eta, \omega + \delta\omega) - F(\eta, \omega) = DF(\eta, \omega)[\delta\eta, \delta\omega] + \mathcal{R}_F, \quad (106)$$

where

$$\|\mathcal{R}_F\| \leq C_F(\|\delta\eta\| + \|\delta\omega\|)^2 \quad (107)$$

Equation (107) is the quadratic remainder bound used only inside the rotational-row mean-value argument. for a tube-uniform second-derivative bound C_F . The linear mean-value form also gives the Lipschitz estimate

$$\|\Delta F\| \leq L_F(\|\Delta\eta\| + \|\Delta\omega\|). \quad (108)$$

Thus the difference of the implemented rotational rows can be written as the triangular Gauss rotational block plus the perturbation $-hA_G\Delta F$, with

the Taylor and Lipschitz controls in Eqs. (106)–(108). Shrinking h_0 , and if necessary the fixed local proof-tube radius, makes

$$h\|A_G\|_2 L_F \leq \frac{1}{2} \quad (109)$$

on the tube. The perturbation is then absorbed into the left-hand side of the triangular inverse estimate, yielding

$$\|\delta\eta\| + \|\delta\omega\| \leq C_1(\|\delta R_\eta\| + \|\delta R_\omega\| + h\|\delta\alpha\|), \quad (110)$$

with C_1 independent of h . This is a genuine Taylor/mean-value proof on the compact tube: Eq. (109) is the small-step hypothesis and Eq. (110) is the absorbed rotational estimate, not a finite-rank numerical probe.

Combining the translational, angular-velocity, and rotational estimates and taking C_{PS2} to dominate the finitely many constants gives

$$\|\delta S\| \leq C_{\text{PS2}}(\|\Pi_{\text{kin}}\delta\mathcal{N}_h^{\text{nd}}\| + h\|\delta A\|) \leq C_{\text{PS2}}(\|\delta\mathcal{N}_h^{\text{nd}}\| + h\|\delta A\|). \quad (111)$$

The proof uses only the non-dynamic map, compact-tube smoothness, the fixed Gauss block, and the row selector in Eq. (111). It does not use the D5 dynamic residual defect or Lemma 39. Therefore it closes the uniform weighted PS2 inverse and no stronger claim. $\square$

Lemma 18 (Finite PS2 linearization diagnostic for P_{state}). *At the solved one-step accepted smooth cylindrical-chain stages for $h \in \{0.04, 0.02, 0.01\}$, the implemented residual Jacobian has the finite weighted PS2 operator*

$$M_h = \begin{bmatrix} D_{S,A}\mathcal{N}_h^{\text{nd}} \\ 0 & hI_A \end{bmatrix}, \quad (\delta S, \delta A) \mapsto (D\mathcal{N}_h^{\text{nd}}[\delta S, \delta A], h\delta A), \quad (112)$$

where $S = (r, \eta, v, \omega)$ has 72 coordinates, $A = (a, \alpha)$ has 36 coordinates, and $\mathcal{N}_h^{\text{nd}}$ has 96 non-dynamic rows. The finite operator in Eq. (112) records full column rank 108/108 for all three probes, a minimum singular value 9.999780×10^{-3} , and a maximum finite state-projection constant 2.019470. This is a finite solved-stage rank diagnostic for the PS2 target; it is not a uniform compact-tube inverse or inf-sup proof and does not close PS2, PS3, P_{state} , any D5 Taylor bound, or PC2.

Lemma 19 (Conditional PS3 conversion for P_{state}). *Assume the weighted PS2 target above holds with a uniform constant on the compact proof tube. If the 96-row non-dynamic residual defect satisfies $\|\delta\mathcal{N}_h^{\text{nd}}\| = O(h^7)$ and the acceleration input satisfies $\|\delta A\| = O(h^6)$, then the state-block perturbation satisfies $\|\delta S\| = O(h^7)$.*

Proof. The weighted PS2 estimate gives

$$\|\delta S\| \leq C_{\text{PS2}}(O(h^7) + h O(h^6)) = O(h^7). \quad (113)$$

Equation (113) records the conditional PS3 conversion and explains why the $O(h^6)$ acceleration-rate obstruction is compatible with the $h\|\delta A\|$ PS2 target. It does not by itself instantiate the actual PS3 lift, because the required residual and acceleration-rate hypotheses still have to be supplied in the lift table; it does not certify any induced Taylor bound, and does not close P_{state} or PC2. $\square$

Lemma 20 (Direct-route state-block corollary; not primitive P_{state} closure). *Assume the aggregate PS2 estimate of Lemma 17, the 96-row non-dynamic certificate, the D5 direct-substitution certificate for the 36 Newton–Euler rows, and the uniform inverse hypothesis in Lemma 39. Then the full 132-row residual route supplies a direct-route state-block corollary only, not a primitive PS3 input for proving P_{state} :*

$$\|\delta S\| = O(h^7), \quad h\|\delta A\| = O(h^7). \quad (114)$$

The route does not use velocity collocation, does not use P_{state} , P_{acc} , or P_λ as an input, and Eq. (114) does not promote a finite rank probe to a compact-tube theorem. It closes only the direct-route PS3 corollary; it does not certify the primitive/Taylor route or any of the 162 induced D5 Taylor subterm bounds.

Proof. Let Z_G be the smooth FullVA lift of the reduced Gauss stage and Z_A the accepted FullVA stage zero. The non-dynamic certificate gives

$$\|\mathcal{N}_h^{\text{nd}}(Z_G)\| = O(h^7) \quad (115)$$

for the 96 non-dynamic rows. Equation (115) is the 96-row non-dynamic input to the direct residual bridge. For each Newton–Euler component row, the D5 direct-substitution certificate expands the implemented row into the closed D1/D2 balance identity, the D3 multiplier-wrench identity, the D4 smooth force/torque input, the D6 row binding, and the lifted-stage balance term. After substituting Z_G , the direct-route residual decomposition has zero residual remainder in each of the 36 dynamic rows. This supplies $R_{\text{dyn}}(Z_G) = 0$, hence the $O(h^7)$ input to Lemma 39; it is not a certification of the primitive 162-subterm Taylor route. In that separate verification route no induced Taylor bounds are certified, the 162 subterms remain under

open primitive interfaces, and the 36 h -weighted acceleration primitive terms remain insufficient. Combining the finite row families gives

$$\|F_{A,h}(Z_G; y)\| = O(h^7) \quad (116)$$

for the full 132-row residual.

Lemma 39 then gives $\|Z_A - Z_G\| = O(h^7)$ in the weighted local stage norm. In particular the state block and weighted acceleration block obey

$$\|\delta S\| + h\|\delta A\| \leq C\|Z_A - Z_G\| = O(h^7). \quad (117)$$

Equivalently, the h -weighted acceleration input required by the aggregate PS2 estimate is supplied by the full residual perturbation itself through Eqs. (116) and (117). The argument never rewrites velocity collocation as

$$\delta A = h^{-1}(A_G^{-1} \otimes I)(\delta V - \rho_V), \quad (118)$$

so it does not assume the P_{state} velocity perturbation that it is trying to control. Avoiding Eq. (118) is why the route is non-circular.

Finally, applying the aggregate PS2 estimate,

$$\|\delta S\| \leq C_{\text{PS2}}(\|\delta \mathcal{N}_h^{\text{nd}}\| + h\|\delta A\|), \quad (119)$$

Equation (119) gives $\|\delta S\| = O(h^7)$. This direct residual-bridge/Kantorovich perturbation route is a direct-route corollary of the full residual certificate. It is not recycled as an independent proof of the primitive/Taylor route, because that route must derive P_{state} , P_{acc} , and P_λ before using them to certify the D5 Taylor subterms. $\square$

Lemma 21 (No direct-route feedback into primitive P_{state}). *The direct-route estimate of Lemma 20 cannot be used as the actual P_{state} primitive in the separate D5 primitive/Taylor route. In that route the P_{state} input must be obtained before the 162 primitive Newton–Euler Taylor subterms are certified, using only the accepted non-dynamic rows, the compact-tube PS2 inverse data, and an acceleration input independent of the state perturbation being estimated. The direct-route estimate instead consumes the full 132-row residual bridge, including the D5 direct-substitution certificate and Lemma 39. It is therefore a theorem-route corollary, not an admissible primitive-route input.*

Proof. The separate primitive/Taylor route has a fixed dependency order: first prove the root lift primitives P_{state} and P_{acc} on the accepted compact branch, then propagate P_λ , and only then bound the primitive D5 subterms. The P_{state} slot in that order is required to be independent of the D5 dynamic residual defect and of the full-map perturbation lemma whose input it would later help certify.

Lemma 20 has a different causality. It first uses the accepted direct PC2 bridge: the 96 non-dynamic row certificate, the 36 dynamic rows supplied as a same-branch base-point zero block at the smooth FullVA lift, and the full-map stage perturbation result. Only after that full residual object is fixed does it read off the state-block and h -weighted acceleration-block estimates. Feeding this output back as the primitive P_{state} input would reverse the proof order and would make the primitive Taylor route depend on the direct route it is meant to be independent from. Hence the direct-route estimate may be cited only as a direct-route corollary for the theorem proof; it leaves actual PS3, P_{state} , the 126 P_{state} -dependent primitive subterms, and the 0/162 primitive Taylor inventory unchanged. $\square$

Lemma 22 (Acceleration-map definition for P_{acc}). *Let the acceleration block be*

$$A = (a_i, \alpha_i)_{i=1}^3 \quad (120)$$

collect the accepted translational and angular stage accelerations for all bodies in the same accepted stage ordering as the FullVA residual. Define $\mathcal{A}_h^{\text{acc}}(A)$ as the ordered vector of acceleration-lift Taylor-row interfaces

$$m_b(\hat{a}_{b,i} - a_b(t_i)), \quad J_b(\hat{\alpha}_{b,i} - \alpha_b(t_i)) \quad (121)$$

for every stage i , body b , and component. Equivalently, the acceleration block is Eq. (120). The interfaces in Eq. (121) give 36 scalar rows: 18 translational acceleration-lift rows and 18 angular acceleration-lift rows. It is exactly the acceleration variable interface used by the D5 term-budget rows $T_{\text{acceleration_lift}}$ and $R_{\text{angular_acceleration_lift}}$. This closes only the PA1 map-definition subproof for P_{acc} . It does not prove an acceleration lift rate, a velocity-collocation implication, a dynamic-row defect bound, or PC2.

Proof. The accepted residual has three stages, two moving bodies, and three Cartesian components per translational acceleration block, giving $3 \times 2 \times 3 = 18$ translational acceleration-lift interfaces. The angular acceleration block

has the same stage-body-component count, giving 18 angular acceleration-lift interfaces. The D5 Taylor-term inventory names these two families as $T_{\text{acceleration_lift}}$ and $R_{\text{angular_acceleration_lift}}$, so the definition above fixes the row ordering and variable interface for the 36 P_{acc} term rows. No rate estimate follows from this definition alone: closing P_{acc} still requires a non-circular proof that the accepted acceleration variables differ from the smooth lifted accelerations by $O(h^7)$, without reading that estimate from the Newton–Euler dynamic balance. $\square$

Lemma 23 (Acceleration-row binding for P_{acc}). *Under the accepted residual ordering verified in D6, the 36 components of $\mathcal{A}_h^{\text{acc}}(A)$ occupy global rows 24–35, 68–79, and 112–123. Within each stage, the row order is body-0 translational, body-0 rotational, body-1 translational, and body-1 rotational, with three Cartesian components per block. The translational components are exactly the D5 term-budget rows $T_{\text{acceleration_lift}}$, and the rotational components are exactly the rows $R_{\text{angular_acceleration_lift}}$. This closes only the PA4 row-ordering/Taylor-term binding subproof for P_{acc} . It does not prove an acceleration lift rate, a velocity-collocation implication, a Taylor bound, or PC2.*

Proof. D6 fixes the dynamic-row layout as a 44-row stage block with dynamic offset 24 and dynamic width 12. Therefore the dynamic rows in stages $s = 0, 1, 2$ are $44s + 24, \dots, 44s + 35$, namely global rows 24–35, 68–79, and 112–123. Its per-stage order lists the translational source rows for body 0, the rotational source rows for body 0, the translational source rows for body 1, and the rotational source rows for body 1. The D5 Taylor-term budget assigns the acceleration-lift translational terms to the two translational Newton-balance blocks and the angular acceleration-lift terms to the two rotational Euler-balance blocks, with the same stage, body, and component indices. Hence every acceleration-lift Taylor row has the same global row, balance block, body, component, and runtime source component in the D5 budget and in the D6 accepted residual ordering. This establishes only row and term binding. It gives no estimate for $\hat{a} - a(t_i)$ or $\hat{\alpha} - \alpha(t_i)$. Thus PA2 remains open; PA3 is the separate non-dynamic input-independence subproof closed in Lemma 24, not an acceleration-rate estimate. $\square$

Lemma 24 (Non-dynamic input boundary for P_{acc}). *The proof route for*

P_{acc} can be restricted to the non-dynamic FullVA row families

$$\begin{aligned} & \text{translational_velocity_weak_defect}, \\ & \text{angular_velocity_weak_defect}, \\ & \text{lower_pair_index3_weak_constraints}, \end{aligned} \tag{122}$$

together with the 96-row non-dynamic certificate. Newton–Euler weak-balance rows are not inputs to this route; Eq. (122) is the allowed row-family list. This closes only the PA3 non-circularity subproof for P_{acc} . It does not prove an acceleration lift rate, a velocity-collocation implication, a Taylor bound, or the primitive-and-Taylor route to PC2; PA2 remains open.

Proof. The accepted stage block has the certified row partition

$$0:6, \quad 6:12, \quad 12:18, \quad 18:24, \quad 24:36, \quad 36:44, \tag{123}$$

with the Newton–Euler weak-balance family occupying only the 24:36 sub-block. The rows used to formulate the lower-pair velocity and acceleration-level FullVA relations occupy the non-dynamic families at offsets 12, 18, and 36 in Eq. (123). These three families are included in the 96-row kinematic row-defect certificate, while the Newton–Euler weak-balance family is explicitly excluded from that certificate. Thus a later PA2 proof may use the certified non-dynamic FullVA relations to derive acceleration lift rates, but it cannot read those rates from the dynamic balance equations whose D5 defect is under proof. The lemma therefore establishes only the non-circular input boundary. $\square$

Lemma 25 (Velocity-collocation obstruction for P_{acc}). *The velocity-collocation rows alone are insufficient to prove the unweighted $O(h^7)$ acceleration-lift primitive P_{acc} . Let δV denote the stacked translational and angular stage-velocity difference between the accepted stage variables and the smooth lifted Gauss stages, let δA denote the corresponding stacked acceleration difference, and let ρ_V be the velocity-collocation row perturbation. For the three-stage Gauss matrix A_G ,*

$$\delta V - h(A_G \otimes I)\delta A = \rho_V, \quad \delta A = h^{-1}(A_G^{-1} \otimes I)(\delta V - \rho_V). \tag{124}$$

Thus an $O(h^7)$ bound on δV and ρ_V gives only an unweighted $O(h^6)$ acceleration bound through Eq. (124). PA2 therefore remains open unless one proves a stronger $O(h^8)$ stage-velocity/velocity-row estimate, an independent non-dynamic inverse that controls acceleration directly, or an explicit P_{acc} regularity assumption.

Proof. The three-stage Gauss matrix is nonsingular, with entries fixed independently of h . Subtracting the lifted velocity-collocation equation from the accepted velocity-collocation equation gives the first displayed identity. Multiplication by $A_G^{-1} \otimes I$ gives the second identity and introduces the factor h^{-1} . Since $\|A_G^{-1}\|$ is a fixed constant, the estimate obtained from velocity-collocation alone is $\|\delta A\| \leq Ch^{-1}(\|\delta V\| + \|\rho_V\|)$. With only $\delta V, \rho_V = O(h^7)$, this is $O(h^6)$. The current D5 Taylor budget uses the unweighted dynamic-row acceleration terms $m_b \delta a$ and $J_b \delta \alpha$, so a bound only on $h\delta A$ is not enough to close the existing P_{acc} primitive. This proves the obstruction and does not close PA2, P_{acc} , or PC2. $\square$

Reference-proof boundary for P_{acc} . This obstruction is exactly why the constrained-BDF proof of Wieloch and Arnold is invoked only as proof-order discipline. Their analysis also exposes step-size-scaled derivative and constraint-difference quantities inside a coupled error recursion; it does not provide a FullVA stage estimate for the unweighted acceleration variables appearing in the present Newton–Euler rows. Importing that reference proof at the primitive level would therefore skip the h^{-1} loss displayed above. The admissible uses of the reference remain the structural ones stated in Table 11: fix the Lie-group discrete equations first, prove the local defect for that fixed object, and then propagate the accepted-branch error. A primitive Taylor proof route would still require an independent unweighted P_{acc} proof, an $O(h^8)$ velocity-stage estimate before inversion, a direct non-dynamic acceleration inverse, or an explicit extra theorem assumption.

PA2 weighted-inverse diagnostic. The finite weighted-inverse diagnostic evaluates the operator $(\delta S, \delta A) \mapsto (DN_h^{\text{nd}}[\delta S, \delta A], h\delta A)$ on the same solved stages used by the PS2 probe. It records bounded finite control of $h\delta A$, separating the weighted acceleration diagnostic from the unweighted acceleration estimate required by the D5 Taylor budget. This does not prove a uniform unweighted acceleration lift. PA2 remains open, P_{acc} remains open, and the primitive-and-Taylor route to PC2 remains open. The finite weighted-inverse constants are reference-boundary diagnostics only: they may guide a separate PA2 proof, but they are not theorem inputs, do not certify an unweighted acceleration lift, and do not change the non-active primitive-route Taylor-bound status.

Lemma 26 (Lower-pair acceleration rows do not close P_{acc}). *The lower-pair acceleration-level rows, even together with the finite weighted-inverse*

PA2 diagnostic, do not by themselves imply the unweighted full-body acceleration primitive P_{acc} . To use these non-dynamic rows as a PA2 closure one would need an h -uniform right inverse, or an equivalent compact-tube inf-sup estimate, from the accepted non-dynamic acceleration-row residuals to every translational and angular stage-acceleration component in the D5 acceleration inventory. No such operator is constructed in this manuscript. Consequently these rows cannot serve as a hidden proof of PA2, P_{acc} , or PC2.

Proof. The P_{acc} primitive used by the D5 Taylor budget is an unweighted estimate for the full stacked translational and angular accelerations appearing in the Newton–Euler rows. The lower-pair acceleration relations constrain constraint-direction combinations of these accelerations, with coefficients determined by the lower-pair Jacobians and with additional terms depending on already chosen position and velocity variables. They are therefore row relations for the mechanism constraints; they are not, without an additional rank and right-inverse certificate, a reconstruction formula for all body acceleration components in the D5 dynamic row inventory.

The finite PA2 diagnostic adds bounded control of the weighted component $h\delta A$. This is compatible with the velocity-collocation obstruction of Lemma 25: after removing the factor h , the available estimate loses one power of h unless a stronger velocity estimate or an independent unweighted acceleration inverse is proved. Hence the current manuscript has no implication of the form

$$\|\delta A\| \leq C(\|N_h^{\text{acc}}\| + \text{already proved state and velocity errors}) \quad (125)$$

Equation (125) records the unavailable unweighted implication: it would need C independent of h and range covering all D5 acceleration components. The lower-pair rows and the weighted diagnostic remain useful diagnostics and possible ingredients for a separate PA2 proof, but they do not close the unweighted P_{acc} assumption used below. $\square$

Lemma 27 (No PS2–PA2 bootstrap closure). *The weighted PS2 estimate for P_{state} and the velocity-collocation estimate for P_{acc} cannot close each other by a small-step bootstrap. Let $x_h = \|\delta S\|$, $a_h = \|\delta A\|$, and let r_h collect the $O(h^7)$ non-dynamic row defects and velocity-row defects on the same compact branch. The available estimates have only the form in Eq. (126):*

$$x_h \leq C_{\text{PS2}}(r_h + ha_h), \quad a_h \leq C_A h^{-1}(x_h + r_h). \quad (126)$$

Substituting the second bound in the first gives Eq. (127):

$$x_h \leq C_{\text{PS2}}(1 + C_A)r_h + C_{\text{PS2}}C_A x_h. \quad (127)$$

The feedback coefficient in Eq. (127) is independent of h , so shrinking the step size does not make this a contraction. Moreover, even if $x_h = O(h^7)$ were supplied by some other non-circular argument, the second estimate in Eq. (126) would give only $a_h = O(h^6)$ from $r_h = O(h^7)$. Thus the current PS2 and PA2 ingredients can at most explain the weighted input $ha_h = O(h^7)$; they do not prove the unweighted P_{acc} primitive, the actual primitive P_{state} lift, or any D5 Taylor subterm bound.

Proof. Equation (103) gives the first inequality in Eq. (126) after the non-dynamic defect terms are collected into r_h . The velocity-collocation loss in Eq. (124) gives the second inequality with a constant C_A depending only on the fixed Gauss matrix and the chosen component norm. Composing the two estimates eliminates the small factor h : the h multiplying a_h in PS2 is cancelled by the h^{-1} in the PA2 velocity-collocation inverse. Therefore the available closed estimates do not contain a small parameter that can absorb the x_h -term in Eq. (127). A valid closure would need an additional proof, such as an $O(h^8)$ velocity-stage estimate, an independent unweighted acceleration inverse, or an explicit retained P_{acc} assumption. No such input is proved here, so the primitive/Taylor route remains non-closing. $\square$

Corollary 1 (Row-level acceleration Taylor bound under P_{acc}). *Let the acceleration-row set be*

$$\begin{aligned} \mathcal{R}_{\text{acc}} = \{ & 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, \\ & 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, \\ & 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123 \} \end{aligned} \quad (128)$$

be the global-row set for the translational and angular acceleration Newton–Euler Taylor components. Assume the acceleration part of P_{acc} gives

$$\|\widehat{a}_{s,b} - a_b(t_s)\| \leq C_a h^7, \quad \|\widehat{\alpha}_{s,b} - \alpha_b(t_s)\| \leq C_\alpha h^7 \quad (129)$$

uniformly for every Gauss stage s and body b . If

$$m_{\text{max}} = \max_b m_b, \quad J_{\text{max}} = \max_b \|J_b\|_2, \quad (130)$$

Equations (129) and (130) are the conditional lift-rate and finite body-constant inputs for this row-level corollary. then every scalar row $r \in \mathcal{R}_{\text{acc}}$ satisfies

$$|e_r^T m_b(\widehat{a}_{s,b} - a_b(t_s))| \leq m_{\max} C_a h^7 \quad (131)$$

on translational rows and

$$|e_r^T J_b(\widehat{\alpha}_{s,b} - \alpha_b(t_s))| \leq J_{\max} C_\alpha h^7 \quad (132)$$

on rotational rows. Thus Eqs. (128)–(132) show that the 36 acceleration Taylor rows satisfy the conditional T3 target $O(h^7)$ row bound under P_{acc} with zero Taylor remainder. This is a conditional row-level consequence only: it does not prove the unweighted acceleration lift, does not close PA2 or P_{acc} , and does not change the PC2 status.

Proof. The acceleration contributions in the expanded Newton–Euler rows are the linear maps $A \mapsto m_b A$ and $A \mapsto J_b A$. Hence subtracting the lifted and exact acceleration terms gives exactly

$$m_b(\widehat{a}_{s,b} - a_b(t_s)), \quad J_b(\widehat{\alpha}_{s,b} - \alpha_b(t_s)), \quad (133)$$

Equation (133) is the exact linear acceleration contribution after subtracting the lifted and exact rows. with no quadratic or higher Taylor remainder. Projection onto a scalar component can only decrease the Euclidean norm, and the finite body set gives the constants $m_{\max}$ and $J_{\max}$. The assumed P_{acc} rates therefore imply the $O(h^7)$ row bounds in Eqs. (131) and (132) for exactly the listed translational and rotational acceleration rows. The proof consumes the P_{acc} lift as an input and does not derive that lift from velocity collocation, the weighted PA2 diagnostic, or the dynamic residual. $\square$

Lemma 28 (KKT multiplier-column interface for P_λ). *Let the accepted three-stage lower-pair multiplier array be*

$$\Lambda_h = \{\lambda_{i,j}\}_{i=1,\dots,3;j=0,1}, \quad \lambda_{i,j} \in \mathbb{R}^4, \quad (134)$$

Equation (134) denotes the lower-pair multiplier variables in the accepted three-stage residual. In each 44-variable stage block, the two 18-variable body blocks are followed by $2 \times 4 = 8$ lower-pair multiplier variables; hence the full Newton vector has 24 multiplier columns. The first two components of $\lambda_{i,j}$ generate the normal force $\lambda_{i,j,0}b_{j,0} + \lambda_{i,j,1}b_{j,1}$ in the joint basis, and the last

two components generate the two axis/twist torque multiplier directions used in the rotational balance.

The D5 multiplier interface therefore exposes the direct wrench terms

$$G_b(q_i)^T \lambda_i, \quad H_b(q_i)^T \lambda_i \quad (135)$$

in the translational and rotational Newton–Euler rows. The direct multiplier-wrench Taylor rows exposed by Eq. (135) are the 36 rows named $T_{\text{multiplier_force_lift}}$ and $R_{\text{multiplier_torque_lift}}$, while the primitive-reduction table records 72 total D5 Taylor rows depending on P_λ . The D3 virtual-work certificate supplies the row-expanded wrench-consistency identity for the same 36 dynamic rows. This closes only the PL1 multiplier-interface subproof for P_λ . It does not prove a multiplier lift rate, does not prove a uniform inf-sup bound, does not certify any Taylor bound. It also does not close the primitive-and-Taylor route to PC2.

Proof. The runtime unpacking is

$$\lambda_{i,j} = x_{44i+36+4j:44i+36+4j+4} \quad (136)$$

The slice in Eq. (136) appears after the body variables in stage i , which is the same column group identified as `lower_pair_lambda`. The translational balance uses the first two components only through the joint-basis normal force, and the rotational balance uses the last two components through the corresponding axis/twist torque moment directions at the proximal and distal lower pairs. Thus the KKT multiplier columns that can affect the Newton–Euler weak rows are explicitly identified before any lift estimate is invoked. The D3 wrench identity checks that these columns produce the expected generalized wrench signs row by row, but that identity is only an algebraic interface check. The PL2 proof below supplies the uniform constrained linearized estimate, and the PL4 proof below supplies only the conditional propagation from state and acceleration lifts to $\lambda_{i,j} - \lambda(t_i) = O(h^7)$; the required input lifts themselves remain open. $\square$

Lemma 29 (Non-circular D3 use for P_λ). *The row-expanded D3 virtual-work identity may be used in the P_λ proof chain only as an algebraic mapping from lower-pair multiplier components to the generalized force and torque entries in the 36 Newton–Euler weak-balance rows. This closes PL3: the D3 wrench-consistency certificate is used non-circularly. It does not prove a multiplier lift rate, does not prove a uniform inf-sup bound, and does not certify a Taylor bound. It also does not close the primitive-and-Taylor route to PC2.*

Proof. The D3 certificate expands the virtual work of the lower-pair reaction forces and torques row by row and checks the same translational and rotational signs as the accepted residual. That check is purely algebraic in the runtime row formulas, joint bases, moment arms, and multiplier slots. It neither compares the implemented stage variables with the lifted exact Gauss stages nor invokes the D5 dynamic residual defect rate. Therefore it can identify the correct wrench map for a separate P_λ estimate without assuming the estimate itself. The PL2 compact-tube inf-sup bound is supplied by the symbolic geometric proof below. The remaining rate boundary is that PL4 can only be instantiated after independent state and acceleration lift inputs are available. $\square$

The finite diagnostic probe `D5_P_LAMBDA_INF_SUP_PROBE` evaluates the accepted dynamic-row multiplier-column block

$$\delta\lambda \mapsto D_\lambda R_{\text{dyn}}(Z_G)\delta\lambda \quad (137)$$

Equation (137) is evaluated on solved one-step stages at $h = 0.04, 0.02, 0.01$. The measured 36×24 block has rank 24 in all three probes, with minimum singular value about 6.14×10^{-1} and maximum condition number about 2.77. This is useful finite evidence that the PL2 target is the right linearized multiplier interface. It is not a uniform compact-tube inf-sup proof, it does not prove a multiplier lift rate, and it does not close P_λ or PC2.

PL2 geometric-margin diagnostic. The finite geometric-margin diagnostic separates the PL2 operator into the stage-local dynamic-lambda block, the translational normal-force subblock, the rotational axis-torque subblock, and their direct-sum diagnostic. On the recorded solved stages at $h = 0.04, 0.02, 0.01$, these finite blocks are full column rank and have positive smallest singular values. These finite margins identify the geometric proof target for the multiplier columns, but they are not a uniform compact-tube inf-sup proof. The symbolic proof below gives that compact-tube lower bound and closes PL2 as a uniform inf-sup subproof. In the primitive-and-Taylor route, the P_λ route and the PC2 route both remain open because the PL4 propagation below is conditional on the still-open state and acceleration lift inputs.

Lemma 30 (PL2 multiplier-column symbolic structure). *For one lower pair, let a be the cylindrical axis, let $B = [b_0 \ b_1]$ be the two normal basis columns*

with $B^T B = I$ and $B^T a = 0$, and write the normal multiplier as $\lambda_N \in \mathbb{R}^2$. The implemented normal-force and smooth-friction subblock has the form

$$F_N(\lambda_N) = B\lambda_N + \varphi(v, \|B\lambda_N\|)a, \quad (138)$$

where φ is the smooth Brown–McPhee scalar on the compact proof tube. Differentiating Eq. (138) gives

$$D_{\lambda_N} F_N = B + ac^T, \quad (B + ac^T)^T (B + ac^T) = I + cc^T, \quad (139)$$

Equation (139) shows that the smooth-friction normal-load derivative adds only a nonnegative Gram correction. for a bounded row vector c . Therefore each joint normal-force column map has smallest singular value at least one; the smooth-friction normal-load derivative cannot reduce this margin. The two-joint translational action–reaction block consequently has the symbolic bound

$$\sigma_{\min} \geq (\sqrt{5} - 1)/2. \quad (140)$$

For an axis-torque block

$$C(a, R, B)\eta = \eta_0 a \times R^T b_0 + \eta_1 a \times R^T b_1, \quad n_B = b_0 \times b_1, \quad (141)$$

one has

$$C^T C = I - c_a c_a^T, \quad (c_a)_i = (Ra) \cdot b_i, \quad \sigma_{\min}(C) = |(Ra) \cdot n_B|. \quad (142)$$

Thus PL2 is reduced to an axis-plane margin: $|(Ra) \cdot n_B| \geq \chi_{\text{axis}} > 0$ on the compact tube for the axis-torque columns by Eqs. (141) and (142). With that margin and the compact-tube upper bounds for distal torque and normal-to-rotational couplings, the recorded block triangular estimate gives a full dynamic-lambda lower bound. The normal-force and smooth-friction part is closed by this symbolic argument; the remaining input for PL2 is the compact-tube axis-plane margin proved in Lemma 31.

Proof. The first two identities use only the orthogonal decomposition $\mathbb{R}^3 = \text{span}\{b_0, b_1\} \oplus \text{span}\{a\}$. Since $B^T a = 0$, the rank-one friction derivative contributes ac^T in a direction orthogonal to the normal basis, and the Gram matrix is $I + cc^T$. Its eigenvalues are at least one. For the two lower pairs, write $u = A_0 x$ and $v = A_1 y$, where each A_j is such a normal-force map. Then

$$\| -u + v \|^2 + \|v\|^2 \geq \frac{3 - \sqrt{5}}{2} (\|u\|^2 + \|v\|^2) \geq \frac{3 - \sqrt{5}}{2} (\|x\|^2 + \|y\|^2), \quad (143)$$

Equation (143) gives the translational lower bound in Eq. (140).

For the axis-torque identity, use $(a \times q_i) \cdot (a \times q_j) = q_i \cdot q_j - (a \cdot q_i)(a \cdot q_j)$ with $q_i = R^T b_i$. Because R is orthogonal and b_0, b_1 are orthonormal, $q_i \cdot q_j = \delta_{ij}$, and $a \cdot q_i = (Ra) \cdot b_i$. The two nonzero singular values are therefore the square roots of the eigenvalues of $I - c_a c_a^T$, whose smaller value is $|(Ra) \cdot n_B|$. The remaining full-block statement is the standard left-inverse estimate for a block triangular operator: the translational normal block controls the normal multipliers, the rotational block controls the axis multipliers once the axis-plane margin is available, and the compact tube bounds the off-diagonal coupling. This lemma records the strict symbolic reduction; the next lemma supplies the required compact-tube margin. $\square$

Lemma 31 (PL2 axis-plane margin on the accepted compact tube). *For each proximal axis-torque pair in the accepted lower-pair chart, let*

$$m_j(z) = |(R_j(z)a_j) \cdot n_j|, \quad n_j = b_{j,0} \times b_{j,1}, \quad (144)$$

where $b_{j,0}, b_{j,1}$ are the fixed normal basis vectors used by the runtime lower-pair multiplier columns. Under Assumption 1, after shrinking the compact proof tube inside the same accepted chart if necessary, there is a constant

$$\chi_{\text{axis}} := \min_{z \in K} \min_j m_j(z) > 0. \quad (145)$$

Thus Eqs. (144) and (145) give the proximal axis-torque blocks a uniform compact-tube lower singular-value bound. Finite probes are not used to establish the margin.

Proof. The previous lemma gives the exact identity

$$\sigma_{\min} C_j(z) = m_j(z) \quad (146)$$

Equation (146) identifies the axis-plane margin with the singular value of the proximal torque block. for each proximal two-column axis-torque block. If $m_j(z) = 0$, then the two axis-torque multiplier columns are linearly dependent and the axis-torque coordinate part of the lower-pair chart loses rank at z . This is precisely the chart transversality failure excluded by Assumption 1: the active lower-pair constraint Jacobian has constant full row rank and the accepted branch is represented in a smooth lower-pair chart. Hence $m_j(z) > 0$ on the accepted smooth branch for every finite proximal axis-torque pair j .

Each m_j is continuous because $R_j(z)$ is smooth in the accepted chart and the basis vectors are fixed. Lemma 9 provides a compact tube K around the accepted branch; shrinking this tube inside the same nondegenerate chart preserves the branch and keeps $m_j > 0$ on K . The minimum over the finite family of continuous positive functions m_j on compact K is therefore positive. This compactness argument uses the symbolic singular-value identity and chart regularity only; the finite singular-value probes are diagnostic and are not used as proof. $\square$

Lemma 32 (PL2 compact-tube reduction). *Let K be the compact smooth proof tube and let $B(z, h) = D_\lambda R_{\text{dyn}}(z, h)$ denote the accepted dynamic-row multiplier-column map restricted to one stage. Let*

$$\gamma_T = (\sqrt{5} - 1)/2, \quad \gamma_R = (2\chi_{\text{axis}}^{-2} + \beta_D^2 \chi_{\text{axis}}^{-4})^{-1/2}, \quad (147)$$

where $\beta_D < \infty$ is the compact-tube bound for the distal axis-torque coupling. If $\beta_N < \infty$ is the compact-tube bound for normal-to-rotational coupling, define

$$\gamma_{\text{PL2}} = (\gamma_T^{-2} + \gamma_R^{-2} + \beta_N^2 \gamma_T^{-2} \gamma_R^{-2})^{-1/2}. \quad (148)$$

Equation (148) combines the translational, rotational, and normal-to-rotational coupling margins into the PL2 constant. Then $\gamma_{\text{PL2}} > 0$, and PL2 has a uniform inf-sup constant on the tube:

$$\|B(z, h)\delta\lambda\| \geq \gamma_{\text{PL2}}\|\delta\lambda\|, \quad (z, h) \in K \times (0, h_0]. \quad (149)$$

The finite PL2 probes are not a uniform compact-tube inf-sup proof; they only identify the normal-force, axis-torque, and smooth-friction perturbation maps. PL2 is closed as a uniform inf-sup subproof by Eqs. (147)–(149). This does not prove the PL4 multiplier-rate propagation, P_λ , or PC2.

Proof. The preceding symbolic-structure lemma gives the translational normal block the lower bound γ_T . Lemma 31 gives the proximal axis-torque lower bound χ_{axis} , while Lemma 9 bounds the distal torque coupling by β_D . The standard two-block left-inverse estimate for that rotational action-reaction block gives γ_R . Applying the same block-triangular estimate to the full dynamic-lambda map, with normal-to-rotational coupling bounded by β_N , gives γ_{PL2} .

All constants in this formula are positive or finite on K , and the finite number of stages and lower pairs lets us take the minimum over the stage-local constants. Hence Eq. (149) holds uniformly for all accepted stage states

in the tube and all sufficiently small h . The proof uses only the symbolic identities, chart transversality, and compact-tube constants; finite probes do not establish the inf-sup constant. $\square$

Lemma 33 (PL4 conditional multiplier-rate propagation). *Let $y = (z, a)$ collect the accepted state and acceleration variables in the lower-pair chart, and let $F(y, \lambda, h)$ be the accepted lower-pair dynamic-row map after the PL1 multiplier-column interface and the PL3 D3 wrench mapping have fixed the multiplier coordinates. On the compact proof tube, assume the PL2 bound*

$$\|D_\lambda F(y, \lambda, h)\delta\lambda\| \geq \gamma_{\text{PL2}}\|\delta\lambda\| \quad (150)$$

with $\gamma_{\text{PL2}} > 0$. Then the local lower-pair multiplier solution branch is uniformly Lipschitz in y :

$$\|\hat{\lambda} - \lambda\| \leq C_{\lambda y}(\|\hat{z} - z\| + \|\hat{a} - a\|). \quad (151)$$

Equations (150) and (151) state the PL2 input and the resulting conditional PL4 propagation estimate. Consequently, if the still-open P_{state} and P_{acc} primitives supply $\|\hat{z} - z\| = O(h^7)$ and $\|\hat{a} - a\| = O(h^7)$, then $\|\hat{\lambda} - \lambda\| = O(h^7)$. This closes only the conditional PL4 propagation step; it does not close P_λ , certify any D5 Taylor term, or close PC2.

Proof. Fix two points on the same accepted lower-pair solution branch inside the compact tube, and write $\delta y = \hat{y} - y$ and $\delta\lambda = \hat{\lambda} - \lambda$. Since both points satisfy the same lower-pair dynamic-row equation in the accepted chart, the first-order Taylor formula with integral remainder gives

$$\begin{aligned} 0 &= F(\hat{y}, \hat{\lambda}, h) - F(y, \lambda, h) \\ &= B(y, \lambda, h)\delta\lambda + D_y F(y, \lambda, h)\delta y + \rho, \\ B &= D_\lambda F. \end{aligned} \quad (152)$$

where compact C^2 bounds give

$$\|\rho\| \leq C_R(\|\delta y\| + \|\delta\lambda\|)^2, \quad \|D_y F(y, \lambda, h)\delta y\| \leq L_y\|\delta y\|. \quad (153)$$

PL2 gives a uniform left-inverse estimate for B , hence

$$\gamma_{\text{PL2}}\|\delta\lambda\| \leq L_y\|\delta y\| + C_R(\|\delta y\| + \|\delta\lambda\|)^2. \quad (154)$$

Equations (152), (153), and (154) are the Taylor identity, compact-tube remainder bound, and pre-absorption estimate for PL4. The absorption is a local-branch condition, not a global multiplier-root selection theorem. Fix a PL4 comparison radius $r_\lambda > 0$, contained in the same compact-tube chart and accepted lower-pair branch, such that

$$\|\delta y\| + \|\delta \lambda\| \leq r_\lambda, \quad C_R(\|\delta y\| + \|\delta \lambda\|) \leq \frac{1}{2}\gamma_{\text{PL2}}. \quad (155)$$

Under Eq. (155), the quadratic term in Eq. (154) is absorbed into the left-hand side, giving

$$\|\delta \lambda\| \leq \frac{2L_y}{\gamma_{\text{PL2}}} \|\delta y\| = C_{\lambda y} \|\delta y\|. \quad (156)$$

Equation (156) is uniform over the finite set of stages and lower pairs once both compared points are already known to remain in the PL4 local branch ball in Eq. (155). Small h enters only through the independent state and acceleration input lifts that place the compared points in that ball; it does not prove global branch selection or exclude remote multiplier roots. Inserting the independent input rates $\|\delta z\| = O(h^7)$ and $\|\delta a\| = O(h^7)$ gives the claimed conditional multiplier rate. The argument uses the Taylor expansion, compact-tube constants, and the PL2 inf-sup constant; it does not use Lemma 39, the D5 dynamic residual defect, direct substitution, finite probes, or the D3 identity as a rate proof. $\square$

Lemma 34 (PL4 closure is not P_λ closure). *Closing the four local PL sub-proofs does not by itself close the multiplier-lift primitive P_λ . The output of PL4 is the conditional implication*

$$\|\delta \lambda\| \leq C_{\lambda y}(\|\delta z\| + \|\delta a\|), \quad (157)$$

on the accepted compact branch and in the fixed FullVA norm. It becomes an actual $O(h^7)$ multiplier estimate only after the state lift $\|\delta z\| = O(h^7)$ and the unweighted acceleration lift $\|\delta a\| = O(h^7)$ have been proved independently for the same branch. Since the current primitive status keeps P_{state} and P_{acc} open, P_λ remains open even though PL4 is available as a conditional propagation implication.

Proof. Lemma 33 proves a Lipschitz dependence of the multiplier coordinates on the already chosen state and acceleration variables. This dependence is an implication from input lift rates to multiplier lift rates; it is not a source

of those input lift rates. Substituting $O(h^7)$ input estimates into Eq. (157) would indeed give $\|\delta\lambda\| = O(h^7)$, but that substitution is admissible only if the two inputs have already been established without using the D5 dynamic residual defect, Lemma 39, the direct PC2 substitution certificate, finite probes, or the D3 wrench identity as a rate proof.

Those independent inputs are precisely the still-open P_{state} and P_{acc} primitives recorded in the D5 primitive inventory. Therefore the PL subproof count 4/4 means that the conditional multiplier propagation mechanism is complete; it does not mean that the actual multiplier primitive is complete. The 72 D5 Taylor subterms that consume P_λ remain uncertified until the two root lifts are closed first. $\square$

Reference-proof boundary for P_λ . The multiplier estimates in the constrained-BDF reference are not imported as a P_λ lift. In that proof the multiplier recursion is tied to the BLieDF multistep defect, hidden-constraint coordinates, and its own start-value assumptions. The present lower-pair multipliers are instead stage unknowns in the 132-row FullVA residual, with their columns fixed by PL1, their compact inf-sup geometry fixed by PL2–PL3, and their rate controlled only conditionally by PL4 from the state and acceleration lifts. Thus the reference proof only motivates the ordering discipline recorded in Table 11: first fix the discrete object and multiplier interface, then prove the local defect for that fixed object, and only then propagate the accepted-branch error. It does not provide the required actual P_λ rate, does not bypass the open P_{state} and P_{acc} inputs, and does not certify any of the 72 primitive Taylor subterms that depend on P_λ .

This table is an obligation map for the separate primitive route: it does not certify actual D5 Taylor subterm bounds and it identifies the independent primitive estimates that would be needed before that separate primitive route could close. None of the five retained primitive rows is allowed to use Lemma 39 or the D5 dynamic residual rate as an input. This keeps the proof non-circular: Table 28 states exactly which independent estimates are still needed before the primitive-and-Taylor route to PC2 can be closed.

Finite dependency certificate for the separate route. The 162-subterm statement below is a finite dependency certificate, not a generic smoothness assertion. The recorded D5 inventory splits into four template families in Eq. (158):

$$\begin{aligned} & 36 \text{ acceleration lift} + 72 \text{ smooth force/torque/friction lift} \\ & + 36 \text{ multiplier-geometry lift} + 18 \text{ gyroscopic bilinear lift} = 162. \end{aligned} \tag{158}$$

Table 28: Retained primitive proof interfaces for D5.

Primitive	Term rows	Missing independent proof	Forbidden shortcut
P_{state}	126	The PS2 weighted inverse and row-injection route are recorded; the required step is the actual PS3 instantiation that converts the 96 certified non-dynamic residual rows, together with an acceleration input independent of the state perturbation being estimated, into $O(h^7)$ pose and velocity lift rates.	Using Lemma 39 or the D5 dynamic residual defect.
P_{acc}	36	Unweighted $O(h^7)$ translational and angular acceleration lift rates in the accepted variable ordering, supplied by one of the admissible PA2 routes: an $O(h^8)$ velocity/velocity-row estimate, an independent non-dynamic inverse that controls acceleration directly, or an explicit P_{acc} theorem assumption.	Reading acceleration rates from dynamic balance.
P_{λ}	72	Instantiate the closed conditional PL4 propagation with independent P_{state} and P_{acc} lift inputs; the PL subproof count is complete only in this conditional sense, and the actual multiplier lift remains open.	Using the D3 wrench identity as a multiplier-rate proof.
P_{geom}	36	Apply the closed accepted-chart geometry reduction below using the pose and multiplier lifts supplied by P_{state} and P_{λ} .	Treating the chart reduction as a pose or multiplier lift proof.
P_{gyro}	18	Apply the closed bilinear reduction below using the angular-velocity lift supplied by P_{state} , then bind it to the rotational rows.	Treating the bilinear estimate as a velocity-lift proof.

Each family has a fixed primitive-input list before any Taylor estimate is read: acceleration rows consume P_{acc} ; smooth force/torque/friction rows consume P_{state} and, for the lambda-dependent friction terms, P_λ ; multiplier-geometry rows consume P_{state} and P_λ ; and gyroscopic rows consume the angular-velocity part of P_{state} . The compact tube primitive supplies the uniform derivative or bilinear constants for all four families. This dependency graph is acyclic in the separate route: P_{state} and P_{acc} are root lift inputs, P_λ is propagated only from those roots, and P_{geom} and P_{gyro} are downstream reductions. Consequently the 162/162 coverage count means that every primitive subterm has an assigned template and primitive-input set. It is template coverage only: the certified count of actual primitive Taylor bounds is 0/162; it does not mean that any of those five open primitive inputs has been proved, and it is not a projection of the accepted 132-row direct residual estimate.

Proposition 1 (Conditional implication order for the separate D5 primitive/Taylor route). *Assume, in addition to the already closed compact-tube primitive P_{tube} , that two independent root-lift estimates are proved on the same accepted compact tube, as in Eq. (159):*

$$\|\delta S\| \leq C_S h^7, \quad \|\delta A\| \leq C_A h^7. \quad (159)$$

Assume further that these two estimates are obtained without using Lemma 39, the D5 dynamic residual defect, the D5 direct-substitution certificate, finite numerical probes, or the D3 wrench identity as a rate proof. Then PL4 gives the conditional multiplier rate P_λ ; the accepted-chart geometry and gyroscopic reductions give P_{geom} and P_{gyro} ; and the finite sum over the 162 D5 primitive Taylor subterms yields Eq. (160),

$$\|R_{\text{dyn}}^{\text{prim}}(Z_G)\| \leq C_{\text{prim}} h^7. \quad (160)$$

Thus the separate primitive/Taylor route has the dependency order in Eq. (161):

$$\begin{aligned} P_{\text{state}}, P_{\text{acc}} &\Rightarrow P_\lambda \Rightarrow P_{\text{geom}}, P_{\text{gyro}} \\ &\Rightarrow 162 \text{ D5 Taylor subterms.} \end{aligned} \quad (161)$$

The proposition is only a conditional sufficiency statement. Its root-lift prerequisites remain unproved in the current manuscript, it does not establish actual primitive Taylor bounds, and it does not make Theorem 1 depend on the primitive/Taylor route.

Proof. The proof is a finite implication chain inside the separate primitive/Taylor sufficiency route. The assumed P_{state} and P_{acc} estimates are exactly the not-yet-established root inputs listed in Table 27. With PL2 and PL3 already fixed in the accepted lower-pair chart, PL4 propagates those root rates to $\|\delta\lambda\| = O(h^7)$, so P_λ would follow conditionally. The row-level smooth force/torque/friction corollary then bounds the 72 smooth-map subterms from P_{state} and, for friction terms, P_λ . The multiplier-geometry chart reduction bounds the 36 geometry-dependent subterms from P_{state} and P_λ , and the gyroscopic bilinear reduction bounds the 18 rotational bilinear subterms from the angular-rate part of P_{state} . The compact-tube primitive supplies the uniform derivative constants for all smooth maps. Applying the recorded Taylor/mean-value template to each of the finite 162 subterms and taking the finite row-scaled dynamic-block norm gives the displayed bound. More explicitly, enumerate the primitive Taylor subterms as τ_i , $i = 1, \dots, 162$, after the row scaling and component ordering used by the D5 inventory have been fixed. Conditional on the retained primitive inputs, the row corollaries provide constants C_i , independent of h , with $\|\tau_i\| \leq C_i h^7$ on the accepted compact tube. The dynamic-block norm is finite dimensional and the row weights are fixed before the estimate is read, so a norm-equivalence constant C_W , also independent of h , gives the aggregate constant $C_{\text{prim}} = C_W \sum_{i=1}^{162} C_i$. This yields $\|R_{\text{dyn}}^{\text{prim}}(Z_G)\| \leq C_{\text{prim}} h^7$.

No inverse projection from the aggregate 132-row residual is used in this argument, and no direct-route corollary is promoted into a primitive estimate. Since the two root estimates are not currently proved, the argument records only the conditional implication order that a separate primitive/Taylor proof for that route must satisfy. $\square$

This proposition records the separate primitive route as a conditional sufficiency map, not as actual primitive-route closure and not as an input to Theorem 1: actual primitive Taylor-term closure is not established on the accepted theorem route. No theorem arrow consumes this proposition unless the root-lift estimates are separately proved on the fixed theorem branch before the theorem constants and admissible window are chosen.

Corollary 2 (Row-level smooth force/torque/friction Taylor bound under P_{state} and P_λ). *Let $\mathcal{T}_{\text{smooth}}$ be the 72 scalar smooth-map D5 subterm slots in*

Eq. (162):

$$T_{\text{external_force_lift}}, \quad T_{\text{friction_force_lift}}, \quad R_{\text{external_torque_lift}}, \quad R_{\text{friction_torque_lift}} \quad (162)$$

over the three Gauss stages, two bodies, and three scalar components. Write $z = (q, v, \omega)$ for the accepted stage state variables used by these maps. Assume the state part of P_{state} gives Eq. (163),

$$\|\widehat{z}_{s,b} - z_b(t_s)\| \leq C_z h^7 \quad (163)$$

uniformly for every stage s and body b , and assume the multiplier part of P_λ gives Eq. (164):

$$\|\widehat{\lambda}_s - \lambda(t_s)\| \leq C_\lambda h^7. \quad (164)$$

On the compact smooth proof tube, let L_{ext} bound the derivatives of the external force and external torque maps with respect to state, and let L_{fric} bound the derivatives of the regularized friction force and torque maps with respect to state and multiplier variables. Then every external force/torque scalar subterm satisfies Eq. (165),

$$|e_r^T(F_{\text{ext}}(\widehat{z}_{s,b}, t_s) - F_{\text{ext}}(z_b(t_s), t_s))| \leq L_{\text{ext}} C_z h^7, \quad (165)$$

and every regularized friction force/torque scalar subterm satisfies Eq. (166):

$$|e_r^T(F_{\text{fric}}(\widehat{z}_{s,b}, \widehat{\lambda}_s) - F_{\text{fric}}(z_b(t_s), \lambda(t_s)))| \leq L_{\text{fric}}(C_z + C_\lambda) h^7. \quad (166)$$

Thus the 72 smooth force, torque, and regularized-friction Taylor subterms have the required $O(h^7)$ row-level bounds under P_{state} and P_λ . This is a conditional consequence only: it does not prove either lift, does not turn the D4 direct-route smoothness certificate into a primitive-rate proof, and does not change the primitive/Taylor PC2 status. It also does not use Lemma 39, direct substitution, or any residual-to-root perturbation estimate as a primitive-bound proof.

Proof. For an external force or torque map, Taylor's formula with integral remainder on the compact tube gives Eq. (167):

$$\begin{aligned} F_{\text{ext}}(\widehat{z}, t) - F_{\text{ext}}(z, t) &= \int_0^1 D_z F_{\text{ext}}(z + \theta(\widehat{z} - z), t) \\ &\quad \times (\widehat{z} - z) d\theta. \end{aligned} \quad (167)$$

The derivative norm is bounded by L_{ext} , so the P_{state} input gives Eq. (165). For the regularized friction maps, apply the same formula to the joint variable (z, λ) . The compact-tube derivative bound L_{fric} and the two conditional lift inputs give Eq. (168):

$$\begin{aligned} \|F_{\text{fric}}(\hat{z}, \hat{\lambda}) - F_{\text{fric}}(z, \lambda)\| &\leq L_{\text{fric}}(\|\hat{z} - z\| + \|\hat{\lambda} - \lambda\|) \\ &\leq L_{\text{fric}}(C_z + C_\lambda)h^7. \end{aligned} \quad (168)$$

Projection onto any scalar component can only decrease the Euclidean norm, and the D5 inventory has exactly 18 external force, 18 friction force, 18 external torque, and 18 friction torque scalar subterm slots. The result therefore holds uniformly over all 72 listed smooth-map subterms. The proof uses only compact smoothness and the conditional lift inputs; it does not use Lemma 39, direct substitution, or finite probes as rate evidence. $\square$

Lemma 35 (Conditional multiplier-geometry chart reduction). *Let $G_b(q)$ and $H_b(q)$ denote the translational and rotational lower-pair multiplier-wrench geometry maps for body b in the accepted Lie chart, and define the wrench maps in Eq. (169):*

$$W_G(q, \lambda) = G_b(q)^T \lambda, \quad W_H(q, \lambda) = H_b(q)^T \lambda. \quad (169)$$

On the compact proof tube from Lemma 9, there are multiplier-geometry constants C_{WG}, C_{WH} such that Eq. (170) holds,

$$\|W_G(\hat{q}, \hat{\lambda}) - W_G(q, \lambda)\| \leq C_{WG}(\|\hat{q} - q\| + \|\hat{\lambda} - \lambda\|), \quad (170)$$

and the same estimate holds for W_H with C_{WH} . Hence the 36 multiplier-geometry Taylor subterms are $O(h^7)$ once the pose and multiplier lifts supplied by P_{state} and P_λ are $O(h^7)$. This records the conditional chart reduction for P_{geom} , but it does not close P_{geom} or PC2 because those lift estimates remain open. Here “conditional chart reduction” means only this downstream implication: the Lipschitz map from the open P_{state} and P_λ inputs to the 36 geometry rows has no further internal chart obligation. It is not a primitive P_{geom} proof and certifies no actual D5 primitive Taylor subterm without those lifts.

Proof. Let $A(q)$ denote either $G_b(q)$ or $H_b(q)$. Add and subtract $A(q)^T \hat{\lambda}$ to get Eq. (171):

$$A(\hat{q})^T \hat{\lambda} - A(q)^T \lambda = (A(\hat{q}) - A(q))^T \hat{\lambda} + A(q)^T (\hat{\lambda} - \lambda). \quad (171)$$

The compact-tube constant lemma gives uniform bounds for A , DA , and $\widehat{\lambda}$. The mean-value theorem then yields Eq. (172):

$$\|A(\widehat{q})^T \widehat{\lambda} - A(q)^T \lambda\| \leq C_W (\|\widehat{q} - q\| + \|\widehat{\lambda} - \lambda\|). \quad (172)$$

The term-budget analysis identifies 18 translational $G_b(q)^T \lambda$ terms and 18 rotational $H_b(q)^T \lambda$ terms. Applying the same estimate to this finite row set only changes the constant. The result is conditional on the independent P_{state} and P_λ lift estimates and cannot be used to prove either estimate. $\square$

Corollary 3 (Row-level multiplier-geometry Taylor bound under P_{state} and P_λ). *Let Eq. (173) define the global-row set for the multiplier-geometry Newton–Euler Taylor components:*

$$\begin{aligned} \mathcal{R}_{\text{geom}} = \{ & 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, \\ & 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, \\ & 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123 \} \end{aligned} \quad (173)$$

Assume the pose and multiplier parts of P_{state} and P_λ give Eq. (174),

$$\|\widehat{q}_{s,b} - q_b(t_s)\| \leq C_q h^7, \quad \|\widehat{\lambda}_s - \lambda(t_s)\| \leq C_\lambda h^7 \quad (174)$$

uniformly for every Gauss stage s and body b . If Eq. (175) holds,

$$\begin{aligned} \|W_G(\widehat{q}, \widehat{\lambda}) - W_G(q, \lambda)\| &\leq C_{WG} (\|\widehat{q} - q\| + \|\widehat{\lambda} - \lambda\|), \\ \|W_H(\widehat{q}, \widehat{\lambda}) - W_H(q, \lambda)\| &\leq C_{WH} (\|\widehat{q} - q\| + \|\widehat{\lambda} - \lambda\|), \end{aligned} \quad (175)$$

and $C_{W,\text{max}} = \max(C_{WG}, C_{WH})$, then every scalar row $r \in \mathcal{R}_{\text{geom}}$ satisfies Eq. (176):

$$\begin{aligned} |e_r^T (W_\star(\widehat{q}_{s,b}, \widehat{\lambda}_s) - W_\star(q_b(t_s), \lambda(t_s)))| &\leq C_{\text{geom}} h^7, \\ C_{\text{geom}} &= C_{W,\text{max}} (C_q + C_\lambda), \end{aligned} \quad (176)$$

where $W_\star$ is W_G on translational multiplier-force rows and W_H on rotational multiplier-torque rows. Thus all 36 multiplier-geometry Taylor rows satisfy the conditional T3 target $O(h^7)$ row bound under the two lift inputs. This is a conditional row-level consequence only: it does not prove either lift, does not close P_{geom} , and does not change the PC2 status.

Proof. Apply Lemma 35 at each listed stage-body pair. The assumed P_{state} and P_λ estimates give Eq. (177):

$$\|W_\star(\hat{q}_{s,b}, \hat{\lambda}_s) - W_\star(q_b(t_s), \lambda(t_s))\| \leq C_{W,\max}(C_q + C_\lambda)h^7. \quad (177)$$

Projection onto any scalar component can only decrease the Euclidean norm. The row inventory contains exactly the listed translational $G_b(q)^T \lambda$ rows and rotational $H_b(q)^T \lambda$ rows for each body at each Gauss stage, so the estimate holds uniformly over all rows in $\mathcal{R}_{\text{geom}}$. $\square$

Lemma 36 (Gyroscopic bilinear reduction). *Let $G(\omega) = \omega \times J\omega$, where the body inertia J is constant. On the compact proof tube from Lemma 9, the gyroscopic contribution is Lipschitz in the angular velocity: if $\|\omega\|, \|\hat{\omega}\| \leq M$, then Eq. (178) holds:*

$$\|G(\hat{\omega}) - G(\omega)\| \leq 2M\|J\| \|\hat{\omega} - \omega\|. \quad (178)$$

Consequently, the 18 rotational Newton–Euler gyroscopic subterms are $O(h^7)$ once the angular-velocity lift supplied by P_{state} is $O(h^7)$. This closes the algebraic reduction for P_{gyro} , but it does not close P_{gyro} or PC2 because P_{state} remains open. Here “closes the algebraic reduction” means only that the bilinear gyroscopic map has no remaining algebraic subproof under the open P_{state} angular-velocity input. It is not a primitive P_{gyro} proof and certifies no actual D5 primitive Taylor subterm without that lift.

Proof. Using bilinearity of the cross product and the constant inertia tensor gives Eq. (179):

$$G(\hat{\omega}) - G(\omega) = (\hat{\omega} - \omega) \times J\hat{\omega} + \omega \times J(\hat{\omega} - \omega). \quad (179)$$

The Euclidean cross-product bound and the compact-tube angular-velocity bound give Eq. (180):

$$\begin{aligned} \|G(\hat{\omega}) - G(\omega)\| &\leq \|\hat{\omega} - \omega\| \|J\| \|\hat{\omega}\| + \|\omega\| \|J\| \|\hat{\omega} - \omega\| \\ &\leq 2M\|J\| \|\hat{\omega} - \omega\|. \end{aligned} \quad (180)$$

The rotational-row inventory contains exactly three gyroscopic components for each body at each of the three Gauss stages, hence 18 rows. Applying the same bound to this finite row set only changes the constant. The result is therefore conditional on the independent angular-velocity lift estimate from P_{state} ; it is not a proof of that lift estimate and cannot be used to close the dynamic residual criterion. $\square$

Corollary 4 (Row-level gyroscopic Taylor bound under P_{state}). *Let Eq. (181) define the global-row set for the rotational gyroscopic Newton–Euler components:*

$$\begin{aligned} \mathcal{R}_{\text{gyro}} = \{ & 27, 28, 29, 33, 34, 35, \\ & 71, 72, 73, 77, 78, 79, \\ & 115, 116, 117, 121, 122, 123 \} \end{aligned} \quad (181)$$

Assume the angular-velocity part of P_{state} gives Eq. (182),

$$\|\widehat{\omega}_{s,b} - \omega_b(t_s)\| \leq C_\omega h^7 \quad (182)$$

uniformly for every Gauss stage s and body b . If the compact tube also gives Eq. (183),

$$\|\omega_b(t_s)\|, \|\widehat{\omega}_{s,b}\| \leq M_\omega, \quad J_{\max} = \max_b \|J_b\|_2, \quad (183)$$

then every scalar row $r \in \mathcal{R}_{\text{gyro}}$ satisfies Eq. (184):

$$\begin{aligned} |e_r^T(G_b(\widehat{\omega}_{s,b}) - G_b(\omega_b(t_s)))| &\leq C_{\text{gyro}} h^7, \\ C_{\text{gyro}} &= 2M_\omega J_{\max} C_\omega. \end{aligned} \quad (184)$$

Thus the 18 gyroscopic Taylor rows satisfy the conditional T3 target $O(h^7)$ row bound under P_{state} . This is a conditional row-level consequence only: it does not prove the angular-velocity lift, does not close P_{gyro} , and does not change the PC2 status.

Proof. Apply Lemma 36 with $\widehat{\omega} = \widehat{\omega}_{s,b}$, $\omega = \omega_b(t_s)$, and $\|J_b\|_2 \leq J_{\max}$. The assumed P_{state} angular-rate estimate gives Eq. (185):

$$\|G_b(\widehat{\omega}_{s,b}) - G_b(\omega_b(t_s))\| \leq 2M_\omega J_{\max} C_\omega h^7. \quad (185)$$

Projection onto any scalar component can only decrease the Euclidean norm. The row inventory contains exactly the listed three rotational components for each body at each Gauss stage, so the estimate holds uniformly over all rows in $\mathcal{R}_{\text{gyro}}$. $\square$

Primitive/Taylor nonclosure invariant. The row-level acceleration, smooth-force, geometry, and gyroscopic corollaries above strengthen the conditional Taylor-reduction record, but they do not certify actual primitive-route Taylor bounds and do not change the primitive route status. They are downstream conditional maps, not roots of the primitive dependency graph. They are

conditional consequences of the still-open lift estimates $P_{\text{state}}, P_{\text{acc}}, P_{\lambda}, P_{\text{geom}},$ and P_{gyro} . Hence the primitive route still has zero actual primitive-route Taylor subterm bounds out of the 162 D5 subterms until those five primitive obligations are proved uniformly on the same compact proof tube. The PC2 stage-residual condition used in Theorem 1 remains discharged by the direct residual-bridge route, which consumes the D5 direct-substitution certificate rather than the open primitive/Taylor route.

Primitive-route one-way implication discipline. The separate primitive implication has the one-way logical form in Eq. (186):

$$\begin{aligned} &P_{\text{state}}, P_{\text{acc}}, P_{\lambda}, P_{\text{geom}}, P_{\text{gyro}}, P_{\text{tube}} \\ &\implies \{162 \text{ D5 primitive Taylor subterms are } O(h^7)\} \\ &\implies R_{\text{dyn}}(Z_G) = O(h^7). \end{aligned} \quad (186)$$

No converse implication is used or available in this manuscript. In particular, the active direct route proves the dynamic-row residual value at Z_G by direct Newton–Euler substitution and then uses the full row-scaled residual inverse to obtain a weighted stage-root perturbation. That output cannot be reinserted as a proof of the primitive inputs. The weighted estimate $h\|\delta A\| = O(h^7)$ does not imply the unweighted P_{acc} estimate $\|\delta A\| = O(h^7)$, and Lemma 20 is a direct-route state-block corollary, not a primitive P_{state} closure. Therefore the conditional row-level reductions below may certify the finite implication from primitive inputs to Taylor subterms, but they do not turn the already proved direct residual bridge into a primitive 162-subterm proof. Actual primitive-route Taylor subterm bounds are not established until the five open primitive obligations are proved independently on the same compact proof tube. The current primitive-status accounting is therefore positive but non-closing: P_{tube} supplies the common compact-tube constant slot, the finite reduction templates cover all 162 recorded D5 subterms, and the remaining proof burden is exactly the five same-object lift/bilinear inputs $P_{\text{state}}, P_{\text{acc}}, P_{\lambda}, P_{\text{geom}}, P_{\text{gyro}}$. Since none of those five inputs is proved on the accepted theorem tuple, the recorded template coverage gives no actual primitive Taylor bound and no primitive-route PC2 certificate. This is a strength statement, not a diagnostic downgrade: the route has a typed finite reduction once its root primitive estimates are supplied, while the accepted theorem uses the direct $96 + 36$ residual bridge already available for the same 132-row object.

Taylor-projection checkpoint. The 132-row residual estimate is an aggregate estimate in the fixed block norm of the row-scaled residual map. It is not a projection certificate for the 162 primitive scalar subterms. To promote an aggregate residual estimate into primitive-route closure one would first need, before invoking the theorem-level local defect, a fixed primitive decomposition operator Π_{prim} , an h -independent primitive projection/right-inverse or norm-equivalence bound on the compact tube, and independent $O(h^7)$ bounds for all primitive lift maps feeding Π_{prim} . No such primitive projection/right-inverse is used in Theorem 1. Hence the aggregate residual bridge can close the T2 residual-defect input while the primitive-route status remains separate from theorem use; this is a one-way proof route, not a hidden projection from theorem order back to primitive Taylor subterms.

Primitive-route non-substitution contract. If a separate T3 primitive/Taylor route is established, its first use in the present proof architecture would be limited to replacing the dynamic-row residual-value certificate inside the full 132-row bridge. It would not by itself replace the T2 full-map inverse/Kantorovich step, the endpoint-closure perturbation, the retained P6 solver-scale condition, the local-to-global stability transfer, or the P7 residual-to-error theorem. Those objects are separate proof slots with their own fixed branch, norm, constants, and scope rules. Conversely, the current direct bridge closes the theorem-level residual value without proving a primitive projection/right-inverse or any of the 162 primitive subterm bounds. Thus T3 closure would be an alternative certificate for one input to the existing theorem chain, not a complete order theorem and not a retroactive proof of the current primitive status.

Reference-proof boundary for the primitive Taylor route. The Wieloch–Arnold constrained-BDF proof motivates the proof-order discipline applied above, but it supplies no primitive FullVA lift estimate. Its local truncation proof is tied to its BLieDF discrete equations, hidden-constraint coordinates, start-value assumptions, and multiplier recursion; it is not an estimate for P_{state} , P_{acc} , P_{λ} , P_{geom} , P_{gyro} , or for a primitive projection Π_{prim} on the 162 D5 subterms. The criterion below is therefore not a borrowed Taylor local-error theorem. It is a FullVA-specific finite implication: after the five primitive inputs are proved in the accepted stage variables and on the same compact proof tube, the listed Taylor templates imply the dynamic-row primitive defect. Until those inputs are proved independently, the reference paper cannot close the primitive/Taylor status map or supply PC2.

Reference-use admissibility test for T3. Any reference-derived estimate proposed for the separate T3 primitive/Taylor route must pass a same-object test before it can enter the primitive inventory: it must be stated for the accepted FullVA stage variables, row-scaled residual norm, lower-pair chart, and compact proof tube used by Theorem 1; it must prove the root lifts P_{state} and P_{acc} without using Lemma 39, the direct D5 residual value, observed grid orders, or finite residual logs; it must propagate P_{λ} , P_{geom} , and P_{gyro} only forward from those root lifts; and it must deliver row-level $O(h^7)$ bounds in the fixed D5 primitive inventory before any finite summation is used. The constrained-BDF estimates do not satisfy this test as direct inputs because they are estimates for BLieDF hidden constraints, multistep increments, and multiplier recursion, not for the present FullVA stage-root variables or the unweighted acceleration primitive. They therefore remain ordering discipline only and do not change the non-active primitive-route Taylor status.

Lemma 37 (Same-object Taylor transport criterion). *A reference-style Taylor or multiplier estimate from another Lie-group DAE scheme can be used in the separate T3 primitive route only after it is transported to the same mathematical object used here. Concretely, before such an estimate may enter the D5 primitive inventory, one must construct h -uniformly bounded identifications from the reference perturbation variables to the accepted FullVA stage variables, from the reference defect components to the fixed D5 primitive inventory, and from the reference norm to the row-scaled proof norm of $F_{A,h}$, all on the same compact Gauss-predictor branch and before any finite-grid or residual-log evidence is used. The transported estimate must then prove the FullVA root lift inputs P_{state} and P_{acc} in those variables; only after those two inputs are independently available may the present P_{λ} , P_{geom} , and P_{gyro} propagation/reduction steps be invoked. Without such a same-object transport, the reference estimate is not a T3 input, cannot certify any of the 162 primitive D5 Taylor subterms, and cannot replace Lemma 40.*

Proof. The T3 route is typed by the objects listed in its inventory: the lifted Gauss stage Z_G , the accepted FullVA stage vector, the fixed lower-pair chart, the row-scaled residual norm, and the compact proof tube used by Theorem 1. A Taylor estimate proved for a different scheme controls a different perturbation vector and a different defect map unless a uniformly bounded identification maps those quantities into this typed FullVA inventory. Because finite-dimensional norm equivalence is available only after the spaces and maps are fixed, the transport constants must be chosen before

the estimate is consumed and cannot be fitted from observed orders, residual histories, or source-policy rows.

After such a transport, the primitive dependency graph still has to be respected. The state and unweighted acceleration lifts are the root inputs for the separate route; the multiplier, geometry, and gyroscopic reductions are downstream same-branch consequences in the present FullVA variables. Therefore a BLieDF hidden-constraint, multiplier-history, or multistep recursion estimate does not become a FullVA primitive bound merely because it has the same formal power of h . In this manuscript no same-object transport map from the constrained-BDF reference variables to the D5 FullVA primitive inventory is constructed. The reference proof therefore remains an ordering discipline only, while the primitive-route status remains at no actual primitive D5 Taylor subterm bounds for a T3 replacement certificate in the accepted theorem route. $\square$

Conditional primitive Taylor implication criterion (non-closure).

The next lemma is the exact one-way criterion for the separate primitive route. It should be read as a finite proof implication, not as an informal route summary: if the five open primitive lift/bilinear estimates are supplied on the same compact tube, then the already listed Taylor templates close all 162 primitive D5 subterms and hence the dynamic-row residual value. The status recorded before this lemma is deliberately asymmetric: one common tube primitive is already available, all 162 conditional template rows are typed, but the certified count of actual primitive Taylor bounds is 0/162 because the five root lift/bilinear estimates remain open. This is only a conditional 36-row dynamic statement: it is not yet a theorem-facing 132-row residual-value certificate, because the latter would still have to assemble the dynamic bound with the same 96-row non-dynamic certificate, row ordering, row scaling, block norm, branch, and endpoint convention used by the theorem. Equally important, the criterion states the open premises explicitly, so the current primitive non-invocation boundary is preserved until those premises are proved without using the direct full-residual bridge as an input.

Lemma 38 (Conditional primitive-obligation implication for D5 (non-closure)). *This separate primitive-route lemma is a conditional finite implication, not a closure certificate, and is not invoked in Theorem 1; it records only what would follow if the five primitive obligations are proved separately. Assume the five remaining primitive obligations $P_{\text{state}}, P_{\text{acc}}, P_{\lambda}, P_{\text{geom}},$ and P_{gyro} hold uniformly on the compact smooth proof tube, and use P_{tube} from*

Lemma 9. Then, only as a conditional implication under those five still-open primitive hypotheses, every D5 Taylor subterm in the 36 Newton–Euler rows is $O(h^7)$, and the lifted-stage dynamic residual satisfies $R_{\text{dyn}}(Z_G) = O(h^7)$ for each expanded component row, using the already closed D1/D2, D3, D4, and D6 inputs. This implication is not an accepted primitive-route closure, not a PC2 input, and not invoked by Theorem 1. All quantities in this conditional implication are evaluated on the same tuple $(h, y, Z_G, F_{A,h}, \|\cdot\|_{\text{proof}})$ and in the same D5 row ordering as the active theorem. Its output is only a 36-row primitive dynamic-block predicate; it becomes theorem-facing only after a separately proved 132-row T3 predicate assembles that block with the same 96-row non-dynamic certificate. No arrow from the direct residual bridge, the T2 Taylor/Kantorovich conclusion, or reported-grid errors is used to establish the five primitive premises.

Proof. Each translational dynamic row has four Taylor subterms and each rotational dynamic row has five. The recorded reduction map assigns every subterm to a finite subset of the primitive obligations. The four Taylor templates are explicit. Linear acceleration terms have no remainder, as shown in Eq. (187):

$$m_b(\hat{a} - a), \quad J_b(\hat{\alpha} - \alpha). \quad (187)$$

The conditional row-level acceleration version is Corollary 1; it consumes P_{acc} and does not prove PA2 or the unweighted acceleration lift. For any smooth force, torque, or regularized friction map F , Taylor’s formula with integral remainder gives Eq. (188):

$$F(\hat{z}) - F(z) = \int_0^1 DF(z + \theta(\hat{z} - z))(\hat{z} - z) d\theta, \quad (188)$$

so the compact-tube derivative bound gives $\|F(\hat{z}) - F(z)\| \leq C_F \|\hat{z} - z\|$. The conditional row-level smooth-map version is Corollary 2: it covers exactly the 72 external-force, friction-force, external-torque, and friction-torque subterms, consumes only the P_{state} and P_λ lift inputs, and does not use the D4 direct-route smoothness certificate as a primitive rate proof. For a multiplier geometry map $A(q)^T \lambda$, use Eq. (189):

$$A(\hat{q})^T \hat{\lambda} - A(q)^T \lambda = A(q)^T (\hat{\lambda} - \lambda) + \left(\int_0^1 DA(q + \theta(\hat{q} - q))(\hat{q} - q) d\theta \right)^T \hat{\lambda}. \quad (189)$$

For the gyroscopic term, use Eq. (190):

$$\widehat{\omega} \times J\widehat{\omega} - \omega \times J\omega = (\widehat{\omega} - \omega) \times J\widehat{\omega} + \omega \times J(\widehat{\omega} - \omega). \quad (190)$$

By Lemma 9, the constants in these four templates are uniformly bounded on the proof tube. The state-, acceleration-, multiplier-, geometry-, and gyroscopic-lift obligations then bound their assigned subterms by Ch^7 . Equivalently, for each dynamic component row r , let $\mathcal{I}_r$ be its finite primitive-subterm index set ($|\mathcal{I}_r| = 4$ for translational rows and $|\mathcal{I}_r| = 5$ for rotational rows). If $\|T_{\ell,h}(Z_G; y)\| \leq C_\ell h^7$ for every $\ell \in \mathcal{I}_r$, then the row defect satisfies Eq. (191):

$$\begin{aligned} \|R_{r,h}^{\text{prim}}(Z_G; y)\| &\leq \sum_{\ell \in \mathcal{I}_r} C_\ell h^7 \\ &= C_{D5,r} h^7, \end{aligned} \quad (191)$$

where $C_{D5,r}$ is independent of h , the reported grid length, backend, and finite diagnostic tolerances. Taking the maximum of the 36 constants $C_{D5,r}$ gives a uniform row-family constant because the dynamic-row inventory is finite. Thus finite primitive-row summation is the only aggregation step in this separate route; it is not an inverse projection from the aggregate 132-row residual estimate back to primitive subterms. The closed D1/D2 balance identities, D3 multiplier wrench identity, D4 smooth-force input, and D6 row-binding input remove the non-lift discrepancy terms, leaving a row-local $O(h^7)$ dynamic defect. $\square$

Lemma 38 is conditional on the five remaining primitive obligations and does not close the primitive-and-Taylor route to PC2. It closes only the finite implication from primitive bounds to the 162 D5 subterm bounds; the state, acceleration, multiplier, geometry, and gyroscopic primitive bounds remain outside the active proof unless they are separately proved on the same compact branch. Thus the Taylor layer now has two separate theorem-facing statements: the active full-map Taylor/Kantorovich lemma used by Theorem 1, and this separate primitive implication criterion for a stricter separate certificate. The first is closed only after two same-object inputs are present together: the direct 132-row residual bridge supplies the residual value $R_h = F_{A,h}(Z_G; y) = O(h^7)$, and the retained compact-tube Taylor/Kantorovich hypotheses supply the inverse, quadratic-remainder, branch-ball, and endpoint-map constants for that same row-scaled residual map.

Without those compact-inverse and remainder interfaces, the direct bridge is only a residual-value certificate, not a stage-root perturbation theorem. The second records only a conditional implication and cannot be spliced into the active theorem as partial evidence or counted as partial closure of the primitive/Taylor route.

Uniform residual-scaling convention. In Lemma 39 and in the full-residual bridge, write the proof residual as $F_{A,h} = SF_{A,h}^{\text{raw}}$, where S is a fixed finite-dimensional row-scaling map chosen with the chart, row ordering, and proof norm. The explicit powers of h appearing in the collocation and FullVA row formulas are part of $F_{A,h}^{\text{raw}}$ before the asymptotic estimates begin; there is no h -dependent left preconditioner or row reweighting chosen from N_h , backend behavior, solver logs, or observed slopes. All norm-equivalence constants associated with S are therefore independent of h . The uniform inverse, second-derivative, and ball constants M, L, r in Lemma 39 are retained P2 compact-tube hypotheses for this fixed scaled residual, not consequences of the D5 row identity, finite rank probes, or the separate primitive/Taylor route.

Lemma 39 (Stage-residual perturbation criterion). *Let K be the retained compact proof-tube state set from Assumption 1. For $y \in K$, let $F_{G,h}(Z; y) = 0$ denote the reduced three-stage Gauss collocation stage equations in the smooth chart and let $Z_G(y, h)$ be their isolated solution. Below, Z_G denotes this $Z_G(y, h)$ for the fixed transition input. Let $F_{A,h}(\cdot; y)$ be the accepted FullVA residual map written in the same local unknowns after the Lie retraction and constraint chart are fixed. If the rows are nonautonomous, the transition input time t_n is fixed before this lemma is invoked and $F_{A,h}$ is a fixed-time-slice abbreviation for F_{A,h,t_n} . In that case Z_G denotes the lifted Gauss stage $Z_G(y, t_n, h)$, all driven rows are evaluated at the stage times $t_n + c_i h$, and the constants M, L, r, C_R, M_E are uniform over the retained time interval as well as over $y \in K$. The Taylor expansion below is only in the stage variables at fixed t_n ; it does not differentiate in time, interpolate between time slices, or absorb a terminal-time correction into $D_Z F_{A,h}$. The Gauss stage is first lifted by Lemma 7 into this same accepted stage-coordinate vector before $F_{A,h}(Z_G; y)$ is evaluated. Thus $F_{G,h}$ is used only to define the reference stage Z_G ; the Taylor expansion and inverse estimate below are taken entirely in the fixed finite-dimensional domain and the row-scaled 132-row codomain of $F_{A,h}$, with the proof norm chosen before the small- h argument. The equation $F_{A,h}(Z; y) = 0$ defines the exact accepted stage root when the local root exists;*

that local existence is the conclusion of the contraction argument below, not a premise. Assume that, uniformly for $y \in K$, on a ball $B_r(Z_G(y, h))$ contained in the compact proof tube, the Jacobian in Eq. (192)

$$J_h := D_Z F_{A,h}(Z_G; y) \quad (192)$$

is invertible with $\|J_h^{-1}\| \leq M$ uniformly in h , and that $D_Z^2 F_{A,h}$ is bounded by L on the same ball, with M, L, r independent of h . Equivalently, for a preselected step window $0 < h \leq \bar{h}$, the same chart, row ordering, row scaling, and proof norm give the compact second-derivative bound in Eq. (193),

$$\sup_{0 < h \leq \bar{h}} \sup_{y \in K} \sup_{Z \in B_r(Z_G(y, h))} \|D_Z^2 F_{A,h}(Z; y)\| \leq L \quad (193)$$

on the retained compact proof tube K . The explicit h -dependence of the Gauss weights and residual row factors is already part of the map $F_{A,h}$ before this supremum is taken; L is not fitted from reported residual logs and is not supplied by the separate 162-subterm primitive Taylor route. Assume also, uniformly for $y \in K$ and $0 < h \leq \bar{h}$, that the accepted residual evaluated at the Gauss stage satisfies Eq. (194),

$$\|F_{A,h}(Z_G; y)\| \leq C_R h^7. \quad (194)$$

In Theorem 1, this hypothesis is discharged by Lemma 40; here it is isolated only to state the perturbation criterion. After possibly shrinking the admissible step window inside $\bar{h}$, choose $h_0 > 0$ so that Eq. (195) holds:

$$2MC_R h_0^7 \leq r, \quad 2M^2 LC_R h_0^7 \leq \frac{1}{2}. \quad (195)$$

Then, for $0 < h \leq h_0$, with the contraction radius defined in Eq. (196),

$$\rho_h = 2M \|F_{A,h}(Z_G; y)\|, \quad (196)$$

there is a unique root Z_A in the contraction ball $Z_G + B_{\rho_h}(0) \subset B_r(Z_G)$. This root is the local accepted root on the Newton branch selected from the Gauss predictor, and it satisfies the explicit stage-residual-to-root estimate in Eq. (197),

$$\|Z_A - Z_G\| \leq C_Z h^7, \quad C_Z = 2MC_R. \quad (197)$$

If the endpoint reconstruction satisfies the corresponding uniform derivative bound in Eq. (198),

$$\sup_{0 < h \leq h_0} \sup_{y \in K} \sup_{Z \in B_r(Z_G(y, h))} \|D_Z \mathcal{E}_h(Z)\| \leq M_E \quad (198)$$

on the same compact tube, then its accepted endpoint differs from the Gauss endpoint by Eq. (199),

$$\|\mathcal{E}_h(Z_A) - \mathcal{E}_h(Z_G)\| \leq C_A h^7, \quad C_A = M_E C_Z. \quad (199)$$

This is a local Newton-branch statement, not a global uniqueness claim for remote nonlinear roots of $F_{A,h}$. The uniform inverse hypothesis is a compact-tube Gauss-predictor-branch condition; finite rank probes are not a compact-tube inverse proof and are not used by this perturbation criterion. Compact-inverse quantifier discipline: the constants M, L, r , the proof norm, and the Gauss-predictor branch are fixed before the residual defect $C_R h^7$ is applied. The proof may shrink h_0 after these constants are fixed, but it does not choose M, L, r , the tube, or the branch from the computed accepted trajectory. The uniform stage-root threshold is chosen once on the compact proof tube; the same contraction radius applies to every retained-tube state and is not selected separately for each reported step. The Taylor expansion in this criterion is the expansion of the full scaled map $F_{A,h}$ with respect to the complete stage vector Z . It therefore uses the single bridged residual value $R_h = F_{A,h}(Z_G; y)$ as input; it does not require, and does not infer, a termwise certification of the 162 primitive Newton–Euler Taylor subterms. **Forward-only Taylor direction.** This is the FullVA analogue of the constrained-BDF consistency step in the reference proof: the exact smooth branch is inserted into a fixed discrete operator, a local defect is proved for that operator, and only then is the defect transported through the local inverse and global error recursion. The analogy is specifically to the reference paper’s Section 6.2 Theorems 2–4 sequence [8]: local truncation errors are formed by inserting the analytic constrained solution into the already specified BLieDF equations, the configuration/velocity error recursion is then formed, and only afterward are the hidden-constraint, multiplier, and convergence estimates used. Here the inserted object is Z_G and the fixed equations are the row-scaled $F_{A,h}$, so the Taylor/Kantorovich inverse acts only after the same-tuple residual value has been proved. This direction is not reversible. A small endpoint error, an observed sixth-order slope, or an accepted Newton residual cannot be used to infer Eq. (194); that residual value must be proved before the contraction map is invoked. Conversely, a future primitive T3 bound would be only a different proof of the same residual-value input for the same 132-row map. It would not replace the compact inverse, the quadratic-remainder bound, the endpoint map, the retained P6 solver-scale condition, or the local-to-global transfer. **Parameter-dependent Taylor-family rule.** The family $h \mapsto F_{A,h}$ is

treated as a parameterized family of finite-dimensional stage maps. For each theorem instance, h , t_n , the row scaling, and the branch chart are fixed first; Taylor's formula is then applied only in the stage variable Z . The uniform bound on $D_Z^2 F_{A,h}$ is a supremum over this predeclared family on $0 < h \leq \bar{h}$, not a derivative with respect to h , not a comparison between residual maps at two step sizes, and not a license to retune row weights or residual formulas with h after inspecting diagnostics. Thus the quadratic remainder Q_h below belongs to the same frozen residual map as R_h , J_h , and the endpoint reconstruction. Single-instance requirement: every object in the displayed hypotheses must belong to the same theorem instance. The branch ball, inverse bound, second derivative bound, row scaling, proof norm, residual value, and endpoint map are fixed together before the estimate is invoked. Replacing one of these objects by a finite rank probe, a residual log in a different norm, a postprocessed endpoint map, or a root found on another nonlinear branch does not give a weaker conclusion of this lemma; it changes the lemma's input data and leaves the perturbation estimate unavailable for that assembled object.

Proof. Write $\Delta = Z - Z_G$ and $R_h = F_{A,h}(Z_G; y)$. Taylor's formula with integral remainder gives Eq. (200), for $\|\Delta\| \leq r$,

$$F_{A,h}(Z_G + \Delta; y) = R_h + J_h \Delta + Q_h(\Delta), \quad (200)$$

where the quadratic remainder is defined in Eq. (201),

$$Q_h(\Delta) = \int_0^1 (1 - \theta) D_Z^2 F_{A,h}(Z_G + \theta \Delta; y) [\Delta, \Delta] d\theta \quad (201)$$

and therefore $\|Q_h(\Delta)\| \leq (L/2)\|\Delta\|^2$. This is the only Taylor remainder used by the perturbation step: the bound is taken on the full implemented residual family after the finite row-scaling convention is fixed, rather than by summing primitive Newton–Euler subterm estimates. Here h is a frozen parameter of the map being expanded. No term in Q_h differentiates the residual with respect to h , changes the stage abscissae, or compares $F_{A,h}$ with $F_{A,\hat{h}}$; all such step-size variation is outside this Taylor remainder and can only enter through the predeclared uniform constants. The zero equation is equivalent to the fixed-point problem in Eq. (202),

$$\Delta = \mathcal{T}_h(\Delta) := -J_h^{-1} R_h - J_h^{-1} Q_h(\Delta). \quad (202)$$

Let $\rho_h = 2M\|R_h\|$. The chosen h_0 gives, for $0 < h \leq h_0$, $\rho_h \leq r$ and $ML\rho_h \leq 1/2$. If $\|\Delta\| \leq \rho_h$, then Eq. (203) shows that the ball maps into itself:

$$\|\mathcal{T}_h(\Delta)\| \leq M\|R_h\| + \frac{ML}{2}\rho_h^2 \leq \rho_h. \quad (203)$$

Moreover, using the integral mean-value form of $Q_h(\Delta_1) - Q_h(\Delta_2)$ on the same ball gives the contraction estimate in Eq. (204),

$$\|\mathcal{T}_h(\Delta_1) - \mathcal{T}_h(\Delta_2)\| \leq ML\rho_h\|\Delta_1 - \Delta_2\| \leq \frac{1}{2}\|\Delta_1 - \Delta_2\|. \quad (204)$$

Thus $\mathcal{T}_h$ is a contraction on $B_{\rho_h}(0)$ and has a unique fixed point Δ_A there. The corresponding zero $Z_A = Z_G + \Delta_A$ satisfies Eq. (205),

$$\|Z_A - Z_G\| = \|\Delta_A\| \leq \rho_h \leq 2MC_R h^7 = C_Z h^7. \quad (205)$$

If $\mathcal{E}_h$ is any smooth Lie-retraction endpoint reconstruction, then Taylor's formula in the stage variables gives Eq. (206),

$$\mathcal{E}_h(Z_A) - \mathcal{E}_h(Z_G) = \int_0^1 D_Z \mathcal{E}_h(Z_G + \theta \Delta_A) \Delta_A d\theta, \quad (206)$$

and the displayed compact-tube derivative bound gives $\|\mathcal{E}_h(Z_A) - \mathcal{E}_h(Z_G)\| \leq M_E C_Z h^7 = C_A h^7$. The contraction ball is the local branch used by the accepted Newton solve; the argument neither requires nor asserts uniqueness of possible roots outside the compact-tube neighborhood. Since M, L, r, C_R are compact-tube constants, the small-step restriction just used is uniform over the retained proof tube and over the reported transition indices; it is not chosen from a solved trajectory, a residual log, or a per-step rank probe. The same fixed residual map and norm are used in R_h, J_h, Q_h , the contraction ball, and the endpoint reconstruction. This same-object condition is what makes the Kantorovich step a proof of a branch-selected local map rather than a post-hoc consistency check assembled from separate diagnostics. $\square$

Kantorovich-radius interpretation. The perturbation criterion is used only through the fixed compact-tube data (M, L, r) and the bridged residual value $R_h = F_{A,h}(Z_G; y)$. With $\beta_h = M\|R_h\|$ and $\rho_h = 2\beta_h$, the small-step restrictions above enforce Eq. (207),

$$\rho_h \leq r, \quad ML\rho_h \leq \frac{1}{2}. \quad (207)$$

Thus the nonlinear Taylor remainder is absorbed inside the same ball on which the inverse and second-derivative constants are valid, and the local root selected by the Gauss predictor satisfies $\|Z_A - Z_G\| \leq 2M\|R_h\|$. Lemma 40 supplies only the scale $\|R_h\| \leq C_R h^7$; it does not select M, L, r or the branch. Those constants and the proof norm are fixed before the direct residual bridge, solver logs, or numerical trajectory are read.

Lemma 40 (Full 132-row residual-defect certificate bridge). *Use the accepted row ordering, finite row-scaling map, and fixed finite-dimensional proof norm for the implemented 132-row stage residual. If the 96-row non-dynamic certificate gives $\|F_{\text{nd},h}(Z_G; y)\| \leq C_{\text{nd}} h^7$ on the lifted smooth Gauss stage and the implemented Newton–Euler rows satisfy the direct same-branch row identity in Eq. (208),*

$$F_{\text{dyn},h}(Z_G; y) = (r_{j,h}^{\text{NE}}(Z_G; y))_{j=1}^{36} = 0 \in \mathbb{R}^{36}, \quad (208)$$

then the full implemented residual satisfies Eq. (209),

$$\|F_{A,h}(Z_G; y)\| \leq C_R h^7 \quad (209)$$

with C_R independent of $y \in K$, h , the reported grid length, backend, and finite diagnostic tolerances. The Z_G appearing in the 96-row certificate, the 36-row Newton–Euler identity, and the full residual evaluation is one fixed lifted Gauss stage with the same stage abscissae, transition input, chart, row ordering, row scaling, proof norm, and endpoint output convention. The bridge is not re-centered at Z_A , at $\tilde{Z}_A$, or at an endpoint-refined state. The row-scaling map is fixed before the asymptotic argument and has h -independent operator norm on this finite row space, so it can change only the constant C_R , not the h^7 power. Any comparison with an unscaled residual norm uses an h -independent finite-dimensional norm-equivalence constant; no hidden h -dependent row weights are introduced beyond the explicit h -factors in the residual formulas. Thus, at the lifted Gauss stage Z_G for this fixed row-scaled residual, the certified 132-row residual-value bridge supplies the stage-residual hypothesis of Lemma 39 for the accepted direct-route formula residual. The bridge discharges only this $C_R h^7$ residual-defect input. At this bridge interface, P3 is used only to identify the implemented residual/Jacobian formula path and fixed row scaling, P4 is used only through the proved 96-row non-dynamic row-local certificate, and P5 supplies the 36 dynamic rows; none of P3/P4 is promoted into a standalone full-132-row theorem-interface discharge or a primitive symbolic-oracle closure. The missing primitive dynamic

symbolic oracle is not a missing input to this bridge. It does not discharge the uniform stage inverse, branch-ball, endpoint/right-inverse, stability/tube-retention, or solver-scale hypotheses, which remain the separate P1/P2/P6 theorem interfaces consumed by the surrounding perturbation argument. The D5 direct-substitution certificate is the row-by-row provenance record for the displayed 36-row identity; the proof-facing input to this lemma is the identity itself on the lifted Gauss stage.

Proof. After the accepted row ordering is fixed, write the block vector as Eq. (210):

$$F_{A,h}(Z_G; y) = (F_{\text{nd},h}(Z_G; y), F_{\text{dyn},h}(Z_G; y)), \quad (210)$$

where $F_{\text{nd},h} \in \mathbb{R}^{96}$ collects the kinematic, collocation, and lower-pair Ful-IVA rows and $F_{\text{dyn},h} \in \mathbb{R}^{36}$ collects the Newton–Euler rows. The finite row scaling is already included in these two blocks; because it is a fixed linear map on a fixed finite-dimensional row space, its operator norm and inverse-equivalence constants are independent of h . The non-dynamic certificate gives $\|F_{\text{nd},h}(Z_G; y)\| \leq C_{\text{nd}}h^7$ in the fixed proof norm after the finite row-scaling map, while the D5 direct-substitution certificate gives Eq. (211),

$$F_{\text{dyn},h}(Z_G; y) = (r_{j,h}^{\text{NE}}(Z_G; y))_{j=1}^{36} = 0. \quad (211)$$

For nonautonomous rows this block vector is evaluated on the same time-indexed residual slice F_{A,h,t_n} . The 96-row block, the 36-row block, the Jacobian used by the perturbation criterion, and the endpoint map therefore share the same transition time and stage abscissae; row families may not be evaluated at different transition times and then assembled into one residual bridge. Finite-dimensional norm equivalence for the fixed 132-row block then gives Eq. (212),

$$\begin{aligned} \|F_{A,h}(Z_G; y)\| &\leq C_{\text{blk}}(\|F_{\text{nd},h}(Z_G; y)\| + \|F_{\text{dyn},h}(Z_G; y)\|) \\ &\leq C_R h^7, \quad C_R = C_{\text{blk}} C_{\text{nd}}. \end{aligned} \quad (212)$$

Thus the $C_R h^7$ hypothesis in Lemma 39 is a finite block-norm consequence of the 96-row $O(h^7)$ certificate plus the 36-row zero dynamic certificate, not an empirical residual fit. The implication is deliberately one-sided. It uses the fixed block projection from the implemented 132-row vector to its $96 + 36$ row families and the finite-dimensional norm equivalence after row scaling; it does not use cancellation between row families, a small assembled residual

to infer smaller primitive subterms, or a dynamic-row identity to improve the non-dynamic row constant. In particular, the zero Newton–Euler block is a base-point identity in the residual vector, not a hidden estimate of the inverse, second-derivative, endpoint, or solver-tolerance constants used later. Direct-route sufficiency boundary: Lemma 39 requires a residual value at the lifted Gauss stage, not a primitive symbolic expansion of every dynamic Taylor term. Once P3 fixes the implemented residual path and row scaling, P4 supplies the 96 non-dynamic rows, and P5 supplies the D5 direct-substitution zero rows, the missing primitive dynamic symbolic oracle is not a missing input to this bridge. The incomplete primitive symbolic route remains a separate primitive-route record and is not used to prove or disprove the displayed $C_R h^7$ defect. This discharge is a pre-perturbation statement on Z_G , not an inference from the accepted numerical trajectory or from small residuals after Newton termination; the latter enter only through Lemma 41 and the retained P6 solver-scale condition. Consequently the bridge is not a compact-tube inverse proof: it does not choose the Newton branch, prove the J_h^{-1} bound in Lemma 39, or close the endpoint, stability, tube-retention, and η_h^{tube} hypotheses. Those interfaces remain explicit conditions around the residual-defect handoff. $\square$

Corollary 5 (Route-separation invariant for the direct residual bridge). *For the theorem invocation, define the active residual predicate in Eq. (213),*

$$\mathbf{R}_{\text{dir}}(h, y) : \quad \|F_{\text{nd},h}(Z_G; y)\| \leq C_{\text{nd}} h^7 \text{ and } F_{\text{dyn},h}(Z_G; y) = 0 \quad (213)$$

on the fixed branch, chart, row ordering, row scaling, and proof norm. The full-residual bridge proves $\mathbf{R}_{\text{dir}}(h, y) \Rightarrow \|F_{A,h}(Z_G; y)\| \leq C_R h^7$, which is the only residual-defect predicate consumed by Lemma 39 and Theorem 1. The predicate is single-object: the non-dynamic rows, dynamic rows, block norm, and endpoint map all refer to the same Z_G ; no row family may be evaluated on a different stage vector and then spliced into this implication. A stricter primitive predicate $\mathbf{R}_{\text{T3}}^{132}(h, y)$ could replace $\mathbf{R}_{\text{dir}}(h, y)$ only after it is proved on the same tuple $(h, y, Z_G, F_{A,h})$. Its primitive part would first have to prove the five state-, acceleration-, multiplier-, geometry-, and gyroscopic-lift obligations and thereby obtain the 36-row dynamic bound; that dynamic bound would then have to be assembled with the same 96-row non-dynamic certificate, row ordering, row scaling, block norm, branch, and endpoint convention to yield

$$\mathbf{R}_{\text{T3}}^{132}(h, y) \implies \|F_{A,h}(Z_G; y)\| \leq C_R^{\text{T3}} h^7. \quad (214)$$

No such 132-row T3 predicate, in the sense of Eq. (214), is a premise in this manuscript's theorem. Therefore the present open status of the primitive/Taylor route neither supplies nor removes a theorem hypothesis: the theorem depends on R_{dir} plus the retained compact-branch, endpoint, stability, and solver-scale interfaces, and it does not infer source-policy, full-TFE, or residual-to-error promotion from either route.

Proof. The statement is a direct consequence of Lemma 40 and Lemma 39. The first lemma supplies the residual value consumed by the second lemma on the fixed implemented residual object $F_{A,h}$, with the row families evaluated at the same lifted Gauss stage and in the same proof norm. A primitive proof of the 36 dynamic rows would still become theorem-facing only after it is assembled with the same 96-row non-dynamic block into a new 132-row residual-value predicate on that object. Since neither lemma maps a full residual estimate back to primitive subterms, reported residual histories, or source-policy outputs, the two route predicates cannot be spliced and neither route closes P6, P7, or external-comparison readiness. $\square$

Lemma 41 (Inexact Newton tolerance scale). *Let Z_A be the exact accepted stage solution of $F_{A,h}(Z; y) = 0$ and let $\tilde{Z}_A$ be the computed Newton iterate. Put $J_A = D_Z F_{A,h}(Z_A; y)$. Assume that, on a local ball $B_{r_A}(Z_A)$ contained in the accepted Newton branch, Eq. (215) holds:*

$$\begin{aligned} \|J_A^{-1}\| &\leq M_A, \\ \|D_Z F_{A,h}(Z_1; y) - D_Z F_{A,h}(Z_2; y)\| &\leq L_A \|Z_1 - Z_2\|, \quad Z_1, Z_2 \in B_{r_A}(Z_A), \\ M_A L_A r_A &\leq \frac{1}{2}, \end{aligned} \tag{215}$$

with constants independent of h . Assume also that $\tilde{Z}_A \in B_{r_A}(Z_A)$ and that the proof-norm residual input satisfies Eq. (216),

$$\|F_{A,h}(\tilde{Z}_A; y)\| \leq \eta_h, \tag{216}$$

and let $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$ denote the full endpoint-output map used after the stage solve, namely Lie endpoint reconstruction followed by the accepted local endpoint-closure map. Assume that this full endpoint-output map satisfies $\|D_Z \mathcal{P}_h\| \leq M_N$ on the same ball. Then Eq. (217) gives the stage and output perturbation bounds:

$$\|\tilde{Z}_A - Z_A\| \leq 2M_A \eta_h, \quad \|\mathcal{P}_h(\tilde{Z}_A) - \mathcal{P}_h(Z_A)\| \leq C_N \eta_h, \quad C_N = 2M_A M_N. \tag{217}$$

Consequently, if the accepted-branch nonlinear solver hypothesis satisfies the asymptotic scale in Eq. (218),

$$\eta_h \leq c_\eta h^7, \quad (218)$$

then the endpoint contribution from inexact Newton is bounded explicitly by Eq. (219),

$$\|\mathcal{P}_h(\tilde{Z}_A) - \mathcal{P}_h(Z_A)\| \leq C_N c_\eta h^7. \quad (219)$$

Here M_N includes the local Lipschitz factor of the endpoint-closure map; the inexact-Newton term is therefore measured between two endpoint-closed maps, not between an unclosed reconstructed endpoint and a closed theorem output. A fixed absolute residual tolerance is a finite-run engineering policy and must be tightened with h to support the asymptotic theorem. This is again a local Newton-branch estimate, not a certificate of global nonlinear solver convergence. Conversely, for a transition already known to remain in the accepted Newton branch ball, the proof-norm stopping rule $\|F_{A,h}(\tilde{Z}_A; y)\| \leq c_\eta h^7$ verifies only the residual-size subcondition of the retained P6 hypothesis for an already branch-certified transition. It still does not prove Newton convergence to the branch, choose the proof norm, or verify the condition on every reported step. The residual inequality and the branch-membership hypothesis are separate load-bearing inputs: the lemma does not infer $\tilde{Z}_A \in B_{r_A}(Z_A)$ from a small residual alone. Residual control outside this ball, or on a different branch, does not control the stage error without the averaged-Jacobian inverse used below.

Branch-ball membership checkpoint. The theorem-level use of this lemma consumes two independent solver facts for every accepted transition: the iterate is certified to remain in $B_{r_A}(Z_A)$, and its residual is bounded in the fixed proof norm by the compact-tube envelope η_h^{tube} . The second fact cannot replace the first. A finite log entry with a small residual, an iteration cap, or a successful line-search flag is therefore not a theorem input unless it is already tied to the same Gauss-predictor branch, the same residual norm, and the same local ball. If this membership certificate is unavailable, the transition is outside the branch-selected map used in Theorem 1; it is not counted as a larger local constant or as a lower-order inexact Newton step.

Equivalently, the accepted proof uses a strong local residual inverse: the averaged Jacobian along the segment from Z_A to $\tilde{Z}_A$ is required to be uniformly invertible by the small-neighborhood perturbation bound. In Theorem 1, the lemma-level residual budget η_h is applied through the compact-tube

upper envelope η_h^{tube} . The single reported-grid maximum is only the trajectory specialization of that envelope, not a replacement for the uniform compact-tube solver hypothesis. For this lemma, the residual norm is the proof residual norm of the accepted 132-row residual after the local chart, row ordering, and finite residual row-scaling convention, including the D6 unweighted scaling convention where applicable, are fixed. Logged implementation residuals specialize η_h only through the accepted implementation-path binding and an h -independent norm equivalence on the compact tube. Without that binding they remain finite diagnostics; they do not introduce a second residual budget, change the proof norm, or prove P6 as a solver theorem. The scalar c_η , residual norm, accepted branch, and constants M_A, M_N are fixed proof data before any Newton log, tolerance sweep, or reported residual ratio is inspected. A run that satisfies a fixed absolute tolerance but violates the compact-tube h^7 residual scale is simply outside this lemma's asymptotic use; post-hoc residual filtering cannot create an accepted transition for the theorem. Likewise, a Newton correction-norm test, relative residual test, iteration cap, line-search success flag, or backend convergence flag is not a substitute for the displayed proof-norm residual inequality. Such a proxy may be used only if, before the run and on the same accepted branch, it is accompanied by a uniform conversion that implies $\|F_{A,h}(\tilde{Z}_A; y)\| \leq c_\eta h^7$ in the fixed 132-row proof norm. Otherwise it is an implementation diagnostic and does not contribute to the $C_N \eta_h$ perturbation bound.

Proof. Set $\delta = \tilde{Z}_A - Z_A$. Since $F_{A,h}(Z_A; y) = 0$, the mean-value form of the residual gives Eq. (220),

$$F_{A,h}(\tilde{Z}_A; y) = \bar{J}_h \delta, \quad \bar{J}_h = \int_0^1 D_Z F_{A,h}(Z_A + \theta \delta; y) d\theta. \quad (220)$$

Write $\bar{J}_h = J_A + E_h$. The Lipschitz bound and $\tilde{Z}_A \in B_{r_A}(Z_A)$ give Eq. (221),

$$\|E_h\| \leq \int_0^1 L_A \theta \|\delta\| d\theta \leq L_A r_A. \quad (221)$$

Hence $\|J_A^{-1} E_h\| \leq 1/2$. By the Neumann perturbation theorem, $\bar{J}_h$ is invertible and $\|\bar{J}_h^{-1}\| \leq 2M_A$. Therefore Eq. (222) follows:

$$\|\tilde{Z}_A - Z_A\| = \|\delta\| \leq 2M_A \|F_{A,h}(\tilde{Z}_A; y)\| \leq 2M_A \eta_h. \quad (222)$$

The full endpoint-output map has $\|D_Z \mathcal{P}_h\| \leq M_N$ on the same neighborhood, hence Eq. (223),

$$\|\mathcal{P}_h(\tilde{Z}_A) - \mathcal{P}_h(Z_A)\| \leq M_N \|\tilde{Z}_A - Z_A\| \leq 2M_A M_N \eta_h = C_N \eta_h. \quad (223)$$

If $\eta_h \leq c_\eta h^7$, the same estimate gives $C_N \eta_h \leq C_N c_\eta h^7$, keeping the nonlinear solve error below the local truncation scale of a sixth-order method with an explicit constant. The averaged-Jacobian inverse is the transfer operator from residual norm to stage-error norm on this accepted local branch. Pointwise Jacobian invertibility alone, or a residual bound away from this ball, is not used as a shortcut. $\square$

Lemma 42 (Reported solver logs are not a P6 proof). *A finite reported-grid solver log, even one satisfying Eq. (224),*

$$\eta_h^{\text{reported}} := \max_{0 \leq n < N_h} \eta_{h,n} = \hat{\eta}_h^{\log} \leq c_\eta h^7 \quad (224)$$

on the recorded accepted trajectory, is not a compact-tube solver envelope and is not a proof of the P6 solver-policy theorem. It can specialize an already fixed P6 theorem instance only after the compact tube, proof residual norm, row scaling, Gauss-predictor branch, branch-ball membership, and constant c_η have all been fixed independently. It cannot calibrate c_η , prove branch membership, prove the averaged-Jacobian inverse, choose the accepted nonlinear root, or turn fixed production tolerances into an asymptotic solver policy.

Proof. Lemma 41 needs two separate inputs for each accepted transition: membership in the local Newton branch ball and the proof-norm residual inequality. The reported quantity η_h^{reported} , denoted $\hat{\eta}_h^{\log}$ in diagnostic tables, records only a maximum over a finite set of already computed trajectory transitions. It is therefore a specialization of the residual-size clause after the theorem instance has selected the branch, norm, row convention, and constant. It is not a quantifier over the compact tube and it does not supply the local inverse that converts residual size to stage error.

Consequently the logical direction is one-way. A predeclared same-branch stopping rule can instantiate the residual-size part of P6 for the transitions it accepts, while the retained compact-tube branch hypotheses still supply membership and local invertibility. Reading the finite log in the opposite direction would choose c_η , the residual norm, or the branch from the evidence that the theorem is meant to bound. That reverse reading is disallowed here. The theorem-level condition remains $\eta_h^{\text{tube}} \leq c_\eta h^7$; the finite reported-grid log remains diagnostic-grade solver information, not a P6 proof. $\square$

Proof-level solver-policy instantiation. The theorem can be invoked with a predeclared proof-level Newton acceptance rule: before the run, fix the accepted branch, proof residual norm, row scaling, c_η , and the compact-tube constants, and accept a transition for the theorem only when the branch-membership hypotheses of Lemma 41 are retained and the proof-norm residual satisfies $\|F_{A,h}(\tilde{Z}_A; y)\| \leq c_\eta h^7$. Under those pre-fixed conditions the residual-size part of P6 is instantiated by the stopping rule itself for the accepted transition set. This does not prove branch membership, averaged-Jacobian invertibility, or all-transition termination; those remain the retained compact-branch solver hypotheses. It also does not promote fixed production tolerances, backend convergence flags, post-hoc residual filtering, or source-policy runs into theorem-level solver-policy proof.

Lemma 43 (Residual-size subcondition under a predeclared P6 instance). *Fix a theorem instance before the run: the accepted Gauss-predictor branch, proof residual norm, row scaling, compact-tube constants, and scalar c_η are fixed, and the branch-ball and strong-local-inverse hypotheses of Lemma 41 are retained. Suppose that for each accepted transition in the reported theorem domain the Newton iterate is accepted only after the predeclared proof-norm residual test in Eq. (225),*

$$\|F_{A,h}(\tilde{Z}_A; y)\|_{\text{proof}} \leq c_\eta h^7 \quad (225)$$

is satisfied. Then the residual-size part of the retained P6 clause is instantiated for those accepted transitions, and Lemma 41 contributes at most $C_N c_\eta h^7$ to the local defect. This verifies only the displayed proof-norm residual inequality on the accepted transition set; it does not promote P6 from a retained compact-tube solver hypothesis to a theorem-level solver-policy certificate unless branch-ball membership, strong local inverse, and the uniform all-transition envelope are proved separately. This lemma does not prove branch-ball membership, strong-local-inverse persistence, all-transition solver termination, global Newton convergence, remote-root exclusion, or any fixed-tolerance asymptotic theorem.

Proof. The stopping rule is stated in the same proof residual norm and with the same constant c_η used by the theorem instance. Therefore, on every accepted transition for which the retained branch and local-inverse hypotheses hold, the inexact-Newton input satisfies $\eta_h \leq c_\eta h^7$. Substituting this inequality into Lemma 41 gives Eq. (226),

$$\|\mathcal{P}_h(\tilde{Z}_A) - \mathcal{P}_h(Z_A)\| \leq C_N \eta_h \leq C_N c_\eta h^7. \quad (226)$$

The conclusion is only a residual-size instantiation on the preselected branch. It uses the branch-membership and local-inverse assumptions rather than proving them, and it makes no statement about a backend fixed tolerance or about solver behavior outside the accepted theorem domain. $\square$

Lemma 44 (Local-to-global transfer on the accepted tube). *Let Ψ_h be a one-step map on the smooth reduced chart from Assumption 1, let K be the compact proof tube, and suppose the exact trajectory on $0 \leq t \leq T$ has distance at least $d_K > 0$ from ∂K . **Time-augmentation convention for the transfer.** If the reduced FullVA flow is nonautonomous because drives, loads, or constraint right-hand sides depend on time, this lemma is applied to the augmented chart in Eq. (227):*

$$\hat{y} = (y, \tau), \quad \dot{\tau} = 1. \quad (227)$$

Then φ_h denotes the autonomous augmented flow, Ψ_h advances $\tau \mapsto \tau + h$ exactly, and the reported theorem norm is obtained by projecting back to the physical reduced variables; the time component contributes zero grid error because both numerical and exact times are t_n on the reported grid. Equivalently, in unaugmented notation $\varphi_h(y)$ abbreviates the two-parameter flow $\varphi_{t+h,t}(y)$ at the fixed reported input time $t = t_n$. This time convention is fixed before the local-defect and Gronwall estimates, so no time-phase mismatch, hidden interpolation, or terminal-time correction is inserted into the transfer. Suppose that, for all $y, \bar{y} \in K$, the local defect input is Eq. (228):

$$\|\Psi_h(y) - \varphi_h(y)\| \leq C_\ell h^{p+1} \quad (228)$$

and the stability input is Eq. (229):

$$\|\Psi_h(y) - \Psi_h(\bar{y})\| \leq (1 + C_s h) \|y - \bar{y}\| \quad (229)$$

for $0 < h \leq h_0$, where $C_s \geq 0$. For the accepted branch-selected map in Theorem 1, the same lemma is used in the restricted-domain form in Eq. (230): if Ψ_h is defined only on the accepted theorem domain $\mathcal{D}_h \subset K$, it is enough that, before the first domain exit, the exact comparison state $Y_n = \varphi_{t_n,0}(y_0)$ and the numerical state y_n satisfy

$$\begin{aligned} y_n, Y_n &\in \mathcal{D}_h \cap K, \\ \|\Psi_h(Y_n) - \varphi_{t_{n+1},t_n}(Y_n)\| &\leq C_\ell h^{p+1}, \\ \|\Psi_h(y_n) - \Psi_h(Y_n)\| &\leq (1 + C_s h) \|y_n - Y_n\| \end{aligned} \quad (230)$$

on the reported transition indices $0 \leq n < N$. This restricted form is the version consumed by the theorem; it does not require an arbitrary extension of the accepted map to all of K . The first-exit argument below closes the compact-tube part of this condition, while branch-ball, endpoint-ball, proof-norm, and P6 solver-envelope membership remain the predeclared admissibility conditions defining $\mathcal{D}_h$. **Synchronized-time stability slice.** In the augmented nonautonomous interpretation, this stability bound is consumed only on the same reported input-time slice: the two arguments are (y, τ_n) and $(\bar{y}, \tau_n)$, with $\tau_n = t_n$. The accepted map advances both time components exactly to $\tau_{n+1} = t_{n+1}$, so the stability estimate controls only the physical reduced-state difference and does not hide a time-registration error. The proof never compares states with different time phases, never fits C_s from time-shifted trajectory data, and never absorbs interpolation or terminal-time mismatch into the stability constant. Equivalently, in unaugmented notation the retained P2 stability input is the one-step Lipschitz bound of the time-indexed map Ψ_{h,t_n} on the physical state at fixed input time t_n . If a sharper negative contraction coefficient is available, enlarge it to a nonnegative upper bound before applying this lemma. Let $t_n = nh$, and let N be any reported grid length with $0 \leq t_n \leq T$ for $0 \leq n \leq N$; in particular, $Nh = T$ may be used when the final time is exactly divisible by h . Let the numerical trajectory be the accepted recurrence in Eq. (231):

$$y_{n+1} = \Psi_h(y_n), \quad 0 \leq n < N, \quad (231)$$

started from the same reduced-chart initial state y_0 used by the exact flow $\varphi_{t_n}(y_0)$. Local-to-global initial-state boundary: the lemma uses zero initial error in the reduced chart. Any unreported initial projection, warm-start, branch-selection, or chart-registration error would add a separate initial-error term to the discrete Gronwall recursion and is not absorbed into the $C_\ell h^{p+1}$ one-step defect or the theorem's h^6 same-initial-state reported-grid estimate. Define the Gronwall factor in Eq. (232):

$$\Gamma_s(T) = \begin{cases} \frac{e^{C_s T} - 1}{C_s}, & C_s > 0, \\ T, & C_s = 0. \end{cases} \quad (232)$$

Uniform-grid instance rule: the constants C_ℓ , C_s , the tube margin d_K , the horizon T , and the resulting $\Gamma_s(T)$ are chosen once for the accepted theorem instance before the reported grid is considered. The proof does not use

transition-dependent constants $C_{\ell,n}$, $C_{s,n}$, or a stepwise tube margin, and it does not replace the recurrence after inspecting a local residual, endpoint diagnostic, or observed order row. If the accepted map is defined only on the theorem domain $\mathcal{D}_h$, this lemma is applied only to the preselected accepted transition set described after the proof; Gronwall controls compact tube retention but does not create new branch-ball, endpoint-ball, or P6 solver-envelope admissibility. Then for h sufficiently small, the numerical trajectory remains in K and satisfies the grid estimate in Eq. (233):

$$\max_{0 \leq n \leq N} \|y_n - \varphi_{t_n}(y_0)\| \leq C_T h^p, \quad C_T = C_\ell \Gamma_s(T), \quad (233)$$

with C_T independent of h . This is an explicit grid-point estimate on the accepted reduced-chart branch and a compact-tube transfer lemma on the reported time grid $t_n = nh$; exact final-time divisibility is not required. The local one-step hypotheses are used only for transitions $0 \leq n < N$; the terminal index N is an output index, not an extra solve or an additional local-map hypothesis. If $Nh < T$, the leftover interval $T - Nh$ is outside this reported-grid estimate; the lemma does not assert an error bound at an unreported terminal time unless that time is added to the reported grid by a separate step. It is not an invariant-region proof away from the accepted branch.

Proof. Let n_* be the first index at which the numerical trajectory would leave K , with $n_* = N + 1$ if no such index exists. For $n < n_*$, the estimates hold in K . Let $\varepsilon_n = y_n - \varphi_{t_n}(y_0)$ and $e_n = \|\varepsilon_n\|$. With the time-augmentation convention above, the semigroup property used here is the autonomous augmented semigroup identity; in unaugmented notation it is the identity $\varphi_{t_{n+1},0}(y_0) = \varphi_{t_{n+1},t_n}(\varphi_{t_n,0}(y_0))$ at the reported grid time. Adding and subtracting $\Psi_h(\varphi_{t_n}(y_0))$ gives Eq. (234):

$$e_{n+1} \leq (1 + C_s h)e_n + C_\ell h^{p+1}. \quad (234)$$

Truncated first-exit recursion checkpoint. At this point the recursion is not yet iterated over the whole reported grid. Set the truncated index in Eq. (235):

$$m_* = \min\{N, n_*\}. \quad (235)$$

For each output index $0 \leq n \leq m_*$, every transition used in the Gronwall sum has index $0 \leq j < n \leq n_*$, hence $j < n_*$. The local defect and stability hypotheses are therefore evaluated only before the hypothetical first

exit, and the proof does not use the local estimates after a hypothetical exit. Iterating the scalar recursion on this truncated index set and using the discrete Gronwall argument in inequality form, with $nh \leq T$, yields Eq. (236) for every $0 \leq n \leq m_*$:

$$e_n \leq C_\ell h^{p+1} \sum_{j=0}^{n-1} (1 + C_s h)^j \leq C_\ell h^p \Gamma_s(T), \quad (236)$$

where the definition of $\Gamma_s(T)$ supplies the limiting case $C_s = 0$. Let $C_T = C_\ell \Gamma_s(T)$. Choose h small enough that $C_T h^p < d_K/2$. First-exit bootstrap domain: the local estimates are used only up to the first possible exit index n_* , and h_0 is then shrunk after the Gronwall constant C_T and the tube margin d_K are fixed. Thus the proof does not assume global tube retention as the conclusion; it proves retention by contradicting the first-exit alternative. The first-exit index is a proof device, not a trajectory failure log. The stability constant C_s , tube margin d_K , Gronwall factor $\Gamma_s(T)$, and C_T are fixed before any reported grid errors, work-precision slopes, or exit diagnostics are inspected; shrinking h_0 after those choices cannot calibrate C_s or the tube margin from a finite run. Then every numerical point covered by the truncated induction stays within distance $d_K/2$ of the exact trajectory. If $n_* \leq N$, the estimate at n_* implies that y_{n_*} is still inside K , contradicting the definition of the first exit. Hence $n_* = N + 1$, and only after this contradiction is the estimate extended to every reported index $0 \leq n \leq N$. This closes the bootstrap and proves the same-initial-state reported-grid $O(h^p)$ estimate on the accepted tube. $\square$

Accepted-domain specialization of the transfer. For Theorem 1, Lemma 44 is not applied to an arbitrary extension of the accepted map from $\mathcal{D}_h$ to all of K . The recurrence is read only on the preselected accepted transition set of Definition 7. The first-exit proof closes the compact-tube component of domain membership: once the local defect and stability estimates have been fixed, the Gronwall bound keeps the numerical and exact comparison states in the retained compact tube for $0 < h \leq h_0$. The other domain components—Gauss-predictor branch selection, branch-ball and endpoint-ball membership, the fixed proof norm, and the P6 scaled solver-envelope condition—are not consequences of Gronwall. They remain the same admissibility conditions on the same reported transition set. Thus the transfer never propagates through an undefined accepted step and never

converts a solver-policy or branch-domain failure into a covered lower-order case.

Table 29: Local-to-global transfer table for Lemma 44 and Theorem 1.

Transfer layer	Input consumed	Output produced	Explicit non-use
Local defect input	Uniform branch-selected local defect $C_{\text{loc}}h^7$ on the accepted compact proof tube.	The lemma input $C_\ell h^{p+1}$ with $p = 6$.	Residual/reaction tables, finite h -sweep fits, fixed production tolerances.
Stability and tube bootstrap	The compact-tube stability scale $1 + C_s h$ and the tube-retention margin d_K .	The accepted numerical trajectory remains in K after shrinking h_0 .	Invariant-region claims away from the accepted branch or remote chart claims.
Discrete Gronwall recursion	The accepted recurrence $y_{n+1} = \Psi_h(y_n)$, started from the same initial reduced-chart state, on reported transition steps $0 \leq n < N_h$.	$C_{\text{red}} = C_{\text{loc}}\Gamma_s(T)$ and the reduced-chart grid-point estimate.	Resampled or postprocessed sequences, unreported initial projection errors, exact final-time divisibility, an extra terminal solve, or an additional local-map hypothesis at T .
Reported-grid output	The reported output indices $0 \leq n \leq N_h$, $N_h h \leq T$, followed by the reporting map in the theorem.	The reduced-chart h^6 bound and the later position-velocity reporting-map consequence.	Unreported leftover interval claims, residual-to-error promotion, or source-policy/full-TFE readiness.

Local-to-global transfer rule: the transfer consumes only the uniform branch-selected local defect, the compact-tube stability scale, and the tube-retention margin before the transfer step. discrete Gronwall on reported transition steps is then applied to the accepted recurrence $y_{n+1} = \Psi_h(y_n)$ from the same initial reduced-chart state, not a resampled or postprocessed sequence, and it consumes zero initial error rather than hiding an initial projection or branch-registration defect. It produces a reduced-chart grid-point estimate on reported output indices; the position-velocity estimate is a later reporting-map consequence. The table does not use residual/reaction tables, finite h -sweep fits, fixed production tolerances, source-policy rows, unreported initial projection errors, an unreported leftover interval, or an extra terminal solve at T . It does not

prove a theorem-level solver-policy theorem, a P7 residual-to-error transfer, source-policy readiness, or full-TFE readiness. Current certificates prove the direct PC2 residual bridge under the retained P6 hypothesis; they do not prove P6 or P7 themselves.

Stability-map binding checkpoint. In the invocation of Lemma 44 inside Theorem 1, the abstract one-step map Ψ_h is the endpoint-closed inexact accepted map $\Psi_h^{\text{G6FVA}} = \mathcal{P}_h(\tilde{Z}_A)$, restricted to the same accepted compact branch and proof norm used in the local-defect estimate. The maps Ψ_h^G , Ψ_h^A , and Ψ_h^E are used only to decompose the one-step local defect before the recurrence is written; their stability is not substituted into the global recursion. The stability constant C_s is therefore the retained same-branch stability scale for the accepted map under the P2 compact-tube and P6 solver-scale interfaces, not a constant borrowed from the Gauss map, exact-stage root, or endpoint-closed exact-stage map. After the four local perturbation terms have produced $\|\Psi_h^{\text{G6FVA}}(y) - \varphi_h(y)\| \leq C_{\text{loc}}h^7$, the Gronwall step uses one recurrence only: $y_{n+1} = \Psi_h^{\text{G6FVA}}(y_n)$. This prevents a hidden map-switch in the local-to-global proof and keeps endpoint closure and inexact Newton inside the same branch-selected map whose grid error is reported.

Lemma 45 (Reporting-map norm consequence on the accepted chart). *Let K be the retained compact proof tube in the reduced chart, and let $\mathcal{R} : K \rightarrow \mathbb{R}^{m_{qv}}$, $\mathcal{R}(y) = (q(y), v(y))$, be the reported position-velocity map. Assume $\mathcal{R}$ is C^1 on a neighborhood of K . Choose once a fixed convex coordinate neighborhood U_K whose closure lies in that chart neighborhood, contains K with positive margin, and satisfies the derivative bound in Eq. (237),*

$$\sup_{y \in U_K} \|D\mathcal{R}(y)\| \leq C_{\mathcal{R}}. \quad (237)$$

Convex chart-neighborhood checkpoint. *The reporting step does not assume that K itself is convex. The line segments used by the integral mean-value formula are taken only in the fixed Euclidean coordinates of U_K , and the local-to-global tube is shrunk, if needed, so that the reported numerical and exact reduced-chart points and all segments between them stay inside U_K . Thus the reporting-map constant is a compact-neighborhood derivative bound chosen before the reported grid, output formatting, or fitted errors are inspected; it is not an atlas switch, not a remote-chart norm equivalence, and not a fitted norm-equivalence constant. Reported-component selection rule: the codomain and component ordering of $\mathcal{R}(y) = (q(y), v(y))$, together*

with the norm weights used for the reported position–velocity estimate, are fixed before the theorem instance and before any numerical table is read. The map does not include accelerations, multipliers, reconstructed reactions, constraint residuals, or stage residual monitors. Adding any of those quantities to the reported output would define a different map $\mathcal{R}_{\text{ext}}$ and would require its own compact-tube derivative bound, admissible chart neighborhood, and local-to-global input estimate; it cannot be obtained by appending columns to the numerical result table after Theorem 1 has been invoked. ***SO(3) reporting convention.*** When a component of q or v contains orientation or angular velocity, the displayed reporting norm is the norm induced by the same local retraction/log chart and component weights fixed above. The estimate is therefore a local bound on the chart-reported rotational coordinates (or on the smooth rotational output already included in $\mathcal{R}$), not an atlas-wide $SO(3)$ geodesic distance, not a statement through a chart cut or antipodal parametrization, and not invariant under a post-hoc change of orientation metric. Replacing the reported orientation metric, switching charts, or comparing a remote attitude representation would require a new smooth output map on the same compact tube and a new derivative constant fixed before the theorem instance. If two reported-grid reduced-chart sequences y_n and $\bar{y}_n$ lie in K and satisfy the reduced-chart input estimate in Eq. (238),

$$\max_{0 \leq n \leq N_h} \|y_n - \bar{y}_n\| \leq C_{\text{red}} h^6, \quad (238)$$

then, on the same reported output indices $0 \leq n \leq N_h$, Eq. (239) follows:

$$\max_{0 \leq n \leq N_h} \|\mathcal{R}(y_n) - \mathcal{R}(\bar{y}_n)\| \leq C_{qv} h^6, \quad C_{qv} = C_{\mathcal{R}} C_{\text{red}}. \quad (239)$$

This is a local chart-to-output norm consequence on the accepted compact tube. It is not a residual-to-error transfer theorem, not a global atlas equivalence, not a remote-branch statement, and not a terminal-time or dense-output estimate outside the reported grid.

Proof. For each reported index n , the segment $\bar{y}_n + \theta(y_n - \bar{y}_n)$, $0 \leq \theta \leq 1$, lies in the fixed convex chart neighborhood U_K by the checkpoint above. The integral mean-value formula in this one fixed Euclidean chart gives Eq. (240),

$$\mathcal{R}(y_n) - \mathcal{R}(\bar{y}_n) = \int_0^1 D\mathcal{R}(\bar{y}_n + \theta(y_n - \bar{y}_n))(y_n - \bar{y}_n) d\theta. \quad (240)$$

Taking norms and the compact-tube derivative supremum yields Eq. (241),

$$\|\mathcal{R}(y_n) - \mathcal{R}(\bar{y}_n)\| \leq C_{\mathcal{R}}\|y_n - \bar{y}_n\|. \quad (241)$$

Maximizing over exactly the same reported indices and substituting the reduced-chart grid estimate gives the displayed $C_{qv}h^6$ bound. The argument uses only the fixed local chart, fixed convex coordinate neighborhood U_K , compact derivative bound, and reported-grid reduced-chart estimate; it does not use residual/reaction tables, source-policy rows, fitted constants, output interpolation, or a chart switch. $\square$

Theorem 1 is therefore a conditional local-defect-to-global-error theorem: sixth order follows from the stated FullVA lift, implemented residual-defect, endpoint-closure, compact-tube stability, and inexact-Newton contracts. The active direct stage-residual proof route, and only that route, is closed by the 96-row non-dynamic certificate plus the D5 direct-substitution certificate for the 36 Newton–Euler rows. The AD-expanded certificate provides implementation-path and derivative-cell closure only for those implemented direct-route identities. This AD-expanded certificate is not the primitive symbolic defect certificate; the primitive dynamic symbolic oracle remains open outside the active direct route. AD-expanded implementation-path boundary: the closed certificate supplies AD-expanded implementation-path/derivative-cell closure for the active direct residual route only; it does not close the primitive dynamic symbolic oracle, source-paper residual replacement, or source-policy package readiness. What remains conditional is not the row-local substitution itself, but the regularity tube, the theorem-level scaled nonlinear-solver hypothesis, and any separate residual-to-error promotion beyond the accepted smooth branch. The numerical h-sweeps below support the scale of the claim, and the solver diagnostics are deliberately finite: a one-step scaled-tolerance probe, a short-trajectory scaled-tolerance probe, and a finite tolerance-regime sweep exercise reported-grid residual maxima on smooth windows. These finite records are the probes referenced by the P6 evidence boundary. Those probes are finite solver-policy diagnostics only: they are not proof inputs, they are not a theorem-level tolerance sweep over every reported trajectory, and they are not a substitute for the explicit $\eta_h^{\text{tube}} \leq c_\eta h^7$ theorem condition. Current certificates prove only the direct PC2 residual bridge under the retained P6 interface; they do not discharge P6 as a theorem-level solver-policy result, and they leave P7 as a separate residual-to-error transfer problem. Tables 23–25 record the row-level

implementation inputs and the remaining solver boundary. For validation scope, no residual-to-trajectory-error transfer theorem is accepted in this manuscript. Residual tables and residual-to-error ratios are diagnostics only. The residual-to-error route remains separate: residual rows for four-link and slider-crank stay mechanism-coverage evidence, not accepted dynamic order rows, until the seven residual-to-error obligations are discharged. Table 30 lists those obligations separately so that the transfer requirements are explicit.

Table 30: Residual-to-error transfer requirements for mechanism residual rows.

Item	Required before residual promotion	Condition state	Manuscript consequence
P7-1	Trajectory-level residual identity relating the fixed residual operator to closed-loop trajectory error on the same reported branch; this is not the stage identity at Z_G	Open	Residual tables do not by themselves define a same-branch trajectory-error residual identity.
P7-2	Residual-consistency rate $O(h^7)$ on that branch, with frozen row scaling	Open	Observed residual magnitudes are not a uniform residual-consistency theorem in a fixed proof norm.
P7-3	Closed-loop DAE stability or inf-sup control for that fixed operator	Open	No uniform stability or inf-sup inverse is accepted for the residual-only rows.
P7-4	Calibrated non-floor-limited residual-to-error estimator in the reported trajectory norm	Open	No estimator is calibrated away from the reference floor in the promoted trajectory norm.
P7-5	Reference-floor exclusion	Open	Floor diagnostics do not promote residual rows to dynamic-order rows.
P7-6	Coarse-first accepted dynamic-order campaign	Open	Four-link, slider-crank, and source-policy residual rows are not accepted dynamic-order campaigns.
P7-7	Manuscript transfer theorem/proof	Open	No theorem in this manuscript transfers those residual bounds to trajectory error.

Residual-to-error boundary: all seven P7 dependency obligations remain open. Residual tables and residual-to-error ratios are diagnostics only; no mechanism-coverage residual row, closed-loop reaction table, common-reference row, or source-policy row is promoted to trajectory-error/order evidence until dynamic residual identity, $O(h^7)$ residual consistency, stability or inf-sup control, calibrated estimator, reference-floor exclusion, coarse-first campaign, and manuscript transfer theorem close together. The closed-loop reaction table remains diagnostic in this P7 sense. The seven items are conjunctive: P7-1–P7-3 are the mathematical

transfer operator requirements, P7-4–P7-6 are the estimator and campaign requirements, and P7-7 is the manuscript theorem requirement. Closing only one group would not promote a residual row. Until all seven are closed together, residual magnitudes remain diagnostics and cannot replace the accepted smooth-branch order proof above. Equivalently, a residual-to-error implication would have to prove a bound of the form in Eq. (242),

$$\|e_h\|_{\text{traj}} \leq C_{\text{P7}} \left(\|\mathcal{R}_h^{\text{mech}}\|_{\text{res}} + \rho_{\text{res}}(h) + \varepsilon_{\text{floor}}(h) + \varepsilon_{\text{est}}(h) \right) \quad (242)$$

with a uniform stability or inf-sup constant on the reported branch, an a priori residual-consistency rate $\rho_{\text{res}}(h)$, an estimator calibrated away from the reference floor, and the same coarse-first dynamic-order campaign. The present residual rows do not supply that implication: small residual norms alone are not a trajectory-error theorem, and residual rows remain diagnostics even when their measured norms are small. The reason is logical rather than empirical: an $O(h^7)$ residual trend, even on the reported branch, is not accepted as an $O(h^6)$ trajectory-error statement unless an h -independent transfer constant, a same-branch residual identity, a reference-floor exclusion, and the manuscript transfer proof are all present. Such a transfer theorem would be a separate P7 boundary theorem; it is not hidden inside the local stage-residual defect, the Newton–Euler row certificate, or the common-reference tables. P7 fixed-operator requirement: before any residual table could be promoted, the proof would have to freeze the reported branch, chart, residual row scaling, trajectory norm, and same-branch residual transfer operator. A fixed linearized inverse must be declared before the residual data are read. In a standard route this would be such an inverse or inf-sup estimate such as Eq. (243),

$$\|\delta z\|_{\text{traj}} \leq C_{\text{P7}} \|D\mathcal{R}_h^{\text{mech}}(z_h^*)\delta z\|_{\text{res}} \quad (243)$$

with C_{P7} independent of h , followed by a nonlinear remainder estimate plus reference-floor and estimator closure on the same branch. The constant is fixed before observed residual data are read; it is not obtained by dividing measured trajectory errors by measured residuals. Without this fixed inverse, residual-size convergence can be a consistency diagnostic but not an order theorem. This is the fixed transfer operator requirement for P7. **No empirical ratio calibration.** Residual-to-error ratios printed for four-link, slider-crank, or source-policy diagnostics are posterior measurements, not transfer constants. They use the same finite reference calculation and reported output norm that a transfer theorem would have to justify, so they

cannot define C_{P7} , choose a mechanism-specific residual weight, remove a reference floor, or select a different trajectory norm after the run. A valid P7 route would need one predeclared operator/norm pair and one h -independent stability or inf-sup constant on the accepted branch before any residual table is read; otherwise the ratio is only a diagnostic consistency number.

P7 residual-to-error operator boundary: the separate P7 theorem is not a scalar residual-size check. It would have to define, on the same reported closed-loop branch, a mechanism residual operator $\mathcal{R}_h^{\text{mech}}$, a trajectory norm $\|e_h\|_{\text{traj}}$ matching the reported dynamic-order rows, and a uniform stability or inf-sup inverse before reference-floor or estimator terms are added. The present four-link and slider-crank rows check constraint closure, reaction consistency, and local residual scale only. They do not identify a same branch residual operator with a uniform inverse, and they do not prove that small reaction or constraint residuals control the trajectory error. P7 fixed-instance rule: any residual-to-error theorem used for promotion must predeclare, before reading the residual table, the residual operator, residual norm, reported trajectory components, trajectory norm, branch selector, reference policy, floor-exclusion rule, estimator form, and stability or inf-sup constant family. Changing any one of these after the four-link, slider-crank, or source-policy residual rows are inspected defines a different diagnostic row, not a promotion of the existing row. In particular, a constraint residual norm, reaction residual norm, common-reference trajectory error, and source-policy timing row cannot be combined after the fact into a single P7 theorem unless a same-branch operator and one h -independent transfer constant bind the entire tuple.

Lemma 46 (Same-codomain obstruction for residual-to-error promotion). *The obstruction is not merely that the present residual tables are finite. They live in a different proof codomain from the trajectory-error conclusion. Let $\mathcal{R}_h^{\text{mech}} : \mathcal{X}_h \rightarrow \mathcal{Y}_h^{\text{res}}$ denote a candidate mechanism residual/reaction diagnostic on a closed-loop branch, and let $E_h : \mathcal{X}_h \rightarrow \mathcal{Y}_h^{\text{traj}}$ denote the reported position/velocity error map used by the dynamic-order claim. A proof of P7 would have to fix, before the data are read, the branch chart, row weights W_h , residual norm, trajectory norm, and reference/floor term ρ_h , and then prove a uniform transfer estimate such as Eq. (244),*

$$\|E_h z - E_h z_h^*\|_{\text{traj}} \leq C_{P7} \|W_h(\mathcal{R}_h^{\text{mech}}(z) - \mathcal{R}_h^{\text{mech}}(z_h^*))\|_{\text{res}} + C_{P7} \rho_h, \quad (244)$$

for all z in the same proof tube, with C_{P7} independent of h and with ρ_h bounded in the same asymptotic scale required by the promoted trajectory

claim. Without this same-codomain transfer, a small constraint residual, a small reaction residual, or a small posterior residual-to-error ratio is an element of $\mathcal{Y}_h^{\text{res}}$, not a bound in $\mathcal{Y}_h^{\text{traj}}$. The four-link, slider-crank, and source-policy residual rows therefore remain mechanism-coverage diagnostics and cannot be substituted into Theorem 1.

Proof. Equation (244) is the missing bridge from the residual diagnostic codomain to the trajectory-error codomain. Without a predeclared same-branch operator, row weighting, trajectory norm, and h -independent constant C_{P7} , the norm of an element in $\mathcal{Y}_h^{\text{res}}$ supplies no inequality in $\mathcal{Y}_h^{\text{traj}}$. Posterior ratios computed from the same finite reference data cannot define this operator or constant, because doing so would choose the transfer after the residual table has already been inspected. Thus residual-size diagnostics can corroborate mechanism coverage, but they cannot enter the local-defect, Gronwall, or reporting-map chain of Theorem 1 as trajectory-error/order evidence. $\square$

Kernel and observability obstruction. The required P7 estimate is also an observability statement, not only a residual-rate statement. Even a preasymptotic $O(h^7)$ trend in $\|\mathcal{R}_h^{\text{mech}}\|$ would not control the reported trajectory components if the linearized residual has an h -uniform near-null direction whose image under E_h is nonzero. A valid P7 proof must therefore fix the branch chart, gauge or quotient for algebraic variables, row weights, and trajectory projection, and then prove that every trajectory-visible perturbation is controlled by the residual operator up to the declared floor/estimator term. Reaction residuals, constraint closure, or posterior residual-to-error ratios do not by themselves exclude such trajectory-visible kernel directions. This observability or inf-sup estimate is exactly the open stability component of P7-3, and it must be fixed before the residual table is read.

The proof uses two deliberately separate implementation routes. The active route is the direct 132-row residual-defect route: the 96 non-dynamic rows have an $O(h^7)$ defect by the kinematic/lower-pair certificate, and the 36 Newton–Euler rows vanish under the D5 direct-substitution certificate on the lifted Gauss stage. This is the route used in Theorem 1. The route-level use is anti-circular in the following narrow sense: the D5 certificate is evaluated at the lifted Gauss stage Z_G and at the implemented residual rows before the local perturbation and local-to-global transfer steps, so it does not use the computed trajectory error or the conclusion of Theorem 1.

Table 31: Direct-route anti-circularity table for Theorem 1.

Proof object	Fixed before	May depend on	Explicit non-dependence
Lifted Gauss stage Z_G	D5 evaluation and local perturbation.	Retained smooth-lift and compact-tube hypotheses.	Inexact Newton iterates, computed trajectory error, or the theorem conclusion.
Implemented residual rows	D5 row substitution and PC2 consumption.	The implemented 132-row formula map, row labels, and runtime AD binding.	Source-paper residual replacement, residual/reaction diagnostics, or source-policy rows.
D5 direct-substitution certificate	Stage-root perturbation, local-defect summation, and local-to-global transfer.	Row-local Newton–Euler identities and the direct Z_G substitution.	Discrete Gronwall, reporting-map consequences, observed h -sweeps, or P7 residual-to-error promotion.
Local perturbation/local defect	The theorem’s grid-point conclusion.	PC1–PC2 plus the retained compact-tube, endpoint-closure, and inexact-Newton interfaces, with retained P6 and PC4 as the solver-scale interface and residual-to-error exclusion.	Separate source-policy/full-TFE readiness or reproducibility package readiness evidence.
Theorem conclusion	No antecedent input; it is the downstream output.	The preceding local-defect, stability, local-to-global, and reporting-map steps.	Feedback into Z_G , D5 identities, P5, or the primitive symbolic route.

Direct-route anti-circularity rule: the direct route is admissible because D5 direct-substitution identities are fixed before local perturbation, local-to-global transfer, and the theorem conclusion; Z_G and the implemented residual rows are fixed at the same pre-perturbation level. The theorem conclusion does not justify row identities; finite h -sweeps, residual/reaction tables, source-policy rows, and the primitive symbolic route are not sources for D5. This table does not prove P1/P2/P6/P7, the dynamic symbolic oracle, the primitive/Taylor route, source-policy/full-TFE readiness, or reproducibility package readiness evidence. The nonuse rule is one-way: row identities feed the theorem, while the theorem’s downstream order estimate cannot feed back into row identity, implementation binding, or D5 closure.

The primitive symbolic Taylor route is retained only as a conditional decomposition of the same dynamic rows into state-, acceleration-, multiplier-, geometry-, and gyroscopic-lift obligations. Because those primitive lift obligations remain separate assumptions, the primitive route is not used to close the theorem. This separation prevents the open primitive symbolic route from weakening the accepted direct proof, and also prevents the direct proof from being misreported as a complete source-paper TFE residual replacement.

The primitive-route status table is one-closed/five-open: P_{tube} is closed, while P_{state} , P_{acc} , P_{λ} , P_{geom} , and P_{gyro} remain open. The D5 term inventory therefore has no certified induced primitive-route Taylor subterm bounds in the active theorem route. Consequently, the theorem does not import a primitive/Taylor bound as PC2 evidence; it consumes the direct 132-row route and carries the compact-tube solver condition separately.

Implementation-route/certificate separation: the accepted proof consumes the implemented 132-row direct route, runtime formula/AD binding, and implementation-path formula certificate only for the stage-residual argument discharged by the direct route. The open primitive symbolic route and the non-active primitive symbolic-oracle completion record remain diagnostic records for separate proof extensions; they are not inputs to the theorem’s stage-residual condition, not contradictions of the direct D5 substitution closure, not source-paper residual replacement, not source-policy/full-TFE readiness, and not reproducibility package readiness evidence. The direct D5 substitution closure is consumed only through this implemented 132-row direct-route proof path.

The proof dependencies used by the theorem are: the PC2 stage-residual condition consumes the 96-row non-dynamic certificate and the 36-row D5

Table 32: Implementation-route/certificate separation table for Theorem 1.

Route or certificate	Theorem role	Closed evidence	Explicit non-use
Active direct residual-bridge route	The stage-residual input used by Theorem 1.	The 96-row $O(h^7)$ non-dynamic certificate plus the 36 Newton–Euler direct-substitution zero rows give the implemented 132-row route.	Primitive/Taylor subterm closure or source-paper residual replacement.
Runtime formula/AD binding	P3 identifies the implemented residual/Jacobian path being solved and differentiated.	Full 132-row formula map and finite AD checks bind the implementation path.	A global source-code equivalence proof or theorem-level solver-policy theorem.
AD-expanded certificate	Closes the implementation-path/derivative-cell identities needed for the active direct proof boundary.	The AD-expanded closure covers the Newton–Euler derivative cells for the implemented direct route.	Primitive symbolic defect certificate or dynamic symbolic oracle completion.
Primitive symbolic route	Diagnostic decomposition of the same 36 dynamic rows into primitive lift obligations.	One primitive obligation is closed and five remain open; no induced primitive-route Taylor subterm bounds are invoked by the theorem.	Not an input to the PC2 stage-residual condition and not a contradiction of the active direct-substitution route.
Source-paper residual replacement	No theorem input.	Kept outside the accepted proof and outside current source-policy readiness.	Full-TFE replacement, source-policy reproduction, or reproducibility package readiness claims.

direct-substitution certificate; P3 consumes only runtime formula/AD binding; P6 consumes the compact-tube branch-solver hypothesis, while finite solver probes attach only to P6 and do not prove the theorem-level solver condition; and P7 records the residual-to-error boundary, while residual tables are recorded only as diagnostics for the open P7 residual-to-error boundary and do not prove a separate residual-to-error transfer theorem. The primitive symbolic route is a separate primitive-route record and is not an input to the PC2 stage-residual condition. The PC2 discharge therefore does not prove P6 or P7; proving either statement would require independent proof. Likewise, an open primitive symbolic route cannot invalidate the accepted direct-substitution closure. For the theorem dependency boundary, finite solver probes attach only to P6. Equivalently, the primitive symbolic route is not an input to the PC2 stage-residual condition.

Scope of theorem. The mathematical claim of Theorem 1 is a conditional order-six theorem under the retained P1–P3 theorem interfaces, the retained P4 binding side, and the P6 solver-scale interface; P4’s 96-row certificate and P5’s direct Newton–Euler route supply the proved residual rows, while P7 is only the open residual-to-error boundary. Equivalently, it does not claim an unconditional theorem without theorem-domain interfaces, a closed $\eta_h^{\text{tube}} \leq c_\eta h^7$ solver-policy condition, fixed-tolerance asymptotic proof, an accepted residual-to-error transfer theorem for mechanism rows, or source-policy/full-TFE package readiness. Any paper-facing shorthand about direct residual-bridge discharge is read only in this direct-PC2 sense: it refers to the same-branch 132-row residual-value bridge, not to the primitive/Taylor lane, the retained P6 solver-policy theorem, the open P7 residual-to-error boundary, source-policy readiness, or a full-TFE replacement.

Auxiliary-evidence scope: theorem inputs are restricted to the retained P1/P2/P3 conditions, the retained P4 binding interface, the P6 solver-scale condition, and the P5 direct Newton–Euler route. Finite solver probes, residual/reaction tables, source-policy rows, and local reproducibility record evidence are diagnostic or reproducibility-boundary evidence only; they are not promoted to theorem-input status, residual-to-error transfer, a theorem-level η_h solver-policy result, or source-policy/full-TFE readiness. The compact theorem-input tuple is exactly P1/P2/P3 conditions, the retained P4 binding interface, the P6 solver-scale condition, and the P5 direct route; all other records in this table remain auxiliary evidence.

Quantifier/domain rule: the theorem quantifiers are ordered as fixed proof data, then h_0 and constants, then all $0 < h \leq h_0$, all retained-tube states y ,

Table 33: Scope of auxiliary evidence for Theorem 1.

Evidence class	Permitted theorem use	Explicit non-use
Accepted direct stage-row route	The 96-row non-dynamic certificate and the 36-row D5 direct-substitution certificate supply the stage-residual input required by PC2 on Z_G .	This is not a primitive symbolic certificate, source-paper residual replacement, or full-TFE stage replacement.
Finite solver probes	They diagnose branch-scale consistency for the retained P6 solver interface.	They do not prove a theorem-level η_h solver-policy theorem and do not turn fixed tolerances into an asymptotic proof.
Residual and reaction tables	They support mechanism coverage and the open P7 residual-to-error diagnostic route.	They do not promote four-link, slider-crank, or source-policy residual rows to trajectory-error/order evidence.
Source-policy and local reproducibility records	They record reproducibility provenance for the narrowed claim and current reproducibility record state.	They do not prove full source-policy reproduction, full source-policy runner package readiness, or external-superiority claims.

and any reported grid $N_h h \leq T$ on the accepted branch. They do not quantify over arbitrary Newton roots, remote charts, residual-only mechanism rows, source-policy rows, unreported terminal times, or local reproducibility record states. The table records the theorem domain; it does not prove a theorem-level solver-policy result, a P7 residual-to-error boundary, source-policy readiness, or full-TFE readiness. The theorem constants as suprema rule is fixed at this point: all theorem constants are compact-tube suprema chosen before any reported branch, observed slope, residual log, or work-precision table is read.

Reader-facing theorem summary. Before the admissibility clauses below, the mathematical content is a standard local-to-global implication. On one fixed compact branch, the implemented 132-row FullVA residual receives a same-branch direct $96 + 36$ residual certificate at the lifted Gauss stage Z_G ; the endpoint closure and inexact Newton solve contribute controlled $O(h^7)$ perturbations; and accepted repeated steps on the same reported grid therefore satisfy the local-defect estimate in Eq. (254), the reduced-grid estimate in Eq. (255), and the reported position-velocity estimate in Eq. (256). The remaining clauses in this subsection fix the branch, norm, row scaling, solver envelope, and non-claims needed to make that implication unambiguous; they are not additional evidence sources and they do not add a second theorem.

Definition 7 (Accepted branch-selected one-step map). *The accepted branch-selected one-step map Ψ_h^{G6FVA} is defined on the accepted subset*

Table 34: Quantifier/domain table for Theorem 1.

Quantifier layer	What is fixed or quantified	Valid domain	Explicit exclusions
Fixed proof data	Fix T , the accepted compact proof tube, the reduced chart, the Gauss-predictor branch rule, the retained P1–P3 contracts, the P4 binding contract, the P6 solver-scale contract, the P5 direct-route certificate, and the P7 output boundary; then choose h_0 and the theorem constants.	The smooth accepted branch specified by Assumption 1.	Finite h -sweep fits, source-policy rows, residual-only mechanism rows, and local reproducibility record state.
One-step estimate	For every $0 < h \leq h_0$ and every y in the retained compact tube satisfying the branch-solver hypothesis.	The accepted branch-selected Gauss6/FullVA one-step map initialized from the Gauss predictor.	Arbitrary Newton initializations, remote nonlinear roots, remote charts, and nonsmooth contact events.
Reported-grid domain qualification	For any reported N_h with $N_h h \leq T$, $t_n = nh$, and transition steps $0 \leq n < N_h$ that satisfy the predeclared accepted theorem-domain conditions in $\mathcal{D}_h$ before the estimate is invoked. Manuscript order rows instantiate the exact-final-time case $N_h h = T$ when reported; the $N_h h < T$ case is only a prefix-grid estimate.	The grid maximum over output indices $0 \leq n \leq N_h$.	Any error claim on the leftover interval $T - N_h h$, dense-output terminal value at T , or extra terminal solve outside the reported transitions. Exact-final-time grids are included as the $N_h h = T$ case but are not required by the theorem.
Solver residual scale	The theorem uses $\eta_h^{\text{tube}} \leq c_\eta h^7$ uniformly on the compact tube; the reported $\eta_h^{\text{reported}} = \max_{0 \leq n < N_h} \eta_{h,n}$ is only the trajectory specialization.	Branch-selected Newton residuals on the same accepted compact-tube branch.	Fixed production tolerances or finite solver probes as asymptotic proof.

of the retained compact proof tube. Its theorem domain is the subset $\mathcal{D}_h$ of retained-tube states for which all branch, local-inverse, endpoint-ball, tube-retention, and scaled solver-residual conditions below hold with the fixed proof norm and constants. More precisely, $\mathcal{D}_h$ is the state projection of a predeclared admissible graph: each accepted input state is paired with the selected Gauss-predictor branch, local stage root, endpoint-output data, proof norm, and solver-residual certificate for that transition. Thus $y \in \mathcal{D}_h$ means that this same input state and its selected transition data satisfy the branch, endpoint, and P6 predicates; P6 is not inferred from state membership alone. For $y \in \mathcal{D}_h$, the map is obtained by: (i) initializing the stage solve from the Gauss predictor in the fixed local chart; (ii) selecting the local root of the implemented 132-row residual in the same chart, with fixed row ordering, finite row scaling, and proof norm, and binding any computed iterate to that local branch; (iii) applying the endpoint-output map $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$, where $\mathcal{E}_h$ is the Lie endpoint reconstruction and $\mathcal{C}_h$ is the accepted local endpoint-closure map; and (iv) accepting the transition only when the computed Newton iterate on that branch has residual satisfying the compact-tube solver hypothesis. The rule is: map is undefined for theorem purposes whenever any branch, tube, or solver-scale condition fails; the theorem makes no claim about that step. A reported trajectory is repeated application only over defined transitions $0 \leq n < N_h$; equivalently, a grid invocation requires $y_n \in \mathcal{D}_h$ for every transition index $0 \leq n < N_h$. The exact comparison states $y(t_n) = \varphi_{t_n}(y_0)$ at which the local defect is evaluated must also be admissible as exact-state inputs for that local-defect comparison. We denote this same fixed input-domain condition by $y(t_n) \in \mathcal{D}_h$ below; it is a statement about the branch-selected map evaluated at the exact state, not a post-run reacceptance test. The compact-tube part is supplied by the same sub-tube/first-exit bootstrap, while the branch and solver-scale parts remain retained theorem conditions for the one-step map evaluated at those inputs; otherwise the term $\Psi_h^{\text{G6FVA}}(y(t_n)) - \varphi_h(y(t_n))$ is not a theorem object. The compact-tube membership is supplied by the first-exit bootstrap after h_0 is fixed, not by an a posteriori deletion of failed steps; the scaled residual part is not supplied by first exit and remains the retained P6 hypothesis. η_h^{reported} is the maximum over those accepted transitions, not an extra terminal solve or remote nonlinear root.

First-exit/P6 separation: Lemma 44 proves compact-tube retention for the accepted numerical and exact comparison states. It does not prove the

scaled residual part of $\mathcal{D}_h$. The P6 condition $\eta_h^{\text{tube}} \leq c_\eta h^7$ is a separate branch-solver hypothesis for each accepted transition; if that condition fails, the transition is outside the theorem instance rather than a lower-order case covered by the same constants.

Common admissible-window checkpoint. The theorem uses one common admissible small-step window, not a collection of per-lemma or per-run windows. After the compact tube, chart, row ordering, row scaling, proof norm, Gauss-predictor branch rule, endpoint-anchor family and right-inverse constants, stage inverse constants, strong residual-inverse constants, stability constant, and c_η have been fixed, define the common window in Eq. (245),

$$h_\star = \min\{h_G, h_{\text{res}}, h_K, h_E, h_N, h_{\text{stab}}, h_{\text{rep}}, \bar{h}\}, \quad (245)$$

where the terms are respectively the windows for the reduced Gauss local defect, the 132-row residual bridge, the Kantorovich stage-root ball, the endpoint-closure ball, the inexact-Newton branch ball, the stability/tube bootstrap, the reporting-map chart neighborhood, and the preselected compact-tube residual-map window. The theorem then chooses $h_0 \leq h_\star$, with the additional explicit Kantorovich restrictions in Eq. (246),

$$2MC_R h_0^7 \leq r, \quad 2M^2 LC_R h_0^7 \leq \frac{1}{2}, \quad (246)$$

and the endpoint/inexact-Newton ball margins required in Lemmas 6 and 41. This is a single same-branch admissibility rule: if one component window fails, the transition is outside Definition 7; it is not covered by a different constant, a different Newton root, or a later diagnostic row. Shrinking h_0 after the proof constants are fixed is allowed, but selecting a new tube, inverse constant, endpoint-anchor family, residual norm, row scaling, or solver-envelope constant after reading a run would define a different theorem instance.

Solver-envelope admissibility criterion. For the fixed theorem data, define the solver-admissible transition set in Eq. (247),

$$\mathcal{S}_h(c_\eta) = \{n : y_n \in K, \tilde{Z}_{A,n} \in B_{r_A}(Z_{A,n}), \|F_{A,h}(\tilde{Z}_{A,n}; y_n)\| \leq c_\eta h^7\}. \quad (247)$$

The theorem is applied only on reported transition indices $\{0, \dots, N_h - 1\} \subset \mathcal{S}_h(c_\eta)$, with the norm, row scaling, branch rule, local ball, and c_η fixed before the run. This criterion is a sufficient trajectory-level verification condition for the residual-size part of P6 once branch-ball membership is also certified; it is

not a postprocessing filter. Shrinking h_0 can satisfy small-step requirements after the constants are fixed, but it cannot turn a fixed absolute tolerance, a residual ratio observed after the run, or a failed branch-ball membership into the compact-tube solver envelope required by Lemma 41.

Accepted-map definition boundary: Ψ_h^{G6FVA} is a branch-selected accepted-domain map tied to the Gauss predictor, the local accepted stage root of the implemented 132-row residual, the fixed proof norm, and the endpoint-output map $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$. It is not a global nonlinear-solver selection rule, not a remote-chart equivalence claim, not a residual-only mechanism-row error theorem, not a source-policy reproduction result, and not a terminal-time correction outside the accepted transition set. The phrase branch-selected accepted-domain map tied to the Gauss predictor means that the map, the local stage root, the endpoint-output map, and the scaled solver-residual condition are one same-branch theorem object.

Local-defect map-chain definition: on every defined accepted transition, the proof compares exactly one chain of maps on the same compact branch: φ_h , the exact reduced flow; Ψ_h^G , the reduced Gauss collocation endpoint map; $\Psi_h^A = \mathcal{E}_h(Z_A)$, the endpoint reconstructed from the exact local root of the implemented 132-row residual before endpoint closure; $\Psi_h^E = \mathcal{C}_h(\Psi_h^A)$, the endpoint-closed exact-stage reference for the inexact Newton perturbation; and $\Psi_h^{\text{G6FVA}} = \mathcal{P}_h(\tilde{Z}_A)$, the endpoint-closed inexact Newton map. All four local-defect terms are measured in the same reduced-chart endpoint norm after the fixed row ordering, finite row scaling, and proof norm have already defined the residual-to-stage perturbation input. The triangle inequality does not change branch, chart, norm, or transition index, and none of its terms is supplied by mechanism residual/reaction tables, source-policy rows, or a terminal-time correction outside the reported transition set.

Algorithm-order consistency: Algorithm 1 computes the inexact stage iterate $\tilde{Z}_A$ first, reconstructs its Lie endpoint by $\mathcal{E}_h$, and then applies the accepted local endpoint closure $\mathcal{C}_h$. The proof uses the same implemented output order in $\Psi_h^{\text{G6FVA}} = \mathcal{P}_h(\tilde{Z}_A)$, with $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$. The auxiliary map $\Psi_h^E = \mathcal{P}_h(Z_A)$ is only the exact-root reference obtained by applying that same endpoint-output map to Z_A ; it is not a reordered algorithm and it is not an endpoint closure performed before Newton. Thus the $C_N \eta_h^{\text{tube}}$ term compares two already endpoint-closed outputs, whereas the $C_E h^7$ term compares the

exact-stage unclosed and closed outputs. The local-defect sum therefore never mixes an unclosed inexact endpoint with a closed exact endpoint.

Non-vacuity boundary. Theorem 1 is an abstract conditional consistency theorem for the nonempty compact branch specified in Assumption 1 and Definition 7. Here nonempty means that the exact comparison states lie in the state projection $\mathcal{D}_h$ of the predeclared admissible graph, and their lifted Gauss-predictor stages and selected transition data lie in the associated local stage chart, compact tube, endpoint ball, proof norm, and P6 solver-envelope predicates for $0 < h \leq h_0$; it is a theorem-domain condition, not a set obtained by deleting failed reported steps after a run. The finite smooth-branch runs and endpoint/solver diagnostics are reported only as finite diagnostics for branch-model relevance on the accepted benchmark, not as theorem evidence for P6. They do not prove global branch selection, remote-root exclusion, or P6 solver-policy theorem. Thus the theorem is not made unconditional by the experiments, and the experiments are not used to define away the theorem domain after the fact.

Direct residual-bridge scope convention. In this proof, direct residual-bridge closure denotes only the residual-bridge/Kantorovich input used by Theorem 1: the 96 non-dynamic implementation rows, the direct D5 36-row Newton–Euler identity at the lifted Gauss stage, and the local perturbation route that consumes them. It does not prove a solver-policy theorem or a P7 residual-to-error boundary, does not complete the separate primitive/Taylor route, and does not establish multiplier/reaction output-order claims, source-policy reproduction, or full-TFE replacement. This is a mathematical scope convention for the direct route; it is not an additional theorem assumption and it does not enlarge the theorem conclusion. The exclusion is part of this theorem instance, not a post-hoc classification that can be changed after reading numerical tables or auxiliary records. Only the direct PC2 residual-value bridge is discharged here. This reader-facing discharge does not close the primitive/Taylor route, P6, P7, source-policy, multiplier/reaction, or full-TFE interfaces.

Theorem input-output reading rule. The theorem is read as one forward implication on a fixed compact branch. Let $\mathcal{I}_h(K)$ collect the retained admissibility interfaces: P1, P2, and P3 on the accepted compact tube; the separate P6 solver-scale interface; and the branch, row-ordering, row-scaling, and proof-norm binding needed to interpret P4. Let $\mathcal{B}_h^{96+36}(Z_G)$ denote the proved P4 96-row non-dynamic certificate plus the P5 direct 36-row identity at the lifted Gauss stage. In this notation P4 is not hidden inside the

retained interface as an assumed defect estimate: its row-local $O(h^7)$ certificate is a discharged input, but it is meaningful only after those admissibility interfaces have fixed the same branch and residual convention used by the theorem. Then the proof chain is Eq. (248):

$$\begin{aligned}
\mathcal{I}_h(K) + \mathcal{B}_h^{96+36}(Z_G) &\implies \|F_{A,h}(Z_G; y)\| \leq C_R h^7 \\
&\implies \|\Psi_h^{\text{G6FVA}}(y) - \varphi_h(y)\| \leq C_{\text{loc}} h^7 \\
&\implies \max_{0 \leq n \leq N_h} \|y_n - y(t_n)\| \leq C_{\text{red}} h^6, \\
&\quad \max_{0 \leq n \leq N_h} \|\mathcal{R}(y_n) - \mathcal{R}(y(t_n))\| \leq C_{qv} h^6.
\end{aligned} \tag{248}$$

No arrow in this chain is used backwards. In particular, observed grid orders, residual tables, finite solver logs, or a discharged direct residual bridge do not prove the retained compact-tube interfaces, turn the solver-scale condition into a separate solver theorem, prove a residual-to-error theorem for reported mechanism rows, or certify the separate primitive/Taylor route. The same instance discipline governs numerical reporting: after $\mathcal{I}_h(K)$, $\mathcal{B}_h^{96+36}(Z_G)$, the branch, proof norm, row scaling, residual envelope, c_η , and h_0 have been fixed, a reported run may only specialize that theorem instance. It cannot retroactively change the theorem branch, norm, row scaling, or residual envelope to make one of the displayed implications true.

Lemma 47 (P6/P7 non-closing arrow contract). *For the theorem instance fixed by the compact branch, residual convention, proof norm, endpoint-output map, and solver envelope, P6 and P7 are forward-only boundary clauses. The P6 clause is an input-domain clause: Eq. (249),*

$$\eta_h^{\text{tube}} \leq c_\eta h^7 \implies C_N \eta_h^{\text{tube}} \leq C_N c_\eta h^7, \tag{249}$$

so it only bounds the inexact-Newton summand after the compact-tube branch, proof norm, row scaling, and residual envelope have been fixed. The reverse arrow is not used: a small reported residual, a tolerance sweep, or an observed sixth-order slope does not prove the compact-tube P6 condition. The P7 clause is an output-exclusion clause: Eq. (250),

$$\begin{aligned}
\max_n \|y_n - y(t_n)\| = O(h^6) &\not\Rightarrow \{\text{residual-only mechanism rows are} \\
&\quad \text{trajectory-error/order evidence}\}.
\end{aligned} \tag{250}$$

Residual and reaction rows therefore remain mechanism-coverage diagnostics until a separate branch-compatible residual-to-error theorem supplies the fixed transfer operator and h -independent constant required in Table 30. These two non-closing arrows are part of the theorem boundary, not additional conclusions.

Proof. The P6 statement is exactly Lemma 41 applied after the accepted branch, residual norm, row scaling, endpoint-output map, branch ball, and residual envelope have already been fixed. That lemma maps the proof-norm residual bound to the inexact-Newton endpoint perturbation; it does not map a finite residual log, tolerance sweep, or observed grid slope back to the compact-tube branch membership, strong local inverse, or h^7 residual envelope required for P6. Reversing that arrow would change the theorem domain after inspecting diagnostics and would therefore be a different theorem instance.

The P7 statement is separate from the local-defect and Gronwall chain. The theorem output is a reduced-chart and reported position-velocity grid estimate for the accepted branch-selected map. A residual or reaction row is an element of a different codomain: it is a mechanism-coverage diagnostic unless a branch-compatible residual-to-error operator with an h -independent constant is proved on the same reported transition set. The seven entries in Table 30 are precisely the required data for such a transfer theorem. Hence P6 cannot be proved by finite solver diagnostics, and P7 cannot be proved by residual or reaction tables without adding the separate P7 boundary-theorem ingredients named above. $\square$

Pre-data theorem-instance discipline. Fix a compact tube K , a reduced chart, the Gauss-predictor branch selector, the row-scaled residual $F_{A,h}$, the proof residual norm, the endpoint-output map, the reporting map, constants c_η and h_0 , and the local inverse, endpoint, stability, and norm-equivalence constants before any solver log, residual table, fitted order, or source-policy row is read. For this fixed theorem instance, Theorem 1 may consume only statements attached to the same instance: the $96 + 36$ row residual bridge at Z_G , the compact-tube inverse and endpoint hypotheses, the branch-selected inexact-Newton envelope $\eta_h^{\text{tube}} \leq c_\eta h^7$, and the local-to-global stability estimate. A reported run can specialize the instance only by showing that its accepted transitions use the predeclared branch, norm, row scaling, and solver envelope. It cannot define these objects after the residual or error tables are inspected.

Argument. The proof of Theorem 1 is a composition of maps with a fixed domain and codomain. The stage-root perturbation uses $D_Z F_{A,h}$ and the residual norm fixed on the compact tube; the endpoint closure uses the predeclared endpoint-output map; the inexact-Newton term uses the same residual norm through Lemma 41; and the Gronwall step uses the fixed reduced-chart stability family. The constants entering the local defect are the pre-data tuple in Eq. (251),

$$C_G, \quad C_A, \quad C_E, \quad C_N c_\eta, \quad (251)$$

are therefore chosen before the reported trajectory is evaluated. If a later diagnostic changes the branch selector, residual row weights, residual norm, trajectory norm, reference policy, floor-exclusion rule, or c_η , the constants above and hence the theorem instance have changed. Such a changed diagnostic can only motivate a separate theorem instance; it is not a proof of the already stated instance.

This pre-data order is exactly what separates the theorem-grade hypotheses from finite diagnostics. Solver logs may verify the recorded residuals for a predeclared P6 instance; they do not create the compact-tube inverse or the h^7 envelope. Residual and reaction tables may verify mechanism coverage for a predeclared residual vocabulary; they do not create the fixed P7 boundary operator or an h -independent residual-to-error constant. Thus no diagnostic row can be moved upstream into the theorem proof unless it is first reformulated as a predeclared theorem assumption or as an independent lemma on the same fixed object.

Lemma 48 (Same-object theorem-composition criterion). *Every lemma used in Theorem 1 is attached to the same mathematical object: the accepted Gauss-predictor branch, the row-scaled 132-row residual $F_{A,h}$, the fixed proof norm, the endpoint-output map $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$, and the compact-tube solver envelope η_h^{tube} . The P4 non-dynamic certificate, the P5 Newton–Euler identity, Lemma 39, Lemma 6, and Lemma 41 are therefore composable only as a jointly admissible tuple whose branch, row convention, proof norm, endpoint map, and solver envelope agree before the constants are chosen. A certificate for a different residual convention, a solver log in a different norm, or a residual/reaction row measured after a different endpoint operation is not an input to the theorem instance. Auxiliary proof records, including the separate D5 primitive/Taylor inventory and the P7 residual-to-error table, may record scope boundaries, diagnostics, or separate-route material only after this tuple*

has been fixed: the separate D5 primitive/Taylor inventory describes a separate sufficient proof route, whereas P7 and residual/reaction tables remain diagnostic material outside the theorem estimate until their own transfer theorem closes. None of these records can choose the tuple, supply an additional constant, or change the codomain of a lemma already used in the composition.

Proof. The proof of Theorem 1 uses the full-residual bridge to supply a defect value for the same map $F_{A,h}$ that appears in Lemma 39; that perturbation lemma then selects a local stage root on the same Gauss-predictor branch and passes that stage root through the same endpoint-output map used by Lemma 6. Lemma 41 perturbs this accepted root only in the fixed proof norm and only under the same compact-tube solver envelope. Hence the branch, row convention, norm, and endpoint map agree before the constants are chosen and before the estimates are composed.

Equivalently, the reference-style C1–C4 slots are admissible only when their outputs live in this same theorem object. The C1 local-error slot may feed the C3 perturbation slot only as the bridged residual value for the declared $F_{A,h}$; the C2 constrained-algebraic slot may feed C3 only as endpoint, right-inverse, and lower-pair data for the same $\mathcal{P}_h$ and compact tube; and C3 may feed C4 only as a one-step map in the same reduced-chart norm used by the stability recurrence. A proof fragment whose natural object is a BLieDF multistep update, a primitive 162-subterm inventory, a residual/reaction diagnostic, or a source-policy output row is therefore a different typed object. It requires an explicit equivalence or transport lemma before it can enter the theorem composition. Without that lemma, the BLieDF fragment is reference guidance, the primitive inventory is separate-route material, and residual or source-policy rows are outside the theorem inputs.

If a P4 row certificate, P5 dynamic identity, endpoint-closure estimate, or solver-residual bound were proved on mutually different stage branches, residual conventions, proof norms, or endpoint maps, the codomain and constant of one estimate would not be the domain data of the next estimate. Such a collection cannot be spliced into a single theorem instance and could not be substituted for the tuple consumed by the theorem without adding a separate equivalence or transfer lemma. This is the present FullVA analogue of the fixed-discrete-object discipline in the Wieloch–Arnold constrained-BDF proof: the reference motivates the order in which objects are fixed and estimates are consumed, while the constants and estimates used here are exactly those of the implemented Gauss6/FullVA residual. $\square$

Theorem reading guide. The theorem below has four layers. First, P1, P2, P3, and P6 fix the domain of the branch-selected statement: smooth lift, compact tube, endpoint map, implementation path, proof norm, stability, and solver residual envelope. Second, P4 and P5 supply the proved residual data on that fixed object: the 96 non-dynamic FullVA rows contribute $O(h^7)$, and the 36 Newton–Euler rows vanish at the lifted Gauss stage Z_G . Third, the stage-root, endpoint, inexact-Newton, and local-to-global lemmas turn that single 132-row residual bridge into an $O(h^7)$ local defect and an $O(h^6)$ same-initial-state reported-grid bound. Fourth, the reporting map converts that bound only to the displayed reduced-chart and position–velocity output; residual, reaction, multiplier, constraint-output, source-policy, and full-TFE replacement claims are not theorem outputs. The separate primitive/Taylor inventory, residual-to-error promotion records, and external-comparison/package records are outside these four layers. The primitive/Taylor propositions in this section are therefore conditional auxiliary implications: they describe what a primitive-route proof for that route must prove, but they cannot be read backward as evidence for P4, P5, P6, the residual bridge, or the output scope consumed by this theorem. In particular, the theorem’s Taylor object is the fixed full residual map $F_{A,h}$ and the increment $Z - Z_G$; the primitive 162-term inventory is a stricter alternative factorization of the dynamic block, not a hidden premise of the direct $96 + 36$ bridge. Equivalently, Theorem 1 consumes T1 and T2 only; T3 is neither a premise nor a consequence of the theorem. The Wieloch–Arnold constrained-BDF proof is invoked as proof-order discipline only for this reading guide: no BLieDF local-error, multiplier, or recursion estimate from that paper is a premise of the theorem below.

Load-bearing proof spine. To separate the theorem’s mathematical core from its boundary guards, the proof below uses only the following five forward steps. S1 applies the classical three-stage Gauss local defect in the reduced FullVA chart. S2 inserts the lifted Gauss stage into the fixed 132-row implemented residual and uses the direct $96 + 36$ bridge to obtain $\|F_{A,h}(Z_G; y)\| \leq C_R h^7$. In that step, the P5 direct substitution identity gives the zero dynamic block at the same lifted Gauss stage and in the same row ordering. S3 applies the same-object Taylor/Kantorovich, endpoint, and inexact-Newton perturbation estimates to obtain the four-term local defect with $C_{\text{loc}} = C_G + C_A + C_E + C_{Nc\eta}$. S4 propagates this local defect on the accepted branch by the first-exit/Gronwall recurrence. S5 applies the fixed reporting map to the same reported grid. P6 contributes only the inexact-Newton

summand in S3 through the retained envelope $\eta_h^{\text{tube}} \leq c_\eta h^7$; P7, the primitive/Taylor T3 route, fixed-tolerance diagnostics, and source-policy records are boundary objects and never feed S1–S5.

Theorem normal form. Equivalently, the result is a numerical-analysis implication: fixed compact branch hypotheses, one same-branch 132-row residual certificate, endpoint and solver perturbation estimates, and accepted one-step stability imply an $O(h^7)$ local defect and an $O(h^6)$ same-initial-state reported-grid position–velocity error. The surrounding interface language identifies which hypotheses and certificates instantiate this implication for the implemented Gauss6/FullVA residual; it is not a second theorem and it does not add conclusions beyond Eqs. (254), (255), and (256). Thus a reader may treat the interface labels as bookkeeping once their provenance has been checked: the mathematical statement remains the standard local-defect plus stability plus reporting-map convergence argument, with the direct FullVA residual bridge supplying the only nonstandard residual-defect input.

Theorem input map. Tables 14 and 16 give the role of the named interfaces consumed by the theorem. Retained inputs are the P1/P2/P3 compact-branch and implementation-domain interfaces, the binding side of P4, and the P6 solver-scale envelope. Discharged residual inputs are only the P4 96-row non-dynamic certificate and the P5 36-row Newton–Euler identity, assembled on the same lifted Gauss stage. Excluded from the theorem inputs are a P7 residual-to-error boundary, the separate primitive/Taylor T3 route, fixed-tolerance diagnostics, source-policy rows, and full-TFE replacement records. These tables specify the theorem inputs for the invocation; they are not an additional proof argument.

Reference-style proof invariant. The Wieloch–Arnold constrained-BDF proof is invoked here only as a proof-order invariant. Its local-error theorem first inserts the exact constrained solution into already fixed discrete equations; its global theorem then propagates those local defects through the appropriate stability and constraint-recursion slots. The corresponding invariant for Theorem 1 is

$$\begin{aligned}
&\text{fixed FullVA residual and branch} \rightarrow \text{same-branch residual defect at } Z_G, \\
&\quad \text{same-branch residual defect} \rightarrow \text{Taylor/Kantorovich local map}, \\
&\quad \text{Taylor/Kantorovich local map} \rightarrow \text{endpoint/solver perturbations}, \\
&\quad \text{endpoint/solver perturbations} \rightarrow \text{same-initial-state reported-grid estimate}.
\end{aligned} \tag{252}$$

Every theorem-facing arrow in Eq. (252) is either proved by the cited lemma

or certificate, or retained explicitly as a P-interface hypothesis, for the present row-scaled Gauss6/FullVA object. No BLieDF local-error constant, BDF zero-stability theorem, hidden-constraint multiplier estimate, or coupled multistep recursion is imported. Conversely, each downstream reported-grid estimate, finite residual diagnostic, source-policy row, and primitive/Taylor status table is not allowed to define an upstream arrow in the chain.

For this theorem invocation the named clauses are fixed before the theorem is used. P1 fixes the smooth lift and chart tube. P2 fixes the compact stage inverse, endpoint raw-defect/right-inverse interface, endpoint ball, and stability/tube-retention data. P3 fixes the implementation path, row ordering, row scaling, and proof norm. P4 supplies only the 96 non-dynamic row-local certificate, and only under the P1–P3 binding. P5 supplies the 36 Newton–Euler rows only through the direct substitution identity on the same lifted Gauss stage. P6 supplies only the branch-selected solver residual envelope. None of these clauses is inferred from observed slopes, residual logs, source-policy rows, or post-hoc filtered solver logs.

Reference technical-slot substitution. The constrained BLieDF proof of Wieloch–Arnold [8] uses three technical slots before its final convergence statement: exact solution insertion into fixed discrete equations to define local truncation errors, a neighbourhood/start-value regime that keeps the Lie-group coordinates and multiplier variables in the admissible proof tube, and a coupled error recursion that propagates the local estimates. In Theorem 1 these slots are not imported; they are replaced by the present FullVA objects. Exact solution insertion becomes the insertion of the smooth Gauss lift Z_G into the fixed row-scaled residual $F_{A,h}$. The neighbourhood/start-value regime becomes the retained accepted-domain conditions on $\mathcal{D}_h$: branch, tube, endpoint ball, proof norm, implementation binding, and P6 solver envelope are fixed before any run is inspected. The start-value component of the BLieDF proof has no multistep prefix analogue here: it is replaced by the same-reduced-chart initial-state clause and by the one-step start-up boundary in the proof of Theorem 1. Any projection, chart-registration, or branch-registration mismatch is therefore an explicit initial-error term outside the local defect, not an unspoken finite-prefix estimate. The multistep coupled recursion becomes the one-step same-initial-state stability and first-exit/Gronwall transfer of Lemma 44. This is a slot-by-slot replacement of proof roles, not a transfer of BLieDF estimates, hidden-constraint multiplier bounds, start-up estimates, or BDF zero-stability constants.

Theorem 1 (Conditional sixth-order error theorem for the accepted branch-selected Gauss6/FullVA map). *Let the theorem instance be fixed before any numerical evidence is read: the compact proof tube K , accepted Gauss-predictor branch, reduced chart, row-scaled implemented residual $F_{A,h}$, proof norm, endpoint-output map, and reporting map are those of Assumption 1 and Definition 7. The retained/discharged partition is part of the theorem statement. For nonautonomous or driven rows this means the time-indexed residual F_{A,h,t_n} , the Gauss stage times $t_{n,i} = t_n + c_i h$, the endpoint map, and the proof norm are fixed on the same transition slice before any estimate or diagnostic row is read; all compact-tube constants used below are uniform over those accepted slices. Retain the P1–P3 theorem interfaces, the binding side of P4, and P6 on this branch: the smooth FullVA lift, uniform stage inverse and derivative bounds, endpoint raw-defect/right-inverse and endpoint-ball conditions, strong local residual inverse for the inexact-Newton perturbation, stability with tube-retention bootstrap, implementation-path binding, and the branch-solver envelope $\eta_h^{\text{tube}} \leq c_\eta h^7$ are fixed as branch-domain hypotheses. P4 is split in this statement: its branch/norm/implementation-binding side is retained as a theorem interface, while its proved 96-row non-dynamic row-local certificate is consumed only below in the single residual-certificate slot together with P5’s 36 Newton–Euler rows. The P6 clause is a residual-norm envelope for the selected Newton iterate in the fixed proof norm; it is not an assumption that the endpoint state, reported trajectory, or grid error is already $O(h^7)$ or $O(h^6)$. Thus P6 is a retained theorem-domain condition, not an unresolved step in the direct residual proof: the theorem proves the order of the accepted map conditional on this envelope and does not claim an automatic Newton-policy theorem. The P7 residual-to-error boundary is not part of these retained branch-domain hypotheses; it is an output nonclaim boundary for residual, reaction, and mechanism-coverage rows. Under these retained branch-domain hypotheses and the active same-branch residual certificate, the accepted map satisfies exactly the local and reported-grid estimates in Eqs. (254)–(256). Residual-certificate slot. For this theorem instance the active residual-value certificate is fixed by Eq. (253):*

$$\mathbf{R}_{\text{active}}(h, y) \equiv \mathbf{R}_{\text{dir}}(h, y), \quad \mathbf{R}_{\text{dir}}(h, y) \implies \|F_{A,h}(Z_G; y)\| \leq C_R h^7. \quad (253)$$

Here $\mathbf{R}_{\text{dir}}$ is the same-branch direct certificate of Corollary 5. The theorem has a single residual-certificate slot. The separate primitive predicate $\mathbf{R}_{\text{prim}}$ is not an additional premise; until it independently proves all five primitive

lift/reduction inputs and all 162 primitive subterm bounds on the same tuple $(h, y, Z_G, F_{A,h})$, it supplies no theorem input. Equivalently, a replacement primitive/Taylor route must prove all five primitive lift/reduction inputs and all 162 primitive subterm bounds before it supplies theorem input; the present route status is exactly Eq. (41). If completed, a future primitive/Taylor certificate may replace R_{active} only after proving a new same-tuple 132-row residual-value bound for the fixed object $(h, y, Z_G, F_{A,h})$; it may not combine with the accepted direct certificate to lower constants, discharge P6, or prove P7. In particular, a primitive/Taylor proof of the 36 Newton–Euler rows would not enter the theorem merely as a 36-row statement. It would first have to be reassembled with the same 96-row non-dynamic certificate, row ordering, row scaling, block norm, branch, and endpoint convention that define the implemented 132-row residual certificate. Thus the theorem-facing Taylor assertion is a full-map assertion: after the single same-tuple residual value for $F_{A,h}(Z_G; y)$ is available, Lemma 39 applies Taylor’s formula to the fixed implemented residual map and the increment $Z - Z_G$. It is not a separate claim that each primitive Newton–Euler subterm has its own $O(h^7)$ bound. This matches the reference proof discipline in which local defects are formed by inserting the smooth constrained solution into fixed discrete equations before any multiplier or global-recursion estimate is used, while the estimates themselves remain those of the present FullVA residual. In the present theorem, R_{dir} is instantiated, not supplemented, by the two discharged residual inputs on the same lifted Gauss stage: P4’s 96-row non-dynamic certificate gives the $O(h^7)$ non-dynamic block, and P5’s direct Newton–Euler substitution identity gives the 36 dynamic rows at Z_G . Consequently, Lemma 40 supplies the residual-value hypothesis $\|F_{A,h}(Z_G; y)\| \leq C_R h^7$ required by Lemma 39. Then there exist $h_0 > 0$ and constants $C_{\text{loc}}, C_{\text{red}}, C_{qv}$, independent of h and the reported grid length N_h , such that for $0 < h \leq h_0$ and any fixed uniform grid $t_n = nh$, $0 \leq n \leq N_h$, with $N_h h \leq T$, whose every transition satisfies the predeclared accepted branch, endpoint-ball, proof-norm, and P6 solver-envelope hypotheses before the estimate is invoked, the accepted branch-selected Gauss6/FullVA one-step map satisfies, uniformly for $y \in \mathcal{D}_h \subset K$, where $\mathcal{D}_h$ is the accepted theorem domain of Definition 7, under the compact-tube scaled branch-solver hypothesis $\eta_h^{\text{tube}} \leq c_\eta h^7$,

$$\|\Psi_h^{\text{G6FVA}}(y) - \varphi_h(y)\| \leq C_{\text{loc}} h^7. \quad (254)$$

Here η_h^{tube} is the a priori compact-tube envelope in the fixed proof norm; the reported $\eta_h^{\text{reported}} = \max_{0 \leq n < N_h} \eta_{h,n}$ may only specialize this envelope after

the accepted branch-selected transitions have been fixed. The corresponding reduced-chart grid values are generated from the same reduced initial state by $y_{n+1} = \Psi_h^{\text{G6FVA}}(y_n)$ on the accepted branch for $0 \leq n < N_h$. For each transition, $y_n \in \mathcal{D}_h$, and the exact-state input used in the local-defect comparison satisfies the corresponding admissibility condition, denoted $y(t_n) \in \mathcal{D}_h$ in the same admissible-graph sense. This means the selected transition data for the exact input also satisfy the retained branch, endpoint, proof-norm, and P6 predicates; it is not an assertion that P6 follows from the exact state alone. The same-initial-state clause is part of the theorem statement: an initial projection, chart-registration, or branch-selection discrepancy would enter the Gronwall recurrence as a separate initial-error term and is not absorbed into the constants C_{red} or C_{qv} . The compact-tube part of this domain is handled by the first-exit tube-retention bootstrap of Lemma 44 after the compact tube and h_0 are fixed. Branch-ball, endpoint-ball, proof-norm, and P6 solver-envelope membership remain retained transition hypotheses on the accepted branch; they are not inferred from the reported-grid estimate. Under these domain conditions, the local estimate in Eq. (254) gives the reduced-grid bound

$$\max_{0 \leq n \leq N_h} \|y_n - y(t_n)\| \leq C_{\text{red}} h^6, \quad (255)$$

and the reported position-velocity map $\mathcal{R}(y) = (q(y), v(y))$ satisfies

$$\max_{0 \leq n \leq N_h} \|\mathcal{R}(y_n) - \mathcal{R}(y(t_n))\| \leq C_{qv} h^6. \quad (256)$$

The map $\mathcal{R}$ in Eq. (256) is only the chart-to-position-velocity reporting map; it is not the P7 mechanism residual operator $\mathcal{R}_h^{\text{mech}}$ used in the residual-to-error certificate template. Equations (255) and (256) are the only same-initial-state reported-grid conclusions asserted by the theorem. The constants may depend on the compact proof tube, T , the regularity, inverse, stability, endpoint-closure, and branch-solver scale constants in Assumption 1, including c_η , but not on h , N_h , the dense/sparse backend, or the fixed production tolerances used in finite diagnostic runs. The quantifier order is fixed in the same sense: the compact tube, branch, proof norm, row scaling, inverse constants, endpoint/right-inverse constants, stability constant, and c_η are chosen first; then h_0 is chosen small enough for all local lemmas and the tube-retention bootstrap to hold; only after this choice may a particular grid $0 < h \leq h_0$ and reported accepted transition set be considered. Grid-scope boundary: the displayed maxima are over the same reported indices

$0 \leq n \leq N_h$, with accepted transitions $0 \leq n < N_h$. Exact-final-time reported order rows instantiate the theorem with $N_h h = T$. If $N_h h < T$, the estimate is only a prefix-grid estimate; the leftover interval $T - N_h h$, any dense-output value, and any separate terminal-time value at T require an additional argument outside this theorem. Undefined steps, branch changes, or residuals measured outside the fixed proof norm are outside the theorem domain rather than cases covered by the constants above.

Scope and non-claims of Theorem 1. Under these compact-branch, endpoint/stability, implementation-binding, and solver-scale conditions above, the accepted branch-selected map is order six only as the same-initial-state reduced-chart bound and the reported position-velocity bound on the same reported grid. The theorem does not prove residual, reaction, multiplier, or constraint-output error estimates; does not prove a P7 residual-to-error transfer; does not supply source-policy reproduction; and it does not turn fixed-tolerance finite runs into an asymptotic solver-policy proof. Reported-grid domain qualification is therefore a specialization rule, not a post-hoc acceptance rule: a reported run may instantiate the theorem only when its transitions already satisfy the predeclared $\mathcal{D}_h$ branch, tube, endpoint, proof-norm, and P6 solver-envelope conditions with the residual norm, row scaling, branch, and c_η fixed before the data are read. Accordingly, the reader-facing shorthand “any reported fixed time grid” means only a reported fixed time grid whose transitions already satisfy those predeclared conditions; it is not an unfiltered or post-hoc reported grid. Only one residual-value certificate is consumed in this theorem invocation. A future primitive/Taylor certificate may replace the accepted direct certificate only by proving a new same-tuple 132-row residual-value bound; it cannot be appended to the direct certificate to lower constants, remove the solver-scale condition, or prove a residual-to-error theorem for reported mechanism rows. Closed-loop residual and reaction-consistency rows therefore remain mechanism-coverage or diagnostic evidence unless a separate P7 residual-to-error theorem is proved for the same residual, branch, norm, and reporting map.

Lemma 49 (No reverse Taylor inference from the reported-grid estimate). *For the theorem instance fixed in Theorem 1, the conclusion in Eq. (257)*

$$\max_{0 \leq n \leq N_h} \|y_n - y(t_n)\| = O(h^6) \quad (257)$$

and any observed finite-grid order, residual history, reaction table, or source-policy row do not imply any of the primitive/Taylor inputs that are kept outside the active route. In particular, they do not construct a primitive projection Π_{prim} , do not prove the five open primitive lift or bilinear estimates $P_{\text{state}}, P_{\text{acc}}, P_{\lambda}, P_{\text{geom}}, P_{\text{gyro}}$, do not certify any of the 162 primitive D5 Taylor subterm bounds, and prove neither P6 nor P7. Hence the separate T3 primitive/Taylor route remains a separate sufficient route unless those primitive inputs and the associated residual-to-root, endpoint, solver, and global-transfer chain are proved on the same compact branch.

Proof. The proof of Theorem 1 is a one-way composition. The predeclared full residual $F_{A,h}$, branch selector, proof norm, row scaling, endpoint-output map, and compact-tube constants determine the local-defect estimate first. The Gronwall step then maps that local defect to a grid error by summing over at most T/h transitions, and the reporting map passes from reduced-chart error to position-velocity error. Neither the Gronwall summation nor the reporting map is an inverse theorem for the stage residual or for the primitive dynamic subterms.

To reverse the conclusion one would need additional data not present in the theorem: an h -independent inverse from reported grid errors back to the accepted stage perturbation, an inverse from the full residual value to a primitive 162-subterm decomposition, and uniform lift estimates for the five primitive inputs listed above. Those are exactly the open primitive-route obligations recorded in the D5 tables. A finite observed order or a small residual table can be consistent with these objects, but it does not prove their existence with the branch, norm, scaling, and constants fixed before the run. Therefore the global $O(h^6)$ output estimate and finite diagnostics cannot be used backward to close the primitive/Taylor route, P6, or P7. $\square$

Proof of Theorem 1. Let φ_h denote the exact reduced constrained flow in the smooth chart from Assumption 1, and let G_h be the three-stage Gauss collocation map applied to that reduced flow. The one-step estimate is local in a retained-tube state y and uses only the compact-tube solver envelope η_h^{tube} . The proof's load-bearing solver object remains this compact-tube envelope. This paragraph records the retained P6 notation before the local-defect sum; it is consumed only as the fourth summand after the direct residual bridge, the full-map Taylor/Kantorovich stage-root perturbation, and the endpoint-closure estimate have been fixed. For each transition in the theorem domain,

the accepted iterate is already on the Gauss-predictor branch ball, the proof residual norm and row scaling are fixed, and the retained P6 envelope gives

$$\|F_{A,h}(\tilde{Z}_A; y)\| \leq \eta_h^{\text{tube}} \leq c_\eta h^7. \quad (258)$$

When Eq. (258) is enforced by a predeclared proof-norm stopping rule, Lemma 43 records only this residual-size instantiation of P6; the theorem proof still retains the branch-ball, local-inverse, and compact-tube hypotheses before using the perturbation lemma. Lemma 41 then consumes exactly this same-branch residual-size input from Eq. (258) and gives

$$\|\mathcal{P}_h(\tilde{Z}_A) - \mathcal{P}_h(Z_A)\| \leq C_N \eta_h^{\text{tube}} \leq C_N c_\eta h^7. \quad (259)$$

Equation (259) is the sole inexact-Newton contribution to the local-defect sum below. Only after a reported accepted transition set has been fixed may the per-transition accepted Newton residuals be recorded as $\eta_{h,n}$, $0 \leq n < N_h$, with reported-grid specialization $\eta_h^{\text{reported}} = \max_{0 \leq n < N_h} \eta_{h,n}$. This is the reported-trajectory specialization of the predeclared envelope; it is not an additional local hypothesis, not a fitted residual sequence, and not fitted from the residual logs. It is the maximum over the accepted branch-selected Newton solves only after the reported transition set has been fixed. Branch consistency and the retained P6 condition then give, for those recorded transitions and only in the fixed proof norm,

$$\eta_{h,n} \leq \eta_h^{\text{tube}} \leq c_\eta h^7. \quad (260)$$

Equation (260) is only a reported-grid specialization of the predeclared compact-tube envelope. The residual norm in this statement is the fixed proof norm of the accepted 132-row residual after the chart, row ordering, and row-scaling conventions have been fixed; implementation logs can specialize the per-transition values $\eta_{h,n}$, and hence η_h^{reported} , only through that same residual map and an h -independent compact-tube norm equivalence. The proof does not use hidden h -dependent row weights, and any logged-norm comparison is through an h -independent finite-dimensional norm-equivalence constant after the explicit residual h -factors have already been defined. Finite logs and scaled-tolerance probes can record a reported-grid diagnostic specialization of the displayed maximum after the branch and proof norm are fixed, but they do not discharge the tube-envelope hypothesis or create an independent solver-policy proof.

Thus the proof does not fit the constants C_N , c_η , or C_{loc} from reported Newton logs, and a small residual in a different norm, branch, row scaling, or terminal solve is not an input to the estimate. Throughout the proof, Ψ_h^{G6FVA} denotes this branch-selected accepted map: Newton is initialized from the Gauss predictor on the same compact-tube branch, and the estimate is not a statement about arbitrary nonlinear roots. Solver-envelope quantifier discipline: the scalar solver-envelope constant and fixed proof norm are chosen before h , N_h , or the finite logs; changing that constant after inspecting residuals changes the theorem assumption and C_{loc} , not the validity of an already accepted proof instance. Classical collocation theory gives a uniform Gauss truncation constant C_G , after shrinking h_0 if needed, such that Eq. (261) holds uniformly for y in the compact trajectory tube [3]:

$$\|\Psi_h^G(y) - \varphi_h(y)\| \leq C_G h^7 \quad (261)$$

Gauss truncation constant boundary: C_G is the analytic collocation constant obtained from C^7 reduced-vector-field bounds on the fixed compact tube. It is fixed before implementation residual diagnostics, endpoint-closure logs, work-precision tables, or observed smooth h -sweep slopes are inspected; those slopes may corroborate the scale but cannot choose C_G , shrink the tube, or select the Gauss-predictor branch. The standard Gauss theorem is applied only after the constrained system has been represented in the smooth reduced chart of Assumption 1. If the reduced field $f(y, t)$ is treated as nonautonomous, the usual autonomous augmentation by $t' = 1$ is used locally. The Lie-group coordinates, lower-pair constraints, and chart-to-configuration norm equivalences enter only through the fixed compact-tube bounds and do not change the h^7 local Gauss defect. No remote chart, nonsmooth event, or global Lie-group order statement is invoked here. **Stage-time binding for nonautonomous rows.** For a reported transition starting at t_n , the proof uses the time-indexed residual F_{A,h,t_n} defined by the method equations with stage times $t_{n,i} = t_n + c_i h$. The lifted Gauss stage is therefore $Z_G(y, t_n)$, obtained by inserting the exact reduced solution and its FullVA lift at those same stage times. Driven constraint right-hand sides, loads, and prescribed kinematic functions are evaluated at $t_{n,i}$ before the residual map, row scaling, proof norm, and Newton branch are fixed for that transition. They are not unknown columns, not endpoint post-corrections, and not values reconstructed from a terminal output or a finite diagnostic table. Thus the local defect compares $\Psi_{h,t_n}^{\text{G6FVA}}(y)$ with $\varphi_{t_n+h,t_n}(y)$ on the same time slice, while the autonomous augmentation above is only a notation for this synchronized

time-indexed one-step statement. **Nonautonomous same-time invariant.** For this transition, every object entering the local defect is keyed by the same input time t_n and the same Gauss stage abscissae $t_{n,i} = t_n + c_i h$. The Gauss lift $Z_G(y, t_n)$, the implemented residual F_{A,h,t_n} , the direct P4/P5 residual certificate, the endpoint map $\mathcal{E}_{h,t_n}$, the endpoint-closure map $\mathcal{C}_{h,t_n}$, and the inexact-Newton perturbation map $\mathcal{P}_{h,t_n}$ are all formed for that same transition. The four-term local-defect sum therefore compares $\Psi_{h,t_n}^{\text{G6FVA}}(y)$ with $\varphi_{t_n+h,t_n}(y)$ without mixing a residual evaluated at another input time, a force value reconstructed from the terminal output, an interpolated drive sample, or a dense-output endpoint. The later Gronwall recursion only concatenates these synchronized one-step maps at $t_n = nh$; it does not insert an additional time-registration or terminal correction term. The SO(3) BCH and adjoint effects are represented only through the retained P2 reduced-chart stability and reporting-map constants chosen for the present compact tube. No BCH constant or Lie-algebra recurrence estimate from Wieloch–Arnold is imported into the bound. This is the local Lie-algebra error-recursion slot corresponding to the Wieloch–Arnold constrained-BDF proof, not an imported estimate or an additional theorem conclusion. Lemma 7 identifies the mathematical FullVA stage rows with the lifted reduced Gauss equations and pointwise constrained dynamics. The accepted proof then evaluates the defined implemented 132-row formula residual at the lifted Gauss stage, after the local chart, row ordering, and row scaling have been fixed. The stage-residual bound $\|F_{A,h}(Z_G; y)\| \leq C_R h^7$ is supplied by Lemma 40: the 96 non-dynamic rows contribute the $O(h^7)$ block and the 36 Newton–Euler rows vanish by the D5 direct-substitution certificate. This is a discharged formula-residual input to the perturbation criterion, not a compact-tube regularity assumption and not an empirical residual fit from the reported trajectory. This is also the precise point at which the Wieloch–Arnold constrained-BDF proof architecture [8] is invoked as an organizational analogue. The transfer is a proof-order transfer, not an estimate transfer: first the discrete Lie-group object is fixed, then the smooth branch is inserted into that fixed object, then a local defect is proved, and only after that is any global error recursion invoked. Lemma 4 identifies the active Taylor object as the full scaled map $F_{A,h}$ in the stage increment $Z - Z_G$. The proof therefore does not import the reference paper’s BLieDF hidden-constraint estimates, BDF $p = k$ theorem, or any primitive FullVA subterm bound; those would be different premises from the 132-row residual bridge used here. The only structural correspondence is the obligation order: fixed discrete object, smooth-branch insertion,

local-defect estimate, and then local-to-global propagation. Concretely, the cited constrained-BDF analogue is the Section 6.2 Theorems 2–4 dependency order: Theorem 2 establishes local truncation estimates, Theorem 3 forms the configuration/velocity global-error recursion, and Theorem 4 uses hidden-constraint difference approximations and a multiplier-error estimate to close the coupled recursion and final order statement after those local estimates are available. The present theorem follows the same dependency order at the level of proof obligations: the same-tuple 132-row residual value is certified before the Taylor/Kantorovich stage-root perturbation is invoked, and only then is the compact-tube stability/Gronwall transfer applied to the reported grid. In the reference proof, the local truncation error is generated by inserting the analytical solution into the already specified BLieDF equations. The FullVA analogue used here is the single insertion of the lifted Gauss branch into the already specified residual $F_{A,h}$. Therefore all later Taylor and Kantorovich constants are constants of this fixed residual object; they are not constants of a reference BDF equation, a primitive D5 factorization, or a table of finite residual diagnostics. **Reference-style insertion checkpoint.** The constrained-BDF proof order fixes the discrete equations first, forms local truncation errors by substituting the exact constrained solution into those equations, and only then derives hidden-constraint, multiplier, and global recursion estimates. The corresponding FullVA checkpoint is the one-way chain in Eq. (262):

$$\begin{aligned}
& Z_G \xrightarrow{\text{insert}} F_{A,h}(Z_G; y), \\
& F_{A,h}(Z_G; y) \xrightarrow{\text{direct bridge}} O(h^7), \\
& O(h^7) \xrightarrow{\text{Taylor/Kantorovich}} \|Z_A - Z_G\| = O(h^7) \xrightarrow{\mathcal{P}_h} O(h^7).
\end{aligned} \tag{262}$$

Every arrow in this checkpoint is an assertion about the implemented Gauss6/FullVA residual and endpoint map fixed before the proof begins. No BLieDF hidden-constraint difference estimate, multiplier recursion, BDF zero-stability norm, or primitive D5 decomposition is a premise of any arrow in this chain, and no later grid error, finite residual row, or source-policy row can be read backward to create such a premise. **Proof-order instantiation.** For this theorem the reference ordering is instantiated in the following forward-only sequence.

1. Fix the discrete object: the accepted Gauss-predictor branch, $F_{A,h}$, row ordering, row scaling, proof norm, and endpoint-output map are

fixed before any estimate or diagnostic row is used. The reference correspondence table is an exposition map for this order of obligations; it is not itself a mathematical input.

2. Prove the local defect on that object: the Gauss $C_G h^7$ defect is combined with Lemmas 40 and 39; the separate primitive 162-term Taylor closure is not used.
3. Apply the constrained and solver interfaces: endpoint closure is handled by Lemma 6, while P6 remains the retained compact-tube condition $\eta_h^{\text{tube}} \leq c_\eta h^7$. Fixed-tolerance logs and finite solver probes do not discharge P6.
4. Propagate and report only after the local defect is fixed: Lemma 44 and Lemma 45 give the grid and position–velocity estimates. P7 residual rows remain nonpromoted diagnostics.

Every constant used after this ordering is supplied by Assumption 1 or by the displayed lemmas on the fixed FullVA object. No constant, Taylor order, or residual estimate is imported from the reference paper, from Table 11, or from a finite diagnostic trajectory.

Direct-route sufficiency. From this point to the Gronwall step the proof uses exactly four positive inputs: the Gauss local defect $C_G h^7$, the direct 132-row residual defect $C_R h^7$ assembled from the 96 non-dynamic rows plus the 36 P5 Newton–Euler rows, the endpoint-closure perturbation $C_E h^7$, and the inexact-Newton perturbation $C_N \eta_h^{\text{tube}}$ with the retained P6 bound. It uses no primitive 162-term Taylor closure, no P7 residual-to-error promotion, no source-policy row, and no fixed-tolerance log as a theorem input. Thus the PC2 discharge below is a direct-route sufficiency statement for the implemented 132-row residual, not a closure of the separate primitive/Taylor route or a promotion of diagnostic residual rows.

Route-exclusivity refinement. These four inputs are consumed as one same-branch proof tuple after the compact tube, branch rule, row scaling, proof norm, and endpoint-output map have been fixed. If a separate T3 primitive/Taylor proof replaces the dynamic-row residual-value certificate, it must first produce its own $C_R^{\text{T3}} h^7$ bound for the same 132-row residual map on that same tuple. It cannot be combined with the accepted direct $C_R h^7$ term to lower constants, remove the retained P6 solver-scale condition, use P7 residual-to-error rows, or turn primitive-route status tables into theorem inputs. Thus the active theorem consumes one residual-value certificate, then one T2 full-map Taylor/Kantorovich perturbation; it never mixes partial

certificates from the direct and primitive routes.

Same-object direct-substitution calculation. For this theorem invocation, write the row-scaled residual at the lifted Gauss stage as the fixed block vector in Eq. (263):

$$R_h = F_{A,h}(Z_G; y) = (F_{A,h}^{\text{nd}}(Z_G; y), F_{A,h}^{\text{dyn}}(Z_G; y)), \quad (263)$$

where the first block contains the 96 non-dynamic rows and the second block contains the 36 Newton–Euler rows in the row ordering fixed by P3. For a nonautonomous transition this is shorthand for

$$R_{h,n} = F_{A,h,t_n}(Z_G(y, t_n, h); y), \quad (264)$$

where both row blocks are evaluated on that same time-indexed residual slice. Equation (264) fixes the time-indexed residual slice used for the direct block calculation. The constants used below are uniform over the reported accepted time slices; no non-dynamic row, Newton–Euler row, Jacobian, or endpoint map is imported from a different t_n or from a terminal-output reconstruction. The P4 certificate gives $\|F_{A,h}^{\text{nd}}(Z_G; y)\| \leq C_{\text{nd}}h^7$, and the P5 direct substitution identity gives

$$F_{A,h}^{\text{dyn}}(Z_G; y) = 0. \quad (265)$$

Hence, for the fixed finite-dimensional block norm,

$$\|R_h\| \leq C_{\text{blk}}\|F_{A,h}^{\text{nd}}(Z_G; y)\| \leq C_R h^7, \quad C_R = C_{\text{blk}}C_{\text{nd}}. \quad (266)$$

Equations (265) and (266) are the direct-route residual certificate consumed by Lemma 39. The phrase fixed block vector denotes this residual vector with this row order, scaling, norm, branch, and stage variable. This is the exact theorem-level residual value inserted into the Taylor/Kantorovich perturbation, not a primitive 162-term estimate and not a residual measured at $\tilde{Z}_A$. With $J_h = D_Z F_{A,h}(Z_G; y)$, the stage-root equation in Lemma 39 is

$$\Delta = -J_h^{-1}R_h - J_h^{-1}Q_h(\Delta), \quad \Delta = Z - Z_G. \quad (267)$$

Here Q_h is the same full-map integral Taylor remainder defined in Eq. (201) and bounded there by $\|Q_h(\Delta)\| \leq (L/2)\|\Delta\|^2$. The constants M, L, r are therefore the retained compact-tube constants of the frozen 132-row residual instance, not constants recomputed from the direct dynamic certificate, the

primitive inventory, or the reported trajectory. On the retained branch, the same compact-tube inverse and quadratic-remainder constants give the accepted fixed point $\Delta_A = Z_A - Z_G$ with

$$\|Z_A - Z_G\| = \|\Delta_A\| \leq 2MC_R h^7 = C_Z h^7. \quad (268)$$

The endpoint reconstruction Lipschitz bound in the same chart then gives

$$\|\Psi_h^A(y) - \Psi_h^G(y)\| = \|\mathcal{E}_h(Z_A) - \mathcal{E}_h(Z_G)\| \leq M_E C_Z h^7 = C_A h^7. \quad (269)$$

Equations (267), (268), and (269) are the same-object Taylor/Kantorovich and endpoint-reconstruction handoff. Thus the direct P4/P5 block substitution and the T2 full-map Taylor/Kantorovich step are one same-object calculation: the block residual value, the inverse J_h^{-1} , the quadratic remainder Q_h , and the endpoint map $\mathcal{E}_h$ all use the same branch, row scaling, proof norm, and stage variable. No constant in this calculation is imported from the reference proof, fitted from a finite trajectory, or borrowed from the separate primitive/Taylor route, summed with that route, or used to partially close that route; the non-active route status remains Eq. (41). In the present proof the dynamic-block certificate slot is occupied by

$$\mathbf{D}_{\text{dir}} : \quad F_{A,h}^{\text{dyn}}(Z_G; y) = 0. \quad (270)$$

A separate primitive certificate would have the template

$$\mathbf{D}_{\text{T3}} : \quad (P_{\text{state}} \wedge P_{\text{acc}} \wedge P_{\lambda} \wedge P_{\text{geom}} \wedge P_{\text{gyro}}) \implies \|F_{A,h}^{\text{dyn}}(Z_G; y)\| \leq C_{\text{dyn}}^{\text{T3}} h^7. \quad (271)$$

Equations (270) and (271) record the accepted direct certificate and the non-active primitive/Taylor replacement template. The latter would be an alternative input to the same full-residual bound, not an additional summand. No instance of $\mathbf{D}_{\text{T3}}$ is available in this manuscript because those five antecedent primitives remain open and the primitive Taylor-bound certificate is not established on the theorem branch. The theorem consumes $\mathbf{D}_{\text{dir}}$ as the proved direct certificate, does not assume $\mathbf{D}_{\text{T3}}$, and never combines the two certificates.

Lemma 39 converts that same-object direct residual defect into the explicit stage-root and endpoint perturbation bound $\|\Psi_h^A(y) - \Psi_h^G(y)\| \leq C_A h^7$. Lemma 6 gives the endpoint-closure contribution under its right-invertibility and raw-defect hypotheses, with the explicit constant $C_E = 4M_{E,\text{ri}}C_{E,\text{raw}}$,

and Lemma 41 keeps the inexact Newton contribution at the same scale under the compact-tube branch-solver hypothesis $\eta_h^{\text{tube}} \leq c_\eta h^7$. More explicitly, let Ψ_h^G be the endpoint map reconstructed from the reduced Gauss stage, let $\Psi_h^A = \mathcal{E}_h(Z_A)$ be the exact branch-selected accepted local stage-root map after Lie endpoint reconstruction and before endpoint closure, let $\Psi_h^E = \mathcal{C}_h(\Psi_h^A)$ be the endpoint-closed exact-stage map, and let $\Psi_h^{\text{G6FVA}} = \mathcal{P}_h(\tilde{Z}_A)$ with $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$ be the endpoint-closed inexact Newton map. The preceding paragraph and lemmas give constants C_G, C_A, C_E, C_N , independent of h, y , and the reported accepted step index $0 \leq n < N_h$, such that, uniformly on the compact proof tube,

$$\begin{aligned} \|\Psi_h^G(y) - \varphi_h(y)\| &\leq C_G h^7, & \|\Psi_h^A(y) - \Psi_h^G(y)\| &\leq C_A h^7, \\ \|\Psi_h^E(y) - \Psi_h^A(y)\| &\leq C_E h^7, & \|\Psi_h^{\text{G6FVA}}(y) - \Psi_h^E(y)\| &\leq C_N \eta_h^{\text{tube}}. \end{aligned} \quad (272)$$

Equation (272) lists the four same-branch local perturbation inputs before they are summed. The constants in these four bounds are uniform in y on the compact proof tube and uniform in the reported accepted step index $0 \leq n < N_h$; the nonlinear-solver term is controlled through the compact-tube branch-solver policy, whose reported-grid specialization is the single accepted-branch maximum $\eta_h^{\text{reported}} = \max_{0 \leq n < N_h} \eta_{h,n}$ over the reported branch-selected Newton steps. Their allowed dependencies are exactly the compact-tube regularity, right-inverse, endpoint-closure, stability, and branch-solver scale bounds listed in Assumption 1; no constant in the local defect sum is chosen from the reported grid length, backend, or fixed production tolerance. The single small-step threshold used in the theorem is the minimum of the windows required by the Gauss truncation estimate, the residual-bridge certificate, the Kantorovich stage-root perturbation, the endpoint-closure right-inverse estimate, the inexact-Newton branch-ball estimate, and the local-to-global tube-retention bootstrap. These windows are all chosen after the compact proof tube, proof norm, row scaling, branch rule, and constants in Assumption 1 have been fixed, and before any reported grid, solver log, separate D5 primitive/Taylor inventory, P7 residual table, or source-policy diagnostic is read. Hence the four local $O(h^7)$ estimates hold simultaneously on the same accepted branch and in the same norm; the proof does not splice estimates that require incompatible step-size restrictions or different nonlinear roots.

Uniform-constant timeline. Equivalently, the theorem has a single quantifier order: first fix the compact tube, chart, row-scaled residual map, proof norm, endpoint map, branch rule, and solver-envelope convention; then

choose the constants in the four local estimates and the one-step stability estimate; then shrink one common h_0 ; only after that may a particular $h \leq h_0$, grid length N_h , and reported trajectory be considered. The constants $C_G, C_R, M, L, r, C_A, C_E, C_N, c_\eta, C_s$ are therefore uniform in the reported step index and independent of the finite grid selected afterward. This is the one-step analogue of the reference constrained-BDF proof step where constants in the error recursion are chosen independently of h and n after the local truncation equations are fixed. Rechoosing any of these constants per reported step, per backend norm, or after seeing a finite diagnostic would define a different theorem instance rather than strengthen this one.

When $\eta_h^{\text{tube}} \leq c_\eta h^7$, define the explicit local-defect constant $C_{\text{loc}} = C_G + C_A + C_E + C_N c_\eta$. The four-term local defect decomposition is therefore

$$\begin{aligned}
\|\Psi_h^{\text{G6FVA}}(y) - \varphi_h(y)\| &\leq \|\Psi_h^G(y) - \varphi_h(y)\| + \|\Psi_h^A(y) - \Psi_h^G(y)\| \\
&\quad + \|\Psi_h^E(y) - \Psi_h^A(y)\| + \|\Psi_h^{\text{G6FVA}}(y) - \Psi_h^E(y)\| \\
&\leq C_G h^7 + C_A h^7 + C_E h^7 + C_N \eta_h^{\text{tube}} \\
&\leq C_{\text{loc}} h^7.
\end{aligned} \tag{273}$$

Equation (273) is the triangle inequality over Gauss truncation, stage-root perturbation, endpoint closure, and inexact Newton, with an explicit uniform constant sum after the compact-tube η_h^{tube} bound. It gives the local estimate stated in Eq. (254). This displayed budget is also the only point where P6 enters the positive order proof: P6 controls the fourth summand, while the direct residual bridge and the full-map Taylor/Kantorovich step control the stage-root summand. Equivalently, the proof has instantiated Lemma 5 on the current transition: the local-defect obligations supply the same-branch input consumed here, and the remaining proof uses only the stability, tube-retention, and reporting-map steps below. No later global estimate, finite run, or diagnostic row feeds back into C_{loc} . **Forward-only Taylor invariant.** No h^6 reported-grid estimate, observed slope, residual table, or finite Newton log is used to define Z_G , choose $F_{A,h}$, select the row scaling or proof norm, prove the D5 direct identity, or calibrate $C_R, M, L, r, C_A, C_E, C_N$. The allowed proof order is fixed object and smooth-branch insertion, then direct residual value at Z_G , then the Taylor/Kantorovich stage-root perturbation, then endpoint and inexact-Newton perturbations, and only then Gronwall and reporting. Thus every Taylor estimate in the accepted route is a pre-global estimate on the fixed implemented equations; a reverse reading from

reported errors or finite diagnostics back to the Taylor inputs would be circular and is excluded from the theorem.

The forward-only exclusion used in the next step is Lemma 49: neither the reported-grid estimate nor any finite diagnostic is used backward to certify primitive/Taylor inputs, P6, or P7.

Proof structure: the load-bearing proof is the four-term uniform local-defect sum followed by the stated local-to-global transfer and reporting-map consequence. The proof-interface tables specify domain and nonuse boundaries; they do not replace the local-defect estimate, choose constants from finite h -sweeps, or promote residual tables into trajectory-error evidence. The proof direction is therefore forward only. The lifted Gauss stage, implemented residual identity, endpoint correction, and solver-envelope constant are fixed before the recurrence for e_n is written; the $O(h^6)$ reported-grid estimate is not used to choose them. The only loss from the local h^7 defect to the same-initial-state reported-grid h^6 estimate is the explicit Gronwall accumulation over at most T/h accepted transitions. The one-step stability scale in Assumption 1 supplies the $(1 + C_s h)$ hypothesis of Lemma 44, and the accepted compact tube supplies the tube-retention bootstrap used there. To make the transfer explicit, define the reduced-chart grid error on the reported indices by

$$\varepsilon_n = y_n - y(t_n), \quad e_n = \|\varepsilon_n\|. \quad (274)$$

Equation (274) defines the only propagated state-error vector and scalar error norm in the theorem. For this theorem invocation the first-exit index is the first reported index at which either the numerical/exact states leave the retained compact tube or the next accepted transition is no longer defined in $\mathcal{D}_h$. The recurrence below is used only before that combined compact-tube/domain exit. Here $\mathcal{D}_h$ denotes the state projection of the pre-declared accepted-domain graph carrying the retained branch-ball, endpoint-ball, proof-norm, and P6 solver-envelope requirements; membership in those non-compact-tube fiber components is not inferred from the Gronwall recursion. The small-step first-exit bootstrap closes the compact-tube component after the constants are fixed; the branch-ball, endpoint-ball, proof-norm, and P6 solver-envelope components are retained admissibility conditions on the same reported transition set and are not proved by the Gronwall recursion. Thus the proof never extends Ψ_h^{G6FVA} outside its accepted theorem domain. **Theorem-domain restriction.** Domain-exit is not a lower-order conclu-

sion: a branch-ball, endpoint-ball, compact-tube, or P6 solver-envelope failure does not create a covered lower-order transition. It means the stated theorem instance is undefined for that transition until a different theorem domain is proved. The first-exit argument is therefore a fixed-domain bootstrap for the compact-tube component: it uses the retained tube margin and stability scale to keep the accepted reported states inside the compact tube for $0 < h \leq h_0$. The other components of $\mathcal{D}_h$ -branch-ball membership, endpoint-ball membership, proof-norm binding, and the P6 solver envelope—remain predeclared transition-domain conditions on the same accepted transition set. They are not proved by the first-exit bootstrap, and they are not recovered by a post-run row deletion rule that would discard failed steps after inspecting a trajectory. On the accepted branch, after the branch-selection and tube-retention hypotheses have been fixed, the local defect and stability estimates give the error recurrence

$$e_{n+1} \leq (1 + C_s h) e_n + C_{\text{loc}} h^7, \quad e_0 = 0, \quad 0 \leq n < N_h. \quad (275)$$

Equation (275) is the scalar Gronwall recurrence for the reduced-chart state-error norm. Its forcing term is only the already proved same-branch $C_{\text{loc}} h^7$ local defect from Eq. (273); raw residual tables, P6 solver logs, and P7 mechanism residual diagnostics are not additional right-hand-side terms in this recurrence. Because e_n is the reduced-chart state-error norm and not a residual norm, this recurrence is not the P7 residual-to-error boundary template in Eq. (242). **Reference-style recursion analogue.** This is the point in the present proof corresponding to the global recursion step in the constrained Lie-group BDF reference. The analogy is structural, not an estimate transfer, and it matches only the proof order in which a proved local defect is propagated. It does not transfer the BLieDF propagated variable set, companion matrices, multiplier recursion, or recursion constants. In the reference proof, Lie-algebra configuration errors, constraint differences, and multiplier estimates are propagated together only after the local truncation defect has been proved for the fixed BLieDF equations. Here the lower-pair algebraic variables and stage multipliers have already been consumed inside the fixed 132-row FullVA residual, the endpoint right-inverse, and the accepted branch map Ψ_h^{G6FVA} . After those same-object inputs have produced the local defect $C_{\text{loc}} h^7$, the only quantity propagated by the theorem is the reduced-chart state-error norm e_n through the displayed one-step stability inequality. There is no hidden multiplier recursion in this step, no reaction-output norm inside e_n , and no residual-to-error operator being applied tacitly. A multiplier

or reaction error theorem would therefore require a separate coupled output recursion with its own branch-compatible constants; it is not a consequence of the scalar/vector Gronwall inequality above. Zero-initial-error rule: the displayed recurrence is the theorem instance with the numerical and exact trajectories started from the same reduced-chart initial state. If an initial projection, chart-registration step, warm-start correction, or branch-selection adjustment introduced $e_0 \neq 0$, the same Gronwall calculation would contain the additional term $\exp(C_s T) \|e_0\|$. That term is not absorbed into $C_{\text{loc}} h^7$ or $C_{\text{red}} h^6$. Thus the theorem's order-six grid estimate is a same-initial-state statement; any reported run with a separate initialization error would need its own initial-error bound before invoking the theorem. One-step start-up boundary: unlike the constrained BLieDF reference proof, there is no multistep start-up layer such as $0 \leq n < 2k - 1$ in the theorem instance above. The accepted map is a one-step map, so the recurrence starts at $n = 0$ from the exact reduced initial state and uses only the single previous accepted state. Any numerical start-up perturbation, projection warm start, or branch-registration mismatch is therefore represented by the initial-error term described above; it is not hidden in the local defect, not treated as a separate finite prefix with its own constants, and not estimated from the first few reported steps. The induction hypothesis is that the numerical and exact states remain in the retained compact tube for the reported transition under consideration; this is the tube-retention induction supplied by Assumption 1, not a condition read from trajectory plots after the run. In this theorem invocation the induction also includes $y_n \in \mathcal{D}_h$ for the transition being estimated and $y(t_n) \in \mathcal{D}_h$ for the exact-state local-defect evaluation, so a branch, endpoint-ball, or solver-scale failure is a theorem-domain exit rather than a lower-order case. Equivalently, the theorem uses the first-exit bootstrap in Lemma 44: the recurrence is applied only until a hypothetical exit from the retained tube, and the small-step threshold is chosen so that the resulting $C_{\text{red}} h^6$ bound stays inside the tube margin while the retained branch/domain conditions remain in force. The first-exit bootstrap closes the compact-tube component under the retained P2 tube-margin and stability hypothesis; branch-ball, endpoint-ball, proof-norm, and P6 solver-envelope membership remain theorem-domain conditions for the accepted transition set. None of these conditions is assumed after inspecting the reported trajectory. Iterating the recurrence before any reported residual or work-precision

table is inspected gives

$$e_n \leq C_{\text{loc}} h^7 \sum_{j=0}^{n-1} (1 + C_s h)^j \leq C_{\text{loc}} h^6 \Gamma_s(T), \quad (276)$$

where the Gronwall factor is

$$\Gamma_s(T) = \begin{cases} (\exp(C_s T) - 1)/C_s, & C_s > 0, \\ T, & C_s = 0. \end{cases} \quad (277)$$

Equations (276) and (277) are the only source of the one-power loss from local h^7 to grid h^6 . Accumulation-factor boundary: the loss of one power of h comes only from accumulating at most $O(1/h)$ reported transition defects over a fixed horizon. The bound uses $N_h h \leq T$ before the constant is chosen, so the geometric sum is bounded by $h^{-1} \Gamma_s(T)$ and the constant C_{red} is independent of the reported step count N_h . It is not fitted from the number of reported steps, an observed work-precision slope, or a residual table. Likewise, C_s , the tube margin, $\Gamma_s(T)$, and C_{red} are inherited from Assumption 1 and Lemma 44 before reported grid errors are read. The count N_h is used only through $N_h h \leq T$; it cannot fit a stability constant or repair a failed tube-retention row. Applying that lemma with $p = 6$, $C_\ell = C_{\text{loc}}$, and the reported grid length denoted by N_h gives the explicit reduced-chart grid constant

$$C_{\text{red}} = C_{\text{loc}} \Gamma_s(T). \quad (278)$$

Writing d_K for the retained distance of the exact reduced trajectory to the compact-tube boundary in Lemma 44, the first-exit closure uses one further pre-data restriction of the common small-step threshold: after C_{red} and d_K are fixed, but before any reported grid is read, shrink h_0 so that $C_{\text{red}} h_0^6 < d_K/2$. Then for every $0 < h \leq h_0$ the bound obtained on the truncated first-exit interval keeps the numerical state strictly inside the retained compact tube, contradicting any compact-tube first exit. This is the only place where the tube margin enters the grid-error proof; it does not repair branch-ball, endpoint-ball, proof-norm, or P6 solver-envelope failures. Equation (278) is the constant used in the reduced-grid bound in Eq. (255) over the same reported time grid $t_n = nh$, $0 \leq t_n \leq T$, with the maximum taken over $0 \leq n \leq N_h$; exact final-time divisibility is not a separate theorem condition. The terminal index N_h is only the last output value; the branch-selected Newton solves and the solver maximum above are indexed by the transition steps

$0 \leq n < N_h$. If $N_h h < T$, the unreported leftover interval $T - N_h h$ is outside the stated grid maximum; no terminal-time error at T is inferred without adding a separate reported step. The final passage to the reported position and velocity errors is the application of Lemma 45; it uses compact chart norm-equivalence, not a new dynamic estimate. On the retained compact tube, the chart-to- (q, v) reporting map

$$\mathcal{R}(y) = (q(y), v(y)) \quad (279)$$

is C^1 with uniformly bounded derivative, say $\|D\mathcal{R}\| \leq C_{\mathcal{R}}$. Define $C_{qv} = C_{\mathcal{R}}C_{\text{red}}$. Hence

$$\max_{0 \leq n \leq N_h} \|\mathcal{R}(y_n) - \mathcal{R}(y(t_n))\| \leq C_{\mathcal{R}} \max_{0 \leq n \leq N_h} \|y_n - y(t_n)\| \leq C_{qv} h^6. \quad (280)$$

Equations (279) and (280) define the reported output map and its compact-chart norm consequence. This step is local to the accepted chart and does not claim a global equivalence across remote charts, nonsmooth contact events, or other branches outside the compact proof tube; for rotational components it is the fixed local retraction/log-chart reporting convention above, not a global $SO(3)$ geodesic-error theorem. $\square$

Scope of the theorem conclusion. The theorem conclusion is limited to the displayed local-defect and same-initial-state reported-grid bounds for the accepted reduced-chart branch-selected map. In particular, residual tables are recorded only as diagnostics for the open P7 residual-to-error boundary; they do not prove P7, do not alter C_{loc} , and do not promote mechanism-coverage residual rows to trajectory-error/order evidence. The theorem does not prove a P7 residual-to-error boundary; the corresponding exclusion rule is explicit: do not promote mechanism-coverage residual rows to trajectory-error/order evidence unless a separate residual-to-error theorem and branch-compatible dynamic-order campaign have been supplied. This P7 boundary is independent of the P6 solver-scale boundary: P6 controls only the inexact Newton summand, while P7 would require a separate residual-to-trajectory transfer operator. The solver-scale clause is a retained hypothesis on the selected Gauss-predictor branch; for a reported trajectory it applies only to transitions whose selected Newton iterate remains on that branch and whose reported-grid maximum $\eta_h^{\text{reported}} = \max_{0 \leq n < N_h} \eta_{h,n}$ satisfies $\eta_h^{\text{reported}} \leq c_{\eta} h^7$ as a trajectory specialization of the retained compact-tube

envelope η_h^{tube} . Equivalently, $\eta_{h,n} \leq c_\eta h^7$ for every accepted reported transition. The theorem does not choose c_η , the residual norm, or the accepted branch from reported residual ratios. If the implementation stops by a fixed production tolerance or a backend residual norm, that log specializes this clause only after the same-branch proof-norm conversion in Eq. (69) and the predeclared $c_\eta h^7$ envelope have been fixed; otherwise the log remains diagnostic and is not a P6 theorem input. In particular, the branch-ball membership certificate and the h^7 residual envelope are checked on the same transition; neither one is inferred from the other. Those objects are fixed before the grid, Newton logs, and tolerance sweeps; if a computed transition fails the branch or scaled-residual condition, the branch-selected map is undefined there rather than lower order. Residual-only mechanism rows, fixed-tolerance logs, and comparison or source-package records remain outside the local-defect estimate and outside the theorem conclusion.

The stage-residual input is the implemented 132-row direct route. The alternative primitive/Taylor decomposition of the Newton–Euler rows is a separate dynamic-block residual-certificate route, not a premise of this theorem. Its five uniform primitive inputs $P_{\text{state}}, P_{\text{acc}}, P_\lambda, P_{\text{geom}}, P_{\text{gyro}}$ are not established on the accepted theorem branch. This is a scope boundary for the separate primitive route, as recorded in Tables 27 and 28 and the accompanying conditional primitive/Taylor implication. The primitive/Taylor route is a non-active replacement-certificate route rather than current theorem evidence; more precisely, it is a separate dynamic-block residual-certificate route. It would become a separate sufficient primitive-route implication for the dynamic-row PC2 input only after those five primitives are independently proved on the same compact branch, proof norm, row scaling, and endpoint/output convention, and the resulting dynamic-row value is reassembled with the same 96-row non-dynamic certificate before being consumed by the unchanged T2 full-map perturbation chain. It is not a term in C_{loc} , not an additional premise of the direct residual-bridge theorem, and the theorem conclusion does not certify it. Equivalently, the theorem uses the implication from the certified direct bridge to the full-residual defect; it does not use the converse implication from theorem order to primitive subterm closure, and it does not require the non-active separate primitive route to be refuted. Finite solver probes, fixed-tolerance production logs, residual/reaction tables, and common-reference or source-policy rows are therefore not post-theorem discharge mechanisms. They may diagnose branch plausibility or record diagnostic scope, but they do not prove a solver-policy theorem, do not prove a P7

residual-to-error boundary, do not complete the primitive/Taylor route, and do not enlarge the theorem beyond the accepted branch-selected map stated above. The same direct residual-bridge scope convention applies throughout this section: it refers only to the residual-bridge/Kantorovich input for this conditional theorem. It does not prove a solver-policy theorem or a P7 residual-to-error boundary, does not complete the primitive/Taylor route, and does not establish source-policy or full-TFE readiness. The conditional primitive/Taylor implication therefore states two distinct mathematical facts, summarized by the status card Eq. (41). Under the named primitive inputs, all 162/162 reduction templates have been assigned to row-level subterms. This is template coverage only; the certified count of actual primitive Taylor bounds is 0/162 without those inputs, so the certificate does not provide an actual primitive-route Taylor bound for a T3 replacement certificate in the accepted theorem route. The first fact shows that the primitive dependency map is complete as a conditional implication. It is not theorem evidence unless the named primitives are independently proved on the same compact branch. The second keeps the separate primitive route out of the accepted theorem until its uniform lift primitives are proved. These two statements are not additive proof evidence: the conditional templates cannot be spliced with the direct residual bridge to strengthen the theorem conclusion or remove a retained interface. A primitive-route theorem would have to prove the five open primitive obligations on the same compact branch—the state and acceleration root lifts, the multiplier lift propagated from them, and the downstream geometry and gyroscopic reductions—and then redo the residual-to-root, endpoint, solver, and global-transfer chain for that route.

Domain of Validity. To invoke Theorem 1, the proof instance must use the same reduced initial state, the same accepted compact-tube branch, the same implemented 132-row residual and proof norm, the same endpoint-output map $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$, the scaled branch-solver condition $\eta_h^{\text{tube}} \leq c_\eta h^7$, and the same reported grid $t_n = nh$, $0 \leq n \leq N_h$, with accepted transitions $0 \leq n < N_h$ and with $y_n \in \mathcal{D}_h$ together with exact-state local-defect admissibility, denoted $y(t_n) \in \mathcal{D}_h$ in the same admissible-graph sense, for each transition input. This common accepted-domain membership is a theorem hypothesis/bootstrap condition, not a filter chosen after a run. No leftover terminal-time value, fixed production tolerance, residual-only mechanism row, source-policy row, or reproducibility package state is part of this invocation. In particular, no reported fixed-tolerance row instantiates P6 unless its branch membership and proof-norm residual maximum satisfy the

stated h^7 envelope on the same accepted transition set. Failure of any domain condition excludes that step or grid from the theorem domain; it does not create an alternative lower-order theorem.

Table 35: Accepted-branch consistency for Theorem 1.

Proof object	Branch identity used	Consistency source	Explicit exclusion
Gauss predictor	Same reduced-chart initial state y and same compact proof tube.	Assumption 1 and the reduced Gauss collocation map G_h .	Remote charts and nonsmooth contact events.
Accepted stage root	Local root selected from the Gauss-predictor ball on the compact tube.	Compact-tube inverse condition and Lemma 39.	Arbitrary Newton roots or global nonlinear solver uniqueness.
Endpoint correction	Endpoint closure applied to that accepted local root before the inexact Newton term.	Lemma 6 and its local ball/right-inverse hypotheses.	Extra terminal solve or unreported leftover interval.
Inexact Newton iterate	Newton iterate remains on the retained branches with $\eta_h^{\text{tube}} \leq c_\eta h^7$.	Lemma 41 and retained P6 branch-solver condition.	Fixed production tolerances as asymptotic proof.
Reported output grid	Reported states y_n are generated by repeated accepted-branch transitions on $0 \leq n < N_h$.	Lemma 44 on the same reported grid.	Residual/reaction tables, source-policy rows, and terminal-time claims at T .

Accepted-branch consistency rule: every theorem bound is tied to the same compact-tube branch selected by the Gauss predictor, the local accepted stage root, endpoint closure, and retained inexact-Newton solver scale. The branch-consistency statement does not claim remote-chart equivalence, arbitrary Newton-root selection, global nonlinear solver uniqueness, fixed-tolerance asymptotic proof, residual-to-error promotion, source-policy reproduction, or terminal-time error at an unreported T .

Nonlinear-solver scale rule: the proof consumes the compact-tube $\eta_h^{\text{tube}} \leq c_\eta h^7$ policy only as a retained P6 theorem condition in the fixed proof residual norm. Implementation logs bind to that norm only through the accepted implementation path and compact-tube norm equivalence. The reported-grid maximum η_h^{reported} and the finite scaled-tolerance probes are branch-scale diagnostics and trajectory specializations only for a P6 instance that has already fixed the tube, branch, norm, and c_η . They are not a theorem-level solver-policy proof, a fixed-tolerance asymptotic proof, global Newton convergence, arbitrary Newton-root selection, a second residual budget, source-policy reproduction, residual-to-error promotion, or reproducibility package readiness evidence. Solver-envelope quantifier discipline: the scalar solver-

Table 36: Nonlinear-solver scale conditions for Theorem 1.

Solver-scale object	Theorem role	Mathematical condition and evidence boundary	Explicit non-use
Compact-tube policy	$\eta_h^{\text{tube}} \leq c_\eta h^7$ uniformly on the retained branch-selected compact tube.	Retained P6 theorem condition used by Lemma 41.	Fixed production tolerances or finite probes as theorem-level proof.
Proof-norm binding	The $F_{A,h}$ -residual is measured in the fixed proof norm of the accepted 132-row residual after chart and row-scaling conventions are set.	Implementation logs can be compared with η_h^{tube} only under the accepted implementation-path binding and compact-tube norm equivalence.	A second residual budget, residual-log calibration of constants, or proof of P6 from logged residuals.
Stopping proxy	Any correction-norm, relative-residual, iteration-count, or backend success criterion used by an implementation.	It must be converted in advance to the same absolute $F_{A,h}$ -residual bound $c_\eta h^7$ on the accepted branch.	Proxy convergence flags, machine-precision floors, or adaptive stopping rules as theorem inputs.
Reported-grid specialization	$\eta_h^{\text{reported}} = \max_{0 \leq n < N_h} \eta_{h,n}$ over accepted branch-selected Newton solves.	Trajectory specialization only after the compact-tube policy, proof norm, branch rule, and c_η have already been fixed.	Using the reported maximum itself to prove the compact-tube policy, choose c_η , admit arbitrary Newton initializations, remote roots, or extra terminal solves.
Finite solver diagnostics	One-step scaled-tolerance probe, short-trajectory scaled-tolerance probe, and tolerance-regime sweep.	Diagnose branch-scale consistency and are attached only to P6.	Closing the compact-tube policy for every retained state or the full reported campaign.
Local-defect absorption	$C_N \eta_h^{\text{tube}} \leq C_N c_\eta h^7$ enters $C_{\text{loc}} = C_G + C_A + C_E + C_N c_\eta$.	Only after the retained compact-tube solver-scale condition is assumed.	Fitting C_N , c_η , or C_{loc} from residual tables or h-sweeps.
Retained solver-policy condition	The theorem is an order-six theorem under the stated compact-tube, endpoint/stability, implementation-binding, and solver-scale interfaces.	Recorded separately from direct stage-residual closure.	Full source-policy archive status, source-policy closure, or residual-to-error promotion.

envelope constant and fixed proof norm are chosen before h , N_h , or finite logs. Changing that constant after logs changes the theorem hypothesis and local-defect constant, not a proof of P6. The accepted branch and the strong-local-inverse ball are fixed at the same time. Reported residual ratios can diagnose finite compatibility with the retained P6 scale, but they cannot calibrate c_η , select a different Newton branch, or turn failed branch-scale rows into theorem inputs.

Table 37: Local-defect decomposition for Theorem 1.

Triangle term	Bound used	Proof source	Explicit non-use
Gauss truncation	$\ \Psi_h^G(y) - \varphi_h(y)\ \leq C_G h^7$.	Classical three-stage Gauss collocation on the smooth reduced flow.	Implemented residual tables or finite-run h-sweeps.
Stage-root perturbation	$\ \Psi_h^A(y) - \Psi_h^G(y)\ \leq C_A h^7$.	Lemma 40 supplies the $C_R h^7$ residual from the 96-row non-dynamic certificate plus the 36-row D5 direct-substitution zero certificate; Lemma 39 then uses the compact-tube inverse/Kantorovich step.	Remote nonlinear roots, primitive dynamic symbolic oracle closure, or source-paper residual reproduction.
Endpoint closure	$\ \Psi_h^E(y) - \Psi_h^A(y)\ \leq C_E h^7$.	Lemma 6 with local right inverse, endpoint-ball margin, and raw endpoint defect.	Global chart equivalence, residual-to-error promotion, or terminal-time extra solve.
Inexact Newton termination	$\ \Psi_h^{\text{G6FVA}}(y) - \Psi_h^E(y)\ \leq C_N \eta_h^{\text{tube}} \leq C_N c_\eta h^7$.	Lemma 41 plus the retained compact-tube branch-solver scale, applied to the full endpoint-output map $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$.	Fixed production tolerances, global Newton convergence, or arbitrary initial guesses.
Local-defect sum	$C_{\text{loc}} = C_G + C_A + C_E + C_N c_\eta$.	Triangle inequality over the four accepted-branch terms above.	Local-to-global grid transfer, source-policy rows, or residual/reaction diagnostics.

Local-defect decomposition rule: the local h^7 statement is obtained only by summing the four displayed accepted-branch terms: Gauss truncation, stage-root perturbation, endpoint closure, and inexact Newton termination under the compact-tube branch-solver scale. The local-defect sum does not use finite-run h-sweeps, residual/reaction tables, source-policy rows, remote nonlinear roots, global Newton convergence, residual-to-error promotion, or full-TFE readiness.

Norm/row-scaling bridge: the proof residual norm, stage-root norm,

reduced-chart norm, and reported q, v norm are not interchangeable. The fixed 132-row residual norm is used only inside the stage-residual perturbation and inexact-Newton estimates. The compact-tube uniform inverse maps that residual norm to the local stage-root norm; the endpoint-output map $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$ maps stage perturbations to the reduced-chart one-step norm, including the local endpoint-closure Lipschitz factor; the local-to-global transfer remains in that reduced-chart norm; and the reporting map applies the separate compact-tube factor $C_{\mathcal{R}}$ only after the reduced-chart grid estimate is proved. All row scalings, chart norms, and norm-equivalence constants are fixed before h_0 and before $C_{\text{loc}}, C_{\text{red}}, C_{qv}$ are chosen. In particular, the row scaling is not reweighted with h, N_h , backend, or residual logs; it cannot move powers of h between residual rows and constants. No residual-log norm, backend norm, mechanism table norm, or fitted work-precision scale is substituted into the four-term local-defect sum.

Table 38: Theorem output scope for Theorem 1.

Output layer	Proven statement	Proof source	Explicit non-output
One-step accepted map	$\ \Psi_h^{\text{G6FVA}}(y) - \varphi_h(y)\ \leq C_{\text{loc}} h^7.$	Four-term local defect decomposition on the accepted compact-tube branch.	Residual/reaction tables, source-policy rows, remote nonlinear roots.
Same-initial-state reduced grid	$\max_{0 \leq n \leq N_h} \ y_n - y(t_n)\ \leq C_{\text{red}} h^6.$	Local-to-global transfer on same-initial-state reported output indices with $N_h h \leq T$.	Unreported terminal-time error, leftover interval, or extra terminal solve.
Reported position-velocity map	$\max_{0 \leq n \leq N_h} \ \mathcal{R}(y_n) - \mathcal{R}(y(t_n))\ \leq C_{qv} h^6.$	The compact chart-to- (q, v) reporting map after the reduced-chart estimate.	Remote chart equivalence, nonsmooth contact events, or residual-to-error promotion.
Method-order sentence	Order six in the same-initial-state reported-grid sense under the retained branch, solver-scale, and compact-tube conditions.	The one-step, reduced-chart, and reporting-map bounds above.	External superiority, full source-policy reproduction, full-TFE readiness, or fixed-tolerance asymptotic proof.

Theorem output scope rule: the theorem outputs are exactly the branch-selected one-step h^7 local defect, the reduced-chart reported-grid h^6 estimate, the reported position-velocity h^6 estimate, and the resulting same-initial-

state reported-grid order-six sentence under the retained conditions. The theorem output scope does not include residual/reaction table promotion, unreported terminal-time error, remote-chart equivalence, external superiority, full source-policy reproduction, full-TFE readiness, or fixed-tolerance asymptotic proof.

Constraint-multiplier and reaction-output fence. The multiplier step in the Wieloch–Arnold constrained-BDF proof is a proof slot for its own hidden-constraint and multiplier-error estimate; it is not a theorem output imported here. In the present FullVA proof, the stage multipliers λ are algebraic unknowns used to close the lower-pair and Newton–Euler rows inside $F_{A,h}$. They enter the accepted theorem only through the 132-row residual bridge, the endpoint-closure lemma, and the compact-tube right-inverse hypotheses. The theorem does not assert an h^6 trajectory estimate for multiplier histories, reconstructed reaction forces, reaction residuals, or constraint residual tables. Those quantities remain mechanism-coverage diagnostics unless a separate multiplier/reaction reporting theorem supplies its own output map, uniform inverse or inf-sup constant, branch/solver envelope, reference policy, and local-to-global recursion. Consequently, small reaction residuals can support consistency of the implemented FullVA rows, but they cannot be read as a residual-to-error promotion or as a multiplier-order claim of Theorem 1.

Reporting-map/norm-equivalence rule: the final reported q, v estimate consumes only the reduced-chart grid error and the compact chart-to- (q, v) C^1 derivative bound on the accepted tube. It is a local reporting-map consequence on the same reported grid, not a residual-to-error transfer theorem, reaction diagnostic, global chart equivalence, remote branch statement, source-policy reproduction, fitted-constant claim, or terminal-time claim at an unreported T .

The reported-grid conclusion is not an interpolation statement. The sequence y_n is compared only with $y(t_n)$ at the reported output indices $0 \leq n \leq N_h$ chosen under $N_h h \leq T$. No bound is asserted for off-grid sampling times, dense output, resampled work-precision curves, or a postprocessed terminal value unless those times are added as reported grid points and the same branch/tube hypotheses are verified there. The constants C_{red} and C_{qv} are selected before the reported trajectory is read and do not depend on output formatting or interpolation policy.

Constant-dependency rule: the theorem constants are compact-tube constants only. They are not fitted from the finite h-sweep, solver

Table 39: Reporting-map/norm-equivalence table for Theorem 1.

Reporting step	Input estimate	Uniform map property	Explicit non-use
Reduced-chart grid	$\max_{0 \leq n \leq N_h} \ y_n - y(t_n)\ \leq C_{\text{red}} h^6.$	Local-to-global transfer on the accepted reduced-chart branch.	Residual/reaction tables, source-policy rows, or sweep fits.
Reporting map definition	$\mathcal{R}(y) = (q(y), v(y))$ on the same compact proof tube.	The accepted chart-to- (q, v) map is C^1 on that tube.	Global atlas equivalence or remote chart switching.
SO(3) orientation convention	Orientation or angular-velocity components inside q, v , when reported.	The same local retraction/log chart and component weights fixed before the theorem.	Atlas-wide geodesic distance, chart cuts, or post-hoc metric changes.
Derivative bound	$\ D\mathcal{R}\ \leq C_{\mathcal{R}}.$	Compact-tube supremum of the reporting-map derivative.	Fitted constants, dense/sparse backend behavior, or fixed tolerances.
Reported position-velocity estimate	$\max_{0 \leq n \leq N_h} \ \mathcal{R}(y_n) - \mathcal{R}(y(t_n))\ \leq C_{qv} h^6.$	$C_{qv} = C_{\mathcal{R}} C_{\text{red}}$ after the reduced-chart estimate.	A new dynamic estimate or residual-to-error transfer theorem.
Reported grid scope	The same output indices $0 \leq n \leq N_h$ and $t_n = nh.$	The reporting map is applied only to the proved grid-point estimate.	Terminal-time claim at an unreported T or an extra terminal solve.

Table 40: Constant-dependency table for Theorem 1.

Quantity	Proof source	Allowed dependencies	Explicit non-dependencies
h_0	Common small-step threshold for the compact tube, Gauss truncation, stage-root inverse, endpoint closure, and branch-selected Newton estimates.	Compact proof tube, regularity margins, local inverse radii, endpoint-ball margin, and branch-selection radius.	Reported grid length N_h , dense/sparse backend, source-policy rows.
C_{loc}	$C_G + C_A + C_E + C_N c_\eta$, the four-term one-step defect sum.	Gauss truncation, direct stage-root perturbation, endpoint closure, strong local residual inverse, and the retained compact-tube solver scale.	Finite h-sweep fits, fixed production tolerances, residual/reaction tables.
C_{red}	$C_{\text{loc}} \Gamma_s(T)$, the reduced-chart grid constant from the local-to-global transfer lemma.	The horizon T , the nonnegative stability constant C_s , and the accepted tube-retention bootstrap.	Exact final-time divisibility, unreported leftover intervals, external comparison rows.
C_{qv}	$C_{\mathcal{R}} C_{\text{red}}$, the reporting-map constant for (q, v) .	The C^1 chart-to- (q, v) map on the retained compact proof tube and the reduced-chart estimate.	Remote charts, nonsmooth contact events, local reproducibility record state.

tolerance table, source-policy comparison, residual/reaction table, or local reproducibility record. The table records analytic constant dependencies; it does not prove a theorem-level η_h solver-policy theorem, a P7 residual-to-error transfer, source-policy readiness, or full-TFE readiness. Constant-dependency rule keyword: compact-tube constants only.

Constant-selection hierarchy: first fix the mechanism data, the reported smooth branch, the compact proof tube, the reduced chart, the Gauss-predictor branch rule, the retained P1–P3 contracts, the P4 binding contract, the P6 solver-scale contract, and the P5 direct-route certificate. Then choose h_0 small enough for the Gauss truncation bound, the local stage inverse, the endpoint right-inverse ball, the inexact-Newton neighborhood, and the tube-retention bootstrap to hold on that same tube. Then choose the theorem constants as suprema or algebraic combinations of those fixed compact-tube bounds. Only after these choices is a reported grid $N_h h \leq T$ considered. No constant is re-fit per step size, per backend, per mechanism-coverage row, or from observed slopes, residual logs, or work-precision tables.

Lemma 50 (Order of the local paper-style TFE target). *The proof-level order target for the $m = 3$ Gauss–Lobatto paper-style TFE formula used in this manuscript is five.*

Proof. The source-paper family is compared through the temporal finite-element formula-order convention $p_{\text{TFE}}(m) = 2m - 1$ in Eq. (34). Substituting $m = 3$ gives $p_{\text{TFE}}(3) = 5$. This is the formula-order comparator used here. It is distinct from the source paper’s reported DAE pendulum observation, where order reduction can make the implemented $m = 3$ run appear fourth order. $\square$

Proposition 2 (Conditional order comparison). *Under Assumption 1 and the current reproducibility contract, Gauss6/FullVA has higher formal order than the local paper-style $m = 3$ Gauss–Lobatto TFE formula target in the narrow order-comparison sense.*

Proof. Theorem 1 gives the conditional route to the sixth-order method claim for Gauss6/FullVA. Lemma 50 gives the fifth-order local paper-style $m = 3$ comparator. This theorem–lemma pair is the formal-order proof of the proposition. The generated h-sweep records observed smooth position and velocity orders 7.161 and 7.066 only as consistency evidence after that comparison;

those finite-window slopes are not proof inputs. The four-example validation criterion is a mechanism-coverage criterion, not a single order theorem: the single- and double-pendulum dynamic FullVA rows provide dynamic order evidence, while the four-link and slider-crank rows are driven closed-loop kinematic/reaction checks whose trajectory differences sit at the roundoff floor and are not interpreted as method-order estimates. Therefore the conditional formal-order comparison is supported only inside the smooth accepted dynamic path specified in Assumption 1 and Theorem 1. The comparison inherits the compact-tube branch selection and solver-scale restrictions of Theorem 1: Newton is initialized from the Gauss predictor on the accepted branch, and the nonlinear residual budget is the theorem-level $\eta_h^{\text{tube}} \leq c_\eta h^7$ policy. It is not a theorem about arbitrary Newton initializations, remote roots, fixed production tolerances, or global solver convergence. The proposition does not compare error constants, wall-clock performance against independently implemented baselines, robustness in the nonsmooth sharp-friction regime, four-link/slider-crank dynamic convergence order, or complete source-paper residual reproduction. Those exclusions define the claim boundary; in particular, the method is not a complete source-paper temporal finite-element residual replacement. $\square$

Corollary 6 (Non-claim). *Proposition 2 does not prove that the implemented nonlinear residual is the complete source-paper temporal finite-element residual.*

Proof. The accepted method solves Eq. (6). A complete source-paper residual reproduction would require replacing every stage row in that solve by accepted paper-derived weak rows and passing the same validation criteria. That stronger criterion remains open. $\square$

Definition 8 (P7 boundary-certificate template). *For a mechanism-coverage row family on a fixed accepted branch, an accepted P7 residual-to-error certificate would be a same-branch tuple*

$$\mathcal{T}_{\text{P7}} = (\mathcal{R}_h^{\text{mech}}, \|\cdot\|_{\text{traj}}, C_{\text{P7}}, \rho_{\text{res}}, \varepsilon_{\text{floor}}, \varepsilon_{\text{est}}, \mathcal{G}_{\text{dyn}}). \quad (281)$$

and this tuple is fixed before the residual table is inspected, with the following properties. Equation (281) fixes the object whose transfer bound is Eq. (282). First, $\mathcal{R}_h^{\text{mech}}$ is the residual operator actually reported for the mechanism rows, with the same branch, time grid, row scaling, and norm used in the

table. Second, $\|\cdot\|_{\text{traj}}$ is the trajectory error norm in which a dynamic-order row would be claimed. Third, C_{P7} is independent of h on the same compact branch. Fourth, $\rho_{\text{res}}(h)$ is an a priori residual-consistency rate for inserting the exact branch into the mechanism residual rows. Fifth, $\varepsilon_{\text{floor}}(h)$ and $\varepsilon_{\text{est}}(h)$ bound, respectively, reference-floor and estimator/calibration errors on the same reported campaign. Sixth, $\mathcal{G}_{\text{dyn}}$ supplies the dynamic identity, closed-loop DAE stability or inf-sup estimate, and output binding needed to connect residual rows to trajectory variables. Seventh, the dynamic-order campaign has a branch-compatible refinement set on which all of these constants and rates are valid. The stability or inf-sup clause is a nullspace exclusion. It must rule out same-branch perturbations that are invisible to the reported mechanism residual but visible in the trajectory norm; otherwise a constraint or reaction residual can be small while the reported position-velocity error is not controlled. Thus the residual operator, row weights, gauge conventions, and trajectory output variables must be fixed together before any residual row is promoted.

The certificate would have to imply Eq. (282), uniformly for all sufficiently small reported step sizes in that campaign,

$$\|e_h\|_{\text{traj}} \leq C_{\text{P7}} \left(\|\mathcal{R}_h^{\text{mech}}\|_{\text{res}} + \rho_{\text{res}}(h) + \varepsilon_{\text{floor}}(h) + \varepsilon_{\text{est}}(h) \right). \quad (282)$$

Definition 8 is a closure criterion, not a theorem premise of Theorem 1: no such tuple is proved or invoked in this manuscript.

Corollary 7 (Residual-to-error transfer boundary). *The four-link and slider-crank residual/reaction rows in the validation section, and the common-reference or source-policy residual rows reported elsewhere, do not become accepted dynamic-order or trajectory-error rows in this manuscript.*

Proof. This exclusion is not a residual-transfer theorem. Those rows are mechanism-coverage evidence for the accepted FullVA residual vocabulary, not residual-to-error evidence for the reported trajectory norm. Promoting them to trajectory-error/order evidence would require the seven residual-to-error obligations in Table 30 to close on the same branch: dynamic residual identity, residual-consistency rate, closed-loop DAE stability or inf-sup control, calibrated estimator, reference-floor exclusion, coarse-first dynamic-order campaign, and a manuscript transfer theorem. Equivalently, the closure would have to instantiate the same-branch tuple in Definition 8. Since that joint criterion remains open, the rows remain diagnostics. Equivalently,

that separate promotion would require an additional same-branch residual operator with a uniform inverse, fixed independently of the observed residual table. More formally, a proof of such a transfer would first have to define a same-branch residual transfer operator on the reported closed-loop branch, a trajectory norm $\|e_h\|_{\text{traj}}$ matching the claimed order rows, and an h -independent constant C_{P7} such that, for all sufficiently small h in the claimed dynamic-order campaign, the transfer bound in Eq. (283) would have to hold:

$$\|e_h\|_{\text{traj}} \leq C_{\text{P7}} \left(\|\mathcal{R}_h^{\text{mech}}\|_{\text{res}} + \rho_{\text{res}}(h) + \varepsilon_{\text{floor}}(h) + \varepsilon_{\text{est}}(h) \right), \quad (283)$$

with the residual identity, residual-consistency rate, floor exclusion, and estimator terms all attached to the same branch. The present residual and reaction tables provide none of this transfer operator: they report small mechanism residuals and consistent reactions, but they do not provide the uniform inverse/stability constant or the reference-floor and estimator closure needed to convert those residuals into trajectory-error order. In particular, they do not exclude residual-null directions on the reported closed-loop branch. A perturbation that changes the reported trajectory while leaving the chosen residual/reaction rows unchanged would violate any uniform P7 transfer estimate, so finite residual smallness is not enough to define the missing inverse. Equivalently, PC4 is not a residual-to-error closure statement. The manuscript retains the scope rule that these rows are not promoted; this retained boundary does not supply any stability, estimator, floor-exclusion, campaign, or manuscript-theorem component of the open P7 route. $\square$

7.1. External comparison boundary

External source-paper comparison is kept outside the main method and proof claims. The accepted evidence below supports the branch-selected Gauss6/FullVA method, the conditional sixth-order theorem, and method-side mechanism coverage. The TFE and public-code source-policy rows are retained only as diagnostic boundary records; they close no source-paper superiority row and no theorem input. The detailed same-test promotion criterion, common-reference tables, and source-policy runner gaps are recorded in Section Appendix A so that the main numerical section is read in the intended evidence order: dynamic-order evidence first, mechanism-coverage evidence second, external-comparison diagnostics last.

Table 41: Proof-dependency map for Proposition 2.

Theorem component	Mathematical input	Role in the claim
Accepted one-step map	132-row residual, one-step algorithm, and endpoint closure in Sections 5–6.	Definition fixed for the theorem and numerical protocol.
Endpoint closure perturbation	Lemma 6: closed endpoint-anchor family, uniform endpoint-ball margin, local right-inverse perturbation, and raw endpoint defect.	Retained; local $O(h^7)$ map perturbation, not a substitute for the stage-row defect certificate.
Sixth-order method path	The mathematical input is the conditional criterion in Theorem 1.	Analytic criterion under the theorem hypotheses; the reported 7.161/7.066 slopes and finite-run scale check are numerical consistency evidence only, not proof closure or solver proof.
Local-to-global transfer	Lemma 44: one-step $O(h^7)$ defect, $1 + C_s h$ stability, and tube retention for $N_h h \leq T$.	Grid-point transfer; constants independent of h, N_h ; no unreported terminal-time error when $N_h h < T$.
Comparator order	Lemma 50 states the local $m = 3$ expected order $2m - 1 = 5$ and separates it from source-paper residual reproduction.	Formal-order comparator only.
Four examples	The four ASME-style examples in Section 9 check the accepted mechanism-coverage policy.	Mechanism coverage; four-link/slider-crank rows are residual/reaction checks, not order-proof rows.
Residual-to-error promotion	No residual-to-error transfer theorem is accepted for residual-only mechanism or source-policy rows.	Excluded output boundary; residual tables are diagnostics, not smooth-branch theorem inputs.
Non-claims	Complete source-paper residual reproduction and full-TFE stage replacement remain excluded from the accepted theorem.	Excluded stronger criterion.
Stage-row perturbation	Lemmas 7–39 plus the 96-row certificate and 36-row direct Newton–Euler identity on Z_G .	Direct 132-row stage-residual input; symbolic defect certificate remains a provenance record, not the route that discharges PC2.
Implementation fidelity	One residual/Jacobian path, 132-row formula-row checks, finite-probe AD checks, and AD-expanded Newton–Euler derivative-cell closure.	Accepted implementation path for the stage-residual proof; not a primitive symbolic defect certificate and not a source-paper residual replacement.
Branch selection	Assumption 1 and Lemmas 39–41 restrict the accepted map to the compact-tube Gauss-predictor Newton branch.	Local branch only; arbitrary Newton initializations, remote roots, and global solver convergence are not claimed.
Inexact Newton error	Lemma 41 states the local tolerance scale; finite probes remain diagnostics.	Theorem condition retained; fixed production runs do not close solver policy.

8. Numerical evidence

The numerical section should be read as a sequence of evidence blocks rather than as a single undifferentiated benchmark table. First, the smooth cylindrical-chain sweep tests the accepted 132-row Gauss6/FullVA stage solve and is finite-run consistency evidence after the conditional order-six theorem; it is not a premise that discharges the compact-tube, endpoint, or solver-scale hypotheses. Second, the four ASME-style mechanisms test the lower-pair FullVA vocabulary across driven, open-chain, and closed-loop settings; only the single and double pendula are used as accepted dynamic-order examples, while the closed-loop examples supply constraint, reaction, and coarse-window trajectory diagnostics only. Third, the source-paper and public-code rows are comparison diagnostics whose role is to expose what remains open in a fair same-test campaign. Finally, the work and solver tables make the evidence reproducible by recording Newton counts, tolerances, and runtime, but they do not by themselves turn the diagnostic baseline rows into superiority claims.

The recorded h-sweep uses $h = \{0.04, 0.02, 0.01\}$ and an $h = 0.005$ reference for the smooth cylindrical-chain benchmark. Table 42 gives the headline order evidence. Reference-solution boundary: every observed order reported from a finite h_{ref} row is a finite-reference consistency diagnostic, not a replacement for the exact-flow comparison used in Theorem 1. The reference step, output norm, and reported grid are fixed before the fit is computed. A finite reference row cannot discharge the compact-tube regularity, endpoint-closure, P6 solver-scale, or P7 residual-to-error boundaries, cannot be used to tune the reported norm after the fact, and cannot exclude a reference floor without a separate refinement or analytic-reference argument.

Table 42: Observed orders for the accepted Gauss6/FullVA path.

Case	Step sizes	Position order	Velocity order
Smooth cylindrical chain	0.04, 0.02, 0.01	7.161	7.066
Sharp friction, coarse sweep	0.04, 0.02, 0.01	1.894	2.683

Table 43: Smooth cylindrical-chain projected-endpoint errors.

h	Steps	Position error	Velocity error	Newton iterations
0.04	2	1.761×10^{-7}	1.007×10^{-5}	12
0.02	4	1.838×10^{-9}	2.954×10^{-8}	24
0.01	8	8.602×10^{-12}	5.609×10^{-10}	48

The slopes 7.161 and 7.066 in Tables 42–43 should not be read as a seventh-order method claim. The theorem above claims sixth order under an $O(h^7)$ local-defect contract. The larger fitted slopes come from a short, smooth, projected-endpoint window in which the leading sixth-order error coefficient is small relative to higher-order terms, while the finest row is already close to reference, nonlinear-tolerance, and floating-point floors. Consequently the paper treats these values as post-theorem consistency evidence for the order-six behavior on the accepted smooth path, not as a proof premise or a new asymptotic order. A wider asymptotic sweep with a compact-tube $\eta_h^{\text{tube}} \leq c_\eta h^7$ nonlinear tolerance would be needed before claiming more than order six, and the closed-loop four-link and slider-crank rows are still not promoted to dynamic order estimates.

Table 45 is retained only as a non-promotion record for the current source-paper comparator status. The paper-style column is an implemented all-row TFE diagnostic using the terminal Lobatto-node output, a consistent initial z_0 policy, and an equilibrated Newton linear solve. Its recorded smooth orders are 4.046 in position and 4.420 in velocity. Because the start policy is not a paper-derived recurrent policy and the full-TFE replacement criterion is not met by the current evidence package, this row is not an accepted fair baseline and is not part of the main order evidence.

For the asymptotic theorem, Table 47 should be read with the inexact-Newton policy of Lemma 41: a proof-level h-sweep must use the nonlinear residual threshold in Eq. (284),

$$\eta_h^{\text{tube}} \leq c_\eta h^7 \quad (284)$$

on the compact-tube branch, or an equivalent update criterion that implies the same stage-solution error. The fixed absolute tolerances in the recorded runs are retained for reproducibility of the reported runs and explain the reported Newton counts; they are not by themselves a derivation of the

Table 44: Order-estimate acceptance policy used in the numerical section.

Evidence row	Step/reference policy	Order information	Manuscript role
Smooth cylindrical chain	$h = 0.04, 0.02, 0.01$ against $h = 0.005$.	Position/velocity slopes 7.161/7.066.	Post-theorem finite-run consistency check for the conditional order-six theorem; not a proof premise or seventh-order claim.
Single pendulum	Accepted dynamic FullVA rows plus coarse public-window rows.	Minimum accepted dynamic order above six; coarse local position order about 6.05.	Accepted dynamic-order example and coarse-first comparison row.
Double pendulum	Method-side FullVA dynamic rows plus coarse public-window rows.	Minimum accepted method-side order above six; coarse public-window position/velocity slopes 7.951/7.042.	Accepted dynamic-order example and coarse-first comparison row.
Four-link and slider-crank	Closed-loop kinematic/reaction rows and strict common-reference coarse rows.	Local coarse trajectory-diagnostic candidates are near six, but floor-limited endpoint rows are not used as asymptotic order estimates.	Mechanism-coverage and work/precision evidence; not accepted asymptotic dynamic-order proof or an external dynamic-order claim.
Paper-style TFE all-row attempt	Same smooth cylindrical-chain steps as the accepted row.	Position/velocity slopes 4.046/4.420.	Internal diagnostic only; not a source-paper same-test row and not an accepted full-TFE replacement.
Public baseline families	Public-code and same-window rows where available.	Useful order/time rows, including nonconvergence rows for some coarse settings.	External-baseline evidence in progress; no external-superiority claim.

Table 45: Non-promoted same-step diagnostic with a TFE-style all-row attempt.

h	Accepted Gauss6/FullVA	Paper-style all-row TFE diagnostic	Diagnostic status
0.04	Position 1.761×10^{-7} ; velocity 1.007×10^{-5} ; 12 Newton.	Position 9.064×10^{-8} ; velocity 1.238×10^{-5} ; 13 Newton.	Not accepted.
0.02	Position 1.838×10^{-9} ; velocity 2.954×10^{-8} ; 24 Newton.	Position 3.378×10^{-9} ; velocity 1.931×10^{-7} ; 21 Newton.	Not accepted.
0.01	Position 8.602×10^{-12} ; velocity 5.609×10^{-10} ; 48 Newton.	Position 3.322×10^{-10} ; velocity 2.701×10^{-8} ; 37 Newton.	Not accepted.

Table 46: Work and solver diagnostics for accepted evidence rows.

Evidence row	Finest work	Runtime (s)	Finest accepted value
Smooth cylindrical chain	$h = 0.01$; 8 steps / 48 Newton	0.199	Position 8.602×10^{-12} ; velocity 5.609×10^{-10} .
Single pendulum	$h = 0.05$; 4 steps / 8 Newton	0.013	Position 7.374×10^{-12} ; max stage residual 3.664×10^{-12} .
Double pendulum	$h = 0.01$; 20 steps / 60 Newton	3.374	Orientation 1.597×10^{-15} ; angular velocity 4.867×10^{-15} .
Four-link closed loop	$h = 0.005$; 40 steps / 164 Newton	0.118	Max dynamics residual 1.338×10^{-13} .
Slider-crank closed loop	$h = 0.005$; 40 steps / 163 Newton	0.122	Max dynamics residual 6.492×10^{-15} .

Table 47: Nonlinear solver tolerances used for the reported evidence rows.

Evidence row	Nonlinear solve	Stopping rule	Finest count
Cylindrical chain	132-row Gauss6/FullVA residual with dense AD Jacobian and dense linear solve; rank diagnostics use tolerance 10^{-10} .	Stop when $\ R\ _2 < 10^{-11}$ or $\ \Delta Z\ _2 < 10^{-11}$; cap 80 Newton iterations.	48 Newton.
Single pendulum	Absolute-coordinate driven FullVA residual with AD Jacobian and dense linear solve.	Stop when $\ R\ _2 < 10^{-11}$ or $\ \Delta Z\ _2 < 10^{-12}$; cap 32 Newton iterations.	8 Newton.
Double pendulum	Imported double-revolute FullVA harness with three Gauss stages, AD Jacobian, and dense linear solve.	Stop when $\ R\ _2 < 10^{-11}$ or $\ \Delta Z\ _2 < 10^{-11}$; cap 64 Newton iterations.	60 Newton.
Four-link closed loop	Local fully constrained kinematic FullVA solve for $\Phi(q, t) = 0$ followed by linear solves for $\Phi_q \dot{q} = \nu$ and $\Phi_q \ddot{q} = \gamma$.	Position Newton stops when $\ \Delta q\ _2 < 10^{-12}$; cap 30 Newton iterations.	164 Newton; max 4/step.
Slider-crank closed loop	Same local closed-loop kinematic FullVA policy as the four-link example.	Position Newton stops when $\ \Delta q\ _2 < 10^{-12}$; cap 30 Newton iterations.	163 Newton; max 4/step.

asymptotic solver-error bound. A solver-scale check of the existing summary residuals records roundoff-scale linear and stage residuals, but the ratios to h^7 grow at the finest reference scale. A separate short-trajectory scaled-tolerance probe records the finite-log reported-grid maximum $\eta_h^{\text{reported}} := \max_{0 \leq n < N_h} \eta_{h,n} = \hat{\eta}_h^{\text{log}} \leq c_\eta h^7$ with $c_\eta = 10^4$ at $h = 0.04, 0.02, 0.01, 0.005$ through $T = 0.08$; all four diagnostic rows and all 30 trajectory steps pass the finite diagnostic filter, and the maximum final residual divided by h^7 is 210.891. This diagnostic pass is not a theorem-level P6 proof. It is a useful finite P6 diagnostic record, broader than a one-step smoke test, and it records reported-window consistency only rather than solver-policy closure. A companion finite tolerance-regime sweep compares fixed 10^{-10} , ch^7 , and ch^8 stopping rules against a local $h = 0.0025$ reference over the same window. The fixed and h^7 policies give position/velocity fits 6.946/6.608, while the h^8 policy gives 6.946/6.608. The two shrinking-target policies are separately recorded as eight finite h -scaled rows and 60 h -scaled trajectory steps, with position/velocity order floor 6.946/6.608. Thus, on this finite diagnostic window, the fixed-tolerance row and the h -scaled rows have the same recorded order floor; this is a finite-window tolerance comparison only, not P6 instantiation or a theorem-level solver-policy proof. The scaled policies exercise the theorem's shrinking tolerance form on this finite window, but the sweep is still not a theorem-level scaled-tolerance proof over every reported trajectory and not a scaled-tolerance proof under a uniform all-transition policy for the reported trajectory. The proof therefore remains explicitly conditional on the $\eta_h^{\text{tube}} \leq c_\eta h^7$ compact-tube policy. The reported-log quantity is only $\eta_h^{\text{reported}} = \max_{0 \leq n < N_h} \eta_{h,n}$, also denoted $\hat{\eta}_h^{\text{log}}$ in diagnostic tables, after the accepted branch, proof norm, row scaling, and c_η have already been fixed; it cannot select the branch, norm, or theorem constant.

Tables 46–47 and Table 45 are solver-scale and diagnostic comparison tables, not accepted fair baseline comparisons. Runtime values are single-machine recorded timings, and the mechanism rows report the accepted finest-step evidence for each validation path. A publication-level comparison still requires accepted implemented baselines and cross-method work/precision curves on the same problems; these tables and Figure 2 only prevent the accepted evidence from being reported without any work, tolerance, Newton-iteration, or source-paper diagnostic context. For the four-link and slider-crank rows, the finest-step residual and reaction entries are read through the P7 boundary stated after Theorem 1: they

demonstrate closed-loop mechanism coverage and consistency of the constructed FullVA rows, but they are not trajectory-order rows unless an accepted residual-to-error transfer is proved.

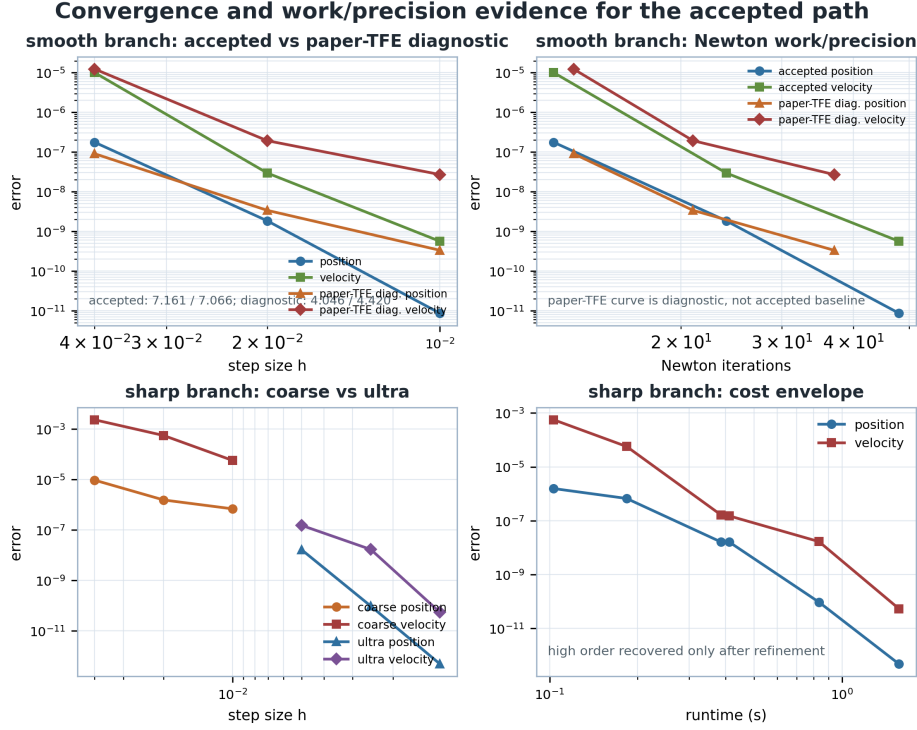


Figure 2: Convergence and work/precision evidence for the accepted Gauss6/FullVA path and the implemented source-paper all-row TFE diagnostic. The top row compares the smooth cylindrical-chain errors and Newton-work/error curves against the diagnostic paper-style all-row attempt; that curve is not an accepted fair baseline. The bottom row contrasts the sharp-friction coarse sweep with the refined cost envelope that recovers high order only at additional runtime.

The sharp-friction coarse sweep is not used as the main order claim. It is reported as a limitation: deeper fixed refinement recovers high order at increased cost, while the practical coarse regime remains order-reduced. The recorded sharp-friction diagnostic shows the practical tradeoff: the coarse $h = \{0.04, 0.02, 0.01\}$ sweep gives only 1.894/2.683 position/velocity order, while ultra refinement over $h = \{0.005, 0.0025, 0.00125\}$ against $h = 0.000625$ recovers 7.521/5.725 all-point position/velocity order. The high-order recovery requires an approximately $7.25\times$ runtime increase relative to the $h = 0.01$

baseline, so the paper keeps this as a practical cost caveat rather than a solved production setting.

9. Four-example mechanism validation

The method-side evidence covers four ASME-style examples, with different accepted roles. This section states the mechanism content used by the validation rows and keeps their evidentiary roles distinct. The examples are not treated as interchangeable benchmark names: each one exercises a different part of the FullVA residual. These validation rows are downstream evidence for method behavior; they do not feed back into the constants, hypotheses, or residual-certificate slot of Theorem 1. In particular, closed-loop residual and reaction rows remain coverage/diagnostic rows unless the separate P7 residual-to-error boundary is discharged.

9.1. Driven single pendulum

The single pendulum is a rigid bar of length 4.0 m, square side length 0.05 m, density 7800 kg/m³, and gravity $g = 9.81$ m/s². The center of mass is 2.0 m from the fixed pivot. The accepted driven trajectory is

$$\theta(t) = \frac{\pi}{2} + \frac{\pi}{4} \cos(2t), \quad (285)$$

Equation (285) defines the accepted driven single-pendulum trajectory, with the corresponding analytic $\dot{\theta}$ and $\ddot{\theta}$ used for reference values. The constraint set consists of a fixed pivot, two DP1 revolute-axis constraints, and a DP1 drive constraint. The strongest accepted row is the absolute-coordinate driven FullVA residual: each Gauss stage solves for orientation, center-of-mass position, velocity, acceleration, angular velocity, angular acceleration, pivot reaction force, axis multipliers, and drive multiplier. Its square residual includes pivot velocity and pivot acceleration rows, two revolute-axis rows, and the acceleration-level drive row.

9.2. Double pendulum

The double pendulum uses two rigid bars with lengths 4.0 m and 2.0 m, the same square side length, density, and gravity as the single-pendulum case, and two revolute joints. The first body is pinned to ground and the second body is pinned to the first body. Friction and external torques are disabled in the accepted method-side comparison. Because the available

legacy rA reference shows an h -independent discrepancy under the coordinate convention change, the accepted reference policy is a nested local Gauss6/FullVA self-reference at $h = 0.005$; the rA comparison is retained only as a diagnostic reference-floor test.

9.3. Four-link closed loop

The four-link example is a driven closed-loop lower-pair graph with three moving rigid bodies and 18 scalar constraints. The constraint inventory is 12 coordinate-difference rows and 6 DP1 rows. Since the graph is fully constrained, the accepted evidence is not a free dynamic convergence claim. At each output time the local FullVA solve enforces Eq. (286),

$$\Phi(q, t) = 0, \quad \Phi_q(q, t)\dot{q} = \nu(q, t), \quad \Phi_q(q, t)\ddot{q} = \gamma(q, \dot{q}, t), \quad (286)$$

then reconstructs multipliers and checks Newton–Euler translational and rotational residuals. The largest recorded dynamics residual is 1.338×10^{-13} .

9.4. Slider-crank closed loop

The slider-crank example is also a three-body fully constrained graph with 18 scalar constraints, but it includes prismatic geometry through DP2 and distance rows. Its inventory is 6 coordinate-difference rows, 7 DP1 rows, 4 DP2 rows, and one distance row. The accepted solve follows the same closed-loop FullVA policy as the four-link mechanism. The largest recorded constraint norm is 1.204×10^{-15} and the largest dynamics residual is 6.492×10^{-15} .

9.5. Coordinate-level mechanism specification

The setup table gives the physical scales; the following coordinate data make the lower-pair graphs reproducible directly from the manuscript. Let $e_x = (1, 0, 0)$, $e_y = (0, 1, 0)$, and $e_z = (0, 0, 1)$. All point vectors s and direction vectors a are body-fixed unless the body is ground; each CD group uses the three coordinate directions e_x, e_y, e_z . Driver functions listed here are the functions actually used by the setup code after model-file overrides.

Single pendulum. The moving body is body 1 and ground is body 0. The ground revolute joint uses CD rows with $s_1 = (-2, 0, 0)$ and $s_0 = (0, 0, 0)$, plus DP1 rows $(a_1, a_0) = (e_x, e_x)$ and (e_y, e_x) . The driven DP1 row uses $(a_1, a_0) = (e_y, -e_z)$ and $f(t) = \cos(\pi/2 + (\pi/4)\cos(2t))$.

Table 48: Four-example validation summary.

Example	Accepted evidence	Role	Headline metric
Single pendulum	Driven absolute-coordinate FullVA residual	Dynamic-order evidence	Minimum observed order 6.024; max stage residual 3.664×10^{-12} .
Double pendulum	Method-side FullVA reference policy	Dynamic-order evidence	Minimum orientation/velocity order 6.089; endpoint velocity constraints below 8.323×10^{-12} .
Four link	Closed-loop kinematic FullVA and reaction dynamics	Mechanism coverage only	Max constraint norm 1.052×10^{-14} ; max dynamics residual 1.338×10^{-13} .
Slider crank	Closed-loop kinematic FullVA and reaction dynamics	Mechanism coverage only	Max constraint norm 1.204×10^{-15} ; max dynamics residual 6.492×10^{-15} .

Table 49: Problem setup values used by the four ASME-style validation examples.

Example	Physical and constraint setup	Time grid and reference policy	Accepted check
Single pendulum	One steel bar, length 4 m, square side 0.05 m, density 7800 kg/m ³ , mass 78 kg, and $J = \text{diag}(0.0325, 104.016, 104.016)$ kg m ² . The graph has one ground revolute joint: CD(3) and DP1(3).	Driven angle $\theta(t) = \pi/2 + (\pi/4) \cos(2t)$, analytic reference, $t_f = 0.2$, and $h = \{0.20, 0.10, 0.05\}$.	Minimum observed order 6.024; max stage residual 3.664×10^{-12} .
Double pendulum	Two steel bars, lengths 4 m and 2 m, same side and density as the single pendulum, masses 78 kg and 39 kg, with two revolute joints and no friction or external torque.	$q(0) = (-\pi/2, 0)$ and regularized angular speeds $(10^{-12}, -10^{-12})$ rad/s. The accepted reference is nested local FullVA at $h = 0.005$; tested $h = \{0.04, 0.02, 0.01\}$.	Orientation and angular-velocity orders 6.101/6.089; endpoint velocity constraint below 8.323×10^{-12} .
Four-link	Three moving bodies with masses 2, 1, 1 kg and inertia diagonals $(4, 2, 0)$, $(12.4, 0.01, 0)$, and $(4.54, 0.01, 0)$ kg m ² . The closed loop has CD(12) and DP1(6).	Driven constraint $\cos(\pi t + \pi/2)$, $t_f = 0.2$, $h = \{0.02, 0.01, 0.005\}$, and local kinematic FullVA reference $h = 0.001$.	Max constraint norm 1.052×10^{-14} ; max dynamics residual 1.338×10^{-13} .
Slider-crank	Crank, rod, and slider masses 0.12, 0.5, 2 kg with inertia diagonals $(10^{-4}, 10^{-5}, 10^{-4})$, $(4 \times 10^{-3}, 4 \times 10^{-4}, 4 \times 10^{-3})$, and $(10^{-4}, 10^{-4}, 10^{-4})$ kg m ² . The graph has CD(6), DP1(7), DP2(4), and D(1).	Driven constraint $\cos(-2\pi t + \pi/2)$, $t_f = 0.2$, $h = \{0.02, 0.01, 0.005\}$, and local kinematic FullVA reference $h = 0.001$.	Max constraint norm 1.204×10^{-15} ; max dynamics residual 6.492×10^{-15} .

Double pendulum. The two links are bodies 1 and 2, with ground body 0. The ground revolute joint for body 1 uses CD rows with $s_1 = (-2, 0, 0)$ and $s_0 = (0, 0, 0)$, plus DP1 rows (e_x, e_x) and (e_y, e_x) . The inter-link revolute joint uses CD rows between body 1 and body 2 with $s_1 = (2, 0, 0)$ and $s_2 = (-1, 0, 0)$. The second axis pair is represented by DP1 rows on body 2 against ground: $(a_2, a_0) = (e_x, e_x)$ and (e_y, e_x) .

Four-link closed loop. The moving bodies are the rotor, link 2, and link 3, numbered 1, 2, and 3; ground is body 0. Revolute joint A between body 1 and ground uses CD rows with $s_1 = (0, 0, 0)$ and $s_0 = (0, 0, 0)$, plus DP1 rows (e_x, e_x) and (e_y, e_x) . Revolute joint D between body 3 and ground uses CD rows with $s_3 = (0, 3.7, 0)$ and $s_0 = (-4, -8.5, 0)$, plus DP1 rows $((0, 0, 1), e_y)$ and (e_y, e_y) . The spherical joint between bodies 3 and 2 uses CD rows with $s_3 = (0, -3.7, 0)$ and $s_2 = (0, -6.1, 0)$. The universal joint between bodies 1 and 2 uses CD rows with $s_1 = (-2, 0, 0)$ and $s_2 = (0, 6.1, 0)$, plus the DP1 row $((0, -0.662, 0.75), e_z)$. The driving DP1 row uses $((-1, 0, 0), e_y)$ and $f(t) = \cos(\pi t + \pi/2)$.

Slider-crank closed loop. The crank, connecting rod, and slider are bodies 1, 2, and 3; ground is body 4. Revolute joint A between ground and crank uses CD rows with $s_4 = (0, 0.1, 0.12)$ and $s_1 = (0, 0, 0)$, plus DP1 rows (e_y, e_x) and (e_z, e_x) . The spherical joint B between rod and crank uses CD rows with $s_2 = (-0.15, 0, 0)$ and $s_1 = (0, 0.08, 0)$. The revolute/cylindrical joint C between rod and slider uses one DP1 row (e_y, e_z) and two DP2 rows with rod point $s_2 = (0.15, 0, 0)$, slider point $s_3 = (0, 0, 0)$, and rod directions e_y and e_z . The translational joint D between slider and ground uses DP1 rows $(-e_x, e_x)$, (e_y, e_x) , and $(-e_x, e_y)$, plus DP2 rows from slider point $s_3 = (0, 0, 0)$ to ground point $s_4 = (0.2, 0, 0)$ with slider directions $-e_x$ and e_y . The distance row constrains the rod point $(0.15, 0, 0)$ and slider point $(0, 0, 1)$ with $f(t) = 1$. The driving DP1 row uses (e_y, e_y) and $f(t) = \cos(-2\pi t + \pi/2)$.

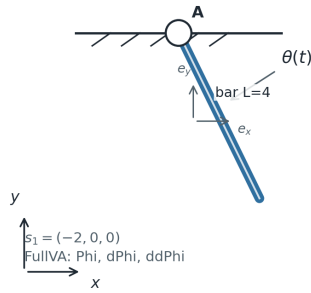
10. Lower-pair constraint formulas

This section makes explicit the scalar constraint rows used by the four example mechanisms. Let body i have center position r_i , rotation matrix $A_i \in SO(3)$, translational velocity v_i , angular velocity ω_i , translational acceleration a_i , and angular acceleration α_i . A marked point and a marked direction fixed in body i are $x_i = r_i + A_i s_i$ and $u_i = A_i a_i$, respectively. The

ASME-style lower-pair validation mechanisms

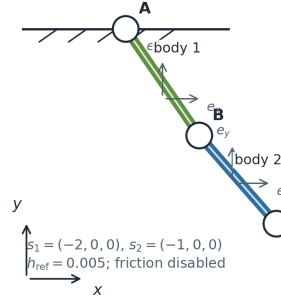
single pendulum

CD(3) + DP1(3); driven theta(t)



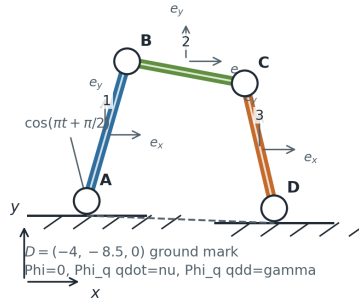
double pendulum

CD(6) + DP1(4); nested FullVA reference



four-link loop

CD(12) + DP1(6); driven closed loop



slider-crank

CD(6) + DP1(7) + DP2(4) + D(1)

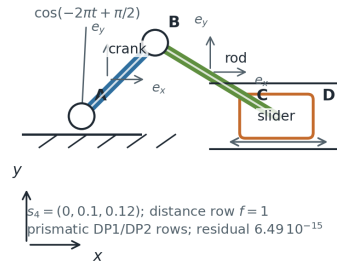


Figure 3: Coordinate-oriented mechanism schematics and lower-pair constraint inventory for the four ASME-style examples. Each panel shows the global coordinate frame, named joints, the principal moving bodies, the driver where applicable, and the CD/DP1/DP2/D constraint vocabulary used by the FullVA rows. Exact body-fixed point and direction vectors are listed in the text immediately preceding the figure.

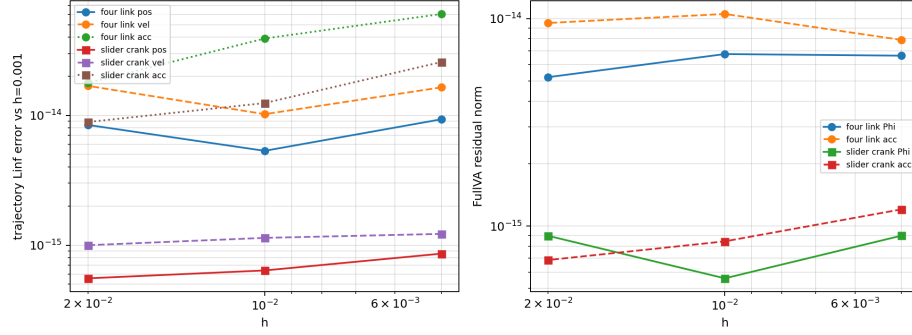


Figure 4: Closed-loop kinematic FullVA mechanism-coverage evidence for the four-link and slider-crank examples.

row families used in the examples are listed in Eq. (290):

$$C_{CD,k}(q, t) = e_k^T (x_j - x_i) - d_k(t), \quad (287)$$

$$C_{DP1}(q, t) = u_i^T u_j - c(t), \quad (288)$$

$$C_{DP2}(q, t) = u_i^T (x_j - x_i) - d(t), \quad (289)$$

$$C_D(q, t) = \|x_j - x_i\|^2 - \ell^2(t). \quad (290)$$

Here CD denotes coordinate difference, DP1 constrains two body-fixed directions, DP2 constrains a body-fixed direction against a point difference, and D is a distance row. Revolute joints are assembled from CD and DP1 rows; the slider-crank prismatic joint additionally uses DP2 and distance rows.

The velocity and acceleration formulas use the rigid-body identities in Eq. (292):

$$\dot{x}_i = v_i + \omega_i \times A_i s_i, \quad \dot{u}_i = \omega_i \times u_i, \quad (291)$$

$$\ddot{x}_i = a_i + \alpha_i \times A_i s_i + \omega_i \times (\omega_i \times A_i s_i), \quad \ddot{u}_i = \alpha_i \times u_i + \omega_i \times (\omega_i \times u_i). \quad (292)$$

Writing $\Delta x = x_j - x_i$, $\Delta \dot{x} = \dot{x}_j - \dot{x}_i$, and $\Delta \ddot{x} = \ddot{x}_j - \ddot{x}_i$, the row derivatives assembled by the FullVA block are Eq. (296) at velocity level

$$\dot{C}_{CD,k} = e_k^T \Delta \dot{x} - \dot{d}_k, \quad (293)$$

$$\dot{C}_{DP1} = \dot{u}_i^T u_j + u_i^T \dot{u}_j - \dot{c}, \quad (294)$$

$$\dot{C}_{DP2} = \dot{u}_i^T \Delta x + u_i^T \Delta \dot{x} - \dot{d}, \quad (295)$$

$$\dot{C}_D = 2\Delta x^T \Delta \dot{x} - 2\ell \dot{\ell}, \quad (296)$$

and Eq. (300) at acceleration level:

$$\ddot{C}_{\text{CD},k} = e_k^T \Delta \ddot{x} - \ddot{d}_k, \quad (297)$$

$$\ddot{C}_{\text{DP1}} = \ddot{u}_i^T u_j + 2\dot{u}_i^T \dot{u}_j + u_i^T \ddot{u}_j - \ddot{c}, \quad (298)$$

$$\ddot{C}_{\text{DP2}} = \ddot{u}_i^T \Delta x + 2\dot{u}_i^T \Delta \dot{x} + u_i^T \Delta \ddot{x} - \ddot{d}, \quad (299)$$

$$\ddot{C}_{\text{D}} = 2\Delta \dot{x}^T \Delta \dot{x} + 2\Delta x^T \Delta \ddot{x} - 2(\dot{\ell}^2 + \ell \ddot{\ell}). \quad (300)$$

10.1. Scalar lower-pair Jacobian entries

The scalar Jacobian rows used by Eq. (30) are obtained by varying the marked points and directions in spatial form. Define the geometry variables by Eq. (301):

$$\rho_i = A_i s_i, \quad u_i = A_i a_i, \quad \Delta x = x_j - x_i. \quad (301)$$

Let ϑ_i be the spatial infinitesimal rotation associated with the stage perturbation $\delta\eta_i$ through Eq. (20). Then

$$\delta\rho_i = \vartheta_i \times \rho_i, \quad \delta u_i = \vartheta_i \times u_i, \quad (302)$$

$$\delta x_i = \delta r_i + \vartheta_i \times \rho_i, \quad \delta \Delta x = \delta x_j - \delta x_i. \quad (303)$$

Equation (303) supplies the point and direction variations used by the scalar lower-pair Jacobian rows. The position-level scalar Jacobian entries are therefore

$$\delta C_{\text{CD},k} = e_k^T \delta \Delta x, \quad (304)$$

$$\delta C_{\text{DP1}} = \vartheta_i^T (u_i \times u_j) + \vartheta_j^T (u_j \times u_i), \quad (305)$$

$$\delta C_{\text{DP2}} = \vartheta_i^T (u_i \times \Delta x) + u_i^T \delta \Delta x, \quad (306)$$

$$\delta C_{\text{D}} = 2\Delta x^T \delta \Delta x. \quad (307)$$

Equation (307) gives the translational columns directly through $\delta r_i, \delta r_j$ and the rotational columns through the map $\delta\eta \mapsto \vartheta$ in Eq. (20). For driven rows, the prescribed functions $d_k(t)$, $c(t)$, $d(t)$, and $\ell(t)$ are fixed during the Newton solve at a stage time and therefore do not add unknown columns.

The velocity-level Jacobian follows by differentiating the velocity rows with the perturbations

$$\delta \dot{x}_i = \delta v_i + \delta \omega_i \times \rho_i + \omega_i \times \delta \rho_i, \quad (308)$$

$$\delta \dot{u}_i = \delta \omega_i \times u_i + \omega_i \times \delta u_i, \quad (309)$$

$$\delta \Delta \dot{x} = \delta \dot{x}_j - \delta \dot{x}_i. \quad (310)$$

Substitution of Eq. (310) into $\dot{C}_{\text{CD}}$, $\dot{C}_{\text{DP1}}$, $\dot{C}_{\text{DP2}}$, and $\dot{C}_{\text{D}}$ gives the exact $K^{\ddot{\Phi},q}$ and $K^{\ddot{\Phi},\dot{q}}$ blocks in Table 4. The acceleration-level Jacobian uses

$$\begin{aligned}\delta\ddot{x}_i &= \delta a_i + \delta\alpha_i \times \rho_i + \alpha_i \times \delta\rho_i \\ &\quad + \delta\omega_i \times (\omega_i \times \rho_i) + \omega_i \times (\delta\omega_i \times \rho_i + \omega_i \times \delta\rho_i),\end{aligned}\quad (311)$$

$$\begin{aligned}\delta\ddot{u}_i &= \delta\alpha_i \times u_i + \alpha_i \times \delta u_i \\ &\quad + \delta\omega_i \times (\omega_i \times u_i) + \omega_i \times (\delta\omega_i \times u_i + \omega_i \times \delta u_i),\end{aligned}\quad (312)$$

$$\delta\Delta\ddot{x} = \delta\ddot{x}_j - \delta\ddot{x}_i. \quad (313)$$

Substituting Eq. (313) into the four $\ddot{C}$ formulas above supplies the $K^{\ddot{\Phi},q}$, $K^{\ddot{\Phi},\dot{q}}$, and $K^{\ddot{\Phi},\ddot{q}}$ blocks. These formulas are the scalar CD/DP1/DP2/D Jacobian entries checked against the dense AD Jacobian on the accepted path.

For any assembled scalar row $\Phi(q, t) = 0$, these formulas are represented in the compact FullVA residual as position, velocity, and acceleration levels in Eq. (314),

$$\begin{aligned}\Phi(q, t) &= 0, \\ \Phi_q(q, t)\dot{q} + \Phi_t(q, t) &= 0, \\ \Phi_q(q, t)\ddot{q} + \frac{d}{dt}(\Phi_q(q, t))\dot{q} + \frac{d}{dt}\Phi_t(q, t) &= 0.\end{aligned}\quad (314)$$

Equivalently, the velocity and acceleration rows are written as $\Phi_q(q, t)\dot{q} = \nu(q, t)$ and $\Phi_q(q, t)\ddot{q} = \gamma(q, \dot{q}, t)$, with nonzero right-hand sides for driven constraints. This is the constraint vocabulary behind the single pendulum, double pendulum, four-link, and slider-crank evidence in Table 48.

The Newton unknown at each collocation stage is ordered as Eq. (315),

$$Z_i = (r_i, \eta_i, v_i, \omega_i, a_i, \alpha_i, \lambda_i), \quad (315)$$

where η_i is the local Lie-algebra rotation increment. The lower-pair Jacobian rows therefore have the block form in Eq. (316),

$$\delta R_{\text{lp}} = R_q \delta q_i + R_{\dot{q}} \delta \dot{q}_i + R_{\ddot{q}} \delta \ddot{q}_i + R_\lambda \delta \lambda_i, \quad (316)$$

with R_q containing CD/DP1/DP2/D position derivatives, $R_{\dot{q}}$ containing the velocity-level blocks, $R_{\ddot{q}}$ containing the acceleration-level blocks, and $R_\lambda = 0$ for pure kinematic rows. The Newton–Euler rows use the same $\Phi_q(q_i, t_i)^T \lambda_i$ columns in their force and torque balances. Thus the paper’s compact notation corresponds to the actual implementation-level row groups: collocation rows, velocity collocation rows, Newton–Euler balance rows, and FullVA lower-pair rows.

11. Diagnostic boundaries and limitations

This section states diagnostic boundaries; it is not a second contribution line and not a prerequisite for the theorem or the accepted validation rows. The method claim can be evaluated from the residual definition, Theorem 1, and Section 8; the diagnostics below are retained only to prevent stronger source-paper, full-TFE, sparse-performance, and external-superiority claims from being read into the main method evidence. The supplementary evidence contains additional diagnostic work toward source-paper temporal finite-element residual replacement. This evidence is useful because it identifies why a stronger reproduction claim is outside the present scope, but it is not needed for Proposition 2. The central diagnostic result is that several endpoint or weak-row variants can close one property at a time, but the reported evidence does not provide a source-free, projection-free, non-terminal-row-replacement 132-row TFE residual that both closes terminal endpoint velocity and preserves the smooth order target.

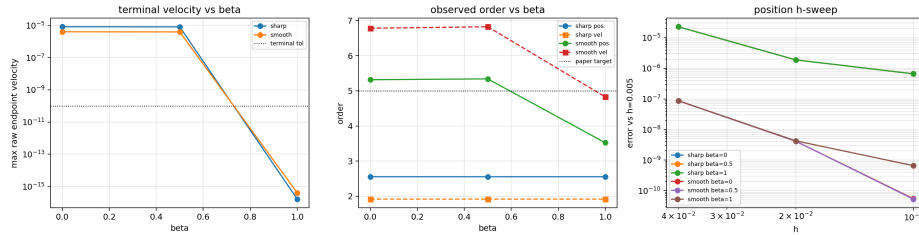


Figure 5: Order/closure blend diagnostic. The tested path separates order preservation from terminal endpoint-velocity closure, which is why independent full-TFE replacement remains open.

Sparse automatic differentiation is also treated as diagnostic engineering evidence rather than a primary paper claim. The accepted sparsity pattern is correct and quantified, but the recorded wall-clock measurements do not show a stable speed win over dense `jacfwd`. The current code should therefore be read as a reference implementation. The recorded small-system pattern has a 132×132 dense dimension, 2637 nonzeros, and 90/60 column/row colors; row-VJP is still 1.20–1.22 times the dense runtime, requiring a 16.9–18.3% row-runtime reduction merely to match dense. No body-count sweep for $N = 2, 4, 8, 16, 32$, memory scaling, conditioning-versus-body-count, or Jacobian/linear-solve breakdown is claimed.

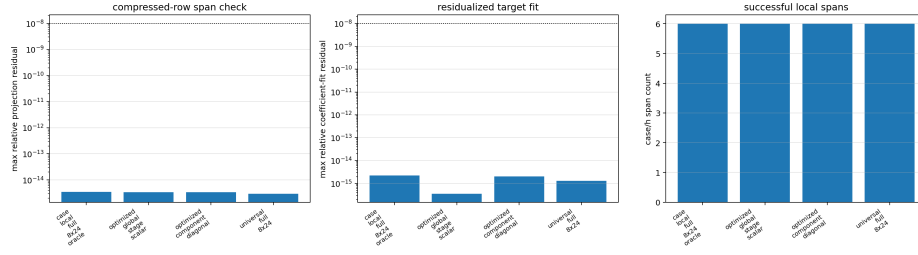


Figure 6: Velocity-compression diagnostic. Local row-space probes quantify the gap between available weak-row features and the terminal bridge needed for accepted full-TFE stage replacement.

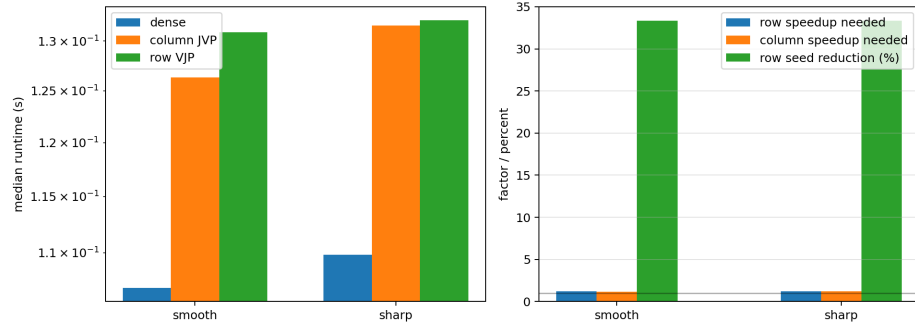


Figure 7: Sparse backend speed-gap diagnostic. The sparse structure is correct and quantified, but dense `jacfd` remains faster in the recorded wall-clock evidence.

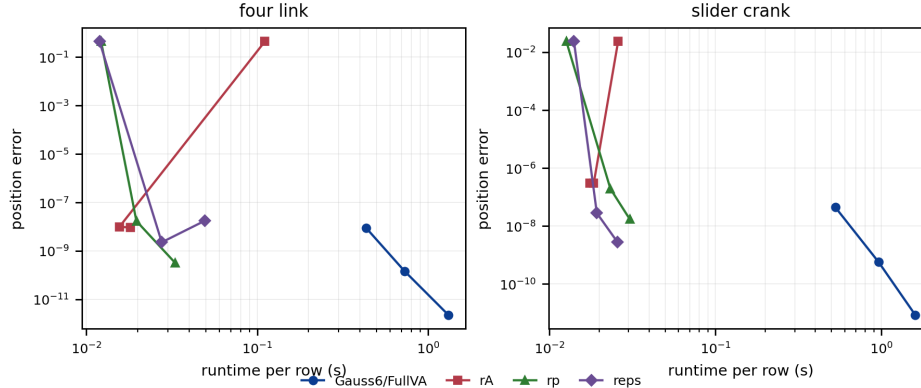


Figure 8: Strict common-reference work/precision rows for bounded diagnostics on the closed-loop four-link and slider-crank mechanisms. Bounded common-reference diagnostic work/precision rows use the same finite accepted-method endpoint reference at $h = 0.0125$ over the coarse $h = 0.1, 0.05, 0.025$ window. The plot removes one fixed-grid diagnostic caveat for these coarse rows, but it is still not an external superiority claim because the broader source-paper suites and manuscript-level review remain open.

Table 50: Diagnostic boundaries separated from the accepted theorem.

Diagnostic boundary	Current evidence	Claim boundary
Independent full-TFE replacement	No accepted 132-row paper-derived weak residual with source-free terminal closure and smooth order recovery.	Boundary retained.
Sparse backend speed	Sparse pattern and colors are quantified, but dense <code>jacfwd</code> is still faster in the recorded wall-clock run.	Backend caveat.
Sharp-friction coarse regime	Coarse sharp-friction sweep is order-reduced; deeper fixed refinement recovers high order at increased cost.	Regime caveat.

Claim boundary and remaining limitations	
accepted method claim	four-example evidence split
accepted Gauss6/FullVA order conditional sixth-order path; observed smooth slopes 7.161/7.066 are finite-window support, not a seventh-order theorem	✓ single dynamic order accepted
accepted mechanism coverage four ASME-style mechanisms are covered; dynamic order is accepted only for single and double pendula	✓ double dynamic order accepted
bounded source-paper comparison local $m=3$ TFE formula target has expected order five; complete source-paper residual replacement is outside the claim	! four-link coarse dynamics / common-ref diagnostics; residual-to-error theorem open
open external superiority not claimed until the same-test public-code campaign and work metrics are complete	! slider-crank coarse dynamics / common-ref diagnostics; residual-to-error theorem open
proof and solver limitations	same-test benchmark limitations
conditional conditional theorem requires regular reduced flow, smooth lift, isolated Newton solve, and stage residual defect $O(h^7)$	open original TFE pendulum source-paper pendulum rows remain to be reproduced on the same policies
direct route implementation oracle runtime layout, AD-expanded oracle, and direct dynamic row closure pass; primitive symbolic lane remains diagnostic	partial 2021 rA/rp/refs suite public baselines and coarse pilots exist; full dynamic work/precision comparison remains open
open nonlinear tolerance proof-level policy is $\eta_h = O(h^7)$; fixed absolute tolerances are finite-run evidence	partial 2022 half-implicit suite bounded pilot is present; full $T=8$ policy and method rows remain open
open residual-to-error route closed-loop residual rows are not promoted to order claims without the missing theorem	open velocity partitioning code path unresolved in the local public-code mirror and metadata search

Figure 9: Claim-boundary and limitation map for the present paper. Green items are accepted claims or accepted evidence rows; orange items are bounded or partial evidence; red items remain open boundaries. The figure summarizes why the manuscript supports a conditional sixth-order Gauss6/FullVA claim under the retained theorem interfaces but still does not claim complete source-paper residual replacement, external same-test superiority, or a closed residual-to-error transfer theorem for the closed-loop mechanisms.

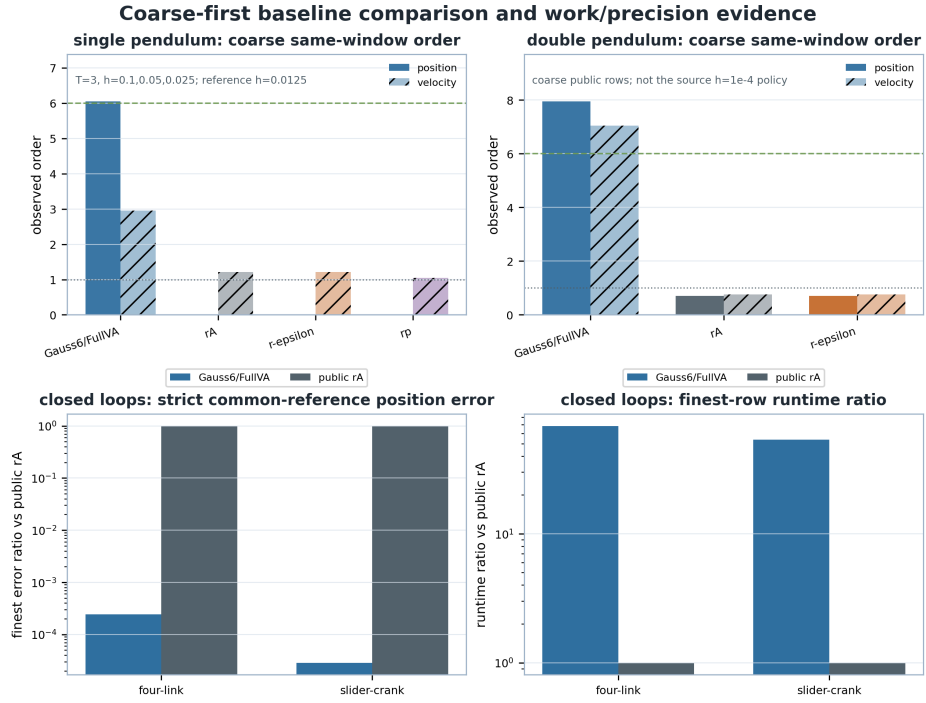


Figure 10: Coarse-first baseline and work/precision diagnostics. The top panels compare observed order on the same coarse window for the single and double pendula. The lower panels use the strict common-reference closed-loop summaries for four-link and slider-crank, with all plotted rows evaluated against the same finite accepted-method endpoint reference at $h_{\text{ref}} = 0.0125$. These panels document coarse-first diagnostics and work/precision availability, but they are not an external-superiority claim and do not replace the unresolved original source-policy campaigns.

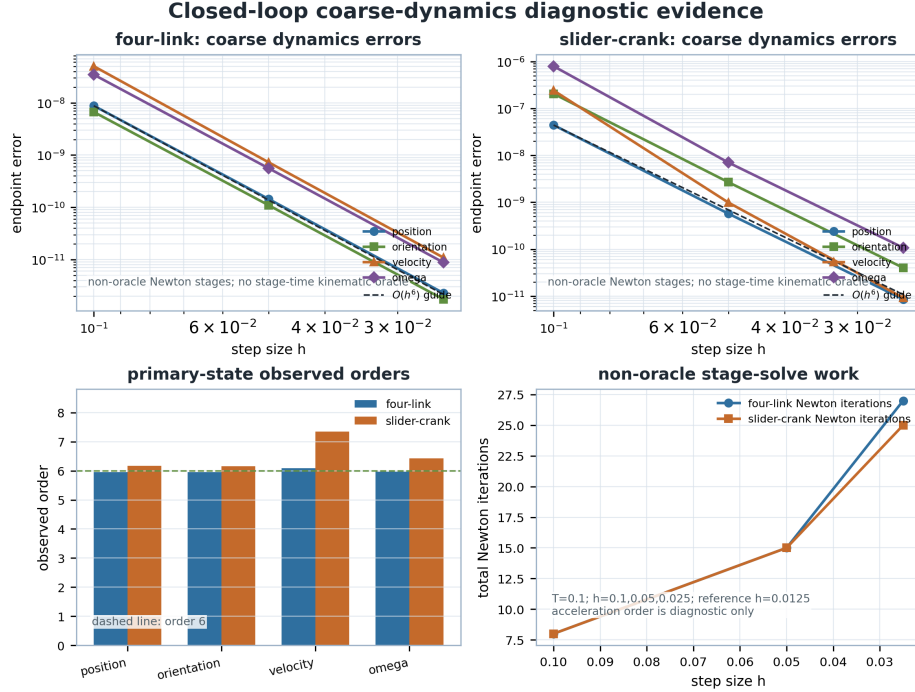


Figure 11: Closed-loop coarse-dynamics diagnostic evidence for the four-link and slider-crank mechanisms. Closed-loop coarse-window trajectory diagnostics plot endpoint position, orientation, linear-velocity, and angular-velocity errors against the $O(h^6)$ guide over $h = 0.1, 0.05, 0.025$ with $h_{\text{ref}} = 0.0125$. The bottom panels summarize the finite-window fitted slopes and the non-oracle Newton stage-solve work. These rows are mechanism-coverage and coarse-window trajectory diagnostics only, not accepted dynamic-order proof; they run without using stage-time kinematic oracles, without making 10^{-4} a default policy, without promoting four-link or slider-crank to accepted dynamic-order rows, and without asserting external same-test superiority. They are finite-window diagnostics: they do not instantiate the residual-to-error implication, do not close P7, and do not promote four-link or slider-crank to accepted dynamic-order examples.

All-method common-reference result matrix

Each cell reports observed velocity order p and finest-step velocity error e .

		Single	Double	Four-link	Slider-crank
Gauss6/FullVA	internal	$p=5.80$ $e=1.2e-12$	$p=6.01$ $e=2.3e-13$	$p=6.08$ $e=1.1e-11$	$p=7.34$ $e=9.2e-12$
HI22 rA	diagnostic	$p=1.00^*$ $e=7.7e-02$	$p=1.05^*$ $e=4.9e-05$	$p=1.02^*$ $e=2.0e-01$	$p=0.94^*$ $e=3.3e-02$
HI22 half	diagnostic	$p=1.84^*$ $e=7.7e-02$	$p=-0.80^*$ $e=3.0e+00$	$p=1.59^*$ $e=2.0e-01$	$p=2.46^*$ $e=3.3e-02$
RA21 rA	diagnostic	$p=0.62^*$ $e=1.3e+01$	$p=1.05^*$ $e=4.9e-05$	$p=1.02^*$ $e=2.0e-01$	$p=0.94^*$ $e=3.3e-02$
RA21 reps	diagnostic	$p=0.62^*$ $e=1.3e+01$	$p=1.05^*$ $e=4.9e-05$	$p=2.89^*$ $e=1.5e-02$	$p=0.94^*$ $e=3.3e-02$
RA21 rp	diagnostic	$p=0.52^*$ $e=1.5e+01$	$p=0.36^*$ $e=5.2e-05$	$p=2.89^*$ $e=1.5e-02$	$p=0.94^*$ $e=3.3e-02$
TFE Newmark	diagnostic	$p=0.90^*$ $e=1.5e+01$	$p=1.79^*$ $e=4.2e-04$	$p=1.87^*$ $e=1.7e-02$	$p=2.05^*$ $e=1.5e-03$
TFE m1	diagnostic	$p=1.02^*$ $e=1.5e+01$	$p=2.00^*$ $e=2.7e-04$	$p=3.87^*$ $e=5.3e-04$	$p=2.48^*$ $e=6.3e-04$
TFE m2	diagnostic	$p=0.05^*$ $e=1.1e+02$	$p=1.87^*$ $e=8.7e-05$	$p=1.02^*$ $e=6.7e-04$	$p=2.29^*$ $e=9.9e-05$
TFE trap.	diagnostic	$p=1.02^*$ $e=1.5e+01$	$p=2.00^*$ $e=2.7e-04$	$p=3.89^*$ $e=5.3e-04$	$p=2.33^*$ $e=7.3e-04$
VP24 rA proxy	diagnostic	$p=0.00^*$ $e=3.2e-11$	$p=1.00^*$ $e=2.2e-03$	$p=-0.00^*$ $e=1.1e-08$	$p=-0.00^*$ $e=1.3e-07$

p >= 5.5
3 <= p < 5.5
1.5 <= p < 3
0.5 <= p < 1.5
p < 0.5

* marks a row with diagnostic/source-policy caveats. Nonlocal rows are common-reference diagnostics only; they are not source-paper superiority claims.
Source: PAPER_NUMERICAL_RESULT_MATRIX.json; $h = 0.1, 0.05, 0.025$; reference $h = 0.0125$; 44/44 method-example cells.

Figure 12: All-method common-reference result matrix for the four validation examples. Each cell reports observed velocity order and finest-step velocity error from the paper numerical result matrix over $h = 0.1, 0.05, 0.025$ with $h_{\text{ref}} = 0.0125$. The local Gauss6/FullVA row is method-side order evidence. All nonlocal rows remain bounded diagnostic rows with open source-policy status, so the figure does not authorize an external source-paper superiority claim.

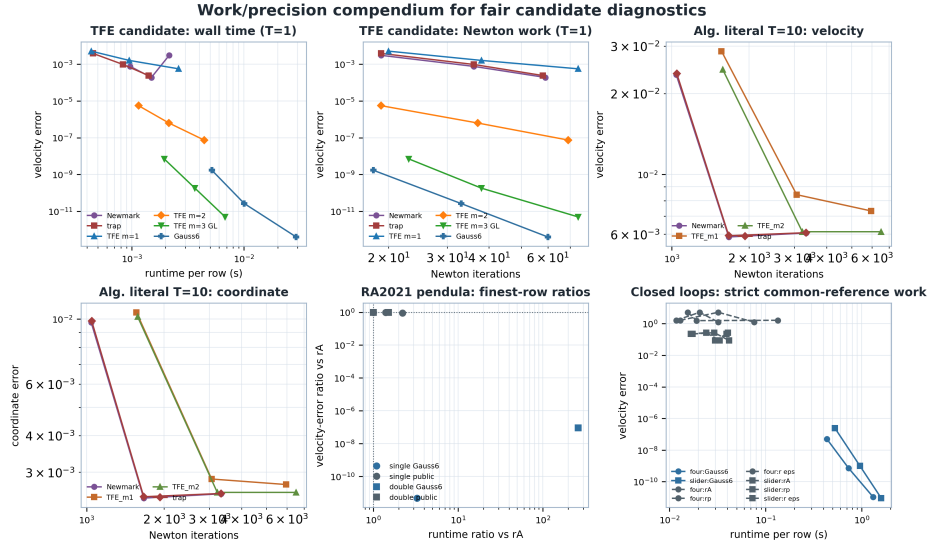


Figure 13: Work/precision compendium for fair candidate diagnostics. The first two panels use the frictionless TFE source-pendulum candidate route on a shared $T = 1$ grid and RK4 reference, plotting velocity error against wall time and Newton work. The Algorithm-1-literal $T = 10$ panels plot velocity and coordinate error against Newton work from the Algorithm-1-literal work/precision diagnostic rows; these fixed- h endpoint rows remain source-policy open because exact- T error sampling and a source-equivalent DAE/method runner are not established. The RA2021 panel summarizes same-window pendulum finest-row runtime and velocity-error ratios relative to public rA rows, and the closed-loop panel collects strict common-reference four-link and slider-crank work/precision rows. These panels consolidate existing evidence for review; they remain candidate/common-reference diagnostics, not source-policy superiority claims.

12. Limitations

Six limitations define the boundary of the claim, ordered from theorem and method conditions to external reproduction boundaries.

1. **Conditional proof boundary.** The accepted stage-residual input is closed for the implemented FullVA direct residual route: the 96 non-dynamic implementation rows are supplied by the non-dynamic certificate and the 36 Newton–Euler dynamic rows are supplied by the D5 direct-substitution certificate. The implementation-path binding is used only to identify the formulas differentiated by the active residual map; a term-by-term primitive verification of the dynamic block is kept as a separate sufficient route and is not an input to the accepted direct proof. Thus the active theorem input is the direct residual bridge: the same-branch residual-value input at the lifted Gauss stage Z_G is proved by that route, giving the stage-residual $O(h^7)$ input used below, while separate primitive or symbolic dynamic-block certificates are not claimed. The theorem remains conditional on the smooth regularity tube and on an inexact nonlinear solve satisfying the stated compact-tube $\eta_h^{\text{tube}} \leq c_\eta h^7$ policy; residual-to-error promotion for additional four-link/slider-crank dynamic-order claims remains outside the accepted theorem. This separation keeps the proof route distinct from broader promotion claims. The direct residual bridge supplies the local-defect input used by the theorem, and the proof certificates supply the accepted direct-route residual/Jacobian binding used by the bridge. Compact-branch regularity, implementation binding, endpoint perturbation, and solver scale remain explicit theorem-domain interfaces. Promoting closed-loop residual or reaction rows to trajectory order would require a separate branch-compatible residual-to-error theorem with a fixed residual operator, trajectory norm, stability or inf-sup inverse, nonlinear remainder control, reference-floor exclusion, and its own dynamic-order campaign. Small constraint, reaction, or mechanism residuals alone do not provide that theorem. In the proof-interface terminology retained in the theorem tables, P7 remains a separate output nonclaim/residual-to-error boundary: residual-only mechanism rows require a separate residual-to-error transfer theorem and are not a theorem premise.
2. **Nonlinear tolerance and endpoint closure.** The asymptotic theorem requires a compact-tube nonlinear residual policy $\eta_h^{\text{tube}} \leq c_\eta h^7$,

while the reported production runs use a fixed finite-run tolerance. The current scaled-tolerance and tolerance-regime probes are finite reported-window diagnostics around this policy; they diagnose this scale but do not close it, prove an asymptotic solver contract, or convert fixed production tolerances into theorem hypotheses. Likewise, endpoint closure is a conditional local $O(h^7)$ perturbation under Lemma 6 and the retained P2 endpoint-ball/right-inverse hypotheses; the raw residual/correction rows do not prove those retained hypotheses and are not hidden source-paper or reference-solution projections. Closing P6 rather than retaining it would require a theorem-level solver result on the same compact tube: the Gauss-predictor branch would have to be selected for every accepted transition, the computed Newton iterate would have to remain in the strong local residual-inverse ball, and the h^7 -scaled residual stopping rule would have to hold uniformly with constants independent of the reported grid and backend. Such a result would discharge the solver-scale interface only; it would still not promote residual-only mechanism rows unless the separate P7 residual-to-error transfer theorem were also proved.

3. **Full source-paper residual replacement.** Independent full-TFE stage replacement is not accepted. The accepted theorem is tied to the Gauss/FullVA residual in Eq. (6); replacing those rows by paper-derived weak rows is a separate method-construction problem.
4. **Performance and scalability.** The sparse automatic differentiation structure is correct and quantified, but dense `jacfwd` remains faster in the recorded wall-clock evidence. The sparse backend is therefore an implementation diagnostic, not a claimed performance improvement. The current implementation is a small-system reference implementation; a production scalability claim would require an $N = 2, 4, 8, 16, 32$ chain benchmark with wall-clock time, Newton iterations, Jacobian-assembly time, linear-solve time, conditioning or rank diagnostics, and memory scaling.
5. **Sharp-friction coarse regime.** The practical coarse sharp-friction regime is order-reduced, although deeper fixed refinement recovers high order at additional cost. The method claim therefore rests on the smooth contract and treats sharp friction as a quantified caveat.
6. **External same-test comparison.** The exact same-test external comparison criterion in Section Appendix A remains open after the public-code baselines and selected/coarse Gauss6/FullVA pilots. The

Chaturvedi–Sandu–Sandu pendulum tests are not source-policy reproduced: the encoded source-shaped candidate smokes and bounded probes remain diagnostic; the Kissel/Negrut suites still lack complete Gauss6/FullVA dynamic order/work rows under the published policies; and the closed-loop surrogate rows do not change that accepted-order boundary. The floor analysis narrows this boundary to velocity/acceleration floor evidence plus a position/reference floor, but it does not close the source-policy dynamic-order comparison. The velocity-partitioning public-code path is also unresolved.

These limitations do not invalidate the narrower conditional formal-order comparison; they define the proof, implementation, and reproduction work still needed before a broader paper-level claim would be justified.

13. Conclusions

This paper has presented a monolithic Lie-group Gauss6/FullVA integrator for lower-pair mechanisms. The method combines $SO(3)$ retractions, lower-pair position/velocity/acceleration rows, and Newton–Euler dynamics in one square stage residual, then reconstructs the endpoint on the accepted branch. Under the stated compact-branch, endpoint/stability, implementation-binding, and solver-scale contracts, this construction gives the conditional sixth-order Lie-group FullVA theorem of Theorem 1. The conditional qualifier fixes the compact branch, endpoint, implementation-binding, and solver-scale domain of the theorem; it is not an empirical replacement for the local-defect proof. The numerical evidence records finite-window smooth position and velocity orders 7.161 and 7.066 as consistency evidence after the order-six theorem has been stated. The single- and double-pendulum rows provide method-side dynamic-order evidence, while the four-link and slider-crank rows provide diagnostic mechanism-coverage evidence, not accepted dynamic-order proof and not residual-to-error evidence, for the same FullVA residual. Against the local order-five $m = 3$ Gauss–Lobatto temporal finite-element target, this is a conditional formal-order comparison, not an external same-test superiority claim.

The proof route is anchored by the implemented 132-row residual bridge: the 96 non-dynamic rows have an $O(h^7)$ certificate on the lifted Gauss stage, and the 36 Newton–Euler rows close by direct substitution with

zero residual on the lifted Gauss stage Z_G . That bridge is then combined with the analytic local-defect decomposition, endpoint perturbation, the retained compact-tube inexact-Newton scale condition, and local-to-global transfer. In that reading, the theorem is a conditional implication for a fixed smooth branch: once the compact branch, endpoint map, implementation binding, and $\eta_h^{\text{tube}} \leq c_\eta h^7$ solver-scale condition are in force, the direct 132-row bridge supplies the $O(h^7)$ residual-defect input; the stage-residual perturbation, endpoint, and retained solver-scale lemmas then supply the $O(h^7)$ local defect and the Gauss stability transfer gives the $O(h^6)$ same-initial-state reported-grid estimate. The finite-window order fits, finite solver probes, and residual/reaction tables are consistency or diagnostic evidence and do not replace those mathematical inputs. The separate primitive/Taylor decomposition of the Newton–Euler rows remains a stricter separate sufficient-route specification and is not used to supply C_{loc} , close the solver-scale interface, close the residual-to-error boundary, or act as a theorem input. The unproved primitive-route inputs are therefore a non-active replacement-certificate boundary for the dynamic-block provenance, not an open algebraic term in the accepted direct residual-bridge proof. Remaining work concerns a separate uniform same-branch solver-policy theorem, residual-to-error promotion for closed-loop residual rows, and external same-test reproduction; these are outside the theorem proved here.

Broader comparison work remains separate from this method result. It should rerun the original pendulum tests and the Kissel/Negrut public-code mechanism suites under identical benchmark definitions, reporting order, error, constraint drift, Newton work, and runtime side by side. The velocity-partitioning code path is part of that separate published-test baseline scope. Complete source-paper temporal finite-element residual replacement and a runner-centered full reproduction package remain outside the accepted claim.

CRedit authorship contribution statement

Jingquan Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing - original draft, Writing - review and editing.

Declaration of competing interest

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Data availability

The journal source archive is the LaTeX archive: it contains the manuscript sources, figures, and declarations. This is an editorial-source archive rather than a full source-policy runner archive. The companion reproducibility package contains the reported-result records and read-only check scripts used for this paper. It is scoped to the accepted method-side evidence and proof-dependency records; external same-test runner reproduction for source papers and full TFE runner comparisons are reserved for a separate package. An archival public repository or DOI can be attached at journal production or revision if the editorial office requires a public deposit.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the author used an AI coding assistant for drafting support, consistency checking, and validation-script editing. The author reviewed and edited the content and takes full responsibility for the published article.

Appendix A. Cross-paper same-test comparison criterion: diagnostic appendix

This subsection is a diagnostic guardrail, not a second contribution claim and not part of the proof chain. It can be read as a promotion rule for separate external source-paper claims after the method definition, proof, and accepted method-side validation have already been fixed. A publication-level source-paper comparison must be made on the same numerical tests used by the source papers, not only on mechanisms with similar names. The criterion is strict: the body parameters, initial conditions, prescribed drivers, friction model, step-size sweep, nonlinear tolerance policy, reference solution, error norm, and work metric must be matched or explicitly justified. Table A.51 records the required external comparisons, the status of this evidence package, and the claim that is allowed before the remaining source-suite decisions close. The current paper uses these rows only to prevent overclaiming: the

method claim, the conditional theorem, and the accepted numerical evidence do not depend on source-policy promotion.

Why the source-policy criterion is separate. The source-policy criterion is required only for a stronger external apples-to-apples statement: that Gauss6/FullVA has been evaluated under the source papers' own model, output, tolerance, reference, and work policies and would then be eligible for a separate source-paper replacement or superiority comparison. It is not a premise of the Gauss6/FullVA residual definition, the retained conditional sixth-order theorem, or the accepted local dynamic-order and mechanism-coverage evidence. Therefore an open source-policy row does not weaken the bounded method/proof claim; it only blocks the broader external same-test and full reproducibility package claims.

Source-policy progress boundary. The detailed source-policy record is boundary evidence only. It records local candidate availability, public-code setup progress, TFE formula diagnostics, and open runner gaps, but no external row is promoted to a source-policy dynamic-order or source-paper superiority claim. The current progress entries are read only as diagnostic inventory: public-code baseline order/timing rows, source-identity checks, coarse double-pendulum self-reference runs, bounded four-link/slider-crank mechanism-diagnostic candidates, and published-formula candidates record implementation progress but do not establish external same-test reproduction. No external source-policy dynamic-order row is accepted, and the local method rows under external source-policy dynamic-order rules remain unpromoted; therefore no source-paper superiority, full source-policy work-precision, or full-TFE replacement claim is made.

Claim-policy implication. The source-policy promotion criterion is interpreted as a logical promotion rule, not as a visual preference over plots. A paper-level external-superiority statement requires the source benchmark, method-set, numerical-control, and work-metric rows in Table A.52 to close for the source-policy campaign being cited. The current evidence record reports no strict external error-claim rows, does not establish any source-policy-closed external same-test error row, and leaves the same-test promotion conditions open. Therefore the common-reference matrix and bounded pilots may be used only as diagnostics and cannot be used as premises in any implemented source-paper superiority claim.

Consequently, the present numerical evidence supports only the method-side Gauss/FullVA order claim and the local order-five TFE formula-target

Table A.51: Required same-test comparisons and current external-suite claim boundary.

External source	Published or code-backed tests	Current evidence	Claim allowed now
Chaturvedi–Sandu–Sandu Lie-group TFE paper [1]	Rigid pendulum without and with joint friction; $m = 1, 2, 3$ TFE; Newmark– β and trapezoidal comparators.	Specification and source-shaped candidate smokes are encoded, including source parameter/output policies and bounded TFE probes; source-policy absolute-coordinate DAE/method-runner rows are outside the current evidence scope.	Formula-order comparator only; no implemented source-paper superiority claim.
Kissel–Taves–Negrut absolute-coordinate $rA/rp/r\epsilon$ suite [11, 12, 13]	Single pendulum, double pendulum, slider-crank, and four-link mechanisms from the public C1/C2 Python code and model files.	Public baseline rows and timing rows are complete. Gauss6/FullVA has single-pendulum public rows, double-pendulum coarse rows, and closed-loop strict-common-reference work/precision rows, but accepted local source-policy dynamic-order evidence is not promoted for any of the four tracked examples.	Coarse-first and common-reference evidence only; no external superiority claim.
Fang–Kissel–Zhang–Negrut half-implicit suite [14]	Double pendulum, slider-crank, four-link, N -pendulum scaling, and slider-crank joint-friction examples.	Bounded $T = 0.1$ rA/rA_{half} pilot complete; full $T = 8$ policy and Gauss6/FullVA rows open.	Bounded pilot only; not a full half-implicit policy comparison.
Kissel–Bakke–Negrut velocity-partitioning Lie-group ODE suite [15]	Single pendulum, double pendulum, four-link, and slider-crank tests reported for velocity partitioning.	The coordinate-partitioning proxy is identified through the implemented rA wrapper; a distinct public VP code path remains unresolved and demoted. The source-code path is not separately reproduced.	Common-reference diagnostic rows only; no source-policy VP superiority claim.

Table A.52: Minimum same-test TFE acceptance protocol before any source-paper superiority claim.

Promotion item	Required evidence	Evidence boundary
Source benchmarks	Reproduce the Chaturvedi–Sandu–Sandu frictionless pendulum and revolute joint with Brown–McPhee friction using the source parameters, horizon, initial data, drivers, and output variables.	Specification and reference-feasibility diagnostics only; source rows not accepted.
Method set	Run source-equivalent TFE $m = 1, 2, 3$, Newmark- β , trapezoidal, and Gauss6/FullVA on the same trajectories.	Local formula/common-reference diagnostics only; no accepted source-policy method matrix.
Numerical controls	Use the same step sizes or a justified common refinement, matched nonlinear tolerance policy, matched reference solution, and the same error norm and constraint-drift reporting.	Open.
Work metrics	Report error versus wall-clock time, Newton iterations, Jacobian assembly time, linear-solve time, and failed-step counts.	Open.
Allowed claim	Only after the rows above pass may the manuscript state a source-paper accuracy or work-precision advantage. Until then, the TFE comparison is a formal-order-level comparison only; diagnostic rows remain non-claim evidence.	No external-superiority claim.

comparison. The method-side evidence consists of the smooth cylindrical-chain benchmark plus the four ASME-style mechanisms in this package; the coverage matrix records 48 rows and 32 completed rows, but no external superiority claim. It does not support a stronger statement that Gauss6/FullVA is better than the published Chaturvedi–Sandu–Sandu implementation or the Kissel/Negrut public-code baselines on their own numerical tests. The extracted public-code benchmark specification and row-level inventory are maintained in the reproducibility appendix. The current evidence state is: specification extracted; 2021 public baselines complete for the already extracted order/timing evidence; 2021 public baseline order/timing evidence closed at 12/12 public order groups and 12/12 timing rows; selected/coarse Gauss6/FullVA pilot rows; a coarse public double-pendulum same-window baseline run; and a single-pendulum coarse same-window tranche. The accepted local Gauss6/FullVA source-policy dynamic-order set is nevertheless empty across the four tracked examples: the double-pendulum row is coarse-first rather than the public 10^{-4} policy, and the four-link/slider-crank public-horizon rows are kinematic/reaction residual rows rather than accepted dynamic-order rows. Because a separate residual-to-error trans-

fer theorem remains open, residual-only rows are not promoted to order claims. The external same-test campaign remains incomplete: full source-policy dynamic-comparison rows remain unaccepted, and the paper makes no external-superiority claim. Equivalently, the source-policy dynamic-order count is nevertheless 0/4 for the external source-paper rows; this count is a progress boundary, not a demotion of the accepted local dynamic-order evidence.

After the coarse common-reference replay was repaired to use fixed public-code times $t_i = ih$, the result package also records a bounded common-reference matrix. All 44 common-reference summary rows pass the row-level policy checks, and the global comparison policy passes 14/14 checks, but a stricter all-example consistency check allows zero paper-level external direct-error claims. Thus the replay is a shared coarse final-state diagnostic, not a reproduction of the source papers' default time horizons, output policies, error norms, or 10^{-4} policies.

Benchmark-policy dictionary. The bounded common-reference policy used in Tables A.54–A.55 fixes $T = 0.1$, $h \in \{0.1, 0.05, 0.025\}$, and $h_{\text{ref}} = 0.0125$ for every runnable method. The single-pendulum row uses the analytic exact final state and a final L^2 error. The double-pendulum row uses the local Gauss6/FullVA $h_{\text{ref}} = 0.0125$ final state and a final L^∞ error. The four-link and slider-crank rows use closed-loop true-dynamic common-reference trajectories at $h_{\text{ref}} = 0.0125$ and then apply the same order/error arithmetic to all runnable methods. Source-policy rows, by contrast, must follow each source paper or public code on its own horizon, time grid, reference solution, error norm, tolerance, and output variables. This dictionary is the reason the common-reference table can be arithmetically reconciled while source-policy external-superiority rows remain open. Table A.54 summarizes the nearest nonlocal bounded common-reference error evidence, and Table A.55 reports the full runnable method-by-example order/error matrix.

We therefore separate two comparison statements. The finite-grid common-reference diagnostic is assembled for arithmetic coverage of the required matrix: all required method labels are accepted, alias-resolved, or source-backed scope-excluded, and the 44-cell table is recomputed from 132 raw rows with zero summary mismatches. The nonlocal cells remain bounded common-reference diagnostics because they do not reproduce every source paper's default horizon, tolerance, output policy, or error norm. These diagnostic comparisons are not promoted to a CMAME paper-level

external-superiority claim. The stronger source-policy statement remains a non-claim, and no common-reference count is promoted to a source-paper superiority claim.

This paragraph separates local-candidate availability from source-policy closure. For the 2021 absolute-coordinate suite, the public-code path, the `body.r/body.dr/body.ddr` output map, and the endpoint time-grid convention have been extracted and checked; in that bounded sense, source identity is now closed only for the diagnostic row map, not for source-policy closure. The remaining gap is promotion of local Gauss6/FullVA rows under the same source-policy dynamic-order contract. Table A.53 records why the existing local candidates are not promoted: the single-pendulum public-grid tranche is floor limited, the double-pendulum row is a coarse self-reference run, and the four-link/slider-crank true-dynamic rows are bounded $T = 0.1$ diagnostics rather than the public $T = 3$ source-policy rows. For the 2022 half-implicit suite, the bounded $T = 0.1$ replay has 24/24 successful rows, while the $T = 8$ coarse replay has 18/24 successful rows and only four of eight complete form-model groups. For the original TFE pendulum suite, bounded runner and endpoint-sensitivity diagnostics exist, but a source-equivalent DAE/friction/output runner has not completed the original rows. Consequently no local method row is accepted under the external source-policy dynamic-order contract, and no source-policy-closed external same-test error row is established.

Table A.53: RA2021 local-candidate promotion boundary.

Example	Local candidate evidence	Why not promoted to RA2021 source-policy row
Single pendulum	Public $T = 3$ row trio at $h = \{10^{-2}, 10^{-3}, 10^{-4}\}$ with $h_{\text{ref}} = 10^{-3}$; finest velocity error is 7.01×10^{-15} .	The row is roundoff/floor limited; observed order is not a stable source-policy dynamic-superiority certificate.
Double pendulum	Coarse $T = 3$ self-reference row with $h = \{0.1, 0.05, 0.025\}$, $h_{\text{ref}} = 0.0125$, and position/velocity order 7.951/7.042.	The step grid and reference do not match the RA2021 double-pendulum policy $h = \{0.01, 0.002, 0.001\}$ with $h_{\text{ref}} = 10^{-4}$.
Four-link	True-dynamic local row at $T = 0.1$, $h = \{0.1, 0.05, 0.025\}$, with position/velocity order 5.955/6.085.	The public $T = 3$ rows are kinematic/reaction checks; the true-dynamic row is not the RA2021 source-policy horizon.
Slider-crank	True-dynamic local row at $T = 0.1$, $h = \{0.1, 0.05, 0.025\}$, with position/velocity order 6.164/7.341.	The public $T = 3$ rows are kinematic/reaction checks; the true-dynamic row is not the RA2021 source-policy horizon.

Table A.54: Bounded common-reference velocity evidence from the current result pack; diagnostic only, not external-superiority evidence.

Example	Local velocity order	Local finest velocity error	Nearest nonlocal method	Nearest nonlocal finest error
Single pendulum	5.801	1.155×10^{-12}	velocity partitioning rA	3.163×10^{-11}
Double pendulum	6.010	2.346×10^{-13}	2021 public rA	4.924×10^{-5}
Four-link	6.085	1.093×10^{-11}	velocity partitioning rA	1.087×10^{-8}
Slider-crank	7.341	9.189×10^{-12}	velocity partitioning rA	1.332×10^{-7}

Table A.55: All runnable common-reference velocity rows. Each entry is observed velocity order / finest-step velocity error on the fixed-grid $h = \{0.1, 0.05, 0.025\}$ sweep with reference $h = 0.0125$; the table is a diagnostic common-reference matrix, not a source-policy superiority table.

Method	Single pendulum	Double pendulum	Four-link	Slider-crank
Gauss6/FullVA	5.801 / 1.155×10^{-12}	6.010 / 2.346×10^{-13}	6.085 / 1.093×10^{-11}	7.341 / 9.189×10^{-12}
HI 2022 rA	1.005 / 7.725×10^{-2}	1.049 / 4.944×10^{-5}	1.015 / 1.970×10^{-1}	0.941 / 3.281×10^{-2}
HI 2022 rA_{half}	1.841 / 7.725×10^{-2}	-0.799 / 2.970	1.590 / 1.970×10^{-1}	2.459 / 3.281×10^{-2}
2021 public rA	0.617 / 1.305×10^1	1.052 / 4.924×10^{-5}	1.015 / 1.970×10^{-1}	0.941 / 3.281×10^{-2}
2021 public re	0.617 / 1.305×10^1	1.052 / 4.924×10^{-5}	2.894 / 1.458×10^{-2}	0.940 / 3.281×10^{-2}
2021 public rp	0.517 / 1.499×10^1	0.361 / 5.241×10^{-5}	2.894 / 1.458×10^{-2}	0.940 / 3.281×10^{-2}
TFE Newmark- β	0.901 / 1.479×10^1	1.787 / 4.246×10^{-4}	1.872 / 1.672×10^{-2}	2.051 / 1.531×10^{-3}
TFE $m = 1$	1.021 / 1.498×10^1	1.996 / 2.708×10^{-4}	3.866 / 5.253×10^{-4}	2.481 / 6.264×10^{-4}
TFE $m = 2$	0.054 / 1.102×10^2	1.870 / 8.711×10^{-5}	1.021 / 6.711×10^{-4}	2.290 / 9.881×10^{-5}
TFE trapezoidal	1.024 / 1.498×10^1	2.000 / 2.708×10^{-4}	3.888 / 5.306×10^{-4}	2.329 / 7.328×10^{-4}
VP 2024 coordinate partitioning rA	0.000 / 3.163×10^{-11}	1.000 / 2.165×10^{-3}	-0.001 / 1.087×10^{-8}	-0.000 / 1.332×10^{-7}

The full common-reference table is arithmetic-consistent but not, by itself, a source-policy reproduction. An independent recomputation from the 132 raw common-reference rows reproduces all 44 summary order/error entries with zero summary mismatches. The remaining concern is interpretive rather than arithmetical: 15 nonlocal method/example rows are flagged for source-policy review, and 10 of those rows have small position error but large velocity error. This pattern is consistent with a velocity-output, state-map, time-grid, or reference-policy mismatch and therefore cannot support an external-superiority claim. Table A.56 records the resulting claim boundary.

The flagged-row count should not be read as the complete set of open external claim rows. All 40 nonlocal method/example rows remain bounded common-reference diagnostics rather than source-policy-closed reproductions; the 15 flagged rows are the anomaly or high-risk subset that requires special attention before any separate external-superiority statement is considered. This distinction keeps the order/error comparison and the source-policy claim separate throughout the paper.

The all-example source-policy check expands this diagnosis from category counts to the actual method/example coverage. It checks all 15 flagged

Table A.56: Source-policy diagnosis for flagged common-reference baseline rows.

Diagnostic category	Rows	Affected examples	Claim-boundary interpretation
Position aligned, velocity mismatched	10	single pendulum, four-link	Likely velocity-output, state-map, or reference-policy mismatch; not a method-failure claim.
Single-pendulum velocity-output policy suspect	7	single pendulum	Original/Taves-Kissel-Negrut/TFE pendulum rows require source setup and velocity-output policy closure.
Near reference floor; order not identifiable	3	single pendulum, four-link, slider-crank	Errors are too close to the shared reference floor for a meaningful observed order statement.
Low order but small error; wider step window needed	4	single pendulum, double pendulum, four-link, slider-crank	The row is diagnostic only until a wider or source-policy step window is reported.
Negative order with large error	1	double pendulum	The fixed-grid replay or selected form requires source-policy recheck before comparison.

summary rows and their 45 raw coarse rows at $h = \{0.1, 0.05, 0.025\}$. The flagged rows cover all four examples: 8 single-pendulum rows, 2 double-pendulum rows, 4 four-link rows, and 1 slider-crank row. Table A.57 records the suite-level closure boundary. A stricter all-method/all-example consistency check then checks the complete 44-cell common-reference table and its 132 raw rows; it keeps all 44 cells as bounded diagnostics but allows zero strict external error-claim rows. In particular, the common-reference replay is valid for the finite-grid diagnostic, but no suite has yet closed the stronger source-policy reproduction required for an external-superiority claim.

For the TFE source evidence, we separate the friction evidence into two levels. The source text identifies the Brown–McPhee velocity-based continuous friction model, reports $\mu_s = 0.3$ and $\mu_d = 0.2$, and uses $h_{\text{ref}} = 10^{-4}$ for source reference solutions; its detailed continuous-friction law is delegated to the source-paper friction references [1, 28]. The model check therefore treats the encoded Brown–McPhee torque law as a candidate published formula structure, not as a source-code-equivalent friction implementation. The transition-velocity, normal-load coupling, friction Jacobian, and DAE multiplier policies remain source-policy boundaries.

Source-policy boundary summary:

Table A.57: All-example source-policy status for the flagged baseline rows.

Suite	Rows	Common-reference row source	Closure boundary
2021 absolute-coordinate public suite	5	Public setup and $rA/rp/r\epsilon$ steppers, fixed-grid replay, and <code>body.r/body.dr/body.ddr</code> output fields.	Source identity is checked; promotion still requires local Gauss6/FullVA source-policy dynamic rows, norm/output binding, runtime/Newton binding, and an independent verification record.
Original TFE pendulum formulas	4	Surrogate common-reference row only: RA2021 public setup with locally installed Newmark and TFE formula proxies, not the source pendulum.	Encode the original pendulum setup, source error norm, Brown–McPhee friction law, output variables, and horizon, or keep these rows demoted.
HI 2022 half-implicit suite	3	Public setup and half-implicit steppers under bounded fixed-grid replay.	Run the full $T = 8$ public policy or explicitly demote the bounded pilot rows.
VP 2024 coordinate partitioning	3	Coordinate-partitioning proxy on the RA2021 public setup.	Resolve the independent velocity-partitioning code path or keep the row as proxy/common-reference evidence only.

Brown–McPhee velocity-based continuous friction model;
 candidate published formula, not source-code-equivalent friction implementation;
 source-policy boundaries;
 Brown–McPhee formula is not fixed;
 source-code-equivalent Brown–McPhee transition law;
 transition velocity v_t ;
 velocity scale in the v/v_t Stribeck factor;
 not the normal-load/multiplier coupling;
 implementation choice, not source-policy evidence;
 candidate-friction DAE trajectory contract records 12 method-grid rows and 56 step-residual rows with nonpositive friction power;
 do not implement a monolithic source-policy DAE runner;
 source-policy runner acceptance contract has six simultaneous parts;
 transition law, including transition velocity;
 velocity-based continuous friction model;
 row-promotion rules binding error, order, runtime, and work metrics;
 monolithic absolute-coordinate DAE time integrator;
 not an accepted source-policy reproduction.
 More concretely, the Brown–McPhee formula is not fixed by μ_s , μ_d , and R alone. It also depends on a transition velocity v_t , the velocity scale in the v/v_t Stribeck factor, and on how the tangential velocity and normal reaction are supplied by the constrained DAE. The source-pendulum text records μ_s , μ_d , R , and h_{ref} , but not v_t , not the normal-load/multiplier coupling, and not the friction-Jacobian convention. Choosing the local surrogate’s Stribeck velocities 0.05, 0.5, or 1.0 would therefore be an implementation choice, not source-policy evidence. A bounded transition-velocity sensitivity check makes this boundary numerical rather than merely documentary: over Stribeck velocities 0.05, 0.5, and 1.0, with 0.5 as the baseline, the $T = 0.024$, $h_{\text{ref}} = 10^{-4}$ candidate DAE check changes the endpoint by at most 2.90×10^{-6} in coordinate and 2.36×10^{-4} in velocity. The check records three sensitivity rows, 36 contract rows, and zero source-policy rows, so the missing transition velocity is treated as a material unresolved source-law parameter rather than an innocuous transcription detail. The positive evidence from the surrogate-friction DAE layer is narrower: the candidate-friction DAE trajectory contract records 12 method-grid rows and 56 step-residual rows, all finite, with candidate DAE residuals below 10^{-9} and nonpositive friction

power. These checks certify local dissipativity and absolute-coordinate residual consistency for the encoded candidate law only. They do not certify a source-code-equivalent Brown–McPhee law, do not implement a monolithic source-policy DAE runner, and do not close any source-policy row.

Consequently, the source-policy runner acceptance contract has six simultaneous parts. A promoted TFE row would need a source-code-equivalent Brown–McPhee transition law, including transition velocity, normal-load coupling, friction Jacobian, and multiplier policy; a monolithic absolute-coordinate DAE time integrator rather than a stepwise residual replay; source grid, output norm, and endpoint/output-sampling binding for $T = 10$ and $h_{\text{ref}} = 10^{-4}$; nonlinear solve, Newton, and tolerance-policy binding; method-runner equivalence for the TFE $m = 1, 2, 3$, Newmark- β , trapezoidal, and local Gauss6/FullVA comparator rows; and row-promotion rules binding error, order, runtime, and work metrics. Equivalently, the present package leaves two documentary blocks open without requiring a new long numerical campaign—the source-code-equivalent Brown–McPhee law and the full- $T = 10$ endpoint/output-sampling policy for endpoint-incompatible step sizes—and four execution-level blocks open: a monolithic source-policy DAE integrator, source-policy method runners for the TFE and one-step comparator rows, a Gauss6/FullVA source-pendulum DAE comparator, and accepted work/precision row binding. Because these blocks are not jointly present, the candidate DAE trajectory evidence remains a diagnostic record of what is missing, not an accepted source-policy reproduction row.

The source check also includes the Appendix-B coefficient formulas for TFE $m = 1$, $m = 2$, and $m = 3$ Gauss–Lobatto against the source equations (40), (41), and (43). The recorded coefficient certificate has three checked rows and zero maximum absolute difference for the extracted source parameter choices. This closes coefficient-transcription evidence only; it does not close source-policy method-runner equivalence or any source-policy reproduction row. Here, Appendix-B coefficient formulas denotes this coefficient-transcription certificate only, not a closed source-policy method runner. The diagnostic chain also includes a full- $T = 10$ coarse formula probe for the Appendix-B $m = 3$ Gauss–Lobatto target. That probe is finite and residual-converged for one checked row and records the formal fifth-order target, but it remains a non-source-policy formula diagnostic because it does not invoke the source $h = 10^{-4}$ reference campaign. Restated as a source-policy boundary, the full- $T = 10$ coarse formula probe targets the Appendix-

B $m = 3$ Gauss–Lobatto target, records a formal fifth-order target, and does not invoke the source $h = 10^{-4}$ reference campaign. The source-reference side is therefore checked separately. A full- $T = 10$ frictionless source-reference feasibility probe integrates the extracted source pendulum with $h_{\text{ref}} = 10^{-4}$ and compares it against a $h = 5 \times 10^{-5}$ check run, using 100000 and 200000 RK4 steps, respectively. The $h_{\text{ref}} = 10^{-4}$ final state is $(\theta, \omega) = (-2.8790156956337563, -1.2836123173308356)$, and the check-run mismatches are 3.82×10^{-14} in coordinate and 2.49×10^{-13} in velocity. This closes only source-reference feasibility for the original pendulum time horizon. It is not source-policy method-runner equivalence for Newmark, trapezoidal, or TFE, it does not cover the Brown–McPhee friction row, and it does not authorize external-superiority claims; no source-policy row is promoted.

The same source-pendulum candidate route now includes a Gauss6/FullVA row on the extracted source output policy. On the frictionless $T = 1$ run with $h_{\text{ref}} = 2.5 \times 10^{-4}$ and $h = \{0.1, 0.05, 0.025\}$, the coordinate, velocity, and Frobenius pairwise orders are (5.98, 6.10), (6.02, 6.03), and (5.98, 6.10), respectively, with maximum Newton residual below 10^{-12} . This is same-test candidate evidence for the Gauss collocation order target on the source pendulum output policy. The candidate frictional row on the same grid shows reduced pairwise orders (about 1.6–2.3), so it is recorded only as a friction-feature diagnostic. Neither row is an absolute-coordinate FullVA DAE source-policy reproduction, and the TFE source-policy closure count remains zero. A separate same-test work/precision diagnostic runs Newmark- β , trapezoidal, TFE $m = 1, 2, 3$, and Gauss6/FullVA on this same frictionless source-pendulum $T = 1$ grid. It records 18 raw rows and six method summaries with runtime and Newton-iteration work proxies; the diagnostic velocity-order floors are about 6.02 for the Gauss6/FullVA candidate and 5.19 for the TFE $m = 3$ Gauss–Lobatto formula target. This diagnostic improves the fair same-grid comparison, but it remains candidate-level evidence: the source-policy method-runner equivalence flag and the external-superiority claim flag are both false.

For the original TFE pendulum suite, the source-model evidence also includes a bounded active-row candidate smoke test. It uses the extracted pendulum parameters and source output/error policy with $h_{\text{ref}} = 10^{-4}$, but only on the short frictionless interval $T = 0.024$ and the source-shaped grid $h = \{0.012, 0.006, 0.003\}$. The row is therefore useful for checking whether the installed Newmark, trapezoidal, and TFE formula runners have coherent local order/error behavior, but it is not the full $T = 10$ original-source cam-

paign and it closes zero source-policy rows. Table A.58 records this bounded diagnostic.

Table A.58: Bounded active-row candidate smoke for the original TFE pendulum rows. The run uses $T = 0.024$, $h_{\text{ref}} = 10^{-4}$, and $h = \{0.012, 0.006, 0.003\}$; it is not a full $T = 10$ source-policy reproduction.

Original-row method	Formula target	Velocity pair orders	Coordinate pair orders	Finest velocity error	Finest coordinate error	Claim boundary
Newmark- β	2	1.973, 1.997	2.060, 2.022	1.650×10^{-10}	2.584×10^{-12}	Bounded candidate smoke only
TFE $m = 1$	1	1.934, 1.915	1.000, 1.000	1.856×10^{-10}	5.742×10^{-7}	Bounded candidate smoke only
TFE $m = 2$	3	4.000, 3.998	3.749, 3.609	6.482×10^{-14}	4.441×10^{-15}	Bounded candidate smoke only
Trapezoidal	2	1.973, 1.997	2.078, 2.028	1.650×10^{-10}	1.987×10^{-12}	Bounded candidate smoke only

The same source-model evidence record also includes a non-heavy full-horizon candidate probe at the original TFE pendulum horizon $T = 10$. This diagnostic uses $h = \{0.1, 0.05, 0.025\}$ and a coarse reference $h_{\text{ref}} = 0.0125$, so it tests full-horizon large-step behavior without invoking the extracted $h_{\text{ref}} = 10^{-4}$ source-policy reference. All four active rows are finite and Newton-residual stable, but the table remains a candidate-runner diagnostic rather than a source-policy reproduction; it closes zero source-policy rows.

Table A.59: Full-horizon coarse candidate probe for the original TFE pendulum rows. The run uses $T = 10$, $h_{\text{ref}} = 0.0125$, and $h = \{0.1, 0.05, 0.025\}$; it does not invoke the source-policy $h_{\text{ref}} = 10^{-4}$ reference.

Original-row method	Formula target	Velocity pair orders	Coordinate pair orders	Finest velocity error	Finest coordinate error	Claim boundary
Newmark- β	2	1.991, 1.998	2.028, 2.007	4.642×10^{-3}	1.956×10^{-3}	Full- T coarse candidate only
TFE $m = 1$	1	1.276, -0.077	2.623, 5.795	6.801×10^{-3}	5.226×10^{-5}	Full- T coarse candidate only
TFE $m = 2$	3	2.927, 2.952	2.694, 2.824	2.375×10^{-6}	3.346×10^{-7}	Full- T coarse candidate only
Trapezoidal	2	1.993, 1.998	2.021, 2.005	3.558×10^{-3}	1.504×10^{-3}	Full- T coarse candidate only

Appendix B. Reproducibility package

The flat LaTeX archive contains the editable Elsevier source, the compiled manuscript, the thirteen figures, and declaration files. The companion

Table A.60: Cross-paper evidence accumulated without treating it as an external superiority result.

Tranche	Rows	Main metric	Interpretation
2021 public $rA/rp/r\epsilon$ baselines	27 + 9 order rows; 12 timing rows	Public $T = 3$ policies; $h = 10^{-2}, 10^{-3}, 10^{-4}$ except the double-pendulum dynamic self-reference sweep $h = 10^{-2}, 2 \times 10^{-3}, 10^{-3}$. $T = 3, h = 10^{-2}, 10^{-3}, 10^{-4}$; finest position/velocity errors $2.584 \times 10^{-14} / 7.012 \times 10^{-15}$.	Baseline execution evidence only.
Gauss6/FullVA single public horizon	3 rows		Public-policy rows closed, but order is reference-floor limited.
Gauss6/FullVA double public horizon, coarse	3 rows	$T = 3, h = 0.1, 0.05, 0.025$; reference $h = 0.0125$; position/velocity orders $7.951 / 7.042$.	Coarse-first order/work evidence, not the public 10^{-4} policy.
Public double coarse same-window baselines	9 rows, 6 ok	$rA/r\epsilon$ at the same $T = 3$, $h = 0.1, 0.05, 0.025$, reference $h = 0.0125$ give position/velocity orders $0.703 / 0.754$; rp fails Newton at all three coarse steps. $T = 3, h = 0.1, 0.05, 0.025$, reference $h = 0.0125$; Gauss6/FullVA position order 6.054 ; public velocity orders about 1.21 .	Large-step baseline evidence; not the source 10^{-4} policy.
Gauss6/FullVA and public single coarse same-window	12 raw rows; 4 summary rows		Coarse-first evidence; no default 10^{-4} run or superiority claim.
Gauss6/FullVA closed-loop public mechanisms	12 rows	Selected and public-horizon residual trios for four-link/slider-crank; finest public residuals $5.136 \times 10^{-13} / 1.113 \times 10^{-14}$.	Residual evidence, not dynamic order/work superiority.
Closed-loop surrogate and floor analysis	2+2 rows	Selected-window residual-to-error ratios for four-link/slider-crank; no source-policy dynamic-order row is accepted; the floor analysis has two velocity/acceleration evidence rows and two position/reference-floor limitations.	Narrows the boundary; not external superiority.
2022 half-implicit pilot	24 rows	Bounded $T = 0.1$, $h = 0.02, 0.01, 0.005$ for rA/rA_{half} on four examples.	Source-family candidate; full $T = 8$ policy remains open.
Closed-loop coarse trajectory diagnostics	6 rows	Four-link primary-state orders $5.955 / 5.955 / 6.085 / 5.971$; slider-crank orders $6.164 / 6.159 / 7.341 / 6.426$ at $T = 0.1, h = 0.1, 0.05, 0.025$, reference $h = 0.0125$.	Local coarse trajectory evidence only; not accepted dynamic-order proof or public superiority.
Bounded common-reference work/precision diagnostic	24 rows	Local Gauss6/FullVA and public $rA/rp/r\epsilon$ rows use the same finite accepted-method endpoint reference; the fixed-reference diagnostic gap count is zero.	Figure 8; no external superiority claim.
Coarse-first readiness diagnostic	4 rows	Single and double pendulum are coarse same-window order/time ready; four-link and slider-crank have local coarse trajectory diagnostics and strict common-reference work/precision rows; the 10^{-4} policy is outside the default evidence package.	Readiness diagnostic, not a superiority claim.
Coverage matrix	48 rows	32 complete, 0 partial; no external-superiority claim.	Diagnostic coverage table, not a paper claim.

repository-facing reproducibility package contains the scripts needed to check the reported tables. The paper-level checks read existing result records unless a full numerical rebuild is explicitly requested; this keeps editorial checks separate from computational campaigns. Companion records encode the accepted claim scope and the non-claims, but the mathematical and numerical arguments in the paper are the authority.

The current reproducibility package is therefore two-tiered. The reviewer-facing minimal candidate is a 10-file package with one 158-line Python replay core; it is a shrink target for the reported-result records and proof-dependency records, not a complete runner-centered source-policy archive. The local accepted-row companion is human-runnable and self-contained for the four local examples and the 12 accepted local rows. A broader external source-policy runner archive would require the 40 source-policy rows, the TFE source-policy runner, and regeneration of both the accepted local rows and any promoted source-policy rows from a standalone runner. That archive is outside the current paper package.

Table B.61: External reproduction and package boundaries retained by the paper package.

Boundary	Current evidence state	Manuscript consequence
External same-test reproduction	External source-policy reproduction has not been established on the source paper tests, and no external rows are treated as source-policy closed.	No source-paper superiority or source-policy direct-error claim is made.
Original TFE runner package	A complete source-policy TFE runner package would need the source pendulum DAE runner, TFE/comparator source-policy method runners, Gauss6 source-policy runner, full-T10 endpoint policy, and accepted work-precision source rows.	TFE rows are formula/common-reference diagnostics, not full-TFE or source-policy runner evidence.
Runner archive boundary	The local accepted-row runner package covers the bounded replay/provenance role for the accepted rows; a full source-policy runner archive is outside this package.	The package supports accepted-row provenance while keeping full source-policy runner evidence outside the accepted theorem and comparison claims.

External reproduction/package boundary: the three rows in Table B.61 are scope boundaries, not theorem hypotheses. They do not change the conditional theorem proof; they keep source-policy reproduction, full-TFE runner evidence, and a full source-policy runner archive outside the accepted theorem and comparison claims. Notation boundary: theorem P6 denotes only the compact-tube solver-scale interface $\eta_h^{\text{tube}} \leq c_\eta h^7$, whereas the sepa-

rate original-TFE source-policy DAE-runner/package gap is an external reproduction boundary, not a theorem condition.

The implementation record identifies `residual_cylindrical_chain`, `R_VALUE`, and `R_JAC` as the implemented FullVA residual and Jacobian path. This supports independent reruns of the reported evidence; it is not a substitute for the proof.

Comparison reconciliation statement.

The finite-grid common-reference diagnostic is arithmetically complete, but the nonlocal cells remain diagnostic comparison records. These rows remain diagnostic scope records and are not promoted to a CMAME paper-level external-superiority claim.

The source package includes the command file for these read-only checks.

Appendix C. Supplementary source package

The Elsevier source package contains the editable `elsarticle` manuscript source, the compiled manuscript PDF, the separate highlights file, the declaration statements, and the thirteen figure files used in this paper. A supporting order-scope record separates accepted dynamic order from four-example mechanism coverage. A flat Editorial-Manager-oriented source folder, `cmame_submission_flat`, is also supplied. In that folder, the submission TeX source and all referenced figure files live at the same directory level, avoiding a dependency on the repository's figure subdirectory. The reproducibility package is therefore two-tiered: a compact 10-file package with one 158-line Python replay core records the bounded local replay, while the full development repository remains provenance for the broader development record. The local accepted-row companion is human-runnable and records 12 accepted local rows, but it is not a complete runner-centered source-policy archive. A full source-policy runner archive is not included; it would require closing the 40 source-policy rows under their source policies. The reviewer-facing runnable evidence for the narrowed current claim is the `cmame_narrowed_repro_bundle` directory. Its `run_narrowed_repro_bundle.py` entry point replays the paper result matrix and launches the four self-contained local accepted-row runners, while recording source-policy external-row closure and the full source-policy runner archive as open boundaries. Thus the local reproducibility bundle supports the present bounded claim boundary, but it is not a source-policy runner archive and does not promote external-superiority rows.

Table B.62: Algorithm-to-source map for the accepted Gauss6/FullVA execution path.

Algorithm block	Source location	Reproduction role
Mechanism setup and examples	<code>v047_cylindrical_chain_pipeline/run_v047.py</code>	Builds the cylindrical-chain data and the four ASME-style example rows used by the accepted method-side evidence.
Residual and Jacobian path	<code>residual_cylindrical_chain, R_VALUE, R_JAC</code>	Defines the 132-row FullVA residual and the single accepted residual/Jacobian path used by Newton.
One-step solve and trajectory loop	<code>gauss_step, integrate, run_case</code>	Implements the Algorithm 1 stage solve, endpoint reconstruction, endpoint closure, and step diagnostics.
Method result records	<code>v047_cylindrical_chain_pipeline/results</code>	Record the h-sweeps, endpoint closure diagnostics, four-example rows, and accepted method-side order/coverage evidence.
Cross-paper comparison records	<code>v048_cross_paper_same_test_benchmarks/results</code>	Store common-reference diagnostics and external-row status; these records do not authorize source-paper superiority claims.
Package-level checks	<code>validate_paper_package.py--latex</code> and <code>validate_pipeline_outputs.py</code>	Rebuild and check the manuscript package and repository inventory without invoking a new full numerical campaign.

Table B.63: Compact reproducibility contract used by the bounded paper package.

Field	Paper-level value
Accepted method	Gauss6/FullVA.
Conditional method order	Sixth-order path of Theorem 1.
Smooth observed orders	Position/velocity 7.161/7.066.
Local method-side coverage examples	single pendulum, double pendulum, four-link, and slider-crank.
Accepted dynamic order rows	Single and double pendulum.
Coverage-only for dynamic order	Four-link and slider-crank; their closed-loop kinematic/reaction rows are not accepted external dynamic-order rows.
Comparator	Local paper-style $m = 3$ Gauss–Lobatto TFE formula target.
Comparator expected order	5.
Numerical comparison scope	Forty-four method/example common-reference order-error cells, with external rows kept outside source-paper superiority.
Runner-package status	The local accepted-row companion is executable for the 12 accepted local rows; a full source-policy runner archive is outside the current package and would require separate source-policy closure and the original-TFE source-policy runner.
Non-claims	No full TFE replacement, no source-paper superiority, and no paper-level direct-error superiority against external methods.

Table B.64: Cross-example and review records retained with the source package.

Record	Role
All-example result matrix	Checks every one of the 44 method/example cells and separates the 15 nonlocal rows that are sensitive to external-row policy.
External-row table	Lists the row-level evidence still needed before any external row can support source-paper superiority.
Case reconciliation record	Separates full-campaign not-run markers from the bounded/coarse evidence already present for the RA2021 and HI2022 families.
Proof and package-boundary record	Keeps proof assumptions, dynamic-row closure requirements, open obligations, all-example numerical sanity, and current package boundary traceable while external reproduction items remain open.

Table B.65: Source-package reproducibility and consistency checks.

Check group	Role
Source-format check	Checks the Elsevier source, PDF, highlights, declarations, clean log, and non-claim scope.
Archive-completeness check	Checks the source archive, required figures, clean logs, and the rule that the full numerical generator is not started by editorial checks.
Order-scope check	Checks that single/double pendulum carry the accepted dynamic-order rows, four-link/slider-crank remain coverage-only for dynamic order, and 10^{-4} is not a default execution policy.
Numerical-matrix check	Checks the 44-cell numerical matrix against the common-reference and reconciliation records.
External-row check	Checks the separated nonlocal rows, zero claim-ready external rows, and the non-default- 10^{-4} execution rule.
Proof-dependency check	Checks proof-label dependency links, theorem-scope assumptions, and open symbolic/primitive-route obligations that are not active direct-PC2 obligations.
Scope-consistency review	Checks that the all-example matrix, proof-dependency record, and stated non-claims use the same claim boundary.
Package check	Runs the paper, source-archive, source-comparison, proof-matrix, four-example, and accepted-method package checks without starting a new full numerical campaign.
Repository-provenance check	Checks the broader repository inventory and records that the full numerical generator was not started by the source-package checks.

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